FERA ফেরা

ALL DOCUMENTS / TEAM EDITION · VERSION 2.0

Complete Team Handbook

Revised project specifications, exact code companions and a fillable meeting workbook

Specification handover · Implementation unverified

Authors

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Issued 11 October 2026 · Asia/Dhaka (UTC+6)

Invented synthetic-data prototype · Interactive team edition

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> Click a section or use the PDF bookmarks. Requirement IDs link to their definitions. External sources are clickable. The meeting workbook contains blank fillable review fields. Application acceptance and pilot evidence remain pending.

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TEAM HANDOVER / V2.0

Team Handover & Meeting Workbook

Start here: decisions, changes, verification and evidence gates · Authors: Sadman Shaharier and Obidit Islam

↗ Contents

Field	Value
Edition	2.0 · 11 October 2026 · Asia/Dhaka (UTC+6)
Authors	Sadman Shaharier; Obidit Islam
Team	Sadman Shaharier · Obidit Islam · Masnun Kabir
Purpose	Revised specifications, development decisions and review evidence for the next team meeting
Release boundary	Invented synthetic-data prototype; real-case pilot readiness is not established
Evidence status	Companion validation completed as described below; application, accounts, evaluations, interviews, actual task progress and spend UNVERIFIED

1 Start here

This package replaces the nine supplied Markdown specifications with V2 editions and updates their three code companions. Original files remain unchanged. Every V2 document names **Obidit Islam** as co-author alongside **Sadman Shaharier**. Existing document owners and team responsibilities are retained; authorship does not imply approval or completed implementation.

The complete handbook combines this workbook with all nine revised documents. The individual PDFs are convenient for discipline-specific review. Editable Markdown and exact code companions are included so the team can implement and maintain the same contracts. This is a specification handover; an application repository was not supplied and is not included.

Read this workbook first, then D1 for scope and rules, D2A for interfaces/schema/authority, D2B for privacy/evaluation/deployment, and D6 for the execution gates. Read D3 and D4 together when implementing screens. D5 is the ID/type index; D7 is the meeting and pitch closing pack. The registry document records assumptions and scope decisions.

2 Document and companion index

File	Team use	Principal V2 change
00_Gap_Check_and_Registry_V2.md	Establish assumptions and scope	Separate planned work from evidence; synthetic-only baseline; official-cutoff uncertainty
D1_PRD_V2.md	Product acceptance and business rules	Required fields, uncertainty, duplicates, role views, lifecycle/count semantics and pilot gates

File	Team use	Principal V2 change
D2_TRD_PartA_V2.md	API, database and authority	Draft metadata/token, scoped submission UUIDs, role DTOs, reliable matching state and atomic pair decisions
D2_TRD_PartB_V2.md	Security, privacy and quality	Full related-payload redaction, paid reservations, valid evaluation and one-database fallback
D3_App_Flow_V2.md	Screen behaviour	Manual completion, role-scoped reads, current-version decisions and explicit referral-recorded labels
D4_Design_Brief_V2.md	UI, Bangla copy and accessibility	Stronger border contrast, narrow officer layouts, shortcut controls and honest labels
D5_Background_Schema_V2.md	Shared types and ID lookup	537 unique IDs; corrected definition locations and revised TypeScript
D6_Implementation_Plan_V2.md	Task/test execution	Rebased gates; 64 tasks, 66 acceptance specifications; corrected dependencies and workload
D7_Closing_Pack_V2.md	Meeting and demonstration narrative	Honest evidence/status claims, timing labels and unresolved decisions
`code/schema_v2.sql`	Fresh demo database	17 tables; required-field, pair, history, redaction and cost controls
`code/fera_types_v2.ts`	Shared TypeScript contract	Distinct kind/state and role shapes; source spans, intervals, tokens and reservation types
`code/tokens_v2.css`	Design token source	Corrected light/dark border colours; retained token names

The three companions are embedded byte for byte in D2A section 7, D5 section 6 and D4 section 16 respectively. Edit the canonical companion and regenerate its embedded copy together. ID stability is preserved; new IDs are API-021–023, ENTITY-015–017, TC-038–066 and TASK-061–064.

3 What changed against the validation report

“Revised” below means the specification now contains the correction. It does not mean the application has passed its acceptance test. Each owner must attach repository/run/decision evidence before closing an implementation finding.

Finding	V2 correction and source	Owner	Remaining evidence
FV-01 · referral and closure	D1 BR-005/012/013; D2A API-010/011; D3; D7: confirmation records a simulated referral; closure records coordination only	Sadman + future safeguarding partner	UI/API transaction test; future identity, guardian, custody, recipient and acknowledgement procedure

Finding	V2 correction and source	Owner	Remaining evidence
FV-02 · phone dedupe	D1 BR-021/FR-022; D2A; TC-038: phone is a weak cue, reports remain visible; reviewed grouping preserves originals	Masnun	Shared-contact hard negatives; detector precision/recall; grouping audit if optional UI built
FV-03 · required/ unknown fields	D1 BR-015/016; D2A types/API; D3/D4: district compulsory; found age unknown allowed; reviewed event bounds	Obidit + Masnun	Runtime validators, manual completion, temporal/provenance negative tests
FV-04 · role permissions	D1 BR-007/024; D2A section 9; D5: owner, staff-own, officer and admin DTOs	Sadman	Every-route ownership/verification tests; no admin rematch candidate/ score leakage
FV-05 · idempotency/ counters	D2A API-001/002/003/007; schema/types: separate opaque token, authoritative draft counter, scoped UUID/hash	Sadman + Obidit	HTTP concurrent replay/conflict test; principal binding and revocation; actual provider dispatch accounting
FV-06 · database invariants	D2A section 7; schema: initial states, required fields, current versions, decision/ referral/audit coupling and immutable history	Sadman	Target PostgreSQL 16 CI, restricted runtime grants, sorted locks and concurrent application decisions
FV-07 · redaction	D1 BR-009; D2A API-016; D2B section 11; schema: direct and related payloads cleared; retained records pseudonymous	Sadman	Application/admin authority, duplicate-note cleanup, provider/export/backup policy; active linked evidence retained
FV-08 · split demonstration DB	D2B deployment; D6/D7: one database per whole journey; isolated local restart on fallback	Sadman	Cloud and offline rehearsals across separate role profiles; tested LAN HTTPS before phone microphone use
FV-09 · matching uncertainty	D1 BR-017/018; D2A adapter; D2B evaluation: national exact baseline, soft geography, unknowns and abstention	Masnun	Frozen embedding space and dev thresholds, hard negatives, no-match results and deployment spike
FV-10 · counts	D1 BR-014/FR-019; D5 DTOs; D6 demo: missing/found rows and confirmed pairs separated	Masnun	One-pair lifecycle count assertions across API, map and table
FV-11 · evaluation	D2B section 17; TASK-064; TC-056/ 057/063: identity-grouped manifest, exact denominators, cache exclusion and failure reporting	Obidit + Masnun	Manifest/hash, frozen run configuration, held-out report and baseline comparisons
FV-12 · provider processing	D1/D3/D4 notices; D2B section 11: synthetic audio/text processed before submission confirmation	Obidit	Actual account/model/endpoint terms, primary and fallback settings; lawful real-use review later
FV-13 · arithmetic/ schedule	D2B section 12; D6; this workbook: USD 4.87 input-equivalent ceiling; 184 h Round 2 and 58 h dependency lower bound	Obidit + Sadman	Actual prices/spend, completed-work inventory, owner availability and revised resource schedule

Finding	V2 correction and source	Owner	Remaining evidence
FV-14 · pitch claims	D1/D5/D7: dated reporting cohort, existing returned/rescued outcomes, coordination need as hypothesis; no partnership/scale guarantee	Sadman	Source/selection-email review; real user/agency discovery; measured product results
FV-15 · deployment fit	D2A adapters; D2B security/deploy; TC-058/059/060/065: pin model/assets, codec tests, CSP/tile rules	Masnun + Obidit	Exact hosting tier/package/runtime test; audio bounds; all offline assets bundled; no bulk OSM caching
FV-16 · screens/ accessibility	D2A API-021/022/023; D3; D4 and tokens; TC-061/062: screen paths and accessible reflow/shortcuts	Sadman	Rendered axe/manual keyboard/zoom/ screen-reader checks; admin list and district drill-through implementation

4 Contracts to freeze at the meeting

4.1 Submission and processing

Missing reports require a complete child age/range, sex selection with unknown allowed, district, non-empty last-seen place, reviewed event interval and reporter contact. Found reports require description, district/place/time interval and staff contact; age may be unknown. Names remain optional. Four follow-up rounds are a conversational limit, not permission to guess remaining required facts; use the accessible manual form.

Draft payload remains on the device before report confirmation. Audio/text can reach declared providers earlier after the processing notice. Server-side draft records contain only principal, expiry, hash, revision and paid-call count. The opaque token is outside CaseDraft. The server checks ownership, expiry and revocation; retries never reset paid-call usage.

Submission uses a client-generated UUID scoped to the authenticated principal/session. The same UUID and canonical reviewed payload returns the same case; different values conflict. A new UUID remains a separate report, even if text/phone matches. Staff and parent submissions both use this contract. Do not turn a text hash or contact phone into a child identity.

4.2 Matching and decisions

Matching begins with national eligible exact retrieval for approximately 200 records. Geography is a soft feature. Unknown found age and overlapping time uncertainty remain visible as uncertainty. Pin 384-dimensional embedding artifact revision, dtype, pooling, L2 normalisation and symmetric query-prefix policy. Qualification and duplicate thresholds are development hypotheses until calibrated and frozen; zero suggestions is a valid result. Similarity scores never mean probability of identity.

Confirmation is one transaction with sorted endpoint locks, current version/role checks, reviewer decision, candidate state, both report states, simulated referral and safe match.confirm audit. Competing pending pairs are voided and all affected reports are reconciled. Rejection records a reason and match.reject audit. Pending candidates cannot survive a report-version change. Reliable queued/running/ complete/failed matching state survives request failures; detached serverless promises are insufficient.

Ordinary terminal edits are locked. A referral row does not prove delivery, recipient acceptance, guardian identity, custody or safe return. Before a real-case pilot, define an audited correction/reopen procedure with two distinct authorised reviewers and preserved superseded history. An isolated synthetic reset is a demo operation, not that correction procedure.

4.3 Authority and privacy

Parents see their own editable report before lock, their own status and available outputs. Staff sees own found entries. Active verified officers see operational evidence/candidates. Admins see aggregate/ metadata views and rematch status only. All reads/actions require server checks and DTO allowlists; debug mode must preserve these boundaries.

Redaction clears names/Latin names, exact age, contact, source URL, source/provenance/hash/vector metadata, notes, snapshots and generated output, including related decision/referral/candidate/duplicate text. IDs, dates, coarse age, district, sex and lifecycle/audit links remain restricted pseudonymous data. Found evidence is retained until the linked missing report closes. Browser exports, provider copies and backups require separate treatment; lazy browser expiry does not guarantee deletion while the browser is closed.

V2 blocks rows marked non-synthetic; a true flag cannot detect real information disguised as synthetic. Keep every fixture invented and prevent real input at the application boundary. Do not enable real public-child news imports. Fixed templates and a review checkbox do not establish legal guardianship or lawful processing.

5 Verification performed for this handover

Check	Result	Practical limit
Revised source coverage	Nine Markdown documents and three companions, plus this workbook	Independent upstream pitch/source PRD and application repository were not supplied
Registry	537 rows / 537 unique IDs; category counts checked	IDs index requirements, not implementation completion
Embedded companions	All three exact matches; SHA-256 hashes recorded	Regenerate both copies after later edits
TypeScript	Strict compile using TypeScript 5.4.5 passed	Standalone contract file only; no runtime JSON/API validation
SQL execution	Fresh schema applied in PostgreSQL 18.3 via PGlite 0.5.8; pgvector 0.8.1	Target PostgreSQL 16 and actual managed-host/ runtime privileges remain untested
SQL behaviours	48 targeted positive/negative checks passed	Sequential companion checks; not network concurrency, API authorisation or a deployed app audit
SQL coverage examples	Required fields, shared-phone reports, unknown found age, endpoint roles, audit-coupled decisions, immutable history, redaction, UUID uniqueness and atomic cost limits	The server still owns principal checks, lock order, safe audit content and provider settlement evidence

Check	Result	Practical limit
Token contrast	V2 border contrast exceeds 3:1 against all three light and dark surfaces	Full component, focus, keyboard, zoom/reflow and assistive-technology validation pending
PDF delivery	Searchable text, rendered diagrams, contents/ bookmarks/links; workbook form fields	Complete and save forms in a compatible PDF viewer; form completion does not alter evidence status automatically

See `verification/schema_checks.json`, `document_checks.json`, `contrast_checks.json` and `pdf_checks.json` for reproducible scope/results. The portable schema check runner and pinned package versions are included. These are companion-file checks; do not present them as 48 application or safeguarding tests passed.

6 Corrected capacity and dependency calculation

The task board contains 64 tasks and 222 estimated hours overall. The Round 2 selection is TASK-001–046, TASK-059/060 and TASK-061–064: **52 tasks, 184 hours**. This adds 12 hours of V2 remediation to the original 172-hour selection. Owner estimates are **Sadman 67 h, Obidit 54 h, Masnun 63 h**.

The declared dependency graph is acyclic. Its longest Round 2 path is **58 hours**:

Sequence	Tasks	Hours
Contract and case foundation	TASK-001 → 061 → 062 → 015	16
Matching and decision UI	TASK-017 → 027 → 028 → 029 → 030	26
Verification and submission	TASK-038 → 042 → 060 → 043 → 044 → 046	16

This is a dependency lower bound. It assumes unlimited independent resources and continuous availability; owner workloads, feedback, model downloads and review queues can make elapsed delivery longer. The original printed path totalled 67 hours and included four undeclared edges, so its approximately 59-hour claim was invalid.

If no work were completed at the start of 11 October, four full days at the assumed ten hours per person per day would offer at most **120 team-hours**, before meetings, interruptions and submission buffer. That assumption is not verified capacity. First inventory actual completed work and remaining estimates; compare each owner's remaining load with actual availability. Reassign or reduce optional scope and state any omitted Must behaviour honestly. Keep the typed/manual journey, review, role enforcement, decision/audit integrity and truthful evaluation ahead of optional import, disaster UI, summary or duplicate-grouping UI.

Actual provider prices and spend remain unknown. The old extraction allowance used 1.32M input + 0.44M output tokens with output weighted four times: 3.08M input-equivalent tokens. USD 15 / 3.08 = **USD 4.87/M**, not USD 5.70/M. This is an allowance calculation, not a provider price quote. Record actual input/output/audio rates and maximum-call reservations. Default cap USD 25; a documented team budget decision may raise it within the USD 30 ceiling. Uncertain paid calls are not silently refunded.

7 Next-meeting agenda and decisions

Minutes	Discussion	Required outcome	Lead
0–10	Actual repo/deploy progress, unfinished tasks and availability	Evidence-backed remaining-work inventory and owner capacity	Sadman
10–25	V2 required fields, token/UUID, role DTOs, decision/referral/count semantics	Agreed canonical contracts; unresolved items get owner/date	All
25–35	Provider/audio/embedding/hosting spike and costs	Tested primary/fallback configuration or explicit manual/demo limitation	Obidit + Masnun
35–45	Evaluation manifest, grouped split, baselines and no-match tests	Frozen development/test boundary and reporting definitions	Masnun + Obidit
45–55	One-database demo, official cutoff/ channel/deliverables and claims	Rehearsal topology and submission checklist with verified organiser instructions	Sadman
55–60	Priorities, handover acknowledgement and next evidence check	Owner/date per unresolved action; no invented completion	All

Bring the actual repository/build link, current board, remaining owner hours, provider account configuration and price sheet, organiser selection email, manifest/ground truth if written, test logs, spend log, and any consented interview notes. Retrieve the original pitch/source PRD if their inherited assumptions remain authoritative.

Use confirmed / changed / unknown for decisions. Do not replace missing answers with “nothing built,” “zero spend,” “no interviews” or “no agency outcomes” without evidence. Record who decided, when, what evidence was examined and the next review date.

8 Evaluation and demonstration handover

Use a canonical manifest of 80 missing + 80 found + 40 additional decoys = 200 records, 60 canonical true pairs, 20 canonical no-match missing queries and 25 duplicate links within that same total. Keep identity-connected and duplicate variants in one split. The intended 40% dev/60% test split may not divide exactly by count; report actual grouped counts. Do not tune on held-out examples.

Freeze model/revision, preprocessing, prefixes, score weights, thresholds, prompt/template and manifest hashes before final evaluation. Force fixtures/cache off for quality runs. Report present-field errors and false additions separately from null agreement, conditional top-1 precision with coverage, Recall@5, clearly defined macro P@5 and aggregate shortlist precision, no-match false suggestions/abstention, duplicate precision/recall/F1, speech WER and critical-field errors. Distinguish gold-field from extracted/ speech-field matching and include baselines/failures. Repeated clips/runs are not new independent samples. Five volunteer timing sessions are exploratory; a 90% completion threshold with five participants needs five completions.

All online roles use one cloud database. All offline roles use one local database in separate browser profiles. Bundle fonts, model, fixture data and district geometry. Use a laptop for the offline microphone baseline; a LAN phone needs tested HTTPS/secure-context access. If cloud fails, restart a fresh local synthetic scenario and disclose the restart; the cloud report does not silently transfer. Optional OSM tiles require current policy/attribution/Referer compliance and cannot be bulk cached for offline use.

Label synthetic data, seeded officer verification, fixture/cache responses, simulated referral recording and compressed video time. A 38-second demonstration containing a longer voice note is edited/time-compressed, not proof of that end-to-end latency. Do not claim CID/MUN integration, real recipient acknowledgement, child return, nationally validated efficacy or results not yet measured.

9 Applying and maintaining the package

Apply `code/schema_v2.sql` to a **fresh isolated synthetic database** with the target pgvector extension installed. It is not a repeatable migration or a V1 upgrade script. Existing/live data needs backup, a reviewed versioned migration, explicit enum/data mapping and rollback rehearsal. Create missing reports initially as submitted then open; construct final fixture states through complete audited lifecycle transactions.

Use separate migration-owner, runtime and restricted redaction credentials. The runtime must not own tables, alter schema, disable triggers or truncate audit. PUBLIC function revocations in the schema are not a deployed privilege configuration; an operator must grant the exact service-role privileges and test them. Passing an admin UUID to redact_case does not authenticate an HTTP caller: server authority checks remain mandatory. Decision/cost helper functions execute inside the caller's transaction and do not replace request-level enforcement.

Keep the companion types at service boundaries and implement strict runtime validators. Convert database numeric/bigint/timestamp/vector values deliberately. Pin tested dependency/model artifacts and commit the real application lockfile. This specification does not bundle a Next.js app or define runnable npm application scripts; D2B/D6 commands are implementation targets.

After a contract edit: update the canonical companion/requirement, all affected API/flow/UI/test rows, D5 registry and companion embeds; rerun scope-appropriate checks; version the evidence and PDF exports. Do not mark TC-001–066 complete merely because the acceptance specifications exist.

10 Sources and evidence boundaries

Public sources below were verified in the preceding validation review. The V2 specifications preserve source distinctions: official sources for rules/technical contracts, reported police figures for scale, and hypotheses for unvalidated workflow need. Missing independent upstream project sources are explicitly marked.

Source	Supported use	Link
Official event page	Track 3, 14 Oct submission date, 24 Oct finale, three-member limit and 5–10-minute finale presentation; exact submission time/ channel/slide/video rules still require selection email	Hack for Humanity

Source	Supported use	Link
Daily Star report, 3 Sep 2026	Jan–Jul reporting cohort: 15,717 reported, 2,038 untraced, 13,679 reported returned/rescued; approximately 13% untraced and 74 reports/day; not a live October count	Police-figure reporting
BSS CID briefing report	Missing Children Cell / MUN authority and 13219; no Fera endorsement or integration	CID briefing
Official Personal Data Protection Act, 2026	Act identity, phased commencement and need for qualified child-data/transfer assessment; no legal compliance conclusion	Official law
Gemini terms; OpenAI data controls	Exact account/endpoint training, retention and human-review terms require review	Gemini · OpenAI
E5 model card; pgvector	384 dimensions/prefixes/normalisation and filtered retrieval behaviour	E5 · pgvector
PostgreSQL constraints	NULL/CHECK behaviour and integrity design	PostgreSQL 16
Hosting/model/speech documentation	Verify actual tier limits, local assets and tested audio formats	Vercel · Transformers.js · Speech
W3C accessibility guidance	Non-text contrast, character shortcuts and reflow tests	Contrast · Shortcuts · Reflow
OSM tile policy	Optional online attribution/Referer; no offline bulk download	Tile policy

The dated cohort disproves a blanket claim that existing agencies record no returned/rescued outcomes. Whether operational tools support Fera's proposed linked coordination workflow remains an interview/partner-discovery hypothesis. Synthetic model results cannot establish safe child identity matching or national effectiveness.

11 Meeting decision worksheet

The PDF edition provides fillable review checkboxes and meeting notes. All fields are blank/unchecked initially; they represent team decisions entered by the reviewer, not pre-approved gates. Use a compatible PDF viewer and save a completed copy with the meeting date. The editable Markdown remains the canonical specification.

Record date/attendees, progress evidence, agreed scope and unresolved actions. Confirm contract freeze, provider/deployment evidence, grouped evaluation manifest, role/redaction tests, one-database rehearsal, honest metrics/claims and official submission instructions. If evidence is missing, leave the corresponding item unchecked and record an owner/date. Real-case use needs the separate legal, safeguarding and partner-operating gate in D2B; this workbook does not approve a pilot.

Review record

Blank initially · Complete in a compatible PDF viewer · Save a meeting copy

Meeting date (YYYY-MM-DD)

Attendees

Progress evidence / repository / report references

- ☐ Required fields, tokens, UUIDs and role DTOs agreed
- ☐ Decision, referral-recorded and count semantics agreed
- ☐ Actual progress and owner availability evidenced
- ☐ Provider/audio/model/hosting configuration tested
- ☐ Grouped manifest and held-out reporting rules frozen
- ☐ Role, redaction and restricted-privilege evidence reviewed
- ☐ One-database online/offline rehearsal completed
- ☐ Metrics, cache and timing claims reviewed
- ☐ Official cutoff, channel and deliverables confirmed

Agreed scope and decisions

Open actions: issue · owner · due date

Recorded by / next review date

Use Latin characters in the editable fields for consistent cross-viewer rendering. Bangla specifications remain searchable and fully rendered. Checking a box records the reviewer's decision; it does not certify application safety or approve real-case use.

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00 / BASELINE / V2.0

Gap Check & Registry

Scope decisions, assumptions and the V2 baseline · Authors: Sadman Shaharier and Obidit Islam

↗ Contents

Field	Value
Version	2.0
Date	11 Oct 2026 (Asia/Dhaka, UTC+6)
Owner	Sadman Shaharier (team lead)
Authors	Sadman Shaharier; Obidit Islam
Status	V2 team handover; implementation status UNVERIFIED
Sources read	S1a `fera_pitch_deck.pdf` (16 slides, Round 1); S1b Claude Doc "Fera — Product Requirements Document (v2)", rev 18, 7 Oct 2026; S2 PROJECT CONTEXT block

Source rank: S1a = S1b > S2 > assumptions. Where S1a and S1b disagree, the newer S1b wins.

V2 release contract

Authors: Sadman Shaharier and Obidit Islam. **Version:** 2.0. **Issued:** 11 October 2026, Asia/Dhaka (UTC+6).

This is the revised team specification, not evidence that the application has been built. Actual repository progress, account configuration, interviews, evaluations, and spend remain **UNVERIFIED** until evidence is attached. V1 defaults are superseded by the V2 rules below; existing requirement IDs are retained.

- This release is a **synthetic-data prototype**. Officer verification and referral delivery are simulated and labelled. The schema rejects records marked non-synthetic; the application must also prevent real reports from entering this release.
- A candidate score is similarity, not identity probability. A reviewer decision does not establish guardian authority or safe reunification. A referral row means **referral recorded**, not delivered or acknowledged.
- Draft payload stays on the device before report confirmation, but providers process audio/text beforehand with a notice. Only expiry/counter/hash metadata and cost reservations are stored server-side before submission.
- Possible duplicates remain visible. Shared phone alone never establishes one child. No physical child-return action is performed by Fera.
- Counts distinguish missing-report rows, found-report rows, and confirmed pairs. They are not a census of unique children or proof of rescue.

- Sources, diagrams, tables, code blocks, and acceptance criteria are part of the handover. The companion code must match the embedded code exactly. Team handover records change closure, evidence gaps, and release gates.

(a) Clarification questions

Only questions whose answers change scope, architecture or the demo. Each has a default. V2 working choices are specified below. Record team approval and evidence; an unanswered account, partner, or organiser question remains pending.

OQ	Question	Why it matters	Default if unanswered	Blocks
OQ-001	Are **voice intake** and **poster + 13219 summary** Must for Round 2?	PRD v2 marks both Should (v2 FR7, FR8). PROJECT CONTEXT lists "Bangla intake → poster + GD" as built and the D6 journey starts with a voice note.	Voice = **Must**, with typed text as the guaranteed path and a cached transcript for the scripted demo note. Poster = **Must** (text poster, no photo). 13219 summary = **Should**.	D1 scope, D6 critical path
OQ-002	Adopt the separate missing/found state machine from PROJECT CONTEXT (draft → submitted → open → matched-pending-review → confirmed → closed/referred)?	PRD v2 uses open / matched / closed / reopened and closes both cases on confirm. Changes BRs, schema enums, map legend.	Use browser draft plus five persisted missing-case states (submitted/open/matched_pending_review/confirmed/closed). Found cases use open → matched_pending_review → referred; the last value means referral recorded only. PRD v2's "reopened" maps to "open".	D1 §13, D2 schema
OQ-003	What did mentors say in session 1 (7–8 Oct) about the environment theme and disaster mode?	PRD v2 rates "judges favour environmental projects" Medium/High risk. The answer decides whether disaster mode gets a demo screen.	Disaster mode stays **Could**: one static screen reusing the found-child form, labelled "planned extension". No need claims.	D1 scope, D6, D7 script
OQ-004	Which accounts and API keys does the team already hold (LLM providers, OpenAI for Whisper, Postgres host, Vercel)? Any free credits?	Sets the default LLM provider, the embedding source and the USD 30 budget estimate.	Provider/account/model/price is PENDING. Only synthetic test data may reach APIs; enforce terms, atomic reservations and actual prices before paid dispatch.	D2 stack, cost model
OQ-005	Finale demo topology: deployed app over venue Wi-Fi, or laptop-local (localhost DB, phone on laptop hotspot)?	Venue internet may fail. Local-first changes DB hosting, embeddings and the fallback design.	Deployed app for Round 2 video. **Single-backend** local stack for the finale fallback. All roles share one local DB; preloaded assets and labelled fixtures support the scripted scenario. Restart the local scenario when switching from cloud.	D2 §3, §14, §15; D6 phase 6

OQ	Question	Why it matters	Default if unanswered	Blocks
OQ-006	What exists today (9 Oct)? Repo skeleton, synthetic set, 50 labelled reports, any user conversations held?	PRD v2 planned build days 7–11 Oct; 5 days remain. The deck also promises 5–10 user conversations.	Actual build and interviews are UNVERIFIED. Inventory the repository and completed tasks on 11 Oct; do not restart completed work or claim interviews without evidence.	D6 phases, D7 risks
OQ-007	Demo auth and roles: real login or a role switcher? Keep PRD v2's extra "helper" role?	PRD v2 roles: parent, helper, staff, officer, admin. PROJECT CONTEXT: parent, officer, shelter staff, admin.	Seeded accounts with protected seeded synthetic login; officer verification **[MOCK]** (seeded `verified=true`). Drop "helper"; a helper files as a parent with relation = "other".	D2 §9, D3 auth flow

Non-blocking open questions (logged now, defaults applied)

OQ	Topic	Conflict or gap	Default applied
OQ-008	Evidence scope	PROJECT CONTEXT says use only 5 figures. Deck and PRD v2 also cite age 0–9 split, MUN Alert "200+ cases / 29 rescued", Teknaf "10 victims".	Use only the 5 PROJECT CONTEXT figures, cited to the originating report; original pitch/ PRD are not supplied.
OQ-009	Accessibility level	PRD v2 targets WCAG 2.1 AA; D4 spec asks 2.2 AA.	Target 2.2 AA. It includes all 2.1 AA criteria, so PRD v2 is still met.
OQ-010	Alert authority wording	Deck slide 4 says only CID's Missing Children Cell can issue alerts, and also that a proposed Cell "never materialised".	Write "CID (via MUN Alert) issues or withdraws alerts". Ask a mentor or CID contact for the correct unit name.
OQ-011	Personal Data Protection Act (2026)	PRD v2 says "passed April 2026". No clauses are in any source.	Cite the Act by name only. Accurately sourced legal references are allowed; unsupported compliance claims are prohibited. Qualified review is required before a pilot.
OQ-012	Retention period for closed cases	PRD v2: "to be set with a partner". Safeguarding requires delete/redact identifying payloads.	Demo: redact identifying payloads on an admin action. Period = UNKNOWN.
OQ-013	LLM/STT provider data terms	PRD v2: "confirm per provider" no-training and retention terms.	Synthetic data only reaches providers before terms are checked.
OQ-014	Persona count	PRD v2 has 4 personas incl. child-protection officer; D1 spec asks for 3.	3 primary personas in full; child-protection caseworker kept as secondary PERSONA-004 (no UI in MVP).
OQ-015	Dedupe priority	PRD v2 puts duplicate merge at Could (v2 FR10). Deck promises a duplicate-detection quality metric in Round 2.	Duplicate **detection + offline metric** = Must. Merge **UI** = Could.

(b) Seed registry

GOALs

Targets are provisional working targets. Tune on development data only, then freeze thresholds/prompts/targets before the held-out test. Synthetic performance does not establish safe real-case identity matching.

ID	Goal	Metric	Target (source)	Set / method
GOAL-001	A parent produces a reviewed Bangla report in the first hour	Time to complete report, assistant vs manual form	≤ 5 min and faster than manual (PRD v2)	Timed tests, ≥ 5 users
GOAL-002	AI extracts case fields correctly from messy Bangla	Field-level extraction accuracy	≥ 85 % on required fields (PRD v2)	50 hand-labelled reports
GOAL-003	The right found-child record appears in the officer's shortlist	Recall@5; precision@1; precision@5 reported	≥ 80 % / ≥ 60 % (PRD v2); P@5 as measured (S2)	Synthetic ~200-record set with ground truth
GOAL-004	Duplicate reports of one child are detected	Duplicate-detection pairwise F1	≥ 0.8 (PRD v2)	Synthetic set with seeded duplicates
GOAL-005	Bangla voice is usable for intake	Word error rate	≤ 30 % (PRD v2)	Recorded test prompts
GOAL-006	Every demo report has an explicit recorded coordination status, visible by district	Seeded cases shown with correct status	100 % (PRD v2)	DB count vs map count
GOAL-007	The live demo does not fail	Rehearsals passing end to end	5 of 5, at least 1 with internet off (internet-off part proposed)	Rehearsal log
GOAL-008	Humans make every match decision	Match-driven status changes with a recorded authorised decision and append-only audit	100 % (PRD v2 FR5)	Audit-log check script
GOAL-009	Results are reported honestly	Metrics slide with failure cases	1 slide, all 5 metrics, ≥ 2 failure examples (PRD v2)	Deck review before the organiser-confirmed 14 Oct cutoff

Legacy map: v2 G1 → GOAL-001, G2 → GOAL-003, G3 → GOAL-006, G4 → GOAL-009, G5 → GOAL-007. v2 secondary goals → GOAL-004, GOAL-005. v2 US/FR/A IDs are remapped in D1.

PERSONAs

Names are fictional, built from published reporting, not interviews (PRD v2 A5).

ID	Persona	One line	Priority
PERSONA-001	Parent/guardian ("Shirin")	Dhaka mother, 30s; son, 13, did not return from madrasa; Bangla-first, weak formal writing, Android browser.	Must
PERSONA-002	Verifying officer ("SI Karim")	Thana duty officer, several missing-child GDs a week; reviews matches on a desktop.	Must
PERSONA-003	Shelter/station staff ("Rumana")	NGO volunteer at a railway-station children's desk; logs found children on a phone.	Must
PERSONA-004	Child-protection caseworker	Intended future referral recipient; V2 records a simulated referral only; no recipient UI or delivery integration.	Should (record only)
PERSONA-005	Disaster-mode shelter volunteer	Logs separated children after floods/cyclones; planned extension, no measured need.	Could (per OQ-003)

MVP feature list

"Demo" = the moment the feature appears in the Round 2 video or finale. Features become FR-IDs in D1.

#	Feature	MoSCoW	Planned implementation / simulation	Why (demo moment or safeguarding)
1	Bangla/mixed/Bangla-numeral **text** intake → case draft + one-at-a-time follow-up questions	Must	Planned; not verified	Demo: parent describes child; draft appears
2	Bangla **voice** intake (STT → transcript for review)	Must*	Planned + cached demo transcript	Demo hero moment; *pending OQ-001
3	Parent review/edit; no submitted case, candidate sharing or export before confirm; provider processing is disclosed earlier	Must	Planned; not verified	Safeguarding: parent reviews before sharing
4	Bangla GD draft from fixed template (LLM may not add facts)	Must	Planned; not verified	Demo: copyable GD text
5	Shareable text poster (no child photo)	Must*	Planned; not verified	Demo: poster; *pending OQ-001
6	13219-ready summary (< 80 words)	Should	Planned if time	Demo add-on; Fera never calls or alerts
7	Found-child logging by staff (form or free text)	Must	Planned; not verified	Demo: station notice enters system

#	Feature	MoSCoW	Planned implementation / simulation	Why (demo moment or safeguarding)
8	Matching: hard filters + embeddings + rules → top-5 with one-line reasons	Must	Planned; not verified	Demo: likely match surfaced
9	Officer review queue, side-by-side evidence, confirm/reject	Must	Planned; not verified	Safeguarding: human confirms every match
10	Simulated referral record on confirm; never imply delivery or auto-return	Must	Planned; not verified	Safeguarding: child-protection referral
11	District map with open/matched/closed counts (icon + text + colour)	Must	Planned; not verified	Demo: case closes on map
12	Role-based access; candidate/other-party matched evidence visible only to verified officers	Must	Planned; verification [MOCK]	Safeguarding: stalker-misuse defence
13	Append-only audit log of match decisions and status changes	Must	Planned; not verified	Safeguarding: auditable decisions
14	Redact identifying payloads from closed/eligible referral-recorded cases (admin action)	Must	Planned (manual trigger)	Safeguarding: delete/redact identifying payloads closed cases
15	Duplicate detection + offline duplicate-detection metric	Must	Planned; not verified	Deck promises the metric in Round 2
16	Evaluation scripts for all 5 metrics	Must	Planned; not verified	Deck: "report as measured, including failures"
17	Demo-safe fallback: cached STT/LLM outputs for scripted inputs; manual form on AI failure	Must	Planned; not verified	Venue internet unreliable
18	Synthetic data seed loader (~200 records, labelled "synthetic")	Must	Planned; not verified	Demo data; no real child data
19	Import invented public-format fixture texts with source metadata	Should	Planned if time	Feeds matching beyond direct reports
20	Duplicate merge UI	Could	—	Metric covered by #15
21	Dashboard filters (age band, date range)	Could	—	Not needed for the core journey
22	Parent case-status view (status only, no match details)	Could	—	Nice to show "being checked"
23	Disaster-mode screen	Could	Static	Per OQ-003
24	Live police data	Won't (build)	[MOCK] = labelled synthetic seed	No data access before a partner agreement
25	Telecom location targeting	Won't (build)	[MOCK] = slide only	Out of scope; no partner

#	Feature	MoSCoW	Planned implementation / simulation	Why (demo moment or safeguarding)
26	Face matching	Won't (build)	[MOCK] = slide only, "future, CID-held records, human review"	No public face search, ever
27	Issuing, withdrawing or broadcasting alerts	Won't (ever)	—	Only CID can (OQ-010)
28	"Runaway" labels, predicting who goes missing	Won't (ever)	—	Safeguarding
29	Native app, SMS/IVR intake	Won't	—	Post-hackathon roadmap

(c) Turn plan

Turn	Unit	Files	Notes
0	Gap check + registry	`00_Gap_Check_and_Registry_V2.md`	This file
1	D1 PRD	`D1_PRD_V2.md`	Applies answers or defaults for OQ-001 – OQ-007
2	D2 TRD Part A (§1–9)	`D2_TRD_PartA_V2.md`	Stack, architecture, components, API, data model
3	D2 TRD Part B (§10–19)	`D2_TRD_PartB_V2.md`, `schema_v2.sql`	Security, privacy, budget estimate, evaluation design
4	D3 App Flow	`D3_App_Flow_V2.md`	All 14 mandatory edge cases
5	D4 Design Brief	`D4_Design_Brief_V2.md`, `tokens_v2.css`	Bangla + English strings
6	D5 Background Schema	`D5_Background_Schema_V2.md`, `fera_types_v2.ts`	May split to Part A/B if the registry overflows
7	D6 Implementation Plan	`D6_Implementation_Plan_V2.md`	Rebased from verified progress on 11 Oct; Round 2 date 14 Oct; exact cutoff pending organiser confirmation
8	D7 Closing Pack	`D7_Closing_Pack_V2.md`	Priorities, risks, timed demo scripts, OQ owners

Time note: specs alone take 9 turns. Use the V2 contract gate before dependent coding. D6 requires actual remaining-work and availability evidence on 11 Oct. The previous nine-turn drafting sequence is historical, not the current delivery schedule.

IDs introduced

- GOAL-001 – GOAL-009 (9)
- PERSONA-001 – PERSONA-005 (5)
- OQ-001 – OQ-015 (15)

IDs referenced

- Legacy PRD v2 IDs (G1–G5, US1–US9, FR1–FR11, A1–A5), cited as "v2 ..." only; remapped in D1.

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D1 / PRODUCT / V2.0

Product Requirements

Product scope, business rules and acceptance requirements · Authors: Sadman Shaharier and Obidit Islam

↗ Contents

Field	Value
Version	2.0
Date	11 Oct 2026 (Asia/Dhaka, UTC+6)
Owner	Sadman Shaharier (team lead)
Authors	Sadman Shaharier; Obidit Islam
Status	V2 team handover; implementation status UNVERIFIED
Sources	Validation report; original D1 source; V2 registry and handover. Original pitch deck/PRD rev 18 were not supplied independently; inherited claims require source checks.

V2 release contract

Authors: Sadman Shaharier and Obidit Islam. **Version:** 2.0. **Issued:** 11 October 2026, Asia/Dhaka (UTC+6).

This is the revised team specification, not evidence that the application has been built. Actual repository progress, account configuration, interviews, evaluations, and spend remain **UNVERIFIED** until evidence is attached. V1 defaults are superseded by the V2 rules below; existing requirement IDs are retained.

- This release is a **synthetic-data prototype**. Officer verification and referral delivery are simulated and labelled. The schema rejects records marked non-synthetic; the application must also prevent real reports from entering this release.
- A candidate score is similarity, not identity probability. A reviewer decision does not establish guardian authority or safe reunification. A referral row means **referral recorded**, not delivered or acknowledged.
- Draft payload stays on the device before report confirmation, but providers process audio/text beforehand with a notice. Only expiry/counter/hash metadata and cost reservations are stored server-side before submission.
- Possible duplicates remain visible. Shared phone alone never establishes one child. No physical child-return action is performed by Fera.
- Counts distinguish missing-report rows, found-report rows, and confirmed pairs. They are not a census of unique children or proof of rescue.
- Sources, diagrams, tables, code blocks, and acceptance criteria are part of the handover. The companion code must match the embedded code exactly. Team handover records change closure, evidence gaps, and release gates.

Changes to earlier documents

V2 revises required fields, duplicate handling, candidate eligibility, draft/submit controls, referral/count semantics, privacy and release assumptions. Original IDs remain stable; the V2 release contract supersedes Turn 0 defaults. See the handover change map for FV-01–FV-16.

1 Document control

Version	Date	Author	Change
1.0	9 Oct 2026	Sadman Shaharier	First issue. Remaps PRD v2 (G1–G5, US1–US9, FR1–FR11) into the suite ID registry.
1.1	9 Oct 2026	Sadman Shaharier	SCREEN-006 priority Could → Should, so the Should story STORY-019 has a screen of equal priority (raised in D3).
2.0	11 Oct 2026	Sadman Shaharier; Obidit Islam	Validation-led contract, safety, evaluation and handover revision

Legacy map: v2 US1 → STORY-002; US2 → STORY-004; US3 → STORY-005; US4 → STORY-008; US5 → STORY-009/010/011; US6 → STORY-013; US7 → STORY-001; US8 → STORY-006/007; US9 → STORY-015. v2 FR1–FR11 → see §11 notes.

2 Executive summary

Item	Summary
What	A Bangla-first web service that turns a parent's voice or text report into a structured missing-child case, matches it against found-child notices and tracks it to a known status.
For whom	Parents and guardians (PERSONA-001), verifying police officers (PERSONA-002), shelter and station staff (PERSONA-003).
Problem	Reports sit in separate GDs, social posts, news, shelters and hospitals. Fera explores whether reviewed intake and linked missing/found reports improve coordination; the unmet operational gap requires partner validation.
Why it matters	15,717 children were reported missing Jan–Jul 2026; 2,038 (13 %) were still untraced in that cohort as reported 3 Sep 2026; 13,679 were reported returned/rescued. See the handover source index.
Differentiator	Coordination and closure between report and rescue. Fera could support a future authorised coordination workflow; it currently has no CID/MUN integration. It never issues alerts.

3 Problem statement

3.1 Current situation

A family notices a child missing. They search, call relatives, visit the thana and post on social media. Found children are held by shelters, stations or hospitals, whose notices stay local. CID controls verified MUN alerts. Fera has no integration with it. Claims about MUN's internal linkage or closure capability remain unverified until a partner process review.

3.2 Pain points

#	Pain point	Who feels it
1	The first hours go to searching and paperwork, not a complete report.	**PERSONA-001**
2	Low-literacy parents struggle to write a formal complaint.	**PERSONA-001**
3	Missing and found notices sit in different places with no link.	PERSONA-002, PERSONA-003
4	The same child is reported several times in different formats.	**PERSONA-002**
5	Case-level linkage and follow-up across participating sources are an unvalidated coordination need; aggregate rescued/returned outcomes are already reported.	PERSONA-002, authorities

3.3 Consequences

Children stay untraced longer. Officers re-key and cross-check by phone. Fera must validate whether participating officers need better case-level linkage and follow-up; aggregate outcomes are already reported.

3.4 Evidence and source limits

- 15,717 children reported missing, Jan–Jul 2026 (Police HQ via The Daily Star, 3 Sep 2026).
- 2,038 still untraced (13 %).
- 97 % aged 10–17.
- About 74 reports per day.
- The absence of a linked operational registry is a hypothesis requiring an authoritative source or partner confirmation.
- MUN Alert: live since 13 Jan 2026; helpline 13219; joined Meta's alert network in May 2026.

3.5 Assumptions about the problem (unverified)

- Assumption A-021: A structured draft shortens the time to file a GD.
- Assumption A-022: Parents will accept AI-drafted text if they can review and edit it.
- Assumption A-023: Low-literacy parents prefer speaking to typing.
- Assumption A-024: Found-child notices exist in public text in enough volume to be worth matching.

- Assumption A-025: Officers will use a ranked review queue if each candidate shows evidence side by side.

Interview status is UNVERIFIED (OQ-006); record the actual count and evidence at the next meeting. Independent user-research evidence was not supplied; these hypotheses remain unvalidated in this handover.

3.6 Boundaries — Fera will not touch

Area	Owner
Issuing, withdrawing or broadcasting alerts	CID via MUN Alert (OQ-010)
Rescue operations, searches, arrests	Police
Placement or return decisions for a found child	Child-protection services
Filing the GD itself	The family at the thana or official online channel

4 Personas

Names are fictional, built from published reporting, not interviews (Assumption A-003).

Field	PERSONA-001 Parent ("Shirin")	PERSONA-002 Verifying officer ("SI Karim")	PERSONA-003 Station staff ("Rumana")
Role	Mother reporting a missing son	Thana duty officer	NGO volunteer, railway-station children's desk
Context	Dhaka, 30s. Son, 13, did not come home from madrasa. Frightened, first hours.	Handles several missing-child GDs a week at a desk.	Meets lone children daily; works from a phone between tasks.
Goals	A complete report and GD draft within the first hour; know the case is being checked.	See ranked candidates with evidence; record a decision in minutes.	Log a found child in under 2 minutes; know it will be matched.
Pains	GD may not be filed; unsure what to do first; posts get lost; no updates.	Incomplete reports; duplicates; cannot search found children elsewhere.	Notices stay local; families never see them.
Technical ability	Uses a mid-range Android phone for calls and social apps. Weak formal writing.	Comfortable with desktop web forms.	Comfortable with phone apps and forms.
Accessibility / language	Bangla-first; may use Bangla numerals; may mix English words; voice preferred.	Bangla and English; keyboard use.	Bangla-first; one-handed phone use.

Field	PERSONA-001 Parent ("Shirin")	PERSONA-002 Verifying officer ("SI Karim")	PERSONA-003 Station staff ("Rumana")
Expected outcome	Reviewed case, GD draft and poster in ≤ 5 min.	Confirms or rejects a candidate in ≤ 2 min with an audit record.	Found-child entry saved and matched in ≤ 2 min.

Secondary personas (no full profile, no UI in MVP):

- PERSONA-004 Child-protection caseworker: intended future recipient; V2 records a simulated referral only, with no actual delivery or recipient interface. Should (record only).
- PERSONA-005 Disaster-mode shelter volunteer: logs separated children after floods or cyclones. Could. No measured need exists.

5 Vision

Every missing-child report in Bangladesh ends with the child's status known.

Today a report starts many separate searches that never meet. Fera joins them into one case, one match queue and one outcome. Humans decide every match; AI only does the reading, structuring and ranking.

6 Goals and success metrics

Targets come from PRD v2, which marks them provisional (Assumption A-002). Timeframe "R2" = by the organiser-confirmed 14 Oct cutoff; "F" = by 24 Oct 2026.

ID	Goal	Metric	Target	Measurement method	Timeframe
GOAL-001	A parent produces a reviewed Bangla report in the first hour	Median time to complete report, assistant vs manual form	≤ 5 min and lower than manual form	Timed tests, ≥ 5 testers (A-010, A-011)	R2 measured; F re-measured
GOAL-002	AI extracts case fields correctly	Field-level extraction accuracy on required fields	≥ 85 %	50 hand-labelled reports	R2
GOAL-003	The right found record appears in the officer's shortlist	Recall@5; precision@1; precision@5	≥ 80 %; ≥ 60 %; reported as measured	Synthetic set, ~200 records, known pairs	R2
GOAL-004	Duplicate reports are detected	Duplicate-detection pairwise F1	≥ 0.8	Synthetic set with seeded duplicates	R2
GOAL-005	Bangla voice is usable	Word error rate	≤ 30 %	20 scripted prompts, ≥ 3 speakers (A-012)	R2
GOAL-006	Every case shows a known status by district	Seeded cases shown with correct status	100 %	Count comparison: register vs map	R2
GOAL-007	The live demo does not fail	Full scripted journey passes	5 of 5 rehearsals; ≥ 1 with internet off (A-017)	Rehearsal log	F
GOAL-008	Humans make every match decision	Match-driven status changes with a logged officer action	100 %	Audit-log check	R2
GOAL-009	Results are reported honestly	Metrics slide	1 slide, all 5 metrics, ≥ 2 failure examples	Deck review	R2

7 Non-goals

- Issuing, withdrawing or broadcasting alerts of any kind.
- Calling 13219, police or families on anyone's behalf.
- Face recognition or any photo-based search.
- Predicting which children will go missing, or why a child left.
- Deciding where a found child goes.
- Real police, CID or MUN Alert data before a partner agreement.
- Native mobile app, SMS or voice-call intake.
- Legal compliance certification under the Personal Data Protection Act (2026).

8 Scope

MoSCoW	Item	One-line reason
Must	Bangla text intake with follow-up questions	Guaranteed path for the core demo journey.
Must	Bangla voice intake	Demo hero moment; low-literacy parents (A-001).
Must	Parent review and edit before submit	Safeguarding BR-004.
Must	GD draft	Demo: "act fast" output.
Must	Text poster without photo	Demo: shareable output (A-001); no photos per BR-006.
Must	Found-child logging by staff	Demo: the "found" side of the match.
Must	Matching: top 5 with reasons	Demo: match surfaced.
Must	Officer review queue, side-by-side evidence, confirm/reject	Safeguarding BR-001.
Must	Referral record and case closure	Safeguarding BR-005; demo: case closes.
Must	District map of statuses	Demo: closure visible.
Must	Role-based access; matched records officer-only	Safeguarding BR-007.
Must	Audit log	Safeguarding BR-010.
Must	Redact identifying payloads from eligible terminal cases	Safeguarding BR-009.
Must	Duplicate detection and its metric	Deck promises the metric in Round 2.
Must	Evaluation report for 5 metrics	Deck promises measured results.

MoSCoW	Item	One-line reason
Must	Manual form fallback and demo-safe mode	Venue internet unreliable; AI can fail.
Must	Synthetic seed data, labelled as synthetic	No real data (A-004).
Should	13219-ready summary	Useful, not needed for the core journey.
Should	Public-text import	Adds sources; not needed for the scripted demo.
Should	Parent edits after submission	Edge case; text fix before submit covers the demo.
Could	Duplicate grouping screen	Detection already meets the metric.
Should	Parent case-status view	Nice to show "being checked".
Could	Disaster-mode screen	Theme link pending OQ-003.
Won't	Alerts, face matching, telecom location targeting, live police data	Out of mandate or no data. Shown only as [MOCK] on slides.
Won't	Dashboard filters, native app, SMS/IVR	Post-hackathon.

9 User journeys

9.1 Primary happy path (core demo)

1. Parent opens Fera; Bangla is default (SCREEN-001).
2. Parent records a 40-second Bangla voice note (SCREEN-002).
3. Transcript appears; parent fixes one word.
4. Fera shows a case draft and asks one missing field: district (SCREEN-002).
5. Parent reviews all fields and confirms (SCREEN-003).
6. Parent copies the GD draft and poster text (SCREEN-004).
7. Staff at a station log a found boy (SCREEN-007).
8. Matching ranks the parent's case first among ≤ 5 candidates.
9. Officer opens the queue, compares evidence and confirms (SCREEN-009, SCREEN-010).
10. A simulated referral is recorded; officer records coordination closure for the missing case (SCREEN-012).
11. The missing-report category changes to Closed; the found report stays Referral recorded, and the pair is counted once (SCREEN-011).

9.2 First-time user

Parent arrives from a shared link. One screen explains in 2 Bangla lines what Fera does and does not do. It states that Fera does not alert police or CID. Parent taps "শুরু করুন" (Start) and lands in intake.

9.3 Returning user

- Parent returns on the same device; their own case and status appear (Should, SCREEN-006). Cross-device return is Won't (OQ-020).
- Officer signs in and lands on the review queue with new candidates first (SCREEN-008, SCREEN-009).

9.4 Error and recovery

Situation	Behaviour
Voice note unintelligible	Ask to retry or type; draft kept.
Extraction fails twice	Manual form opens, pre-filled with any extracted fields (SCREEN-005).
Network drops mid-upload	Text and audio kept on device; retry button.
Officer opens a record they cannot see	"Not found" screen; attempt logged (SCREEN-016).
Officer rejects a match	Candidate removed; case returns to open if no other candidates.

9.5 Empty state

- Review queue with no candidates: "No likely matches yet. New reports are checked automatically."
- Map with no cases in a district: district shown with 0 counts, not hidden.

9.6 Offline or speech/AI service unavailable

- Typed text and the manual form always work.
- Demo-safe mode returns stored outputs for scripted demo inputs (FR-027).
- Live parents see: voice unavailable, please type, with the reason in plain Bangla.

10 User stories

Screens are reserved here and fully specified in D3.

STORY-001 · **Must** · **Report by voice** As a parent who cannot write well, I want to speak my report in Bangla so that I get a case without typing.

- Preconditions: microphone permission granted; voice note 5–120 s.
- AC1: Given a 5–120 s Bangla voice note, when uploaded, then an editable transcript appears within 20 s (p95).
- AC2: Given an unintelligible note, when processed, then the user is asked to retry or type, and no draft is created from it.
- AC3: Given no microphone permission, when the parent taps record, then the text box is focused and a Bangla notice explains why.
- Dependencies: DEP-001. Screens: SCREEN-002.

STORY-002 · Must · Report by typed text As a parent, I want to describe my child in Bangla, English or a mix, so that I get a structured case without a form.

- Preconditions: none.
- AC1: Given 20–2,000 characters including Bangla numerals, when submitted, then a draft with extracted fields appears within 15 s (p95).
- AC2: Given "বয়স ১৩", when extracted, then age = 13.
- AC3: Given a field not stated in the text, then it is empty, never guessed.
- Dependencies: DEP-002. Screens: SCREEN-002.

STORY-003 · Must · Answer follow-up questions As a parent, I want to be asked only for what is missing, one question at a time, so that I finish without a long form.

- AC1: Given a draft missing a required field (BR-015), when shown, then exactly one Bangla question asks for one missing field.
- AC2: Given 4 rounds without an answer, then the review screen opens with the field marked "unknown".
- Screens: SCREEN-002.

STORY-004 · Must · Review before anything is shared As a parent, I want to check and correct the draft so that wrong details are never shared.

- AC1: Given a draft, when the parent edits a field and confirms, then the submitted case equals the edited values in 100 % of test runs.
- AC2: Given no confirmation, then nothing is visible to staff or officers, matched or exported.
- AC3: Given the same principal and submission UUID retried with identical reviewed values, then exactly one case exists and the reporter receives it. Reusing that UUID for a changed payload conflicts; a different UUID is a separate report, optionally flagged for review.
- Screens: SCREEN-003.

STORY-005 · Must · Get a GD draft As a parent, I want ready-to-file GD text so that the thana can register the case quickly.

- AC1: Given a submitted case, when the parent taps "জিডি খসড়া" (GD draft), then Bangla text with child name or "অজানা" (unknown), age, last-seen place, date-time and reporter phone appears and can be copied.
- AC2: The text contains no fact absent from the confirmed fields.
- Screens: SCREEN-004.

STORY-006 · Must · Get a shareable poster As a parent, I want a poster so that neighbours and groups can help look.

- AC1: Given a submitted case, the parent reviews description/place/date, chooses whether to include contact phone, sees 13219 and acknowledges sharing before copy/download. Export cannot be remotely recalled.
- AC2: The poster has no photo field and no alert wording (BR-002, BR-006).
- Screens: SCREEN-004.

STORY-007 · Should · Get a 13219-ready summary As a parent, I want a short summary to read on a 13219 call.

- AC1: Given a submitted case, then a Bangla summary of ≤ 80 words appears.
- Screens: SCREEN-004.

STORY-008 · Must · Log a found child As station staff, I want to log a found child quickly so that families elsewhere can be matched.

- AC1: Given the found-child screen, when staff enter a description, place and time and confirm, then a found case is opened within 15 s and matching runs.
- AC2: Median entry time ≤ 2 min in timed tests.
- Screens: SCREEN-007.

STORY-009 · Must · See ranked candidates As a verifying officer, I want ranked candidates with reasons so that I check the likeliest first.

- AC1: Given eligible reports, when matching runs, then zero to five qualified current candidates appear with similarity, available evidence and unknowns. Eligibility alone never forces a suggestion; district is a soft feature.
- AC2: Candidates that fail BR-017 never appear.
- Screens: SCREEN-009.

STORY-010 · Must · Compare and confirm a match As an officer, I want both records side by side so that I decide on evidence.

- AC1: Given a candidate, when opened, then missing and found fields appear in two aligned columns with agree/differ/unknown markers per field.
- AC2: Given the officer confirms, then the missing case becomes confirmed, the found case becomes referred, and a referral and an audit entry exist.
- Screens: SCREEN-010.

STORY-011 · Must · Reject a candidate As an officer, I want to reject a wrong candidate so that it stops wasting time.

- AC1: Given a reject with a reason, then the candidate leaves the queue and the case returns to open if no other candidate is pending.
- AC2: The same pair is not suggested again unless either record is edited.
- Screens: SCREEN-010.

STORY-012 · Must · Refer and close As an officer, I want a confirmed match to create a referral and let me close the case so that the outcome is recorded.

- AC1: Given a confirmed missing case with a referral, when the officer selects an outcome and closes, then the case is closed and the map count changes within 5 s of reload.
- AC2: Closing is impossible without a referral record.
- Screens: SCREEN-012.

STORY-013 · Must · See the district picture As an officer, I want a district map of statuses so that I see where cases are open.

- AC1: Given seeded report rows, the map/table separately shows missing Open/Matched/Closed, found Open/Matched/Referral recorded and confirmed-pair counts. Each equals the corresponding register query and the counting basis is visible.
- AC2: Each status shows an icon, a text label and a colour.
- Screens: SCREEN-011.

STORY-014 · Must · Detect duplicate reports As an officer, I want duplicate reports of one child flagged so that I do not review the same child twice.

- AC1: Given two records meeting BR-021, then both show "সম্ভাব্য একই শিশু" (possible same child) and are linked.
- AC2: Flagged links remain separate searchable reports. Only an explicit reviewed consolidation may change grouping, while preserving access to every report.
- Screens: SCREEN-012.

STORY-015 · Should · Import synthetic public-format fixtures As an admin, I want to import invented news/alert-format fixtures so that the synthetic prototype can test intake from varied sources.

- AC1: Given invented fixture texts and example.invalid source URLs, each reviewed record is tagged synthetic and follows required-field/state rules. Real public-child material is rejected in V2.
- Screens: SCREEN-013.

STORY-016 · Must · Redact identifying payloads from eligible terminal cases As an admin, I want to redact identifying payloads closed cases so that personal details are not kept longer than needed.

- AC1: Given closed missing cases and eligible referral-recorded found cases whose linked missing report is closed, when payload redaction runs, then name, phone, free text and sub-district place are removed and identifying payloads and related notes are redacted; IDs and aggregate fields remain under restricted access as pseudonymous records.
- Screens: SCREEN-013.

STORY-017 · Must · Audit decisions As an admin or officer, I want every decision logged so that any outcome can be traced.

- AC1: Given any confirm, reject, close, edit, redact identifying payloads or denied access, then an entry with actor, role, action, record, time and reason exists.
- AC2: Entries cannot be edited or deleted through the product.
- Screens: SCREEN-012.

STORY-018 · Must · Continue when AI fails As a parent, I want a plain form if the assistant fails so that I can still report.

- AC1: Given two failed extraction attempts or an AI service error, then the manual form opens within 2 s, pre-filled with any extracted fields.
- Screens: SCREEN-005.

STORY-019 · Should · Edit after submission As a parent, I want to correct my case after submitting so that the match uses right details.

- AC1: Given a case in submitted, open or matched-pending-review, when edited and confirmed, then a new version is stored, pending candidates are voided and matching re-runs.
- AC2: Given a confirmed or closed case, then edit is not offered.
- Screens: SCREEN-006.

STORY-020 · Should · See my case status As a parent, I want to see my case status without match details.

- AC1: Status shows one of Open / Being checked / Closed with date; no found-child data appears.
- Screens: SCREEN-006.

STORY-021 · Could · Disaster-mode logging As a shelter volunteer after a flood, I want to log a separated child so that families can be matched.

- AC1: The screen reuses found-child logging and is labelled "পরিকল্পিত সম্প্রসারণ" (planned extension).
- Screens: SCREEN-015.

STORY-022 · Must · Measure and report results As the team, we want one command to compute all 5 metrics so that the deck reports measured results.

- AC1: Given the labelled sets, when run, then a report lists all 5 metrics, the set sizes, the run date and ≥ 2 failure examples.
- Screens: none (report file).

STORY-023 · Must · Run the demo without internet As the demo operator, I want the scripted journey to work offline so that venue Wi-Fi cannot break it.

- AC1: Given demo-safe mode and network disabled, when the scripted inputs are used, then the full §9.1 journey completes.
- AC2: Seeded records show a "synthetic" label.
- Screens: all Must screens.

11 Functional requirements

ID	The system shall…	Pri	Story	Acceptance test	Notes
FR-001	accept typed reports of 20–2,000 characters in Bangla, English or mixed script.	Must	STORY-002	TC-002	v2 FR1
FR-002	accept voice notes of 5–120 s and return an editable Bangla transcript.	Must	STORY-001	TC-001	v2 FR7; Must per A-001
FR-003	normalise Bangla digits and derive explicit Asia/Dhaka event intervals anchored to the recorded intake clock; ambiguous time requires review.	Must	STORY-002	TC-002	"গতকাল বিকেলে" → yesterday 16:00 ±2 h, flagged approximate
FR-004	extract case fields from text, leaving unstated fields empty and never guessing.	Must	STORY-002	TC-002	v2 FR1; BR-008, BR-015
FR-005	ask at most four one-at-a-time follow-ups, then require accessible manual completion of unresolved mandatory fields.	Must	STORY-003	TC-003	
FR-006	show every extracted field as editable and require an explicit confirm.	Must	STORY-004	TC-004	v2 FR2
FR-007	keep drafts out of matching, staff/officer views and exports until confirmed.	Must	STORY-004	TC-004	BR-004
FR-008	generate a Bangla GD draft from a fixed template using confirmed fields only.	Must	STORY-005	TC-005	v2 FR3; A-008
FR-009	generate a reviewed no-photo poster with 13219, a separate sharing acknowledgement and optional contact inclusion.	Must	STORY-006	TC-006	v2 FR8; A-001
FR-010	generate a 13219-ready summary of ≤ 80 words.	Should	STORY-007	TC-007	v2 FR8
FR-011	let staff log a found child by form or free text and review before saving.	Must	STORY-008	TC-008	v2 FR4

ID	The system shall...	Pri	Story	Acceptance test	Notes
FR-012	rank up to five qualified candidates nationally with evidence reasons and similarity scores; allow an empty shortlist.	Must	STORY-009	TC-009	v2 FR4; BR-018
FR-013	exclude candidate pairs that fail the hard filters in BR-017.	Must	STORY-009	TC-009	
FR-014	list cases with pending candidates for verified officers, newest candidate first.	Must	STORY-009	TC-009	v2 FR5
FR-015	show missing and found records side by side with agree/differ/unknown per field.	Must	STORY-010	TC-010	
FR-016	atomically record a verified candidate decision, current endpoint states, simulated referral recorded and safe audit; no receipt or custody claim.	Must	STORY-010, STORY-012	TC-010	BR-001, BR-005
FR-017	on reject with a reason, remove the candidate and apply BR-012 and BR-019.	Must	STORY-011	TC-011	
FR-018	record coordination_recorded closure after confirmed/referral-recorded state, or resolved_outside_fera from open; neither proves safe reunification.	Must	STORY-012	TC-010	
FR-019	show separate missing-report, found-report and confirmed-pair district counts with accessible icon, text and colour.	Must	STORY-013	TC-012	v2 FR6; BR-014
FR-020	restrict candidates and matched records to verified officers per BR-024.	Must	STORY-009	TC-022, TC-023	BR-007
FR-021	write an append-only audit entry for every action listed in STORY-017.	Must	STORY-017	TC-016	BR-010
FR-022	flag possible duplicates while keeping every unreviewed report independently discoverable; shared phone alone never suppresses a report.	Must	STORY-014	TC-013	Detection only
FR-023	provide optional reviewed reversible duplicate grouping/dismissal with profile evidence, reason and preserved original reports.	Could	STORY-014	TC-013	v2 FR10
FR-024	import invented fixture texts in public-report format with a synthetic label and source metadata; real public-child import is disabled in V2.	Should	STORY-015	TC-014	v2 FR9

ID	The system shall...	Pri	Story	Acceptance test	Notes
FR-025	redact direct and related identifying payloads only from eligible terminal reports; retain active linked evidence.	Must	STORY-016	TC-015	BR-009
FR-026	open a pre-filled manual form after 2 failed extractions or an AI service error.	Must	STORY-018	TC-017	
FR-027	in demo-safe mode, return stored outputs for scripted demo inputs with no network.	Must	STORY-023	TC-021	A-017
FR-028	report frozen held-out metrics with explicit denominators, abstention/coverage, cache exclusion and failure examples.	Must	STORY-022	TC-020	
FR-029	let parents edit a case after submission per BR-025, keeping all versions.	Should	STORY-019	TC-018	
FR-030	show parents their own case status and authorised edit/output availability without candidate or other-party evidence.	Should	STORY-020	TC-019	BR-007
FR-031	idempotently return one case per principal and submission UUID; identical retry returns it, a changed payload conflicts, a new UUID remains a distinct report.	Must	STORY-004	TC-031	
FR-032	load synthetic seed records and label them "সিনথেটিক" (synthetic) on every screen.	Must	STORY-023	TC-021	BR-026
FR-033	default parent screens to Bangla with an English toggle on every screen.	Must	STORY-002	TC-002	
FR-034	block generated or displayed text containing alert wording or "runaway" terms.	Must	STORY-005, STORY-006	TC-024, TC-025	BR-002, BR-003
FR-035	treat officer accounts as verified only when an admin marks them so. [MOCK: seeded accounts]	Must	STORY-009	TC-023	A-006
FR-036	provide a disaster-mode logging screen labelled as a planned extension.	Could	STORY-021	TC-035	OQ-003
FR-037	sign in staff, officers and admins with a role; parents need no account.	Must	STORY-008, STORY-009	TC-023	A-001

Mocked capabilities (not built; slide or placeholder only):

Capability	Status	Where shown
Live police data	[MOCK] synthetic seed labelled "synthetic"	Seed data, all screens
Officer verification	[MOCK] seeded `verified` flag	FR-035
Telecom location targeting	[MOCK] slide only	Deck roadmap
Face matching (CID-held records, human review)	[MOCK] slide only	Deck roadmap

12 Non-functional requirements

ID	Category	Requirement	Measure
NFR-001	Performance	Parent intake screen usable on a mid-range Android over 4G	Interactive ≤ 3 s
NFR-002	Performance	Text → draft	≤ 15 s p95
NFR-003	Performance	120 s voice note → transcript	≤ 20 s p95
NFR-004	Performance	Matching for one case vs 500 records	≤ 5 s
NFR-005	Performance	District map with 1,000 cases	≤ 2 s
NFR-006	Performance	Concurrent users (A-020)	50 users, error rate < 1 %
NFR-007	Reliability	Scripted demo journey with network off	5 of 5 rehearsals
NFR-008	Reliability	Draft survives network loss; fallback form after AI failure	0 lost drafts in tests; form ≤ 2 s
NFR-009	Security	Encrypted transport; no secrets reach the browser; rate limits 10/min/user, 30/min/IP on AI actions	Inspection + test
NFR-010	Privacy	No payload in logs/analytics; synthetic audio/text reaches declared providers before submit with notice; endpoint/account review remains pending	Log review: 0 phone numbers or names
NFR-011	Accessibility	WCAG 2.2 AA; tap targets ≥ 44 × 44 px; text contrast ≥ 4.5:1	Automated + manual audit, 0 AA failures on Must screens

ID	Category	Requirement	Measure
NFR-012	Usability	Parent stories completed unaided; any field corrected in 1 tap	≥ 90 % completion, ≥ 5 testers
NFR-013	Compatibility	Android browser at 360 px width; current desktop browsers; Bangla renders with no missing glyphs	Manual check on 2 phones, 2 desktop browsers
NFR-014	Maintainability	AI provider switched by configuration only; setup from README	0 code changes; setup ≤ 15 min
NFR-015	Scalability	Register supports all 64 districts and 30,000 cases/year (74/day × 365 ≈ 27,000) without data-model change	Load test with 30,000 synthetic rows
NFR-016	Cost	Total external AI spend from 9 Oct to 24 Oct	≤ USD 30; ≤ 6 AI calls per case
NFR-017	Auditability	Audit entries immutable through the product	100 % of decisions logged; edit/delete attempts fail

13 Business rules

13.1 Safeguarding rules (non-negotiable)

ID	Rule	FRs	TCs
BR-001	AI retrieves/ranks only. Active verified officers record candidate decisions; identity and guardian verification are external pilot gates.	FR-016, FR-017, FR-020	TC-010, TC-023, TC-046
BR-002	No issuing, withdrawing or broadcasting alerts. Outputs are reviewed drafts; no CID integration is claimed.	FR-008, FR-009, FR-010, FR-034	TC-024, TC-025
BR-003	No runaway labels, inferred absence reasons, or rescue promises.	FR-004, FR-034	**TC-024**
BR-004	No submitted child payload, matching, other-party visibility, or export before reporter review/confirmation. Providers process intake after notice; pre-submit DB holds metadata only.	FR-006, FR-007	TC-004, TC-060
BR-005	Confirm creates a simulated internal referral record. referred means referral recorded; no receipt, custody, return, or safe reunification is inferred.	FR-016, FR-018	TC-010, TC-047
BR-006	No face search, photos, identity documents, exact coordinates, or physical return action in this release.	FR-009, FR-011, FR-020	**TC-026**

ID	Rule	FRs	TCs
BR-007	Candidates and other-party matched evidence are officer-only. Parent gets own scoped editable payload/ status; staff gets own found payload/status. Admin gets aggregate/redacted metadata only.	FR-020, FR-030, FR-035	TC-019, TC-022, TC-044, TC-045
BR-008	Only specified report fields are collected. Evidence quotes refer to reviewed source text; deterministic derivations retain source spans; unknowns remain unknown.	FR-004, FR-011	TC-027, TC-060
BR-009	Admin redacts identifying payloads only after closure; found evidence is retained while linked missing case is unclosed. Retained IDs/date/district are pseudonymous, not irreversible anonymity.	FR-025	TC-015, TC-050
BR-010	Decisions, state changes, edits, payload redaction and denied access have safe append-only audit events. Denials after rollback are logged separately; unknown case FK is null.	FR-021	TC-016, TC-051, TC-052
BR-011	No unsupported legal compliance claim. Accurate primary-law references are permitted. Real-data processing requires qualified partner/legal/provider review.	FR-025, NFR-010	TC-037, TC-066

13.2 Case state transitions

BR-012 **Missing report.** Browser draft is not a persisted case. Initial persisted state is submitted; submitted → open; open → matched_pending_review when at least one qualified candidate exists; matched_pending_review → open when none remain; matched_pending_review → confirmed only with an authorised decision and simulated referral; confirmed → closed for coordination_recorded; open → closed for resolved_outside_fera. Initial confirmed/closed insertion is prohibited. Unknown mandatory fields move to manual completion, not silent guessing.

BR-013 **Found report.** Initial state open; open ↔ matched_pending_review; matched_pending_review → referred only in the confirmation transaction. The UI says “রেফারেল নথিভুক্ত (সিমুলেশন)” / “Referral recorded (simulation)”. closed_at on referred is the coordination terminal timestamp, not a rescue timestamp.

Ordinary terminal edits and backward transitions are blocked. Before a pilot, a correction/reopen procedure must require two distinct verified reviewers, evidence, a reason, superseded decision linkage, and a preserved audit. This workflow is a pilot gate, not implemented by the V2 demo. Resetting an isolated synthetic dataset is permitted and labelled; it is not a real-case correction mechanism.

BR-014 **Counts.** Missing Open = submitted/open; Missing Matched = matched_pending_review/ confirmed; Missing Closed = closed. Found Open = open; Found Matched = matched_pending_review; Found Referral recorded = referred. Confirmed pairs are counted once from confirmed candidate rows. Never sum missing+found rows and call the result unique children. All 64 district rows, including zeros, are shown. Flagged duplicates remain report rows; label the counting basis.

13.3 Validation

BR-015 **Required submission fields.** Missing: age or complete age range, sex selection (unknown allowed), district, non-empty last-seen place, event interval, and reporter phone. Found: description, district, place, event interval, staff contact; age may be unknown. Name is optional. District is mandatory for V2 and is confirmed through an accessible manual selector if extraction cannot resolve it. No field is guessed to unblock submit.

BR-016 **Field rules.** Missing ages 0–17. Exact age yields age_min = age_max = age_years; a range has both endpoints and no conflicting exact age. Found estimated ranges may extend to 25 to represent uncertainty; wholly adult profiles are not automatically treated as children. Store event instants with timestamptz and serialise explicitly in Asia/Dhaka. Exact events cannot be future; approximate bounds must be consistent with the intake time, and ambiguous/future periods are clarified. Phone format is 11 digits starting 01 after digit normalisation; this does not verify ownership. District codes 1–64 are internal identifiers, not official geocodes.

13.4 Match logic

BR-017 **Eligibility.** Opposite kinds; sex equal or unknown; age ranges overlap within ±2 years when both known, otherwise unknown is allowed with less evidence; found upper time ≥ missing lower time; live open/matched versions; neither record payload-redacted/confirmed/closed/referred. Evaluate uncertainty before pruning. Eligibility is necessary but not sufficient to suggest a candidate.

BR-018 **Ranking.** Retrieve eligible records nationally using exact filtered search for the ~200-record baseline. District/adjacency is a soft bonus, never a gate after five weak local results. L2-normalised embeddings share a pinned space/revision/prefix policy. Use clamped cosine and available-feature weight normalisation; missing feature values are null, not fabricated agreement. Calibrate qualification/abstention and dedupe thresholds on dev data, freeze before test, display at most five qualified candidates, and allow zero. Until calibrated, label ranking experimental and require explicit uncertain evidence. Score "মিলের মাত্রা" is not identity probability.

BR-019 **Rejections.** A rejected pair is not re-suggested at the same report versions. Editing creates new versions and invalidates old pending candidates. Display prior rejection history to the officer.

BR-020 **One-to-one by report.** At most one confirmed candidate per endpoint. The transaction locks endpoints in sorted UUID order, rechecks versions/authorisation, confirms one pair, voids other pending pairs, and reconciles each affected report's queue state. Duplicate report rows do not establish unique child identity.

BR-021 **Duplicates.** Shared phone creates only a weak flagged cue and cannot independently yield confirmed_same or suppress a report. Compatible profiles may create a possible-duplicate cue using dev-calibrated similarity, district and time evidence. Human review may confirm a link with profile evidence and a reason; preserve every original report and make consolidation reversible. V2 detection and pairwise F1 are Must; merge/group UI is Could. Flagged reports are always independently discoverable.

13.5 Generation, limits, permissions, edits

BR-022 **Outputs.** Fixed templates use only reporter-reviewed fields plus fixed instructions and 13219. Poster export has a separate preview/share acknowledgement and contact inclusion choice. Downloads cannot be remotely recalled; no photo does not mean no sensitive information. Cache deterministic output by case/type/version rather than generating five identical copies.

BR-023 Limits. Audio 5–120 s and ≤2 MB; codec/bitrate must satisfy both, and duration is verified server-side. Text 20–2,000 characters; at most four conversational follow-ups, then manual completion. At most six dispatched paid transcribe/extract calls per draft, including retries/fallbacks; authoritative metadata and atomic reservation enforce this across replay/concurrency. Retain the 10/min principal and 30/min network limits with HMAC-hashed network keys and short expiry. Fail closed for new paid calls if limit/budget state is unavailable; manual intake remains usable. Draft metadata expires after 24 h; parent sessions after 30 days with revocation. Device draft expiry is checked on access, not a claim of guaranteed physical deletion.

BR-024 Permissions. Parent: own missing creation/read/edit/output only. Staff/unverified officer: own found creation/read and found-only intake. Verified officer: candidate review/decision/closure, operational case list, aggregate dashboard and safe audit. Admin: redacted metadata, aggregate dashboard, safe audit, seeded verification [MOCK], redaction, synthetic import and rematch status; no candidate detail or decisions. Public health/auth responses expose no child payload. Every route checks ownership/role server-side; the client never supplies trusted authority.

BR-025 Edits. Owner edits only submitted/open/matched_pending_review with expected_version. Increment version, snapshot the old reviewed payload, void pending pairs, update affected states, and queue bounded reliable rematching. Terminal edits are rejected; payload redaction is a separate restricted action. Pilot correction procedure is required before accepting real cases.

BR-026 Synthetic release. Every demo/seed/import record is synthetic, with a visible label; real public child reports are not imported in V2. Model-quality evaluation disables fixtures and caches. Mock authority, cached outputs, compressed video time, and lack of integration are disclosed. No measured result is invented.

14 Assumptions

ID	Assumption
A-001	OQ-001 – OQ-007 unanswered; Turn 0 defaults apply: voice and poster Must, 13219 Should; separate missing/found state machine; disaster mode Could; one free LLM key plus paid speech key; deployed app for Round 2 and a laptop-local setup for the finale; nothing built and 0 interviews on 9 Oct; seeded staff/officer/admin logins, no "helper" role.
A-002	Numeric targets are provisional and revised after the first evaluation run on 11 Oct.
A-003	Personas are fictional, built from published reporting.
A-004	No real police, CID or MUN Alert data is available before the finale.
A-005	Parents use a smartphone browser with a microphone; feature phones are out of scope.
A-006	Officer verification is a flag set by an admin on seeded accounts [MOCK].
A-007	A referral is a recorded entry with an agency name; no external integration.

ID	Assumption
A-008	The GD draft uses a generic Bangla template; the official online GD field list is UNKNOWN (OQ-016).
A-009	The parent reads the 13219 summary on their own call; Fera never contacts 13219.
A-010	Usability testers are adult volunteers role-playing a parent, not real affected parents (OQ-019).
A-011	The manual-form baseline is the same testers filling a plain form with the same fields.
A-012	The speech test set is 20 scripted Bangla prompts recorded by ≥ 3 speakers.
A-013	The team compiles a neighbouring-district list; if absent, search goes same district, then national (OQ-017).
A-014	Duplicate thresholds (±1 year, 72 h) are provisional and tuned on the synthetic set.
A-015	A 0–100 similarity score is understandable to officers if labelled as similarity.
A-016	Drafts expire after 24 h.
A-017	Demo-safe mode covers only scripted inputs for ≥ 3 named synthetic cases.
A-018	Hosting and database run on free tiers; the USD 30 budget covers AI services only.
A-019	Public reference data gives names and centroids for all 64 districts (DEP-004).
A-020	50 concurrent users is a PRD v2 target; realistic demo load is ≤ 10.
A-021	A structured draft shortens the time to file a GD.
A-022	Parents accept AI-drafted text if they can review and edit it.
A-023	Low-literacy parents prefer speaking to typing.
A-024	Public found-child notices exist in enough volume to be worth matching.
A-025	Officers use a ranked queue if each candidate shows evidence side by side.

15 Dependencies

ID	Dependency	Owner	Needed by
DEP-001	Speech-to-text service with Bangla support	Obidit	11 Oct
DEP-002	At least two interchangeable language-model services	Obidit	10 Oct
DEP-003	Text-similarity capability that also works offline for scripted inputs	Masnun	11 Oct
DEP-004	District reference data: 64 districts, Bangla/ English names, centroids	Masnun	11 Oct
DEP-005	Synthetic set, ~200 records, known pairs, duplicates, decoys	Masnun	11 Oct
DEP-006	50 hand-labelled reports	Masnun, Obidit	11 Oct
DEP-007	Public news and alert texts (public pages only)	Masnun	12 Oct
DEP-008	Free-tier hosting and database	Sadman	10 Oct
DEP-009	Finale kit: laptop, Android phone, mobile hotspot	Sadman	23 Oct
DEP-010	Mentor or organiser answers to OQ-003 and OQ-010	Sadman	13 Oct
DEP-011	Base map display; offline fallback without map tiles	Masnun	12 Oct

16 Risks and mitigations

L = likelihood, I = impact (Low / Medium / High / Severe).

ID	Risk	L	I	Mitigation	Owner
RISK-001	Core journey unfinished by 14 Oct official cutoff (time pending)	M	H	Cut order: disaster mode → 13219 → import → voice (typed path stays)	Sadman
RISK-002	Bangla speech recognition too poor	H	M	Typed path is guaranteed; transcript editable; WER reported honestly	Obidit
RISK-003	AI invents or misreads fields	M	H	Empty-not-guess rule; parent review; officer review	Obidit
RISK-004	AI service outage, quota or spend over USD 30	M	H	Second provider; manual form; spend tracked daily	Obidit
RISK-005	Venue internet fails at the finale	M	H	Demo-safe mode; local setup; recorded video backup	Sadman
RISK-006	Matching below targets	M	M	Tune rules on synthetic set; report as measured	Masnun
RISK-007	Synthetic set too easy, inflating metrics	M	M	Include decoys and near-duplicates; label results "synthetic"	Masnun
RISK-008	Stalker or trafficker misuse	L	Severe	BR-007, rate limits, audit of denied access	Sadman
RISK-009	False match sends a child into danger	L	Severe	BR-001 human confirm; BR-005 referral before any return	Sadman
RISK-010	Judges favour environment-themed projects	M	H	Disaster-mode slide; ask mentors (OQ-003)	Sadman
RISK-011	"Why not just MUN Alert?"	H	M	Position Fera as the intake, matching and closure layer that feeds it	Sadman
RISK-012	Overclaiming on metrics, interviews or law	M	H	Show only measured results; 0 interviews stated as 0; no legal claims	Sadman
RISK-013	Bangla text renders badly on low-end phones	L	M	Test on 2 phones; embed a Bangla-capable font	Masnun

ID	Risk	L	I	Mitigation	Owner
RISK-014	Instructions hidden in imported public text steer the AI	M	M	Treat imported text as data only; parent/officer review; import is Should	Obidit

17 Open questions

ID	Question	Status / default
OQ-001	Voice and poster Must for Round 2?	Default applied: both Must (A-001)
OQ-002	Adopt the separate missing/found state machine?	Default applied: BR-012, BR-013
OQ-003	Mentor view on environment theme / disaster mode?	Default: Could (FR-036)
OQ-004	Which accounts and keys exist?	Default: A-001; costs UNKNOWN until D2
OQ-005	Finale demo topology?	Default: local-capable finale (FR-027)
OQ-006	What exists today; interviews held?	Default: nothing; 0 interviews
OQ-007	Demo login and roles?	Default: FR-037, no helper role
OQ-008	Use figures beyond the 5 in PROJECT CONTEXT?	Default: no (§3.4)
OQ-009	WCAG 2.1 vs 2.2?	Default: 2.2 AA (NFR-011)
OQ-010	Correct name of the CID unit that issues alerts?	Default: "CID via MUN Alert"
OQ-011	Personal Data Protection Act (2026) specifics?	No clauses used; legal review before pilot
OQ-012	Retention period for closed cases?	UNKNOWN; manual redact identifying payloads
OQ-013	AI providers' training and retention terms?	Synthetic data only until checked

ID	Question	Status / default
OQ-014	Persona count?	3 primary + PERSONA-004, PERSONA-005
OQ-015	Dedupe priority?	Detection Must, merge screen Could
OQ-016	Official online GD field list and format?	UNKNOWN; generic template (A-008)
OQ-017	Source for neighbouring-district adjacency?	Team-compiled (A-013)
OQ-018	Which child-protection agency name should a referral show?	UNKNOWN; demo shows "শিশু সুরক্ষা সেবা" (child-protection services)
OQ-019	Can usability tests ethically include real affected parents?	Default: no (A-010)
OQ-020	Should parents return to a case from another device?	Default: Won't this release

18 Definition of done

A requirement is done when all hold:

1. Its acceptance test (TC) passes on the deployed Round 2 build.
2. Bangla is the default text; English toggle works; Bangla numerals accepted.
3. Loading, empty, error and success states exist for its screen.
4. Any [MOCK] is labelled on screen and in the deck.
5. No personal data appears in logs.
6. Linked BRs pass their TCs.
7. No open AA accessibility failure on its screen.

Round 2 readiness requires demonstrated Must behaviour, an honest evaluation report and organiser-confirmed deliverables submitted through the confirmed channel before the official cutoff. A 5–8-slide deck and 1–2-minute video are internal working defaults pending the selection email. If scope is reduced, label omissions and update the release checklist; do not claim full Must completion.

19 Traceability matrix

TC IDs are reserved here; steps are written in D6.

GOAL	STORY	FR	SCREEN	TC
GOAL-001	STORY-001 – STORY-006	FR-001 – FR-009, FR-031, FR-033	SCREEN-001 – SCREEN-004	TC-001 – TC-006, TC-031
GOAL-002	STORY-002, STORY-022	FR-004, FR-028	**SCREEN-003**	TC-002, TC-020
GOAL-003	STORY-008, STORY-009, STORY-010	FR-011 – FR-015, FR-028	SCREEN-007, SCREEN-009, SCREEN-010	TC-008, TC-009, TC-020
GOAL-004	STORY-014, STORY-022	FR-022, FR-028	**SCREEN-012**	TC-013, TC-020
GOAL-005	STORY-001, STORY-022	FR-002, FR-028	**SCREEN-002**	TC-001, TC-020
GOAL-006	STORY-012, STORY-013	FR-016, FR-018, FR-019	SCREEN-011, SCREEN-012	TC-010, TC-012
GOAL-007	STORY-018, STORY-023	FR-026, FR-027, FR-032	SCREEN-005 + all Must screens	TC-017, TC-021
GOAL-008	STORY-010, STORY-011, STORY-017	FR-016, FR-017, FR-020, FR-021, FR-035, FR-037	SCREEN-008, SCREEN-010, SCREEN-012	TC-010, TC-011, TC-016, TC-022, TC-023
GOAL-009	**STORY-022**	**FR-028**	— (deck)	**TC-020**
Safeguarding	**STORY-016**	**FR-025**	**SCREEN-013**	TC-015, TC-037
Safeguarding	STORY-004, STORY-005, STORY-006	**FR-034**	**SCREEN-004**	TC-024, TC-025, TC-026, TC-027
Support	**STORY-007**	**FR-010**	**SCREEN-004**	**TC-007**
Support	**STORY-015**	**FR-024**	**SCREEN-013**	**TC-014**
Support	STORY-019, STORY-020	FR-029, FR-030	**SCREEN-006**	TC-018, TC-019
Support	**STORY-021**	**FR-036**	**SCREEN-015**	**TC-035**
Support	**STORY-014**	**FR-023**	**SCREEN-014**	**TC-013**
Rules	—	BR-012 – BR-021	**SCREEN-016**	TC-029, TC-030, TC-032
NFRs	—	NFR-001 – NFR-017	all	TC-028, TC-033, TC-034, TC-036

Reserved screens (specified in D3)

ID	Screen	Pri
SCREEN-001	Welcome and language	Must
SCREEN-002	Intake chat (voice/text)	Must
SCREEN-003	Case review	Must
SCREEN-004	Case submitted and outputs	Must
SCREEN-005	Manual case form	Must
SCREEN-006	Parent case status and edit	Should
SCREEN-007	Found-child log	Must
SCREEN-008	Sign-in	Must
SCREEN-009	Review queue	Must
SCREEN-010	Match review (side by side)	Must
SCREEN-011	District dashboard	Must
SCREEN-012	Case detail and audit trail	Must
SCREEN-013	Admin data tools	Must
SCREEN-014	Duplicate review	Could
SCREEN-015	Disaster mode	Could
SCREEN-016	Not found / not allowed	Must

Reserved test cases (written in D6)

ID	Covers	ID	Covers
TC-001	Voice intake	TC-020	Evaluation report
TC-002	Text intake, mixed script, Bangla numerals	TC-021	Offline scripted demo
TC-003	Follow-up questions	TC-022	Unauthorised access to matched record
TC-004	Review before share	TC-023	Non-officer cannot confirm
TC-005	GD draft fields, no invented facts	TC-024	No alert or "runaway" wording
TC-006	Poster, no photo	TC-025	Outputs labelled as drafts

ID	Covers	ID	Covers
TC-007	13219 summary ≤ 80 words	TC-026	No photo upload or face search
TC-008	Found-child log	TC-027	Data minimisation
TC-009	Top-5 candidates, hard filters	TC-028	No personal data in logs
TC-010	Confirm → referral → close	TC-029	One-to-one confirm lock
TC-011	Reject → open, not re-suggested	TC-030	Field validation
TC-012	Map counts equal register	TC-031	Duplicate submission guard
TC-013	Duplicate flag and link	TC-032	Illegal state transitions rejected
TC-014	Public import with source URL	TC-033	Accessibility audit
TC-015	Payload redaction	TC-034	Performance targets
TC-016	Audit completeness	TC-035	Disaster-mode screen
TC-017	Manual form fallback	TC-036	AI spend tracking
TC-018	Edit after submission	TC-037	No legal compliance claims
TC-019	Parent status view hides match data		

IDs introduced

- STORY-001 Report by voice
- STORY-002 Report by typed text
- STORY-003 Answer follow-up questions
- STORY-004 Review before sharing
- STORY-005 GD draft
- STORY-006 Shareable poster
- STORY-007 13219 summary
- STORY-008 Log a found child
- STORY-009 See ranked candidates
- STORY-010 Compare and confirm
- STORY-011 Reject a candidate

- STORY-012 Refer and close
- STORY-013 District picture
- STORY-014 Detect duplicates
- STORY-015 Import synthetic public-format fixtures
- STORY-016 Redact identifying payloads from eligible terminal cases
- STORY-017 Audit decisions
- STORY-018 Continue when AI fails
- STORY-019 Edit after submission
- STORY-020 See my case status
- STORY-021 Disaster-mode logging
- STORY-022 Measure and report results
- STORY-023 Run demo without internet
- FR-001 Typed intake
- FR-002 Voice intake
- FR-003 Numeral and date normalisation
- FR-004 Field extraction, no guessing
- FR-005 Follow-up questions
- FR-006 Editable review and confirm
- FR-007 Drafts isolated
- FR-008 GD draft
- FR-009 Poster
- FR-010 13219 summary
- FR-011 Found-child logging
- FR-012 Top-5 ranking with reasons
- FR-013 Hard filters
- FR-014 Review queue
- FR-015 Side-by-side evidence
- FR-016 Confirm action
- FR-017 Reject action
- FR-018 Close with outcome
- FR-019 District map
- FR-020 Officer-only matched records
- FR-021 Append-only audit
- FR-022 Duplicate detection
- FR-023 Duplicate grouping screen
- FR-024 Public-text import
- FR-025 Payload redaction
- FR-026 Manual form fallback
- FR-027 Demo-safe mode

- FR-028 Evaluation report
- FR-029 Edit after submission
- FR-030 Parent status view
- FR-031 Duplicate submission guard
- FR-032 Synthetic seed and label
- FR-033 Bangla default UI
- FR-034 Banned-wording filter
- FR-035 Officer verification [MOCK]
- FR-036 Disaster-mode screen
- FR-037 Role sign-in
- NFR-001 Intake load time
- NFR-002 Extraction time
- NFR-003 Transcription time
- NFR-004 Matching time
- NFR-005 Map load time
- NFR-006 Concurrency
- NFR-007 Offline demo reliability
- NFR-008 Draft survival and fallback time
- NFR-009 Security basics and rate limits
- NFR-010 Privacy in logs and AI sharing
- NFR-011 WCAG 2.2 AA
- NFR-012 Usability completion
- NFR-013 Compatibility
- NFR-014 Maintainability
- NFR-015 Scalability
- NFR-016 AI cost cap
- NFR-017 Audit immutability
- BR-001 Human confirms every match
- BR-002 No alerts
- BR-003 No "runaway" label
- BR-004 Review before share
- BR-005 Referral, not return
- BR-006 No face search or photos
- BR-007 Officer-only matched records
- BR-008 Data minimisation
- BR-009 Redact identifying payloads from eligible terminal cases
- BR-010 Append-only logging
- BR-011 PDPA in mind, no claims
- BR-012 Missing-case state machine

- BR-013 Found-case state machine
- BR-014 Map status grouping
- BR-015 Required fields
- BR-016 Field validation
- BR-017 Match hard filters
- BR-018 Ranking rules
- BR-019 Rejected pairs not re-suggested
- BR-020 One-to-one confirm
- BR-021 Duplicate rule
- BR-022 Generated text facts only
- BR-023 Limits
- BR-024 Permissions
- BR-025 Edits after submission
- BR-026 Synthetic/public labels
- A-001 – A-025 (one line each in §14)
- DEP-001 Speech service
- DEP-002 Two language-model services
- DEP-003 Text similarity, offline-capable
- DEP-004 District reference data
- DEP-005 Synthetic set
- DEP-006 Labelled set
- DEP-007 Public texts
- DEP-008 Hosting and database
- DEP-009 Finale kit
- DEP-010 Mentor answers
- DEP-011 Base map
- RISK-001 – RISK-014 (one line each in §16)
- OQ-016 GD format
- OQ-017 District adjacency source
- OQ-018 Referral agency name
- OQ-019 Ethics of testing with real parents
- OQ-020 Cross-device return
- SCREEN-001 – SCREEN-016 (reserved, one line each in §19)
- TC-001 – TC-037 (reserved, one line each in §19)

IDs referenced

- GOAL-001 – GOAL-009 (00_Gap_Check_and_Registry)
- PERSONA-001 – PERSONA-005 (00_Gap_Check_and_Registry)

- OQ-001 – OQ-015 (00_Gap_Check_and_Registry)

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D2A / ENGINEERING / V2.0

Technical Requirements · Part A

Architecture, APIs, data model and authority · Authors: Sadman Shaharier and Obidit Islam

↗ Contents

Field	Value
Version	2.0
Date	11 Oct 2026 (Asia/Dhaka, UTC+6)
Owner	Sadman Shaharier (lead/integration)
Authors	Sadman Shaharier; Obidit Islam
Status	V2 team handover; implementation status UNVERIFIED
Split	Part A = §1–§9 (this file). Part B = §10–§19 plus `schema_v2.sql`, which must equal §7 here byte for byte.
Inputs	D1_PRD_V2.md; V2 registry; validation report; schema_v2.sql and fera_types_v2.ts.

V2 release contract

Authors: Sadman Shaharier and Obidit Islam. **Version:** 2.0. **Issued:** 11 October 2026, Asia/Dhaka (UTC+6).

This is the revised team specification, not evidence that the application has been built. Actual repository progress, account configuration, interviews, evaluations, and spend remain **UNVERIFIED** until evidence is attached. V1 defaults are superseded by the V2 rules below; existing requirement IDs are retained.

- This release is a **synthetic-data prototype**. Officer verification and referral delivery are simulated and labelled. The schema rejects records marked non-synthetic; the application must also prevent real reports from entering this release.
- A candidate score is similarity, not identity probability. A reviewer decision does not establish guardian authority or safe reunification. A referral row means **referral recorded**, not delivered or acknowledged.
- Draft payload stays on the device before report confirmation, but providers process audio/text beforehand with a notice. Only expiry/counter/hash metadata and cost reservations are stored server-side before submission.
- Possible duplicates remain visible. Shared phone alone never establishes one child. No physical child-return action is performed by Fera.
- Counts distinguish missing-report rows, found-report rows, and confirmed pairs. They are not a census of unique children or proof of rescue.

- Sources, diagrams, tables, code blocks, and acceptance criteria are part of the handover. The companion code must match the embedded code exactly. Team handover records change closure, evidence gaps, and release gates.

Changes to earlier documents

V2 freezes adapter, API, role DTO, draft-token, idempotency, matching-state, decision-transaction and schema contracts. API-021/022 close missing admin/district paths; API-023 specifies optional duplicate resolution. ENTITY-015–017 support counters and reservations. These changes implement the validation findings; the application remains unverified.

1 Technical summary

Item	Value
Shape	One Next.js App Router app in TypeScript: UI and API route handlers in a single deployable.
Data	PostgreSQL 16 with pgvector. One `cases` table holds missing and found records.
AI use	Speech-to-text (1 call per voice note) and field extraction (1–2 calls, ≤ 2 more for unparsed answers). Everything else is deterministic.
Matching	SQL hard filters → pgvector cosine on local embeddings → rule bonuses → template reason. No LLM in ranking.
Offline	Local embeddings, template outputs and a demo-safe cache let the scripted journey run with no internet (FR-027).
Budget	≤ 3 paid AI calls per typical case (NFR-016 cap = 6). Usage estimate against USD 30 is in Part B §12.

2 Stack

Layer	Choice	One-line justification
Frontend	Next.js App Router, React, TypeScript (strict), Tailwind CSS	Deck stack; one codebase for mobile parent and desktop officer screens.
Backend	Next.js route handlers, Node.js runtime	Same deployable as UI; Node runtime needed for audio parsing and local embeddings.
Validation	Zod	One schema validates requests, AI output and forms (PRD v2).
Database	PostgreSQL 16 + pgvector (exact filtered cosine baseline; optional HNSW)	Relational state machine plus vectors in one store (DECISION-002).
DB access	`pg` driver + `pgvector` helper + hand-written SQL	`schema_v2.sql` stays the single source; no ORM migration drift.

Layer	Choice	One-line justification
Auth	Auth.js credentials provider (staff, officer, admin); signed anonymous cookie (parents)	PRD v2 choice; parents need no account (D1 FR-037).
Speech-to-text	Whisper API behind `SpeechProvider` adapter	Deck choice; adapter allows swap or cache.
LLM	`LLMProvider` adapter with Gemini, OpenAI and Claude implementations; model IDs in env	No vendor lock-in (deck, PRD v2). Default provider: OQ-004.
Embeddings	Transformers.js running a local multilingual sentence model (candidate: multilingual-e5-small, 384-d)	Free, offline, no per-call cost (Assumption A-026, A-027).
Map	Leaflet + OpenStreetMap tiles; GeoJSON markers without tiles offline	Deck choice; free (DECISION-009).
Poster image	Browser canvas render to PNG	No server image library; Bangla font already loaded in the page (DECISION-015).
Hosting	Vercel (Round 2); laptop `next start` + local Postgres (finale fallback)	Free tier; local copy survives venue Wi-Fi failure (DECISION-010).
Managed DB	Free-tier Postgres host with pgvector	Host choice open (OQ-021, Assumption A-034).
Unit/ integration tests	Vitest	Fast TypeScript tests for normaliser, matcher, state guard.
End-to-end + a11y	Playwright + @axe-core/playwright	Scripted journey and WCAG checks in one runner.
Logging	pino JSON logs with a redaction list	Structured logs; names and phones redacted (NFR-010).
Evaluation	`tsx` scripts in `/eval`	One command per metric (Part B §17).

2.1 LLM adapter contract

The authoritative interfaces are in D5 section 6 and fera_types_v2.ts. ExtractInput contains kind, sourceText, normalisedText and explicit nowIso. Evidence quotes/spans refer to sourceText, never to a silently changed numeric spelling. The normaliser preserves a source-span map and records deterministic derivations. Unknown defaults and metadata are exempt from evidence; every non-null extracted/derived fact requires a supported span or an explicit reviewed manual field.

1. Parse strict runtime schemas; reject unknown keys. One repair retry is allowed and consumes a reserved paid call.
2. Check quote/span integrity, field/value support and normaliser derivation; absent/contradictory evidence becomes unknown for user review. A substring match alone is not semantic verification.
3. Providers have no tool execution, no URL fetching and no authority to decide identity or status. Input is delimited data; test prompt injection and contradictory statements.
4. Timeout/fallback each has its own budget reservation and draft counter increment. Propagate AbortSignal; uncertain provider usage retains its reservation until reconciled.

5. Provider/model/account/data terms are pending team evidence. Only synthetic inputs may reach primary or fallback endpoints in V2.

2.2 Speech adapter contract

SpeechProvider accepts Uint8Array, MIME and Bangla hint. Convert at the endpoint boundary if a provider SDK expects Buffer. Accept tested WebM/Opus, MP4 or WAV only within actual size/duration limits; Ogg requires verified endpoint support or explicit transcoding. Verify codec/duration server-side. A character-script threshold is a heuristic, not proof of language or intelligibility. Transcript remains editable; silence/noise/dialect/number tests are required.

2.3 Embedding adapter contract

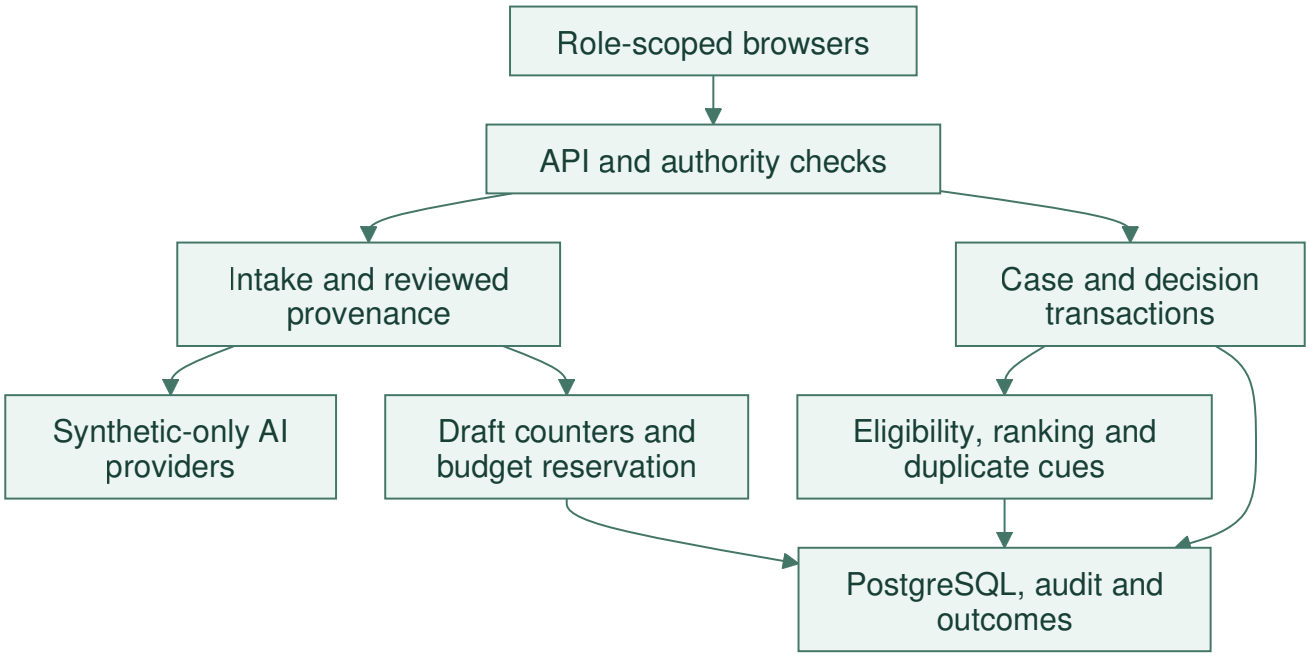
EmbeddingProvider returns exactly 384 finite values per text, L2-normalised; reject empty/zero/incorrect vectors. Pin model artifact revision, dtype, pooling, prefixes and normalisation in embedding_space. For symmetric report-to-report comparison, use the documented query prefix consistently on both sides. Exact filtered retrieval is the baseline. Switching space requires pausing matching, re-embedding every live eligible case, validating coverage, then atomically activating the new space. Never mix spaces; budget/terms review is required for an API fallback.

3 Architecture

3.1 System context

Actor / system	Interaction
Parent (mobile browser)	Creates missing cases; gets outputs. No account.
Staff (mobile/desktop)	Creates found cases.
Officer (desktop)	Reviews candidates, confirms/rejects, closes; views map.
Admin (desktop)	Verifies officers [MOCK], imports, redact identifying payloadss.
Speech provider	Receives audio; returns transcript.
LLM providers (2)	Receive normalised case text; return extraction JSON.
Map tile server	Serves base tiles to officer browsers only.
CID / MUN Alert	No integration. Families use 13219 themselves (D1 A-009).

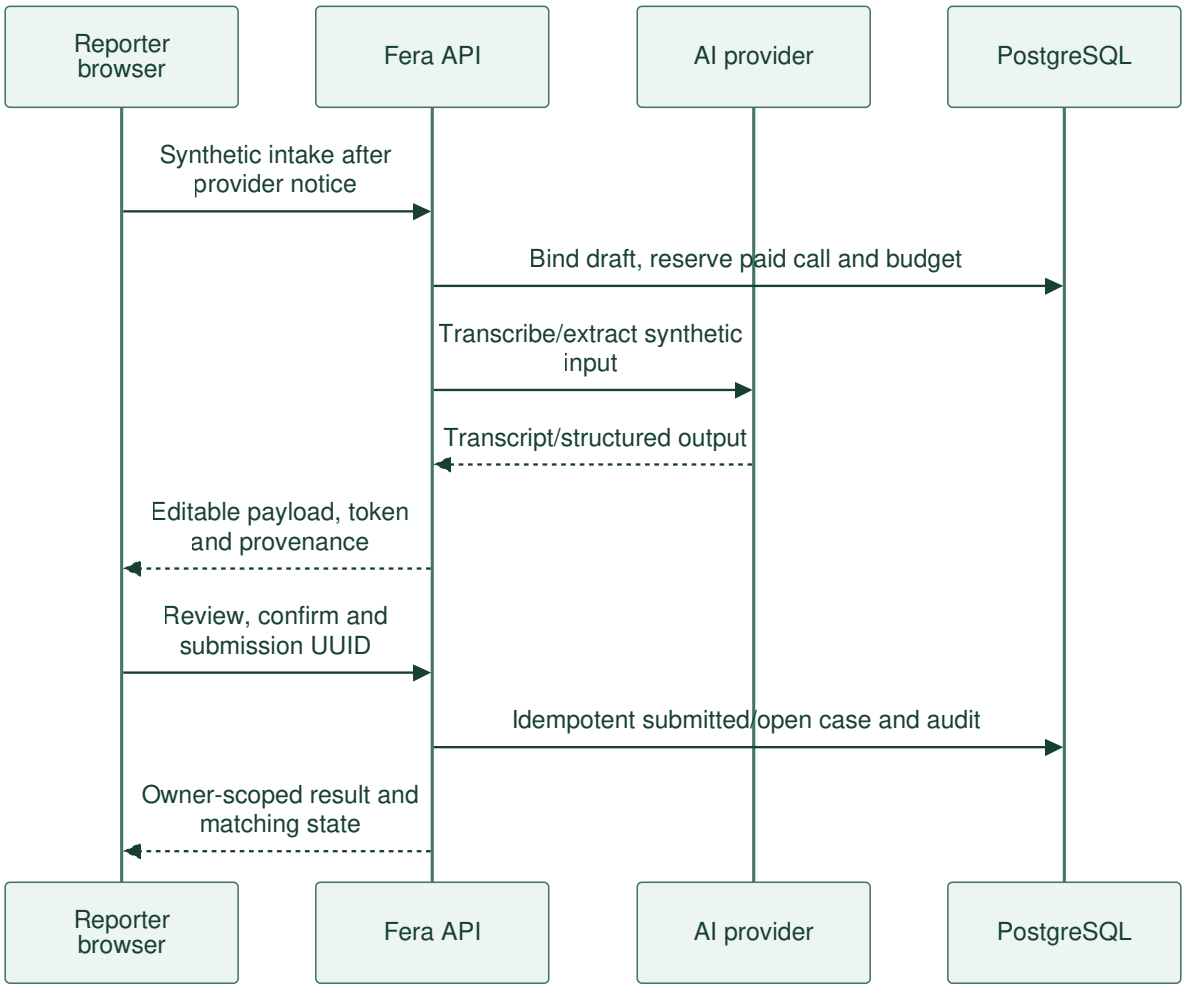
3.2 Architecture



V2 specification diagram · see adjacent rules for authority and evidence limits

Trust boundaries: device payload; signed-in or anonymous principal; server enforcement; database role; external provider. Optional public map tiles receive browser requests; the offline baseline uses bundled district outlines and a table.

3.3 Sequence: intake and submission



V2 specification diagram · see adjacent rules for authority and evidence limits

Candidate decisions are a separate transaction: authenticate and lock candidate/endpoints in sorted order; recheck versions/eligibility; record decision, confirmed pair, both statuses, simulated referral and safe audit; void competitors and reconcile affected queue states. Closure records a coordination outcome, not rescue. These operations have no external custody effect.

3.4 Data flow

1. Input arrives at an API route as audio, text or a form.
2. Audio → COMP-006 → transcript. Parent edits transcript client-side.
3. Text → COMP-008 normaliser (Bangla numerals, dates, phone) → COMP-007 extraction → evidence check → draft.
4. Missing required fields → templated Bangla question (COMP-009). Answers parsed by COMP-008 first; LLM only if unparsed.
5. Draft payload is held on the device before confirm. Only principal/expiry/hash/counter/budget metadata reaches the DB; outside providers process synthetic intake after a notice (BR-004).
6. Confirm → COMP-014 inserts `submitted`, embeds, runs COMP-013, sets `open`, runs COMP-012, writes audit.
7. Officer decisions go through COMP-014; the database trigger rejects illegal transitions.

8. Dashboard separates missing reports, found reports and confirmed pairs; all 64 districts appear, including zero rows (BR-014).

3.5 Trust boundaries

ID	Boundary	Trusted for	Not trusted for	Controls
TB1	Parent device	Holding its own draft	Any input content, session tampering	Zod validation, signed cookie, rate limits
TB2	Staff/officer device	Authenticated role actions	Role claims from client	Server-side role and `verified` check per request
TB3	Fera server	All business rules	—	Secrets in env only; state guard; audit
TB4	Third-party AI APIs	Returning text/ JSON	Correctness; data retention (OQ-013)	Zod parse, evidence check, synthetic data only until OQ-013 closes
TB5	Database host	Storage	Network exposure	TLS connection string; no public access; app role without DDL

3.6 Single points of failure

SPOF	Effect	Mitigation
Database	Whole app down	Local Postgres copy for finale; seed in < 2 min
Hosting region	App down	Laptop-local `next start` (DECISION-010)
Speech provider	No voice	Typed path; cache for scripted notes
LLM providers	No extraction	Fallback provider; manual form (FR-026); cache for scripted text
Venue network	Live calls fail	Phone on own mobile data; demo-safe mode
Demo laptop	No officer screens	Second team laptop with the same local setup

4 Component design

ID	Component	Responsibility	Inputs	Outputs	Depends on	Failure behaviour
COMP-001	Parent client	Screens SCREEN-001–006; draft in `localStorage`	User text, audio, edits	API calls	COMP-003, COMP-004	Keeps draft on network loss; retry button (NFR-008)
COMP-002	Staff/officer/ admin client	SCREEN-007–016	Session, user actions	API calls	COMP-004, COMP-018	Shows E-xxx message; no optimistic decisions
COMP-003	Voice recorder	Capture 5–120 s via MediaRecorder; show timer	Mic stream	Audio blob (webm/opus)	Browser mic, secure origin	No permission or no API → focus text box (E-022)
COMP-004	API layer	Route handlers; Zod; auth guard; rate limit; request ID	HTTP requests	JSON responses, error envelope	COMP-005, COMP-021	Uniform error envelope; never leaks stack traces
COMP-005	Auth and session	Auth.js credentials; parent cookie	Credentials, cookies	Session `{role, verified, userId}` or `{parentSessionId}`	DB `users`, `parent_sessions`	Expired → E-019 (parent) or E-002 (staff)
COMP-006	Speech adapter	Transcribe; script check; cache lookup	Audio, mime	Transcript	STT provider, COMP-017	Timeout 25 s → E-014; empty/ garbled → E-010; down → E-011
COMP-007	LLM adapter	Extraction via contract §2.1; retry; fallback	Normalised text	`ExtractOutput`	LLM providers, COMP-017	Invalid twice → E-012; down → E-013
COMP-008	Normaliser	Bangla numerals → ASCII; dates/time words → BST ISO; phone; district lookup (bn/en/alias); romanise names	Raw text, fields	Normalised text and fields	`districts` table, alias list	Unknown token left as text; never invents
COMP-009	Intake orchestrator	Merge extraction + answers; evidence check; required-field check; templated question; AI call counter (≤ 6)	Text, prior draft, answer	Draft, missing fields, next question	COMP-006–008	Counter reached → review screen with "unknown" fields

ID	Component	Responsibility	Inputs	Outputs	Depends on	Failure behaviour
COMP-010	Output generator	GD, poster text, 13219 summary from templates	Confirmed case	Bangla text	**COMP-016**	Missing field → "অজানা"; banned word → E-016
COMP-011	Embedding service	Build case description string; embed	Case fields	`vector(384)`	Transformers.js model files	Model load failure → case saved `open` without vector; matching uses rules only; logged
COMP-012	Matching engine	Hard filters (SQL), cosine (pgvector), bonuses, top 5, template reason	Case ID	`match_candidates` rows	DB, COMP-011	No candidates → case stays `open`
COMP-013	Duplicate detector	Apply D1 BR-021; write `duplicate_links`	New case	Links	DB	Errors logged; case still opens
COMP-014	Case service + state guard	Create, edit (versions), transitions, referral, close, redact identifying payloads	Commands	DB rows, audit	DB trigger, COMP-015	Illegal transition → E-005; transaction rollback
COMP-015	Audit logger	Append `audit_log` rows inside the same transaction	Action, actor, case	Row	DB	Audit write failure aborts the action
COMP-016	Wording guard	Block alert wording and "runaway" terms in Fera-generated text and labels	Generated text	Pass / E-016	Term list (bn + en)	Blocked text never shown; logged
COMP-017	Demo-safe cache	Fixtures for scripted inputs: transcript, extraction, embeddings	SHA-256 of normalised text or audio fixture ID	Stored output, `cached: true`	`/fixtures/*.json`	Miss → live call (or E-011/E-013 when offline)
COMP-018	Dashboard and map	Aggregate counts; Leaflet map; table view	Stats API	Map + accessible table	Leaflet, tiles, district centroids	Tiles fail → markers on blank background; table always shown
COMP-019	Admin tools	Seed loader (CLI), public import, redact identifying payloadsr, verify officer [MOCK]	Files, commands	Cases, updates	**COMP-014**	Import row error → skipped row reported; batch continues

ID	Component	Responsibility	Inputs	Outputs	Depends on	Failure behaviour
COMP-020	Evaluation harness	Compute 5 metrics; write report	Labelled sets	`eval/report-<date>.md`	COMP-007–013	Any metric error → report marks it "not measured"
COMP-021	Rate limiter	Fixed-window counters in Postgres	Key (session/ user/IP), action	Allow / E-007	DB `rate_limits`	DB error → allow and log (fail-open only for limiter)

Wording guard scope (COMP-016): it checks Fera-generated text and UI labels. A parent's own words in `circumstances` are stored as given but never copied into generated outputs.

5 API specification

5.1 Conventions

- Base path `/api`. JSON in and out, UTF-8. Times ISO 8601 with `+06:00`.
- Every response carries header `x-request-id`.
- Case data responses send `Cache-Control: no-store`.
- Error envelope (all endpoints):

```
{ "error": { "id": "E-012", "key": "LLM_INVALID_OUTPUT", "message_bn": "…", "message_en": "…", "retryable": false, "fallback": "manual_form" } }
```

5.2 Error IDs used by the API

Full user-facing messages and recovery actions are in Part B §13.

ID	Key	HTTP
E-001	VALIDATION_FAILED	400

ID	Key	HTTP
E-002	UNAUTHENTICATED	401
E-003	FORBIDDEN_ACTION	403
E-004	NOT_FOUND (also returned for records the caller may not see)	404
E-005	ILLEGAL_TRANSITION	409
E-006	SUBMISSION_PAYLOAD_CONFLICT	409
E-007	RATE_LIMITED	429
E-008	AUDIO_TOO_LARGE	413
E-009	AUDIO_LENGTH_OUT_OF_RANGE	422
E-010	SPEECH_UNINTELLIGIBLE	422
E-011	SPEECH_UNAVAILABLE	503
E-012	LLM_INVALID_OUTPUT	422
E-013	LLM_UNAVAILABLE	503
E-014	AI_TIMEOUT	504
E-015	UNSUPPORTED_LANGUAGE	422
E-016	OUTPUT_BLOCKED	422
E-017	REFERRAL_REQUIRED	409
E-018	EDIT_LOCKED	409
E-019	SESSION_EXPIRED	401
E-020	OUTPUT_REQUEST_LIMIT_RESERVED	429
E-021	INTERNAL	500
E-022	MIC_UNAVAILABLE (client-only, no HTTP)	—
E-023	AI_BUDGET_CAP (≤ 6 calls per draft reached)	429

5.3 Auth labels

Label	Meaning
`parent`	Valid signed parent-session cookie (created on first intake call)
`staff`	Auth.js session, role `staff`, or role `officer` with `verified=false`

Label	Meaning
`officer`	Auth.js session, role `officer`, `verified=true` [MOCK verification]
`admin`	Auth.js session, role `admin`
`public`	No session needed

5.4 Shared types

D5 section 6 and `fera_types_v2.ts` are authoritative. Runtime request/response validation is required in the application; TypeScript alone does not validate provider JSON or HTTP bodies. There is no shared unrestricted CaseView for every role.

Contract	Required boundary
CaseDraft	Reviewed editable missing/found payload; no signed token, AI counter or session metadata inside the draft
ExtractResponse / TranscribeResponse	Opaque draft token and expiry beside the payload; server-owned call count
OwnerCaseView / StaffOwnCaseView	Own payload for review/edit before lock; status/outputs afterward; no other-party evidence
OfficerCaseView	Verified active officer operational projection; contact access and reads audited
AdminCaseView	Identifier, status, district and redaction metadata only; no child payload, candidates or similarity scores
DecisionRequest	Discriminated confirm/reject; candidate plus expected endpoint versions; reject needs reason; confirm needs review checklist
DecisionResponse	Recorded simulated referral, explicit endpoint state/version; no receipt or custody claim
DashboardCounts	Missing-report counts, found-report counts and confirmed-pair count kept separate

Database numeric/bigint/timestamp/vector values require explicit boundary conversion. Pin tested dependencies and maintain strict validation examples alongside the app. The supplied TS file is a specification companion, not implementation of these validators.

5.5 Endpoints

API-001 · POST /api/intake/transcribe · Must

Field	Value
Purpose	Editable synthetic voice transcript; FR-002
Auth	Parent for missing; staff/unverified officer/verified officer for own found draft
Request	Multipart audio, kind, optional draft_token, duration_s hint, optional fixture_id in protected DEMO_MODE only

Field	Value
Response	TranscribeResponse: transcript, verified duration, language hint, cached, draft_id, draft_token, ai_calls_used
Validation	Server codec/size ≤2 MB and duration 5–120 s; synthetic/provider notice; ownership/token expiry; atomic paid-call reservation. Same-Origin CSRF check applies to multipart.
Failure	E-007–E-011/E-014/E-015/E-023; editable typed/manual path; no silent cached result

API-002 · POST /api/intake/extract · Must

Field	Value
Auth	Parent missing; staff/officer found only
Request	kind; text OR answer; prior CaseDraft and draft_token when continuing; reject expired/foreign/ replayed revision updates
Response	ExtractResponse includes opaque token, draft ID/revision/expiry, draft, missing_fields, next_question, calls_used and cached
Validation	Strict runtime schema; supported raw-source spans and derivations; six dispatched paid transcribe/ extract calls per draft; all retries/fallbacks count
Follow-up	One question at a time, at most four rounds; unresolved required values move to the manual form with explicit blocking fields
Errors	E-001/E-007/E-012/E-013/E-014/E-015/E-019/E-023

API-003 · POST /api/cases · Must

Field	Value
Auth	Parent session owner
Request	CreateMissingRequest: reviewed missing draft, confirmed=true, client_submission_id UUID, optional draft_token
Response	201 new OwnerCaseView; 200 identical UUID retry returns the same ref and replayed=true; outputs_available and matching_state
Validation	BR-015/016, server synthetic-only mode, own principal, canonical payload hash, same UUID/ different payload → E-005
Transaction	Unique(session, submission UUID); save reviewed payload as submitted then open, audit; output is owner-scoped. A same-hash/time-window warning never replaces UUID idempotency.
Matching	Bounded synchronous matching or durable queued state with retry via API-014; never an unawaited serverless background promise
Errors	E-001/E-005/E-006/E-007/E-019/E-021

API-004 · GET /api/cases/:id · Must

Field	Value
Auth	Ownership or verified operational role checked server-side
Response	Parent: OwnerCaseView including own editable draft only before terminal lock; staff: StaffOwnCaseView for own entry; verified officer: OfficerCaseView; admin: AdminCaseView metadata only
Privacy	No parent/staff candidate details or other-party report fields; no admin candidate ID/score or child payload
Errors	E-004 indistinguishable missing/forbidden; safe denial audit with nullable case FK

API-005 · PATCH /api/cases/:id · Should

Field	Value
Auth	Owning parent
Request	Reviewed draft, confirmed=true, expected_version
Response	OwnerCaseView with version+1 and current matching state
Transaction	Lock, verify editable state/version, snapshot, increment, void old pending pairs, reconcile affected reports, queue/retry rematching and audit
Errors	E-001/E-004/E-005/E-018; terminal reports locked

API-006 · POST /api/cases/:id/outputs · Must GD/poster; Should helpline

Field	Value
Auth	Owning parent
Request	type gd/poster/helpline; poster contact inclusion chosen in separate share preview
Response	OutputResponse; reviewed fixed template, draft label and ≤80-word helpline summary
Storage	Unique(case,type,version) returns cached deterministic output. Output for obsolete versions is never served as current.
Privacy	Download acknowledgement explains child/phone/location exposure and lack of remote recall
Errors	E-004/E-016/E-007; E-020 reserved

API-007 · POST /api/found-cases · Must

Field	Value
Auth	Staff or officer; owner is server-derived user ID
Request	Reviewed found draft, confirmed=true, client_submission_id UUID; optional own draft_token
Response	StaffOwnCaseView plus matching_state; no candidate details; officer can separately read queue

Field	Value
Validation	Description/place/district/event/contact required; age may be unknown; no photo
Idempotency	Unique(created_by_user_id,submission UUID), canonical hash, same UUID/different payload conflict; synthetic label server-owned
Errors	E-001/E-002/E-005/E-006/E-007

API-008 · GET /api/review-queue · Must

Field	Value
Auth	Active verified officer
Query	district?, limit 1–50, cursor?
Response	Own authorised operational queue with OfficerCaseView and up to five current qualified candidates; cursor; uncertain components visible
Rules	Flagged duplicates stay discoverable; no threshold-qualified candidates → explicit empty state
Errors	E-002/E-003

API-009 · GET /api/candidates/:id · Must

Field	Value
Auth	Active verified officer only
Response	CandidateDecisionView, raw supported field comparison, contradictions/unknowns, score components and versions
Privacy	No identity probability or safe return claim; authorised officer contact is not logged
Errors	E-002/E-004; missing and forbidden identical

API-010 · POST /api/candidates/:id/decision · Must

Field	Value
Auth	Active verified officer; author ID ignored if client supplied
Request	Discriminated DecisionRequest: reject requires reason 3–500; confirm requires reviewed checklist and both expected versions
Confirm transaction	Sorted endpoint locks, live version/eligibility/verification checks; decision, candidate confirmed, missing confirmed, found referred with closed_at, simulated referral, safe audit; void competitors and reconcile all affected reports
Reject transaction	Reasoned decision and rejected candidate; reconcile states when last pending candidate disappears; same-version rejection not re-suggested
Response	DecisionResponse with referral_state=recorded and referral_is_simulated=true

Field	Value
Errors	E-001/E-002/E-004/E-005; conflict returns current safe status; retry never creates a second decision

API-011 · POST /api/cases/:id/close · Must

Field	Value
Auth	Active verified officer
Request	outcome coordination_recorded or resolved_outside_fera; reviewed note 3–500; expected_version
Validation	coordination_recorded requires confirmed pair and its simulated referral; resolved_outside_fera requires open missing report. Both are synthetic coordination outcomes.
Transaction	Lock, record status/outcome/closed_at and officer audit; no physical child action
Response	Ref, closed status and coordination timestamp
Errors	E-001/E-004/E-005/E-017

API-012 · GET /api/dashboard/stats · Must

Field	Value
Auth	Verified officer or admin
Response	DashboardStatsResponse; counting_basis=report_rows_separated_by_kind; missing_reports, found_reports, confirmed_pairs; all 64 district rows including zeros
Privacy	Only aggregates; no child/source/contact payload
Count rule	One confirmed pair contributes one pair count; referral updates found category; missing closure changes only missing category
Errors	E-002/E-003

API-013 · GET /api/cases/:id/audit · Must

Field	Value
Auth	Verified officer operational access; admin safe audit metadata
Response	Timestamp, actor role, internal actor ID/display if authorised, action and allowlisted safe details
Privacy	No report/free-text/provider payload, phone/name, unmasked requested URL or raw network address
Errors	E-002/E-004

API-014 · POST /api/match/:caseId · Must

Field	Value
Auth	Verified officer, admin or trusted internal call
Response	RematchResponse only: case_id and matching_state. Candidate details are available separately through officer-only API-008/009.
Logic	Idempotent bounded work; model-space check; durable queued/running/complete/failed status; retry on known failure
Errors	E-004/E-005; terminal/redacted ineligible

API-015 · POST /api/admin/import · Should; synthetic samples only

Field	Value
Auth	Admin on protected synthetic deployment
Request	1–50 invented test texts with example.invalid source URLs and explicit synthetic annotation; kind and fixture intake time
Validation	No real public child reports in V2; no URL fetch; prompt injection is data; model terms and budget enforced
Response	Imported/skipped counts and flagged duplicate counts; no child/candidate payload
Errors	E-001/E-002/E-003/E-023

API-016 · POST /api/admin/anonymise · Must; legacy route name

Field	Value
Auth	Admin route plus restricted redaction DB role
Request	Exactly one of case_ids or all_eligible; explicit typed action confirmation
Response	redacted, skipped_not_eligible; mode=pseudonymisation; external_copies_removed=false
Logic	Call redact_case: direct identifiers, exact age, related outcome/referral/decision/candidate notes, payload provenance/hashes, vectors/model metadata, snapshots and outputs cleared; audit IDs/ aggregate fields retained
Eligibility	Missing closed; found referred only when its confirmed missing report is closed; otherwise preserve evidence
Disclosure	Legacy anonymised_at records redaction time. Backups, browser exports and provider copies follow separate retention policies.
Errors	E-001/E-002/E-003/E-018

API-017 · POST /api/admin/users/:id/verify · Must [MOCK]

Field	Value
Auth	Admin, synthetic deployment only

Field	Value
Request	verified boolean
Response	ID, officer role, verified flag; no password hash
Rules	Real-world onboarding/identity verification is a pilot gate; verification revocation checked per officer request
Errors	E-001/E-003/E-004

API-018 · GET /api/my-case · Should

Field	Value
Auth	Parent session owner
Response	Owner report refs/status/updated/editable; own details via API-004 only. No candidates or found details.
Dependency	SCREEN-006 and owner edit are Should together; their minimal owner-read API-004 is Must
Errors	**E-019**

API-019 · GET /api/health · Must

Field	Value
Auth	Public, no data
Response	App version, DB/embedding readiness, synthetic/demo mode; no credentials/provider raw error
Use	Readiness evidence, not a disclosure of sensitive topology

API-020 · /api/auth/ · Must*

Field	Value
Auth	Public sign-in; seeded accounts protected from public paid-API abuse
Request	Credentials username/password
Response	8-hour credential session; httpOnly, Secure on production HTTPS, SameSite=Lax, __Host- path rules
Errors	Generic E-002; E-007 after five failures/min hashed network key
Local	Separate browser profiles for staff/officer; local HTTP cookie/mic behaviour tested or use laptop-only scenario

API-021 · GET /api/admin/users · Must for admin verification screen

Field	Value
Auth	Admin
Query	Cursor and limit 1–50
Response	ID, username, display_name, role, verified, is_active; no hash, secrets or child data
Covers	FR-035, SCREEN-013

API-022 · GET /api/cases · Must for district drill-through

Field	Value
Auth	Active verified officer
Query	District, kind, status-group, cursor, limit 1–50
Response	OfficerCaseView rows with no candidate leakage to other roles
Covers	FR-019, SCREEN-011/012

API-023 · POST /api/duplicates/:id/decision · Could

Field	Value
Auth	Active verified officer
Request	DuplicateDecisionRequest: confirmed_same/dismissed, reason and profile_evidence_reviewed
Validation	Same-kind endpoints, explicit profile evidence; same_phone alone cannot confirm_same; no deletion or automatic hidden report
Response	Link status and safe audit ref; grouping preserves original access
Covers	FR-023, SCREEN-014

5.6 Transaction and DTO acceptance

Route handlers use the D5 type shapes and runtime validators, explicit field allowlists, request IDs, no-store and same-origin checks. SQL constraints are defence in depth, not an authorisation substitute. Return a stable 200 replay for the same submission UUID; no duplicate rows. Canonical submission hashes include reviewed values/intervals but no device-specific timestamps. Denials never expose existence or fail because the requested case FK does not exist. Always recheck the live report versions before a decision.

6 Data model

6.1 Entities

ID	Entity	Table	Purpose
ENTITY-001	User	`users`	Staff, officer and admin accounts
ENTITY-002	ParentSession	`parent_sessions`	Anonymous parent identity for own cases
ENTITY-003	Case	`cases`	Missing and found records
ENTITY-004	CaseVersion	`case_versions`	Edit history snapshots
ENTITY-005	MatchCandidate	`match_candidates`	Ranked pairs
ENTITY-006	ReviewDecision	`review_decisions`	Officer confirm/reject
ENTITY-007	Referral	`referrals`	Child-protection referral record
ENTITY-008	DuplicateLink	`duplicate_links`	Possible same child
ENTITY-009	AuditEntry	`audit_log`	Append-only trail
ENTITY-010	District	`districts`	64 districts, names, centroid
ENTITY-011	DistrictNeighbour	`district_neighbours`	Adjacency for search widening
ENTITY-012	GeneratedOutput	`generated_outputs`	GD/poster/summary texts and regeneration count
ENTITY-013	AiCall	`ai_calls`	AI usage and cost tracking, no PII
ENTITY-014	RateLimitBucket	`rate_limits`	Fixed-window counters
ENTITY-015	IntakeDraft	`intake_drafts`	Principal/expiry/hash and paid-call count; no child payload
ENTITY-016	AiBudget	`ai_budget`	One committed/reserved cost cap
ENTITY-017	AiReservation	`ai_reservations`	Upper bound, settlement and uncertain usage

6.2 Key fields (ENTITY-003 Case)

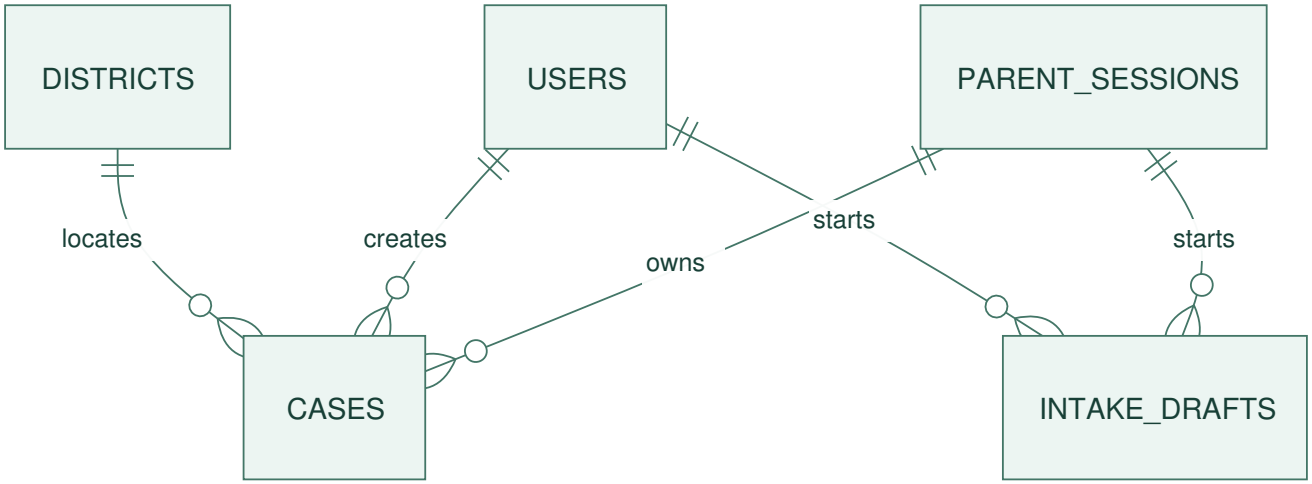
Fields	Type / requirement	V2 interpretation
id, ref_code	UUID; unique M/F code	Stable report reference, not child identity

Fields	Type / requirement	V2 interpretation
kind, status, version	Kind, persisted state, positive version	Immutable kind; forward audited state transitions; optimistic edit/decision checks
source, source_url, is_synthetic	Enum, optional fixture URL, true	Public real-child imports disabled; synthetic input guard required beyond this flag
reporter_session_id / created_by_user_id	Parent session / staff author	Scoped ownership; no owner authority from client body
client_submission_id, submission_payload_hash	UUID + SHA-256 required for parent/staff	Unique per principal; application returns stable retry/conflict response
raw_text, child_name, child_name_latin	Optional bounded strings	Reviewed source and names; no invented name; identifying payload
age_years, age_min, age_max	Missing: complete 0–17 exact/ range; found: unknown or 0–25 estimate	Exact age equals both endpoints; found unknown is valid
sex	male/female/other/unknown	Explicit unknown remains uncertain evidence
height_cm, clothing, marks, circumstances	Optional bounded facts; found description required	Never fill absent facts merely to increase similarity
event_at, event_at_lower, event_at_upper, event_at_approx	Reviewed display instant plus consistent bounds	Exact event has equal bounds; approximate eligibility uses intervals
district_code, place_text, contact_phone	Required non-redacted submission values	Internal district 1–64, non-empty place, normalised 01 phone; not verified contact ownership
thana, reporter_relation, gd_number	Optional reviewed facts	Subdistrict/provenance minimisation; no document image
provenance	JSON field-span/derivation mapping	Case payload after submit only; no free notes in safe audit
embedding, embedding_model, embedding_revision, embedding_space	Optional 384-vector with pinned space metadata	Reliable matching can fail/queue without losing saved report; never mix spaces
matching_state, matched_at, match_error_id	queued/running/complete/failed, time, safe error	Bounded retry; no detached serverless promise
input_hash	Optional deterministic input hash	Not a child identity or the submission idempotency key
outcome, outcome_note, closed_at	Coordination outcome and terminal time	Closure is not rescue; notes are identifying payload and are cleared by redaction
anonymised_at	Legacy timestamp name	Payload redaction/pseudonymisation, not proven irreversible anonymity
created_at, updated_at	Timestamps	Encode explicit timezone at API boundary

No precise per-case latitude/longitude (DECISION-013). The authoritative column/check definitions are in section 7. Redaction clears identifiers, source/provenance/hash/vector metadata, notes, snapshots and generated outputs; district/date/status and coarse age remain restricted pseudonymous data.

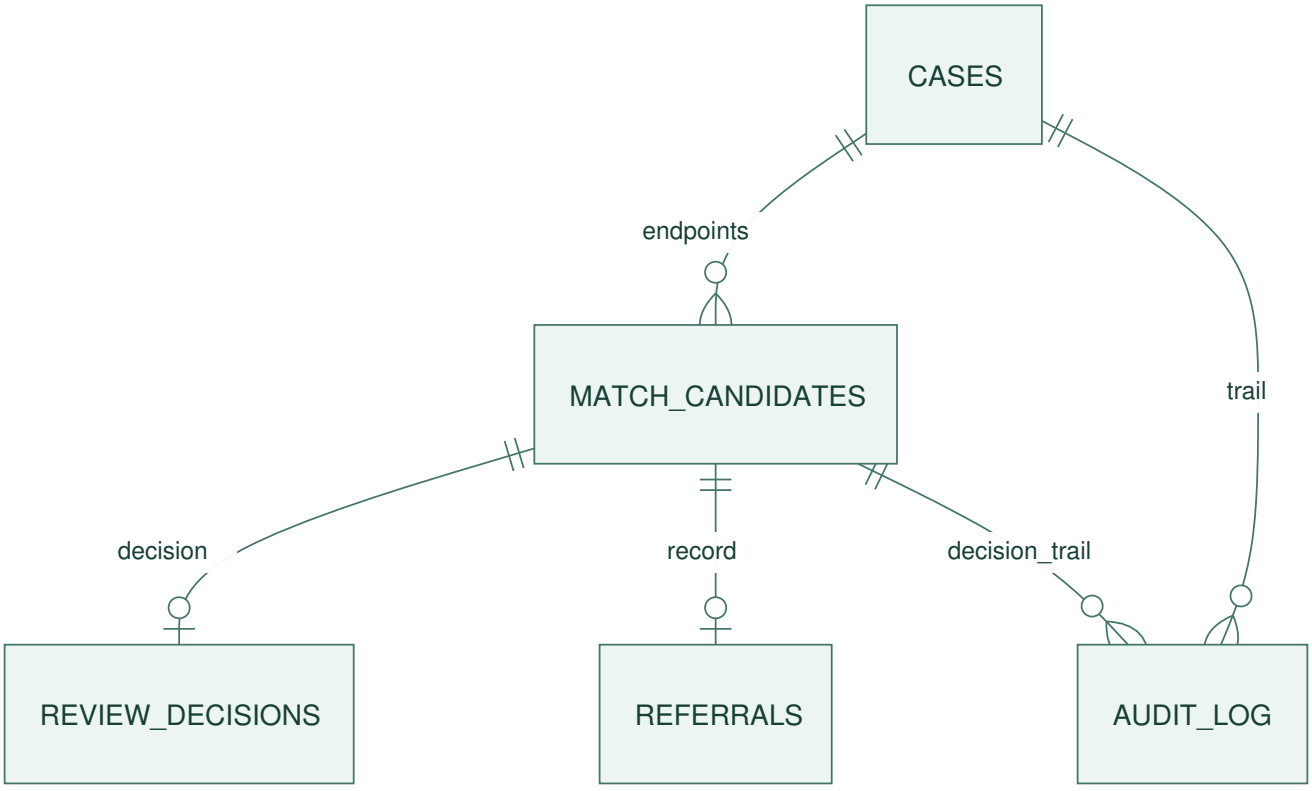
6.3 Relationships

Intake and ownership relationships



V2 specification diagram · see adjacent rules for authority and evidence limits

Candidate decision and audit relationships



V2 specification diagram · see adjacent rules for authority and evidence limits

Additional relationship	Cardinality / rule
District → district_neighbours	One-to-many outgoing rows; no self-neighbour
Case → case_versions / generated_outputs	One-to-many; versioned snapshots/templates, cleared on redaction
Case ↔ duplicate_links	Each link references two ordered same-kind endpoints; reports retained
User → review_decisions / referrals	One-to-many authors; decision author must be active verified officer at decision time
User / parent_session → audit_log	Optional actor references; safe audit remains after payload redaction
Intake draft → ai_reservations / ai_calls	One-to-many metadata and usage links; no child payload in draft metadata
Reservation → ai_calls	At most one usage row per reservation
Case → ai_calls	Optional case link after submission
ai_budget singleton	Global reserved/committed cap enforced transactionally; no per-child identity

6.4 Indexes and constraints (summary)

Table	Index / constraint	Purpose
cases	`(kind, status, district_code)`	Hard filters, map counts
cases	`(event_at)`	Time filter
cases	scoped submission UUID unique	Race-safe idempotency
cases	HNSW `embedding vector_cosine_ops`	Nearest neighbours
cases	CHECKs: age ranges, kind/status pairs, closed ⇒ outcome, payload-redacted ⇒ PII null	Integrity
cases	Trigger `enforce_case_transition`	D1 BR-012, BR-013 at DB level
match_candidates	Unique `(missing_case_id, found_case_id, missing_version, found_version)`	BR-019 via versions
match_candidates	Partial unique on each side `WHERE status='confirmed'`	BR-020 one-to-one
audit_log	Triggers block UPDATE/DELETE/TRUNCATE; restricted runtime grants	**NFR-017**
duplicate_links	`CHECK (case_a < case_b)`, unique pair	No double links

7 Database schema (PostgreSQL 16 + pgvector >= 0.8.1)

The fresh-database companion schema_v2.sql is exact. It is not an in-place V1 migration. Non-synthetic rows are rejected; application input safeguards remain required. The runtime role must not own/alter tables. The validation report distinguishes the portable PostgreSQL execution check from the still-required target-host/role/concurrency checks.

```
-- Fera schema V2.0 - D2A section 7. PostgreSQL 16, pgvector ⩾ 0.8.1.
-- Authors: Sadman Shaharier; Obidit Islam. Date: 11 October 2026, Asia/Dhaka.
-- Fresh synthetic-demo database only. This is not an in-place V1 migration.
-- Rows marked non-synthetic are blocked by cases_synthetic_release_only.
-- Application runtime must use restricted credentials, never the schema owner.
BEGIN;

CREATE EXTENSION IF NOT EXISTS vector;

-- Enums ---------------------------------------------------------------
CREATE TYPE case_kind        AS ENUM ('missing', 'found');
CREATE TYPE case_status      AS ENUM ('submitted', 'open', 'matched_pending_review', 'confirmed',
'closed', 'referred');
CREATE TYPE sex_t            AS ENUM ('male', 'female', 'other', 'unknown');
CREATE TYPE source_t         AS ENUM ('parent', 'staff', 'public_import', 'synthetic');
CREATE TYPE user_role        AS ENUM ('staff', 'officer', 'admin');
CREATE TYPE candidate_status AS ENUM ('pending', 'confirmed', 'rejected', 'voided', 'superseded');
CREATE TYPE decision_t       AS ENUM ('confirm', 'reject');
CREATE TYPE outcome_t        AS ENUM ('coordination_recorded', 'resolved_outside_fera');
CREATE TYPE duplicate_status AS ENUM ('flagged', 'confirmed_same', 'dismissed');
CREATE TYPE matching_state_t AS ENUM ('queued', 'running', 'complete', 'failed');
CREATE TYPE output_t         AS ENUM ('gd', 'poster', 'helpline');
CREATE TYPE actor_t          AS ENUM ('parent', 'staff', 'officer', 'admin', 'system');

-- Shared trigger: updated_at ------------------------------------------
CREATE FUNCTION set_updated_at() RETURNS trigger LANGUAGE plpgsql AS $$
BEGIN
  NEW.updated_at := now();
  RETURN NEW;
END $$;

-- Reference data ------------------------------------------------------
CREATE TABLE districts (
  code        smallint PRIMARY KEY CHECK (code BETWEEN 1 AND 64),
  name_en     text NOT NULL UNIQUE,
  name_bn     text NOT NULL UNIQUE,
  division_en text NOT NULL,
  lat         numeric(8,5) NOT NULL CHECK (lat BETWEEN 20 AND 27),
  lng         numeric(8,5) NOT NULL CHECK (lng BETWEEN 88 AND 93)
);

CREATE TABLE district_neighbours (
  district_code  smallint NOT NULL REFERENCES districts(code),
  neighbour_code smallint NOT NULL REFERENCES districts(code),
  PRIMARY KEY (district_code, neighbour_code),
  CHECK (district_code <> neighbour_code)
);

-- Identities ----------------------------------------------------------
CREATE TABLE users (
```

```
  id            uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  username      text NOT NULL UNIQUE CHECK (username ~ '^[a-z0-9._-]{3,32}$'),
  display_name  text NOT NULL CHECK (length(display_name) BETWEEN 2 AND 80),
  password_hash text NOT NULL,
  role          user_role NOT NULL,
  verified      boolean NOT NULL DEFAULT false,
  district_code smallint REFERENCES districts(code),
  is_active     boolean NOT NULL DEFAULT true,
  created_at    timestamptz NOT NULL DEFAULT now(),
  updated_at    timestamptz NOT NULL DEFAULT now(),
  CHECK (role = 'officer' OR verified = false)
);
CREATE TRIGGER users_updated_at BEFORE UPDATE ON users
  FOR EACH ROW EXECUTE FUNCTION set_updated_at();

CREATE TABLE parent_sessions (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  created_at   timestamptz NOT NULL DEFAULT now(),
  last_seen_at timestamptz NOT NULL DEFAULT now(),
  expires_at   timestamptz NOT NULL DEFAULT (now() + interval '30 days'),
  revoked_at   timestamptz,
  CHECK (expires_at > created_at)
);

-- Draft metadata (no transcript or case payload before submission) -----
CREATE TABLE intake_drafts (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  session_id uuid REFERENCES parent_sessions(id),
  user_id uuid REFERENCES users(id),
  kind case_kind NOT NULL,
  created_at timestamptz NOT NULL DEFAULT now(),
  expires_at timestamptz NOT NULL DEFAULT (now() + interval '24 hours'),
  payload_hash text CHECK (payload_hash ~ '^[0-9a-f]{64}$'),
  ai_calls_used smallint NOT NULL DEFAULT 0 CHECK (ai_calls_used BETWEEN 0 AND 6),
  revision integer NOT NULL DEFAULT 1 CHECK (revision ⩾ 1),
  CHECK ((session_id IS NOT NULL) <> (user_id IS NOT NULL)),
  CHECK (expires_at > created_at)
);

-- Cases ----------------------------------------------------------------
CREATE TABLE cases (
  id                  uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  ref_code            text NOT NULL UNIQUE CHECK (ref_code ~ '^[MF]-[0-9A-Z]{6}$'),
  kind                case_kind NOT NULL,
  status              case_status NOT NULL,
  source              source_t NOT NULL,
  source_url          text,
  is_synthetic        boolean NOT NULL DEFAULT true,
  reporter_session_id uuid REFERENCES parent_sessions(id) ON DELETE SET NULL,
  created_by_user_id  uuid REFERENCES users(id),
  raw_text            text CHECK (length(raw_text) ⩽ 4000),
  child_name          text CHECK (length(child_name) ⩽ 80),
  child_name_latin    text CHECK (length(child_name_latin) ⩽ 80),
  age_years           smallint CHECK (age_years BETWEEN 0 AND 17),
  age_min             smallint CHECK (age_min BETWEEN 0 AND 25),
  age_max             smallint CHECK (age_max BETWEEN 0 AND 25),
  sex                 sex_t NOT NULL DEFAULT 'unknown',
  height_cm           smallint CHECK (height_cm BETWEEN 40 AND 220),
  clothing            text CHECK (length(clothing) ⩽ 300),
```

```
  marks               text CHECK (length(marks) ⩽ 300),
  circumstances       text CHECK (length(circumstances) ⩽ 500),
  event_at            timestamptz,
  event_at_approx     boolean NOT NULL DEFAULT false,
  event_at_lower      timestamptz,
  event_at_upper      timestamptz,
  provenance          jsonb NOT NULL DEFAULT '{}',
  embedding_model     text,
  embedding_revision  text,
  embedding_space     text,
  matching_state      matching_state_t NOT NULL DEFAULT 'queued',
  matched_at          timestamptz,
  match_error_id      text,
  place_text          text CHECK (length(place_text) ⩽ 200),
  district_code       smallint REFERENCES districts(code),
  thana               text CHECK (length(thana) ⩽ 80),
  reporter_relation   text CHECK (length(reporter_relation) ⩽ 40),
  contact_phone       text CHECK (contact_phone ~ '^01[0-9]{9}$'),
  gd_number           text CHECK (length(gd_number) ⩽ 40),
  embedding           vector(384),
  client_submission_id uuid,
  submission_payload_hash text CHECK (submission_payload_hash ~ '^[0-9a-f]{64}$'),
  input_hash          text CHECK (input_hash ~ '^[0-9a-f]{64}$'),
  version             integer NOT NULL DEFAULT 1 CHECK (version ⩾ 1),
  outcome             outcome_t,
  outcome_note        text CHECK (length(outcome_note) ⩽ 500),
  closed_at           timestamptz,
  anonymised_at       timestamptz,
  created_at          timestamptz NOT NULL DEFAULT now(),
  updated_at          timestamptz NOT NULL DEFAULT now(),
  CONSTRAINT cases_synthetic_release_only CHECK (is_synthetic),
  CHECK ((age_min IS NULL) = (age_max IS NULL)),
  CHECK (age_min IS NULL OR age_min ⩽ age_max),
  CHECK (age_years IS NULL OR (age_min IS NOT NULL AND age_min = age_years AND age_max =
age_years)),
  CHECK (kind = 'found' OR age_min IS NULL OR age_min ⩽ 17),
  CHECK (event_at_lower IS NULL OR event_at_upper IS NULL OR event_at_lower ⩽ event_at_upper),
  CHECK (event_at IS NULL OR (event_at_lower IS NOT NULL AND event_at_upper IS NOT NULL
         AND event_at BETWEEN event_at_lower AND event_at_upper)),
  CHECK (event_at_approx OR event_at IS NULL OR (event_at = event_at_lower AND event_at =
event_at_upper)),
  CHECK (embedding IS NULL OR (embedding_model IS NOT NULL AND embedding_revision IS NOT NULL AND
embedding_space IS NOT NULL)),
  CHECK (source <> 'parent' OR anonymised_at IS NOT NULL OR
        (kind = 'missing' AND reporter_session_id IS NOT NULL AND client_submission_id IS NOT NULL
         AND submission_payload_hash IS NOT NULL)),
  CHECK (source <> 'staff' OR anonymised_at IS NOT NULL OR
        (created_by_user_id IS NOT NULL AND client_submission_id IS NOT NULL
         AND submission_payload_hash IS NOT NULL)),
  CHECK (anonymised_at IS NOT NULL OR
        (event_at IS NOT NULL AND event_at_lower IS NOT NULL AND event_at_upper IS NOT NULL
         AND district_code IS NOT NULL AND contact_phone IS NOT NULL AND place_text IS NOT NULL AND
length(place_text) ⩾ 1)),
  CHECK (kind <> 'missing' OR anonymised_at IS NOT NULL OR age_min IS NOT NULL),
  CHECK (kind <> 'found' OR anonymised_at IS NOT NULL OR
        length(trim(concat_ws(' ', circumstances, clothing, marks))) ⩾ 3),
  CHECK (kind = 'found' OR age_max IS NULL OR age_max ⩽ 17),
  CHECK (kind = 'missing' OR status NOT IN ('submitted', 'confirmed', 'closed')),
  CHECK (kind = 'found' OR status <> 'referred'),
```

```
  CHECK ((ref_code LIKE 'M-%') = (kind = 'missing')),
  CHECK (source <> 'public_import' OR anonymised_at IS NOT NULL OR source_url IS NOT NULL),
  CHECK (source <> 'synthetic' OR is_synthetic),
  CHECK ((status IN ('closed', 'referred')) = (closed_at IS NOT NULL)),
  CHECK (status <> 'closed' OR outcome IS NOT NULL),
  CHECK (anonymised_at IS NULL OR status IN ('closed', 'referred')),
  CHECK (anonymised_at IS NULL OR (child_name IS NULL AND child_name_latin IS NULL
         AND contact_phone IS NULL AND raw_text IS NULL AND place_text IS NULL AND embedding IS
NULL
         AND height_cm IS NULL AND clothing IS NULL AND marks IS NULL AND circumstances IS NULL
         AND thana IS NULL AND reporter_relation IS NULL AND gd_number IS NULL AND source_url IS
NULL
         AND age_years IS NULL AND outcome_note IS NULL AND reporter_session_id IS NULL
         AND created_by_user_id IS NULL AND input_hash IS NULL AND submission_payload_hash IS NULL
         AND client_submission_id IS NULL AND provenance = '{}'::jsonb
         AND embedding_model IS NULL AND embedding_revision IS NULL AND embedding_space IS NULL)),
  UNIQUE (reporter_session_id, client_submission_id),
  UNIQUE (created_by_user_id, client_submission_id)
);
CREATE TRIGGER cases_updated_at BEFORE UPDATE ON cases
  FOR EACH ROW EXECUTE FUNCTION set_updated_at();

CREATE INDEX cases_kind_status_district_idx ON cases (kind, status, district_code);
CREATE INDEX cases_event_at_idx             ON cases (event_at);
CREATE INDEX cases_session_hash_idx         ON cases (reporter_session_id, input_hash);
-- Optional index; V2 demo matcher uses exact filtered search first.
CREATE INDEX cases_embedding_hnsw_idx       ON cases USING hnsw (embedding vector_cosine_ops);

-- State machine guard (D1 BR-012, BR-013) ------------------------------
CREATE FUNCTION enforce_case_transition() RETURNS trigger LANGUAGE plpgsql AS $$
DECLARE step text;
BEGIN
  IF TG_OP = 'INSERT' THEN
    IF (NEW.kind = 'missing' AND NEW.status <> 'submitted') OR
       (NEW.kind = 'found' AND NEW.status <> 'open') THEN
      RAISE EXCEPTION 'invalid initial case status' USING ERRCODE = 'check_violation';
    END IF;
  ELSE
    IF NEW.kind <> OLD.kind THEN
      RAISE EXCEPTION 'case kind is immutable' USING ERRCODE = 'check_violation';
    END IF;
    IF OLD.anonymised_at IS NOT NULL AND
       (to_jsonb(NEW) - 'updated_at') IS DISTINCT FROM (to_jsonb(OLD) - 'updated_at') THEN
      RAISE EXCEPTION 'redacted case is immutable' USING ERRCODE = 'check_violation';
    END IF;
    IF OLD.status IN ('confirmed','closed','referred') AND NEW.anonymised_at IS NULL AND
       (to_jsonb(NEW) - ARRAY['status','closed_at','outcome','outcome_note','updated_at']) IS
DISTINCT FROM
       (to_jsonb(OLD) - ARRAY['status','closed_at','outcome','outcome_note','updated_at']) THEN
      RAISE EXCEPTION 'terminal report fields are locked' USING ERRCODE = 'check_violation';
    END IF;
    IF OLD.status IN ('closed','referred') AND NEW.anonymised_at IS NULL AND
       (to_jsonb(NEW) - 'updated_at') IS DISTINCT FROM (to_jsonb(OLD) - 'updated_at') THEN
      RAISE EXCEPTION 'terminal outcome is locked' USING ERRCODE = 'check_violation';
    END IF;
    step := OLD.status::text || '>' || NEW.status::text;
    IF NEW.status <> OLD.status AND NOT (
      (OLD.kind = 'missing' AND step = ANY (ARRAY[
      'submitted>open','open>matched_pending_review','matched_pending_review>open',
```

```
      'matched_pending_review>confirmed','confirmed>closed','open>closed'])) OR
      (OLD.kind = 'found' AND step = ANY (ARRAY[

'open>matched_pending_review','matched_pending_review>open','matched_pending_review>referred']))
    ) THEN
      RAISE EXCEPTION 'illegal transition % for % case', step, OLD.kind USING ERRCODE =
'check_violation';
    END IF;
    -- Redaction may alter only identifying fields; it may not rewrite the lifecycle.
    IF NEW.anonymised_at IS NOT NULL AND OLD.anonymised_at IS NULL AND
       (NEW.status <> OLD.status OR NEW.outcome IS DISTINCT FROM OLD.outcome OR
        NEW.closed_at IS DISTINCT FROM OLD.closed_at OR NEW.kind <> OLD.kind OR
        NEW.ref_code <> OLD.ref_code OR NEW.district_code IS DISTINCT FROM OLD.district_code OR
        NEW.event_at IS DISTINCT FROM OLD.event_at OR NEW.sex <> OLD.sex) THEN
      RAISE EXCEPTION 'redaction must preserve aggregate lifecycle fields' USING ERRCODE =
'check_violation';
    END IF;
  END IF;
  IF NEW.anonymised_at IS NULL AND NEW.event_at_lower > clock_timestamp() THEN
    RAISE EXCEPTION 'event interval starts in future' USING ERRCODE = 'check_violation';
  END IF;
  RETURN NEW;
END $$;
CREATE TRIGGER cases_transition_guard BEFORE INSERT OR UPDATE ON cases
  FOR EACH ROW EXECUTE FUNCTION enforce_case_transition();

CREATE TABLE case_versions (
  id        uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  case_id   uuid NOT NULL REFERENCES cases(id),
  version   integer NOT NULL CHECK (version ⩾ 1),
  snapshot  jsonb,
  edited_by actor_t NOT NULL,
  edited_at timestamptz NOT NULL DEFAULT now(),
  UNIQUE (case_id, version)
);

-- Matching ------------------------------------------------------------
CREATE TABLE match_candidates (
  id              uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  missing_case_id uuid NOT NULL REFERENCES cases(id),
  found_case_id   uuid NOT NULL REFERENCES cases(id),
  missing_version integer NOT NULL CHECK (missing_version ⩾ 1),
  found_version   integer NOT NULL CHECK (found_version ⩾ 1),
  score           numeric(5,2) NOT NULL CHECK (score BETWEEN 0 AND 100),
  rank            smallint NOT NULL CHECK (rank BETWEEN 1 AND 5),
  reason_bn       text NOT NULL CHECK (length(reason_bn) ⩽ 200),
  reason_en       text NOT NULL CHECK (length(reason_en) ⩽ 200),
  components      jsonb NOT NULL,
  status          candidate_status NOT NULL DEFAULT 'pending',
  created_at      timestamptz NOT NULL DEFAULT now(),
  decided_at      timestamptz,
  CHECK (missing_case_id <> found_case_id),
  CHECK ((status = 'pending') = (decided_at IS NULL)),
  UNIQUE (missing_case_id, found_case_id, missing_version, found_version)
);
CREATE INDEX match_candidates_missing_status_idx ON match_candidates (missing_case_id, status);
CREATE INDEX match_candidates_found_status_idx   ON match_candidates (found_case_id, status);
CREATE UNIQUE INDEX match_candidates_one_confirmed_missing ON match_candidates (missing_case_id)
WHERE status = 'confirmed';
```

```
CREATE UNIQUE INDEX match_candidates_one_confirmed_found   ON match_candidates (found_case_id)
WHERE status = 'confirmed';

CREATE TABLE review_decisions (
  id           uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  candidate_id uuid NOT NULL UNIQUE REFERENCES match_candidates(id),
  officer_id   uuid NOT NULL REFERENCES users(id),
  decision     decision_t NOT NULL,
  reason       text CHECK (length(reason) ⩽ 500),
  decided_at   timestamptz NOT NULL DEFAULT now(),
  officer_verified_at_decision boolean NOT NULL DEFAULT true CHECK (officer_verified_at_decision),
  CHECK (decision = 'confirm' OR (reason IS NOT NULL AND length(reason) BETWEEN 3 AND 500))
);

CREATE TABLE referrals (
  id              uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  candidate_id    uuid NOT NULL UNIQUE REFERENCES match_candidates(id),
  found_case_id   uuid NOT NULL REFERENCES cases(id),
  missing_case_id uuid NOT NULL REFERENCES cases(id),
  agency_name     text NOT NULL DEFAULT 'শিশু সুরক্ষা সেবা' CHECK (length(agency_name) ⩽ 120),
  referred_by     uuid NOT NULL REFERENCES users(id),
  referred_at     timestamptz NOT NULL DEFAULT now(),
  record_state    text NOT NULL DEFAULT 'recorded' CHECK (record_state = 'recorded'),
  is_simulated    boolean NOT NULL DEFAULT true CHECK (is_simulated),
  note            text CHECK (length(note) ⩽ 500)
);

CREATE TABLE duplicate_links (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  case_a      uuid NOT NULL REFERENCES cases(id),
  case_b      uuid NOT NULL REFERENCES cases(id),
  score       numeric(5,2) NOT NULL CHECK (score BETWEEN 0 AND 100),
  rule        text NOT NULL CHECK (rule IN ('profile', 'same_phone')),
  status      duplicate_status NOT NULL DEFAULT 'flagged',
  created_at  timestamptz NOT NULL DEFAULT now(),
  resolved_by uuid REFERENCES users(id),
  resolved_at timestamptz,
  reason      text CHECK (length(reason) ⩽ 500),
  CHECK (case_a < case_b),
  CHECK (rule <> 'same_phone' OR status <> 'confirmed_same'),
  CHECK ((status = 'flagged') = (resolved_at IS NULL)),
  CHECK ((status = 'flagged') = (resolved_by IS NULL)),
  CHECK (status = 'flagged' OR (reason IS NOT NULL AND length(reason) ⩾ 3)),
  UNIQUE (case_a, case_b)
);

-- Outputs, audit, usage, limits ----------------------------------------
CREATE TABLE generated_outputs (
  id           uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  case_id      uuid NOT NULL REFERENCES cases(id),
  type         output_t NOT NULL,
  text_bn      text CHECK (length(text_bn) ⩽ 3000),
  case_version integer NOT NULL,
  created_at   timestamptz NOT NULL DEFAULT now()
);
CREATE UNIQUE INDEX generated_outputs_version_idx ON generated_outputs (case_id, type,
case_version);
CREATE INDEX generated_outputs_case_type_idx ON generated_outputs (case_id, type);
```

```
CREATE TABLE audit_log (
  id                bigint GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  at                timestamptz NOT NULL DEFAULT now(),
  actor_type        actor_t NOT NULL,
  actor_user_id     uuid REFERENCES users(id),
  actor_session_id  uuid REFERENCES parent_sessions(id),
  action            text NOT NULL CHECK (action ~ '^[a-z_.]{3,60}$'),
  case_id           uuid REFERENCES cases(id),
  candidate_id      uuid REFERENCES match_candidates(id),
  details           jsonb NOT NULL DEFAULT '{}'::jsonb,
  request_id        text
);
CREATE INDEX audit_log_case_idx ON audit_log (case_id, at);

CREATE FUNCTION audit_log_append_only() RETURNS trigger LANGUAGE plpgsql AS $$
BEGIN
  RAISE EXCEPTION 'audit_log is append-only' USING ERRCODE = 'insufficient_privilege';
END $$;
CREATE TRIGGER audit_log_no_truncate BEFORE TRUNCATE ON audit_log
  FOR EACH STATEMENT EXECUTE FUNCTION audit_log_append_only();
CREATE TRIGGER audit_log_no_update_delete BEFORE UPDATE OR DELETE ON audit_log
  FOR EACH ROW EXECUTE FUNCTION audit_log_append_only();

-- Atomic global budget reservations; failed/uncertain provider calls still count.
CREATE TABLE ai_budget (
  id boolean PRIMARY KEY DEFAULT true CHECK (id),
  cap_usd numeric(10,6) NOT NULL DEFAULT 25 CHECK (cap_usd > 0 AND cap_usd ⩽ 30),
  committed_usd numeric(10,6) NOT NULL DEFAULT 0 CHECK (committed_usd ⩾ 0),
  reserved_usd numeric(10,6) NOT NULL DEFAULT 0 CHECK (reserved_usd ⩾ 0),
  CHECK (committed_usd + reserved_usd ⩽ cap_usd)
);
INSERT INTO ai_budget (id) VALUES (true);
CREATE TABLE ai_reservations (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  draft_id uuid REFERENCES intake_drafts(id),
  reserved_usd numeric(10,6) NOT NULL CHECK (reserved_usd ⩾ 0),
  actual_usd numeric(10,6) CHECK (actual_usd ⩾ 0),
  state text NOT NULL DEFAULT 'reserved' CHECK (state IN ('reserved','settled')),
  created_at timestamptz NOT NULL DEFAULT now(),
  settled_at timestamptz,
  CHECK ((state = 'settled') = (settled_at IS NOT NULL)),
  CHECK ((state = 'settled') = (actual_usd IS NOT NULL))
);

CREATE TABLE ai_calls (
  id            bigint GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  at            timestamptz NOT NULL DEFAULT now(),
  draft_id      uuid REFERENCES intake_drafts(id),
  case_id       uuid REFERENCES cases(id),
  reservation_id uuid UNIQUE REFERENCES ai_reservations(id),
  request_id    text,
  operation     text NOT NULL CHECK (operation IN ('transcribe', 'extract', 'embed')),
  provider      text NOT NULL,
  model         text NOT NULL,
  duration_ms   integer NOT NULL CHECK (duration_ms ⩾ 0),
  input_units   integer,
  output_units  integer,
  est_cost_usd  numeric(10,6) NOT NULL DEFAULT 0 CHECK (est_cost_usd ⩾ 0),
  cache_hit     boolean NOT NULL DEFAULT false,
```

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  ok            boolean NOT NULL,
  error_id      text CHECK (error_id ~ '^E-[0-9]{3}$')
);
CREATE INDEX ai_calls_at_idx ON ai_calls (at);

CREATE TABLE rate_limits (
  key          text NOT NULL,
  window_start timestamptz NOT NULL,
  count        integer NOT NULL DEFAULT 0 CHECK (count ⩾ 0),
  PRIMARY KEY (key, window_start)
);

-- Relationship and decision checks are deferred until the transaction is complete.
CREATE FUNCTION check_pair_integrity() RETURNS trigger LANGUAGE plpgsql AS $$
DECLARE c match_candidates; m cases; f cases; d review_decisions; r referrals;
        cid uuid;
BEGIN
  IF TG_TABLE_NAME = 'cases' THEN
    SELECT * INTO m FROM cases WHERE id=NEW.id;
    IF EXISTS (SELECT 1 FROM match_candidates p JOIN cases x ON x.id=p.missing_case_id
        JOIN cases y ON y.id=p.found_case_id
        WHERE (p.missing_case_id=m.id OR p.found_case_id=m.id) AND p.status='pending'
        AND (p.missing_version<>x.version OR p.found_version<>y.version
          OR x.status NOT IN ('open','matched_pending_review')
          OR y.status NOT IN ('open','matched_pending_review'))) THEN
      RAISE EXCEPTION 'case update leaves an ineligible or stale pending candidate' USING
ERRCODE='check_violation';
    END IF;
    IF m.status='matched_pending_review' AND NOT EXISTS (
        SELECT 1 FROM match_candidates p WHERE p.status='pending'
          AND (p.missing_case_id=m.id OR p.found_case_id=m.id)) THEN
      RAISE EXCEPTION 'pending-review case requires a pending candidate' USING
ERRCODE='check_violation';
    END IF;
    IF m.status = 'confirmed' AND NOT EXISTS (
      SELECT 1 FROM match_candidates WHERE missing_case_id=NEW.id AND status='confirmed') THEN
      RAISE EXCEPTION 'confirmed case requires confirmed candidate' USING
ERRCODE='check_violation';
    END IF;
    IF m.status = 'referred' AND NOT EXISTS (
      SELECT 1 FROM match_candidates WHERE found_case_id=NEW.id AND status='confirmed') THEN
      RAISE EXCEPTION 'referral-recorded case requires confirmed candidate' USING
ERRCODE='check_violation';
    END IF;
    IF m.status='closed' AND m.outcome='coordination_recorded' AND NOT EXISTS (
      SELECT 1 FROM match_candidates WHERE missing_case_id=NEW.id AND status='confirmed') THEN
      RAISE EXCEPTION 'coordination closure requires confirmed pair' USING
ERRCODE='check_violation';
    END IF;
    RETURN NULL;
  END IF;
  IF TG_TABLE_NAME='match_candidates' THEN cid:=NEW.id;
  ELSE cid:=NEW.candidate_id; END IF;
  SELECT * INTO c FROM match_candidates WHERE id=cid;
  SELECT * INTO m FROM cases WHERE id=c.missing_case_id;
  SELECT * INTO f FROM cases WHERE id=c.found_case_id;
  IF m.kind <> 'missing' OR f.kind <> 'found' THEN
    RAISE EXCEPTION 'candidate endpoint kinds are invalid' USING ERRCODE='check_violation';
  END IF;
```

```
  IF c.status='pending' AND (m.version<>c.missing_version OR f.version<>c.found_version OR
      m.status NOT IN ('open','matched_pending_review') OR f.status NOT IN
('open','matched_pending_review') OR
      m.anonymised_at IS NOT NULL OR f.anonymised_at IS NOT NULL) THEN
    RAISE EXCEPTION 'pending candidate must use eligible current versions' USING
ERRCODE='check_violation';
  END IF;
  SELECT * INTO d FROM review_decisions WHERE candidate_id=cid;
  SELECT * INTO r FROM referrals WHERE candidate_id=cid;
  IF d.id IS NOT NULL AND ((d.decision='confirm' AND c.status<>'confirmed') OR
      (d.decision='reject' AND c.status<>'rejected')) THEN
    RAISE EXCEPTION 'decision must agree with candidate status' USING ERRCODE='check_violation';
  END IF;
  IF c.status='rejected' AND (d.id IS NULL OR d.decision<>'reject') THEN
    RAISE EXCEPTION 'rejected candidate requires rejection decision' USING
ERRCODE='check_violation';
  END IF;
  IF r.id IS NOT NULL AND (r.missing_case_id<>c.missing_case_id OR r.found_case_id<>c.found_case_id
OR
      c.status<>'confirmed' OR r.referred_by<>d.officer_id) THEN
    RAISE EXCEPTION 'referral must describe its confirmed candidate and author' USING
ERRCODE='check_violation';
  END IF;
  IF c.status='confirmed' AND (d.id IS NULL OR d.decision<>'confirm' OR r.id IS NULL OR
      m.version<>c.missing_version OR f.version<>c.found_version OR
      m.status NOT IN ('confirmed','closed') OR f.status<>'referred') THEN
    RAISE EXCEPTION 'confirmed pair requires decision, referral record, and both final states'
USING ERRCODE='check_violation';
  END IF;
  IF d.id IS NOT NULL AND NOT EXISTS (SELECT 1 FROM audit_log a
      WHERE a.candidate_id=cid AND a.actor_type='officer' AND a.actor_user_id=d.officer_id
        AND a.action=('match.' || d.decision::text)) THEN
    RAISE EXCEPTION 'review decision requires matching officer audit event' USING
ERRCODE='check_violation';
  END IF;
  RETURN NULL;
END $$;
CREATE CONSTRAINT TRIGGER candidates_integrity AFTER INSERT OR UPDATE ON match_candidates
  DEFERRABLE INITIALLY DEFERRED FOR EACH ROW EXECUTE FUNCTION check_pair_integrity();
CREATE CONSTRAINT TRIGGER decisions_integrity AFTER INSERT OR UPDATE ON review_decisions
  DEFERRABLE INITIALLY DEFERRED FOR EACH ROW EXECUTE FUNCTION check_pair_integrity();
CREATE CONSTRAINT TRIGGER referrals_integrity AFTER INSERT OR UPDATE ON referrals
  DEFERRABLE INITIALLY DEFERRED FOR EACH ROW EXECUTE FUNCTION check_pair_integrity();
CREATE CONSTRAINT TRIGGER cases_integrity AFTER INSERT OR UPDATE ON cases
  DEFERRABLE INITIALLY DEFERRED FOR EACH ROW EXECUTE FUNCTION check_pair_integrity();

CREATE FUNCTION verify_decision_author() RETURNS trigger LANGUAGE plpgsql AS $$
BEGIN
  IF NOT EXISTS (SELECT 1 FROM users WHERE id=NEW.officer_id AND role='officer' AND verified AND
is_active) THEN
    RAISE EXCEPTION 'decision author must be an active verified officer' USING
ERRCODE='insufficient_privilege';
  END IF;
  RETURN NEW;
END $$;
CREATE TRIGGER decisions_author BEFORE INSERT ON review_decisions
  FOR EACH ROW EXECUTE FUNCTION verify_decision_author();

CREATE FUNCTION lock_decision_history() RETURNS trigger LANGUAGE plpgsql AS $$
```

```
BEGIN
  IF TG_OP='UPDATE' AND NEW.reason='[REDACTED]' AND
     (to_jsonb(NEW)-'reason')=(to_jsonb(OLD)-'reason') THEN RETURN NEW; END IF;
  RAISE EXCEPTION 'decision history is immutable; use an audited correction workflow before a
pilot'
    USING ERRCODE='insufficient_privilege';
END $$;
CREATE TRIGGER decisions_history BEFORE UPDATE OR DELETE ON review_decisions
  FOR EACH ROW EXECUTE FUNCTION lock_decision_history();

CREATE FUNCTION protect_confirmed_candidate() RETURNS trigger LANGUAGE plpgsql AS $$
BEGIN
  IF OLD.status='confirmed' THEN
    IF TG_OP='UPDATE' AND NEW.reason_bn='[REDACTED]' AND NEW.reason_en='[REDACTED]'
       AND NEW.components='{}'::jsonb AND
       (to_jsonb(NEW)-ARRAY['reason_bn','reason_en','components'])=
       (to_jsonb(OLD)-ARRAY['reason_bn','reason_en','components']) THEN RETURN NEW; END IF;
    RAISE EXCEPTION 'confirmed pair history is locked except payload redaction' USING
ERRCODE='check_violation';
  END IF;
  RETURN CASE WHEN TG_OP='DELETE' THEN OLD ELSE NEW END;
END $$;
CREATE TRIGGER confirmed_candidate_lock BEFORE UPDATE OR DELETE ON match_candidates
  FOR EACH ROW EXECUTE FUNCTION protect_confirmed_candidate();

CREATE FUNCTION lock_referral_history() RETURNS trigger LANGUAGE plpgsql AS $$
BEGIN
  IF TG_OP='UPDATE' AND NEW.note IS NULL AND
     (to_jsonb(NEW)-'note')=(to_jsonb(OLD)-'note') THEN RETURN NEW; END IF;
  RAISE EXCEPTION 'referral history is immutable except note redaction' USING
ERRCODE='insufficient_privilege';
END $$;
CREATE TRIGGER referrals_history BEFORE UPDATE OR DELETE ON referrals
  FOR EACH ROW EXECUTE FUNCTION lock_referral_history();

CREATE FUNCTION check_duplicate_integrity() RETURNS trigger LANGUAGE plpgsql AS $$
DECLARE a cases; b cases;
BEGIN
  SELECT * INTO a FROM cases WHERE id=NEW.case_a;
  SELECT * INTO b FROM cases WHERE id=NEW.case_b;
  IF a.kind<>b.kind THEN RAISE EXCEPTION 'duplicate reports must have same kind' USING
ERRCODE='check_violation'; END IF;
  IF NEW.status<>'flagged' AND NOT EXISTS (
      SELECT 1 FROM users WHERE id=NEW.resolved_by AND role='officer' AND verified AND is_active)
THEN
    RAISE EXCEPTION 'duplicate resolution requires verified officer' USING
ERRCODE='insufficient_privilege';
  END IF;
  RETURN NEW;
END $$;
CREATE TRIGGER duplicate_integrity BEFORE INSERT OR UPDATE ON duplicate_links
  FOR EACH ROW EXECUTE FUNCTION check_duplicate_integrity();

-- Reserve upper-bound provider cost before dispatch. Identity/expiry is checked in API.
-- No refunds to the per-draft count: dispatched retries/fallbacks each consume a call.
CREATE FUNCTION reserve_ai_call(p_draft uuid, p_upper_usd numeric) RETURNS uuid LANGUAGE plpgsql AS
$$
DECLARE rid uuid;
BEGIN
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```
  IF p_upper_usd IS NULL OR p_upper_usd⩽0 THEN RAISE EXCEPTION 'positive cost upper bound
required'; END IF;
  -- Global lock order: budget first, then draft; settlement uses budget then reservation.
  PERFORM 1 FROM ai_budget WHERE id=true FOR UPDATE;
  IF p_draft IS NOT NULL THEN
    UPDATE intake_drafts SET ai_calls_used=ai_calls_used+1,revision=revision+1
      WHERE id=p_draft AND expires_at>now() AND ai_calls_used<6;
    IF NOT FOUND THEN RAISE EXCEPTION 'draft expired or six-call limit reached' USING
ERRCODE='check_violation'; END IF;
  END IF;
  UPDATE ai_budget SET reserved_usd=reserved_usd+p_upper_usd
    WHERE id=true AND committed_usd+reserved_usd+p_upper_usd⩽cap_usd;
  IF NOT FOUND THEN RAISE EXCEPTION 'AI budget exhausted' USING ERRCODE='check_violation'; END IF;
  INSERT INTO ai_reservations(draft_id,reserved_usd) VALUES(p_draft,p_upper_usd) RETURNING id INTO
rid;
  RETURN rid;
END $$;

CREATE FUNCTION settle_ai_call(p_reservation uuid,p_actual_usd numeric) RETURNS void LANGUAGE
plpgsql AS $$
DECLARE r ai_reservations;
BEGIN
  PERFORM 1 FROM ai_budget WHERE id=true FOR UPDATE;
  SELECT * INTO r FROM ai_reservations WHERE id=p_reservation FOR UPDATE;
  IF NOT FOUND THEN RAISE EXCEPTION 'reservation not found'; END IF;
  IF r.state='settled' THEN
    IF r.actual_usd IS DISTINCT FROM p_actual_usd THEN RAISE EXCEPTION 'settlement conflicts with
prior result'; END IF;
    RETURN;
  END IF;
  IF p_actual_usd IS NULL OR p_actual_usd<0 OR p_actual_usd>r.reserved_usd THEN
    RAISE EXCEPTION 'actual cost exceeds reserved upper bound';
  END IF;
  UPDATE ai_budget SET reserved_usd=reserved_usd-
r.reserved_usd,committed_usd=committed_usd+p_actual_usd WHERE id=true;
  UPDATE ai_reservations SET state='settled',actual_usd=p_actual_usd,settled_at=now() WHERE
id=p_reservation;
END $$;

-- Clearing identifying payloads is pseudonymisation, not irreversible anonymity.
CREATE FUNCTION redact_case(p_case uuid,p_admin uuid) RETURNS void LANGUAGE plpgsql AS $$
DECLARE rec cases;
BEGIN
  IF NOT EXISTS (SELECT 1 FROM users WHERE id=p_admin AND role='admin' AND is_active) THEN
    RAISE EXCEPTION 'redaction requires active admin' USING ERRCODE='insufficient_privilege';
  END IF;
  SELECT * INTO rec FROM cases WHERE id=p_case FOR UPDATE;
  IF NOT FOUND OR rec.status NOT IN ('closed','referred') THEN RAISE EXCEPTION 'case not eligible
for redaction'; END IF;
  IF rec.kind='found' AND EXISTS (SELECT 1 FROM match_candidates mc JOIN cases m ON
m.id=mc.missing_case_id
      WHERE mc.found_case_id=p_case AND mc.status='confirmed' AND m.status<>'closed') THEN
    RAISE EXCEPTION 'retain found evidence until linked missing report is closed';
  END IF;
  IF rec.anonymised_at IS NOT NULL THEN RETURN; END IF;
  UPDATE cases SET child_name=NULL,child_name_latin=NULL,age_years=NULL,
    age_min=CASE WHEN age_min IS NULL THEN NULL ELSE 2*(age_min/2) END,
    age_max=CASE WHEN age_max IS NULL THEN NULL ELSE least(25,2*(age_max/2)+1) END,
    height_cm=NULL,clothing=NULL,marks=NULL,circumstances=NULL,contact_phone=NULL,
```

```
    raw_text=NULL,place_text=NULL,thana=NULL,reporter_relation=NULL,gd_number=NULL,source_url=NULL,
    outcome_note=NULL,reporter_session_id=NULL,created_by_user_id=NULL,input_hash=NULL,
    submission_payload_hash=NULL,client_submission_id=NULL,embedding=NULL,

embedding_model=NULL,embedding_revision=NULL,embedding_space=NULL,provenance='{}',anonymised_at=now
()
    WHERE id=p_case;
  UPDATE case_versions SET snapshot=NULL WHERE case_id=p_case;
  DELETE FROM generated_outputs WHERE case_id=p_case;
  UPDATE match_candidates SET reason_bn='[REDACTED]',reason_en='[REDACTED]',components='{}'
    WHERE missing_case_id=p_case OR found_case_id=p_case;
  UPDATE review_decisions SET reason='[REDACTED]' WHERE candidate_id IN (
    SELECT id FROM match_candidates WHERE missing_case_id=p_case OR found_case_id=p_case);
  UPDATE referrals SET note=NULL WHERE missing_case_id=p_case OR found_case_id=p_case;
  UPDATE duplicate_links SET reason='[REDACTED]' WHERE case_a=p_case OR case_b=p_case;
  INSERT INTO audit_log(actor_type,actor_user_id,action,case_id,details)

VALUES('admin',p_admin,'case.payload_redacted',p_case,'{"mode":"pseudonymisation","external_copies_
removed":false}');
END $$;

REVOKE UPDATE,DELETE,TRUNCATE ON audit_log FROM PUBLIC;
REVOKE ALL ON FUNCTION redact_case(uuid,uuid) FROM PUBLIC;
REVOKE ALL ON FUNCTION reserve_ai_call(uuid,numeric),settle_ai_call(uuid,numeric) FROM PUBLIC;
-- Hosting operator must grant service functions to separate restricted runtime roles.
-- No application role may own tables, create/alter schema, disable triggers, or truncate.
COMMIT;
```

7.1 Applying the companion schema

Apply `schema_v2.sql` only to a fresh synthetic-demo database. It is not an idempotent deployment script or an in-place V1 migration. Back up and review any existing database; a versioned migration and rollback rehearsal are required before changing a live environment. Load missing reports initially as submitted, then open them; load found reports as open. Terminal demo states must be constructed through complete audited transactions, never by inserting a confirmed/closed row directly. Keep all seeded names, phones, reports and identities fictional.

Deferred pair checks run at transaction completion: a confirmed pair requires current versions, one verified decision, the corresponding match.confirm audit event, a matching simulated referral and both final endpoint states. Rejection requires match.reject audit. Pending reports cannot retain stale candidates or a pending-review state without a pending candidate. Confirmed candidate/decision/referral history is immutable except prescribed payload redaction. HTTP authority, lock order, UI state reconciliation and durable retry still require application tests.

8 State management

Concern	V2 rule
Device draft	Reviewed source, parsed fields and provenance on device only; clear on submit/discard; expiry checked on access, with clear-device control. No audio persisted.
Server draft metadata	intake_drafts principal/expiry/hash/revision/counter, no transcript or case payload; delete expired metadata under a documented cleanup task.

Concern	V2 rule
Paid-call counter	Atomic server update and budget reservation before every dispatched transcription/extraction/repair/fallback. Signed token binds ID/principal/expiry; old tokens cannot reset usage.
Global budget	Reserve a cost upper bound in one transaction; settle actual usage; retain uncertain reservations; fail closed for paid dispatch. Manual forms remain usable.
Submission	Scoped UUID unique + canonical payload hash; identical replay returns same case, altered payload conflicts.
Matching	queued/running/complete/failed, bounded request or durable retry; no detached serverless promises.
Concurrency	Sorted endpoint locks, candidate lock, live versions, verified officer, transaction retries for known conflicts. Reconcile other cases after candidate invalidation.
UI	No optimistic submit/decision/closure/redaction. AbortSignal/cancellation is propagated; provider usage is accounted for even when UI stops waiting.
Cache	Case/queue/stats no-store; deterministic outputs by version; local assets pinned; demo fixtures excluded from quality evaluation.
Model switch	Pause matching, re-embed eligible data, validate dimension/space coverage, activate atomically.

9 Authentication and authorisation

9.1 Roles

Role	How obtained	Session
parent	First intake call sets a signed cookie `fera_ps` (HMAC of `parent_sessions.id`)	30 days; draft separately 24 h
staff	Admin-created account; role `staff`	Auth.js JWT cookie, 8 h
officer	Admin-created; `verified=true` set by admin [MOCK]	Auth.js JWT cookie, 8 h
admin	Seeded account	Auth.js JWT cookie, 8 h

An officer with `verified=false` gets staff permissions only (D1 BR-024).

9.2 Permission matrix

Action/API	Parent	Staff / unverified officer	Verified officer	Admin
API-001/002 intake	Own missing	Own found	Own found	None
API-003 / API-007 create	Missing only	Found only	Found only	None

Action/API	Parent	Staff / unverified officer	Verified officer	Admin
API-004 read	Own OwnerCaseView	Own StaffOwnCaseView	Operational OfficerCaseView	AdminCaseView metadata
API-005/006/018	Own edit/output/status	None	None	None
API-008/009/010/011/022	None	None	Allowed	None
API-012 aggregate stats	None	None	Allowed	Allowed
API-013 safe audit	None	None	Operational	Metadata only
API-014 rematch	None	None	Status only	Status only
API-015/016/017/021	None	None	None	Synthetic admin tools
API-023 duplicate resolution	None	None	Could	None
API-019/020	Public health/auth only	Public health/auth only	Public health/auth only	Public health/auth only

9.3 Access-control rules

Every handler validates principal, ownership and role on the server. Officer active/verified state is read on every operational request and rechecked in decision transactions. Record reads use SQL ownership filters plus explicit allowlist DTOs. Staff may read their own found payload; neither staff nor parents receive other-party found details or candidate data. Admin never receives candidate scores/references or child payload and cannot decide or close. E-004 is identical for absent/forbidden resources; audit uses a null FK when the requested ID is absent, with safe identifier details only. Client authority, synthetic flags, owner IDs and verification claims are ignored.

9.4 Sessions, tokens and CSRF

Parent sessions have server-side expires_at/revoked_at; signed cookies bind an opaque session. Draft tokens bind draft ID/principal/expiry; their embedded counters are not authoritative. Credential sessions expire at 8 hours. Production cookies use httpOnly/Secure/SameSite=Lax, __Host- prefix, Path=/ and no Domain. Local secure-context behaviour is tested; offline fallback uses one local backend and separate browser profiles.

State-changing routes require same-origin validation and explicit route content types: JSON for normal APIs, multipart for API-001. Auth.js handles its auth-route protections; other routes still enforce their own checks. Password hashes and provider keys are never shipped in the package. Restrict seeded demo access and paid endpoints; do not expose one-click privileged public login.

9.5 Account recovery

Role	Recovery
staff, officer	Admin resets password and re-sets `verified`

Role	Recovery
admin	Re-seed from environment on the server
parent	None across devices (D1 OQ-020). Same device keeps access while the cookie lives.

Decisions (Part A)

ID	Decision	Section
DECISION-001	Single Next.js App Router app (UI + API), TypeScript strict	§2
DECISION-002	PostgreSQL 16 + pgvector; raw SQL via `pg`; `schema_v2.sql` is the migration	§2, §7
DECISION-003	Auth.js credentials for staff/officer/admin; signed anonymous cookie for parents	§9
DECISION-004	LLM used for extraction only, behind a fixed JSON contract, temp 0, 1 retry, 1 fallback provider	§2.1
DECISION-005	Speech via `SpeechProvider`; Whisper API default; cache for scripted notes	§2.2
DECISION-006	Local embeddings (Transformers.js, 384-d) by default; API embeddings as switchable fallback	§2.3
DECISION-007	Follow-up questions, GD, poster, 13219 summary and match reasons from templates, not LLM	§4
DECISION-008	Matching = SQL hard filters + pgvector cosine + rule bonuses	§4
DECISION-009	Leaflet with OSM tiles online; tile-less markers plus table offline	§2
DECISION-010	Vercel for Round 2; laptop-local copy for the finale; phone uses its own mobile data	§3.6
DECISION-011	Rate limits stored in Postgres (serverless-safe)	§4
DECISION-012	No optimistic updates for any state-changing action	§8
DECISION-013	No per-case coordinates; district centroid only	§6
DECISION-014	Draft held in the browser; nothing persisted before confirm	§3.4, §8
DECISION-015	Poster PNG rendered in the browser via canvas	§2

Assumptions (Part A)

ID	Assumption
A-026	The candidate local model (multilingual-e5-small, 384-d) gives usable Bangla similarity; verified in the 11 Oct evaluation. Its documented input prefixes are applied.
A-027	The local model loads within the hosting function's size and time limits; if not, `EMBED_PROVIDER=api` and re-embed (RISK-015, OQ-023).
A-028	Match score = 100 × (0.55 cosine + 0.15 name + 0.15 clothing + 0.10 district + 0.05 age closeness); weights tuned on the synthetic set.
A-029	Duplicate "high description similarity" = No fixed raw-cosine identity threshold. Calibrate on identity-grouped dev hard negatives, then freeze before test.
A-030	Android Chrome MediaRecorder output (webm/opus) is accepted by the speech provider; verified 10 Oct (OQ-022).
A-031	Browsers allow microphone access only on HTTPS or localhost. The finale phone reaches the deployed HTTPS app over its own mobile data.
A-032	District codes 1–64 are Fera-internal, not official geocodes.
A-033	The team builds a Bangla→Latin romanisation table for name similarity.
A-034	A free-tier managed Postgres with pgvector >= 0.8.1 (HNSW) is available (OQ-021).
A-035	Time words map to fixed hours: সকাল 08:00, দুপুর 13:00, বিকেল 16:00, সন্ধ্যা 18:30, রাত 21:00, all flagged approximate ±2 h.
A-036	2 MB is enough for a 120 s webm/opus note from Android Chrome.

Open questions (Part A)

ID	Question	Default
OQ-021	Which free-tier managed Postgres host (with pgvector HNSW) will the team use? Limits UNKNOWN.	Pick on 10 Oct; must pass `CREATE EXTENSION vector` and HNSW index creation
OQ-022	Speech provider's accepted formats, size limit and per-minute price? UNKNOWN in sources.	Verify from provider docs on 10 Oct; record in Part B §12
OQ-023	Hosting function size and duration limits for the local embedding model? UNKNOWN.	10 Oct spike; fallback A-027

Risks (Part A)

ID	Risk	L	I	Mitigation	Owner
RISK-015	Local embedding model too large or slow for serverless hosting	M	M	Spike on 10 Oct; switch to API embeddings and re-embed	Masnun
RISK-016	Phone microphone blocked when the app is served over plain HTTP in local mode	M	M	Phone uses deployed HTTPS app; local fallback shows parent flow on laptop with the scripted note	Sadman

IDs introduced

- COMP-001 Parent client
- COMP-002 Staff/officer/admin client
- COMP-003 Voice recorder
- COMP-004 API layer
- COMP-005 Auth and session
- COMP-006 Speech adapter
- COMP-007 LLM adapter
- COMP-008 Normaliser
- COMP-009 Intake orchestrator
- COMP-010 Output generator
- COMP-011 Embedding service
- COMP-012 Matching engine
- COMP-013 Duplicate detector
- COMP-014 Case service and state guard
- COMP-015 Audit logger
- COMP-016 Wording guard

- COMP-017 Demo-safe cache
- COMP-018 Dashboard and map
- COMP-019 Admin tools
- COMP-020 Evaluation harness
- COMP-021 Rate limiter
- API-001 Transcribe
- API-002 Extract
- API-003 Create missing case
- API-004 Read case
- API-005 Edit case
- API-006 Generate output
- API-007 Create found case
- API-008 Review queue
- API-009 Candidate detail
- API-010 Decision
- API-011 Close case
- API-012 Dashboard stats
- API-013 Case audit trail
- API-014 Re-run matching
- API-015 Public import
- API-016 Redact identifying payloads
- API-017 Verify officer [MOCK]
- API-018 Parent status
- API-019 Health
- API-020 Auth routes
- ENTITY-001 User
- ENTITY-002 ParentSession
- ENTITY-003 Case
- ENTITY-004 CaseVersion
- ENTITY-005 MatchCandidate
- ENTITY-006 ReviewDecision
- ENTITY-007 Referral
- ENTITY-008 DuplicateLink
- ENTITY-009 AuditEntry
- ENTITY-010 District
- ENTITY-011 DistrictNeighbour
- ENTITY-012 GeneratedOutput
- ENTITY-013 AiCall
- ENTITY-014 RateLimitBucket

- DECISION-001 – DECISION-015 (one line each in "Decisions")
- E-001 – E-023 (one line each in §5.2)
- A-026 – A-036 (one line each in "Assumptions")
- OQ-021 Postgres host
- OQ-022 Speech provider limits and price
- OQ-023 Function limits for local model
- RISK-015 Local embeddings on serverless
- RISK-016 Phone mic over HTTP

IDs referenced

- FR-001 – FR-037, BR-001 – BR-026, NFR-008, NFR-010, NFR-016, NFR-017 (D1)
- A-009, A-016 (D1)
- OQ-004, OQ-013, OQ-020 (00, D1)
- SCREEN-001 – SCREEN-016 (D1, reserved)

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D2B / QUALITY / V2.0

Technical Requirements · Part B

Security, privacy, costs, evaluation and deployment · Authors: Sadman Shaharier and Obidit Islam

↗ Contents

Field	Value
Version	2.0
Date	11 Oct 2026 (Asia/Dhaka, UTC+6)
Owner	Sadman Shaharier (lead/integration)
Authors	Sadman Shaharier; Obidit Islam
Status	V2 team handover; implementation status UNVERIFIED
Split	Part A = §1–§9 (includes §7 schema). Part B = §10–§19. Companion `schema_v2.sql` is a byte-identical copy of Part A §7.
Inputs	D1 and D2 Part A V2.0; validation report; V2 evaluation, privacy, cost and release gates.

V2 release contract

Authors: Sadman Shaharier and Obidit Islam. **Version:** 2.0. **Issued:** 11 October 2026, Asia/Dhaka (UTC+6).

This is the revised team specification, not evidence that the application has been built. Actual repository progress, account configuration, interviews, evaluations, and spend remain **UNVERIFIED** until evidence is attached. V1 defaults are superseded by the V2 rules below; existing requirement IDs are retained.

- This release is a **synthetic-data prototype**. Officer verification and referral delivery are simulated and labelled. The schema rejects records marked non-synthetic; the application must also prevent real reports from entering this release.
- A candidate score is similarity, not identity probability. A reviewer decision does not establish guardian authority or safe reunification. A referral row means **referral recorded**, not delivered or acknowledged.
- Draft payload stays on the device before report confirmation, but providers process audio/text beforehand with a notice. Only expiry/counter/hash metadata and cost reservations are stored server-side before submission.
- Possible duplicates remain visible. Shared phone alone never establishes one child. No physical child-return action is performed by Fera.
- Counts distinguish missing-report rows, found-report rows, and confirmed pairs. They are not a census of unique children or proof of rescue.

- Sources, diagrams, tables, code blocks, and acceptance criteria are part of the handover. The companion code must match the embedded code exactly. Team handover records change closure, evidence gaps, and release gates.

Changes to earlier documents

V2 revises provider notices, pseudonymous redaction, append-only audit privileges, cost reservations, evaluation denominators and a single-database offline topology. Cached demos are excluded from quality results. Existing error IDs remain stable.

10 Security

10.1 Controls

Area	V2 control
Input	Strict runtime schemas on body/query/provider output; NFC; source-span mapping; reject unknown keys and illegal kind/status pairs.
SQL	Parameterised queries; separate migration, runtime and redaction credentials; runtime never owns or alters tables.
Prompt	Text is data, no tools/URL fetching, constrained JSON and supported provenance. Substring presence alone does not validate a fact.
Audio	Tested WebM/MP4/WAV, actual size ≤2 MB and verified duration 5–120 s; codec-aware MIME parsing; Ogg only after verified support/transcoding.
Secrets	Server env, no public keys, no passwords in package; argon2id; secret scanning and committed lockfile.
Paid limits	Server-owned draft counters and HMAC-hashed network keys; atomic budget reservation before dispatch; fail closed for paid calls, keep manual forms available.
Logs	Allowlist request ID, operation/model, duration, costs, success and safe error ID. Never log bodies, transcript, provenance, provider raw errors, names, phones or free notes.
Audit	UPDATE/DELETE/TRUNCATE blocked; safe actor/resource/version/action only. Unknown resource FK null. Rejected transaction audit is written after rollback.
DTOs	Owner/staff-own/officer/admin allowlists; admin rematch returns status only; candidate details officer-only.
CSRF	Same-Origin for all state changes; JSON on standard routes and multipart on transcription; credential cookie and server expiry/revocation checks.
Headers	Test Next.js nonce/hash CSP with hydration/auth/map/poster. Bundled map baseline: self-only images. If OSM tiles enabled, allow canonical tile.openstreetmap.org and required attribution/Referer.
Referrer	Bundled map: no-referrer acceptable. OSM mode: origin-only compatible Referer policy, reviewed privacy notice. Do not suppress required Referer.

Area	V2 control
Demo access	DEMO_ONLY=true; restrict seeded privileged accounts and paid routes. DEMO_MODE fixtures are labelled and never enabled in quality evaluation.
Dependencies	Pinned lockfile, audit and deployment-specific checks; do not claim a dependency scan verifies the application.

10.2 OWASP Top 10 (2021) check

#	Item	Fera exposure	Control	Test
A01	Broken Access Control	Parent or staff reading matched records	Server-side `requireRole`; SQL ownership filters; E-004 for hidden; `verified` read from DB	TC-022, TC-023
A02	Cryptographic Failures	Phones and names in transit or at rest	HTTPS only; managed DB encryption at rest (host-dependent, OQ-021); argon2id passwords; HMAC-signed cookies	Header check in TC-033 run
A03	Injection	SQL; prompt injection via report or import text	Parameterised SQL; data-only prompt framing; schema-only output; evidence check	TC-030, TC-014
A04	Insecure Design	AI auto-confirming matches; alert issuance	No code path from AI to state change; BR-001; no alert feature	TC-010, TC-025
A05	Security Misconfiguration	Debug output, permissive CORS	No CORS (same origin); errors without stack traces; `DEMO_ONLY=true`; protected fixture mode enabled only for labelled demo runs	Manual config review
A06	Vulnerable and Outdated Components	npm supply chain	Lockfile; `npm audit`; pinned major versions	CI audit step
A07	Identification and Authentication Failures	Weak passwords; session theft	Min 10 chars; argon2id; httpOnly Secure cookies; 8 h expiry; brute-force limit	**TC-023**
A08	Software and Data Integrity Failures	Tampered draft token; altered audit	HMAC draft token; append-only audit trigger	**TC-016**
A09	Security Logging and Monitoring Failures	Undetected misuse	Audit of decisions and denied access; pino logs with request IDs	TC-016, TC-022
A10	Server-Side Request Forgery	`source_url` on import	URLs stored only, never fetched by the server	Code review; TC-014

11 Privacy, retention and pilot gate

Only invented synthetic records are permitted in V2. Public availability does not remove a child's privacy interests; real news/alert imports remain outside this release.

System/data	V2 treatment	Real-case gate
Child case fields	Reviewed fields only; no photos, ID documents or precise coordinates	Partner-defined minimisation and lawful basis
Before submit	Payload stays on device; server metadata only	Consent/authority and device threat review
Provider audio/text	Synthetic only, with processing notice before recording/sending	Account/endpoint retention, training, processor and transfer review
Draft storage	Clear on submit/discard; check 24-hour expiry on access; clear-device control	Do not promise physical removal while browser is closed
Audio	In-memory/transit only; not in localStorage or DB	Endpoint terms, actual retention and incident duties
Redaction	Direct identifiers, exact age, source URL, owner/hash/ provenance/vector metadata, outcome/referral/ decision/candidate notes, snapshots and outputs cleared	Field-specific retention and correction/ evidence requirements
Retained record	IDs, age band, sex, district, dates/status and safe audit links under access control	Pseudonymisation; residual linkage/re-identification risk assessment
Found evidence	Redact only after linked missing case is closed	Partner evidence-preservation/correction window
External copies	Downloads, provider copies and backups follow separate policy	Withdrawal/deletion propagation and backup expiry
Map requests	Bundled geometry baseline; optional tiles disclose viewport/request metadata and IP	Approved provider/policy and privacy notice
Poster export	Preview phone/place/age, choose contact inclusion, acknowledge public sharing and no remote recall	Appropriate authority and safer institutional contact
Access	Own scoped views; operational verified-officer evidence; admin metadata	Real officer onboarding, assignment/ supervision and revocation

The checkbox “আমি যাচাই করেছি” verifies accuracy, not legal guardianship. “Nothing shared before confirm” means no submitted-register/candidate sharing; provider processing happens earlier and is disclosed. Retention duration and receiving agency remain partner decisions; they are not invented defaults.

The official Personal Data Protection Act, 2026 is Act 63, dated 10 April 2026, with phased commencement. Its child-data and transfer provisions require qualified assessment before real-case use. Accurate sourced legal references are allowed; the product makes no compliance claim. Official law.

Unpaid Gemini service terms permit improvement/human review uses and caution against personal/sensitive input. OpenAI data treatment varies by endpoint and settings. “No training” is not a universal zero-retention promise. Verify the exact account, model, endpoint and fallback, using Gemini terms and OpenAI data controls. Only synthetic data is allowed until a future release passes the pilot gate.

Before a pilot: responsible agency; approved identity/guardian/safeguarding checks; authorised consent/lawful basis; legal/provider/transfer review; verified onboarding; referral acknowledgement; correction/reopen procedure; retention schedule; access review; breach response; backup/restore test; operational staffing. A data agreement alone does not establish safe operation or national scale.

12 Performance targets and AI budget

12.1 Performance

Target	Value	Measured by
Parent intake first load, mid-range Android, 4G	LCP ≤ 3 s; route JS ≤ 170 KB gzip	Lighthouse mobile profile, 3 runs median
Non-AI API p95	≤ 500 ms (case read, queue, decision, stats)	Playwright timing + server logs
Transcription, 120 s note	≤ 20 s p95 (NFR-003)	`ai_calls.duration_ms`
Extraction	≤ 15 s p95 (NFR-002)	`ai_calls.duration_ms`
Voice note (60 s) → draft shown	≤ 35 s p95 end to end	E2E timer
Matching, one case vs 500 records	≤ 5 s (NFR-004); target ≤ 1 s on 200	`eval/perf.ts`
Map stats, 1,000 cases	≤ 2 s (NFR-005)	`eval/perf.ts`
Concurrent demo users	10 realistic; 50 design (A-020)	k6-style script `eval/load.ts`, error rate < 1 %

12.2 AI usage estimate against USD 30

Provider prices are not in any source: UNKNOWN (OQ-004, OQ-022). The plan below fixes usage volumes and the maximum unit price the budget allows.

Use	Speech (min)	Extraction calls	Basis (Assumption A-037)
Development and testing	60	300	3 devs, 9–13 Oct
WER evaluation	90	0	60 unique clips × 30 s × 3 repeated runs; repeats are not new independent samples
Extraction evaluation	0	220	50 reports × 4 runs × 1.1 retries
Matching evaluation (extracted-field mode)	0	400	200 records × 2 runs

Use	Speech (min)	Extraction calls	Basis (Assumption A-037)
Public import	0	50	Should only
Timing study + rehearsals + video + finale	40	130	5 testers, 10 rehearsals, 3 takes
Total	**190 min**	**1,100 calls**	

Cost line	Allocation	Max unit price that fits
Speech	USD 5	≤ USD 0.026 per audio minute
Extraction (≈ 1,200 input + 400 output tokens per call → 1.32 M in, 0.44 M out)	USD 15	≤ USD 4.87 per 1 M input-equivalent tokens, counting output at 4 × input
Embeddings	USD 0	Local model (DECISION-006)
Reserve	USD 10	Re-runs, API embeddings if RISK-015 fires
Total	**USD 30**	Atomic reservation stop at USD 25; raising toward the USD 30 ceiling requires a documented team budget decision

Rule: at the 11 Oct contract gate, Obidit records real prices from provider pages in `docs/costs.md`. If prices exceed allocation, reduce repeated runs or optional import, retain independent sample coverage and failure reporting, and recalculate actual input/output rates separately. 1.32 + 4×0.44 = 3.08M input-equivalent tokens; 15/3.08 = USD 4.87/M. A reservation upper bound covers maximum tokens/duration; uncertain calls are not refunded silently.

13 Error catalogue

ID	Key	User message (Bangla / English)	Developer log	Recovery
E-001	VALIDATION_FAILED	কিছু তথ্য ঠিক নেই। চিহ্নিত ঘরগুলো দেখুন। / Some details need fixing. Check the marked fields.	`warn` route, Zod issue paths	Highlight fields; keep input
E-002	UNAUTHENTICATED	আবার সাইন ইন করুন। / Please sign in again.	`info` route	Redirect to SCREEN-008, return after
E-003	FORBIDDEN_ACTION	এই কাজের অনুমতি আপনার নেই। / You don't have permission for this action.	`warn` user ID, role, route	Show SCREEN-016
E-004	NOT_FOUND	তথ্যটি পাওয়া যায়নি। / We couldn't find that.	`warn` + `access.denied` audit if signed in	SCREEN-016 with link home

ID	Key	User message (Bangla / English)	Developer log	Recovery
E-005	ILLEGAL_TRANSITION	কেসটির অবস্থা বদলে গেছে। পাতাটি আবার খুলুন। / This case has changed. Reload the page.	`warn` from→to, case ID	Reload latest state
E-006	SUBMISSION_PAYLOAD_CONFLICT	এই জমা নম্বরের তথ্য বদলেছে। তথ্য যাচাই করে আবার জমা দিন। / This submission key has different details. Review before submitting again.	`warn` safe request/key conflict, no payload	Preserve draft; review new submission action
E-007	RATE_LIMITED	{s} সেকেন্ড পরে আবার চেষ্টা করুন। / Please try again in {s} seconds.	`warn` key, count	Countdown, then enable button
E-008	AUDIO_TOO_LARGE	রেকর্ডিং অনেক বড়। ২ মিনিটের কম রেকর্ড করুন। / The recording is too large. Keep it under 2 minutes.	`info` bytes	Re-record or type
E-009	AUDIO_LENGTH_OUT_OF_RANGE	রেকর্ডিং ৫ সেকেন্ড থেকে ২ মিনিটের মধ্যে হতে হবে। / Recordings must be 5 seconds to 2 minutes.	`info` duration	Re-record
E-010	SPEECH_UNINTELLIGIBLE	কথাগুলো পরিষ্কার শোনা যায়নি। আবার বলুন বা লিখে দিন। / We couldn't hear that clearly. Try again or type instead.	`info` transcript length, script ratio	Re-record or focus text box
E-011	SPEECH_UNAVAILABLE	এখন ভয়েস কাজ করছে না। লিখে জানান। / Voice isn't working right now. Please type.	`error` provider, status	Text path
E-012	LLM_INVALID_OUTPUT	সহকারী তথ্য গুছাতে পারেনি। নিচের ফর্মটি পূরণ করুন — আপনার লেখা রাখা আছে। / The assistant couldn't organise this. Please fill the form — your text is kept.	`error` provider, Zod issues (no text)	SCREEN-005 pre-filled
E-013	LLM_UNAVAILABLE	সহকারী এখন সাড়া দিচ্ছে না। ফর্মে তথ্য দিন। / The assistant isn't responding. Please use the form.	`error` both providers, status	**SCREEN-005**
E-014	AI_TIMEOUT	অনেক সময় লাগছে। আবার চেষ্টা করুন বা ফর্ম ব্যবহার করুন। / This is taking too long. Try again or use the form.	`warn` operation, 25 s	Retry once, then SCREEN-005
E-015	UNSUPPORTED_LANGUAGE	বাংলা বা ইংরেজিতে বলুন বা লিখুন। / Please speak or write in Bangla or English.	`info` script ratios	Re-enter
E-016	OUTPUT_BLOCKED	এই লেখাটি তৈরি করা যায়নি। আবার চেষ্টা করুন। / We couldn't create this text. Please try again.	`error` output type, matched term ID	Regenerate; report to team
E-017	REFERRAL_REQUIRED	রেফারেল ছাড়া কেস বন্ধ করা যাবে না। / A referral is needed before closing.	`warn` case ID	Confirm a match first

ID	Key	User message (Bangla / English)	Developer log	Recovery
E-018	EDIT_LOCKED	মিল নিশ্চিত হওয়ার পর কেস বদলানো যায় না। / This case can't be edited after a match is confirmed.	`info` case ID, status	Use the stated case-support process; 13219 remains the external MUN helpline, not a direct Fera-officer contact
E-019	SESSION_EXPIRED	আপনার খসড়ার সময় শেষ। নতুন করে শুরু করুন। / Your draft has expired. Please start again.	`info` session age	Start new intake
E-020	OUTPUT_REQUEST_LIMIT_RESERVED	Reserved for future output-abuse limits; not an active V2 regeneration counter. Identical current-version templates are reused.	Safe rate-limit metadata only	Reuse current output; no five-generation rule
E-021	INTERNAL	কিছু একটা ভুল হয়েছে। আবার চেষ্টা করুন। / Something went wrong. Please try again.	`error` stack (server only), request ID	Retry; draft kept
E-022	MIC_UNAVAILABLE	মাইক্রোফোন চালু করা যায়নি। আপনি লিখে জানাতে পারেন। / The microphone isn't available. You can type instead.	client `console.warn` only	Focus text box
E-023	AI_BUDGET_CAP	সহকারীর সীমা শেষ। বাকি তথ্য ফর্মে দিন। / The assistant's limit is reached. Add the rest in the form.	`warn` calls used or spend total	SCREEN-005
E-024	NETWORK_OFFLINE	ইন্টারনেট সংযোগ নেই। আপনার তথ্য রাখা আছে — সংযোগ ফিরলে আবার পাঠান। / No internet connection. Your details are saved — send again when you're back online.	client only	Retry button; draft kept in browser

14 Observability

Item	Specification
Logs	pino JSON to stdout: `time`, `level`, `request_id`, `route`, `role`, `status`, `duration_ms`, `error_id`. PII redacted (§10.1).
AI metrics	One `ai_calls` row per call: operation, provider, model, duration, units, estimated cost, cache hit, error ID.
Spend report	`npm run spend` prints today's and total spend vs `AI_SPEND_CAP_USD`. Run daily (TASK-059).
Health	API-019 `/api/health`: DB ping, embedding model loaded, demo mode flag, git SHA.
Debug mode	Verified officers may inspect safe matching components on SCREEN-010; admins receive request/model/cache operational metadata only, never candidates, scores or child payload. Debug does not bypass DTO/role checks.

Item	Specification
Demo-safe fallback	`DEMO_MODE=1` loads `/fixtures/demo/*.json` (transcripts, extraction outputs and embeddings for 3 scripted cases). Lookup key: SHA-256 of normalised text, or `fixture_id` for the 3 scripted audio files. Any cached result shows a small "ক্যাশ" (cached) badge so judges are never misled.
Alerts	None automated. Lead checks `/api/health` and `npm run spend` before each demo.

15 Deployment

15.1 Environment variables

Name	Example	Required
`DATABASE_URL`	`postgres://fera:…@host:5432/fera?sslmode=require`	✓
`AUTH_SECRET`	32+ random bytes, base64	✓
`PARENT_COOKIE_KEY`, `DRAFT_SIGNING_KEY`	32+ random bytes each	✓
`LLM_PROVIDER`, `LLM_MODEL`	`gemini` + model ID from OQ-004	✓
`LLM_FALLBACK_PROVIDER`, `LLM_FALLBACK_MODEL`	second provider	✓
`GEMINI_API_KEY` / `OPENAI_API_KEY` / `ANTHROPIC_API_KEY`	as held	per provider
`STT_PROVIDER`, `STT_MODEL`	`whisper` + model ID	✓
`EMBED_PROVIDER`, `EMBED_MODEL`, `EMBED_CACHE_DIR`	`local`, candidate model, `./models`	✓
`AI_SPEND_CAP_USD`	`25`	✓
`DEMO_ONLY`	`true`; blocks real-case intake	✓
`DEMO_MODE`	`0` for quality evaluation; `1` only for labelled scripted fixtures	✓
`SEED_ADMIN_USER`, `SEED_ADMIN_PASSWORD`	for seeding only	seed

15.2 Local setup (≤ 15 min working target, NFR-014)

The following commands are an implementation target, not delivered application code. The handover contains specification companions only. Apply the schema to a fresh isolated DB; define the npm scripts and README in the actual repository, then test setup time.

```
git clone <repo> fera && cd fera
cp .env.example .env.local            # fill keys
```

```
docker compose up -d db               # postgres 16 + pgvector image (pin and verify exact image/
extension versions at contract gate)
psql "$DATABASE_URL" -v ON_ERROR_STOP=1 -f code/schema_v2.sql
npm ci
npm run model:fetch                   # downloads embedding model to ./models
npm run seed -- --set demo            # districts, neighbours, users, ~200 synthetic records
npm run dev                           # http://localhost:3000
```

15.3 Build and deploy

Step	Command / action
Check	`npm run lint && npm run typecheck && npm test`
Build	`npm run build`
Deploy (Round 2)	Push to `main` → Vercel production deploy. Preview deploy per pull request.
Database	Apply `schema_v2.sql` once on the managed host; seed with `npm run seed -- --set demo` using the production URL.
Smoke test	`npm run smoke -- --base https://<prod-url>` (health, sign-in, one extract, map stats)
Finale laptop	`DEMO_MODE=1 npm run build && npm start`; local DB seeded; phone uses the deployed HTTPS URL over mobile data (A-031).

15.4 Rollback

Layer	Method
App	Re-deploy the previous green commit; tag `r2-freeze` on 14 Oct and `finale` on 23 Oct.
Schema	`schema_v2.sql` is version 1.0 only. Any change ships as `migrations/NNN_*.sql` with a matching down script.
Data	`npm run seed -- --reset --set demo` restores demo data in < 2 min.

15.5 Seed and demo data

Set	Contents
`districts`	64 rows with Bangla/English names and centroids (DEP-004)
`neighbours`	Team-compiled adjacency (A-013)
`users`	1 admin, 2 verified officers, 1 unverified officer, 2 staff — seeded passwords from env
`synthetic`	~200 records, all `is_synthetic=true`, incl. 3 named demo cases (D6 §9)
`fixtures/demo`	Cached outputs for the 3 scripted inputs

16 Trade-offs

Decision	Alternatives	Reason	Cost	Migration path
DECISION-004 LLM adapter, extraction only	Single vendor SDK throughout	No lock-in; fallback on outage; pick best Bangla extractor after eval	~4 h extra code; 3 prompt variants to test	Add a provider class; set env
DECISION-002 pgvector in Postgres	Separate vector DB service	One store, one backup, transactional with case state; 200–30,000 rows is small	HNSW tuning limited; vector search shares DB CPU	Export embeddings; swap COMP-012 query layer
DECISION-008 Embeddings + rules	Pure LLM pairwise judging	Deterministic, explainable, offline, ~0 cost; LLM over 200×200 pairs breaks budget and NFR-004	Misses some semantic nuance	Add an optional LLM re-rank of top 5 behind a flag
DECISION-006 Local embeddings	API embeddings	Free, offline for finale	Model size/cold start risk (RISK-015)	`EMBED_PROVIDER=api` + `npm run reembed`
DECISION-007 Template outputs	LLM-written GD, poster, reasons	Cannot add facts (BR-022); offline; 0 calls	Less natural wording	Add LLM "polish" behind flag with fact-diff check
DECISION-010 Deployed app + local copy	Deployed only	Venue Wi-Fi risk	Two environments to seed	Single env after the hackathon
DECISION-016 Dev/ test split for matching evaluation	Tune and score on the same set	Avoids inflated metrics (RISK-007)	Smaller test set	Larger set later
DECISION-017 Hard spend cap in app	Watch provider dashboards only	Budget is fixed at USD 30	One table + one check	Raise cap via env

17 Evaluation design and reporting

Metric	Set	Definition	Script
Extraction GOAL-002	50 unique labelled reports; identity/variant-grouped dev/test; exact source spans	Present-field accuracy, false additions/ omissions, per-field precision/recall; null agreement reported separately, never headline accuracy. Phone exact, exact time tolerance separate from approximate interval correctness.	eval:e xtract
Matching GOAL-003	Manifest: 80 missing + 80 found + 40 additional decoys = 200 total; 60 canonical true pairs, 20 canonical no-match missing queries; 25 duplicate relationships within the same 200	Identity/duplicate-connected-group split 40% dev/60% test; all variants stay together. Hit/ Recall@5 on true-pair queries; top-1 accuracy and coverage; query-level P@5 and separately named aggregate shortlist precision.	eval: match

Metric	Set	Definition	Script
No-match and uncertainty	Include unknown age, overlapping times, shared contacts, wrong district, similar names and contradictions	Report false-candidate/above-threshold suggestion rates, abstention coverage, true-pair misses; minimum qualification threshold frozen on dev.	eval: match
Duplicates GOAL-004	25 declared same-identity duplicate links; explicit different-child shared-phone negatives	Pairwise precision/recall/F1 plus false consolidation/hiding count. Detection only; no claimed merge accuracy.	eval:d edupe
Speech GOAL-005	20 prompts × 3+ speakers = 60 unique clips; noise/dialect/Banglish slices; repeated runs labelled	WER=(S+D+I)/N may exceed 100%; tokenisation documented; unique count distinct from repeats; phone/age/place/time errors and speech→extraction errors.	eval:w er
Timing GOAL-001	5+ adult volunteers, invented scenarios, paired/counterbalanced methods	Median/range and completion numerator/ denominator; help and order effects recorded; five-person 90% threshold requires 5/5. Exploratory, not population efficacy.	eval:ti me
Human/ audit GOAL-008	Known seeded lifecycle plus concurrent and denied actions	Every terminal coordination event has a verified recorded decision where needed and a safe audit; identity verification is not inferred.	eval:a udit

Manifest fields: record_id, kind, identity_id, duplicate_group_id, true_partner_ids, no_match, decoy_type, source_variant, split, speaker_id where applicable, and ground_truth_revision. Assert total=200 and declared link/query counts. The 40 decoys are additional to the 160 canonical reports; duplicate variants/ links are included within the 200, not an extra uncounted set. Group splitting may alter the exact 40/60 count: report actual counts rather than force identity leakage. Validate an independent test manifest before any tuning.

Ground truth is written and hashed before matcher tuning. Tune prompts/weights/thresholds only on dev. Freeze model revision, preprocessing, embedding space, score components, qualification/dedupe thresholds, prompt/template hashes, and goals before held-out runs. Evaluation enforces DEMO_MODE=0 and cache_hit=false; a fixture hit invalidates the relevant quality run.

Baselines: manual form timing; rules-only extraction where feasible; rules-only matching; embeddings-only and combined ranking. Run gold-field and speech/extracted-field modes. Separate live AI latency/ spend from cached demo timing. Record model/account mode, sample/query count, SHA, date, hardware/deploy, denominators, uncertainty and at least two failures. Approximately 36 true-pair queries may remain from 60 pairs at a 60% split; do not claim national accuracy from this small synthetic test.

Synthetic record text, outcome labels and fixture names do not imply successful rescue or operational officer adoption. If a promised metric cannot be measured, show “not measured” with reason and next action; never a target as an achieved value.

18 Technical acceptance criteria

#	Criterion	Check
1	Every Must FR passes its TC on the protected synthetic deployment; unmet items are disclosed and block the claimed feature.	Playwright run report
2	`schema_v2.sql` applies cleanly to an empty Postgres 16 + pgvector database.	CI job
3	Illegal state transitions fail at the DB trigger even when called directly.	**TC-032**
4	`audit_log` UPDATE, DELETE and TRUNCATE fail; runtime role cannot alter/disable protections.	**TC-016**
5	Parent/staff responses contain no candidates or other-party found-case fields; staff own-entry and parent own-edit DTOs are tested.	TC-019, TC-022
6	Logs from a full E2E run contain 0 phone numbers and 0 child names (regex scan).	**TC-028**
7	Scripted journey completes with network disabled in `DEMO_MODE=1`.	**TC-021**
8	`npm audit --omit=dev`: 0 high/critical.	CI
9	axe plus manual keyboard, Bangla screen reader, short-cut disabling, 320px reflow/ 400% viewport and 200% text zoom; no unresolved AA issue in required flows.	**TC-033**
10	Atomic cap defaults USD 25; any reviewed increase stays ≤ USD 30; reservations and actual settled spend are reported.	`npm run spend`
11	All §12.1 targets met or the miss reported in the deck.	`eval/perf.ts`
12	Evaluation report contains all 5 deck metrics with failure examples.	**TC-020**

19 Engineering task breakdown

TASK IDs, owners, dependencies and done criteria are in D6 §5. The full board totals 222 h; the specified Round 2 subset totals 184 h. Actual remaining effort requires a completed-work inventory.

Area	TASKs	Hours
Setup and spikes	TASK-001 – TASK-005	11
Frontend	TASK-008, TASK-021, TASK-022, TASK-024, TASK-025, TASK-026, TASK-027, TASK-028, TASK-029, TASK-030, TASK-031, TASK-032	47
Backend	TASK-006, TASK-007, TASK-015, TASK-020, TASK-023, TASK-059	24
Database and data	TASK-009, TASK-010, TASK-011, TASK-019	16
AI pipeline	TASK-012, TASK-013, TASK-014, TASK-016, TASK-017, TASK-018, TASK-033	31

Area	TASKs	Hours
Integration and polish	TASK-034, TASK-035, TASK-036	8
Testing and evaluation	TASK-037 – TASK-041, TASK-043, TASK-060	24
Deployment	TASK-042, TASK-053	6
Demo and submission	TASK-044, TASK-045, TASK-046, TASK-054, TASK-055, TASK-058	20
Should / Could	TASK-047 – TASK-052, TASK-056, TASK-057	23
V2 contract, schema, budget and manifest remediation	TASK-061 – TASK-064	12
All tasks	**64 tasks; full task-board estimate**	**222**

Assumptions (Part B)

ID	Assumption
A-037	Usage volumes in §12.2 (190 audio minutes, 1,100 extraction calls, ~1,200 input + 400 output tokens per call).
A-038	Output tokens cost about 4 × input tokens for budgeting; replace with verified real prices at the 11 Oct contract gate.
A-039	The canonical manifest totals 200 reports: 80 missing + 80 found + 40 decoys; 60 canonical true pairs, 20 no-match missing queries and 25 duplicate links within that same total.

IDs introduced

- E-024 NETWORK_OFFLINE (client only)
- DECISION-016 Dev/test split for matching evaluation
- DECISION-017 Hard AI spend cap in app
- A-037 AI usage volumes
- A-038 Output/input token price ratio
- A-039 Synthetic set composition

IDs referenced

- E-001 – E-023, DECISION-002, DECISION-004, DECISION-006 – DECISION-008, DECISION-010, API-019, COMP-012, COMP-021 (D2 Part A)
- FR-025, BR-001, BR-006, BR-011, BR-022, NFR-002 – NFR-005, NFR-014, NFR-016, A-010 – A-013, A-020, A-031 (D1, D2 Part A)

- GOAL-001 – GOAL-005, GOAL-008 (00)
- OQ-004, OQ-011 – OQ-013, OQ-021, OQ-022 (00, D1, D2 Part A)
- RISK-007, RISK-015 (D1, D2 Part A)
- DEP-004 (D1)
- SCREEN-005, SCREEN-008, SCREEN-010, SCREEN-016 (D1)
- TC-010, TC-014, TC-016, TC-019 – TC-023, TC-025, TC-028, TC-030, TC-032, TC-033 (D1)
- TASK-001 – TASK-060 (D6)

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D3 / FLOWS / V2.0

Application Flows

Role journeys, screen states and error recovery · Authors: Sadman Shaharier and Obidit Islam

↗ Contents

Field	Value
Version	2.0
Date	11 Oct 2026 (Asia/Dhaka, UTC+6)
Owner	Sadman Shaharier
Authors	Sadman Shaharier; Obidit Islam
Status	V2 team handover; implementation status UNVERIFIED
Inputs	D1 and D2 Parts A/B V2.0; V2 role projections and state/decision contracts.
Notation	Screen [S-xxx] = SCREEN-xxx; action ACT-xxx; decision D-xxx; API call API-xxx; error E-xxx. UI components COMP-022 – COMP-050 are specified in D4.

V2 release contract

Authors: Sadman Shaharier and Obidit Islam. **Version:** 2.0. **Issued:** 11 October 2026, Asia/Dhaka (UTC+6).

This is the revised team specification, not evidence that the application has been built. Actual repository progress, account configuration, interviews, evaluations, and spend remain **UNVERIFIED** until evidence is attached. V1 defaults are superseded by the V2 rules below; existing requirement IDs are retained.

- This release is a **synthetic-data prototype**. Officer verification and referral delivery are simulated and labelled. The schema rejects records marked non-synthetic; the application must also prevent real reports from entering this release.
- A candidate score is similarity, not identity probability. A reviewer decision does not establish guardian authority or safe reunification. A referral row means **referral recorded**, not delivered or acknowledged.
- Draft payload stays on the device before report confirmation, but providers process audio/text beforehand with a notice. Only expiry/counter/hash metadata and cost reservations are stored server-side before submission.
- Possible duplicates remain visible. Shared phone alone never establishes one child. No physical child-return action is performed by Fera.
- Counts distinguish missing-report rows, found-report rows, and confirmed pairs. They are not a census of unique children or proof of rescue.

- Sources, diagrams, tables, code blocks, and acceptance criteria are part of the handover. The companion code must match the embedded code exactly. Team handover records change closure, evidence gaps, and release gates.

Changes to earlier documents

V2 brings parent, staff, officer and admin screens into agreement with the V2 role projections and APIs; adds manual completion, abstention, stale-decision and referral-recorded states. SCREEN-006 retains its Should priority from the earlier change history.

1 Navigation map per role

Role	Entry	Reachable screens	Home after action
Parent (no account)	Shared link or `/`	S-001 → S-002 → S-003 → S-004; S-005 (fallback); S-006 (Should); S-015 (Could)	S-004
Staff	`/staff` → S-008	S-007, S-016	S-007
Officer (verified)	`/officer` → S-008	S-009, S-010, S-011, S-012, S-014 (Could), S-016	S-009
Officer (unverified)	`/officer` → S-008	Same as staff (D1 BR-024)	S-007
Admin	`/admin` → S-008	S-013, S-011, S-012 (read), S-016	S-013

Persistent elements: COMP-022 header with "বাংলা | English" toggle (ACT-030) on every screen; COMP-023 "সিনথেটিক" (synthetic) badge on any seeded record.

2 Screen inventory

SCREEN-001 Welcome and language · Must

Attribute	Value
Purpose	Explain Fera in 2 lines and start intake
Entry / exit	`/` or shared link → S-002
Components	COMP-022, COMP-050 consent notice, primary button
Primary action	ACT-001 "শুরু করুন" / Start
Secondary	ACT-030 language toggle; link "১৩২১৯-এ কল করুন" / Call 13219 (`tel:13219`)
Data required	None

Attribute	Value
States	Loading: none (static). Empty: n/a. Error: n/a. Success: navigates to S-002
Permissions	Public
Accessibility	Single H1; consent text ≥ 16 px; start button 48 px tall

Copy: "ফেরা আপনার কথা থেকে নিখোঁজ শিশুর রিপোর্ট তৈরি করে। ফেরা পুলিশ বা সিআইডি-কে সতর্কবার্তা পাঠায় না।" / "Fera turns what you say into a missing-child report. Fera does not send alerts to police or CID."

SCREEN-002 Intake chat (voice/text) · Must

Attribute	Value
Purpose	Collect the report by voice or text; ask missing fields one at a time
Entry / exit	S-001, or draft restore → S-003 when D-004 = complete or rounds = 4; → S-005 on E-012/E-013/E-023
Components	COMP-024 chat bubble, COMP-025 voice record button, COMP-026 transcript editor, COMP-027 question prompt, text box, COMP-046 loading, COMP-048 cached badge
Primary action	ACT-002 record voice or ACT-003 type, then ACT-005 send
Secondary	ACT-004 edit transcript; ACT-006 answer question; ACT-014 open form; ACT-015 discard
Data required	Draft from `localStorage` if present
API	API-001, API-002
States	Loading: "শুনছি..." / Listening..., "বুঝে নিচ্ছি..." / Understanding...; at 10 s "এখনও কাজ চলছে, একটু অপেক্ষা করুন" / Still working, please wait. Empty: prompt "আপনার শিশুর কথা বলুন বা লিখুন" / Tell us about your child, by voice or text. Error: E-001, E-007, E-008 – E-011, E-014, E-015, E-022, E-024. Success: next question or move to S-003
Permissions	Parent (cookie created on first API call)
Accessibility	Recorder announces state via `aria-live="polite"` ; timer readable; transcript is the caption for the audio; text path always visible

Follow-up question templates (COMP-009):

Field	Bangla / English
age	"{নাম}-এর বয়স কত?" / How old is {name}?
sex	"শিশুটি ছেলে না মেয়ে?" / Is the child a boy or a girl? (buttons: ছেলে / মেয়ে / অন্য / জানি না)
district	"{নাম}-কে শেষবার কোন জেলায় দেখা গিয়েছিল?" / In which district was {name} last seen?
event time	"শেষবার কবে, কখন দেখা গিয়েছিল?" / When was {name} last seen?
phone	"আপনার মোবাইল নম্বর দিন, যাতে খবর পেলে যোগাযোগ করা যায়।" / Your mobile number, so you can be contacted.

SCREEN-003 Case review · Must

Attribute	Value
Purpose	Parent checks and edits every field before submit (D1 BR-004)
Entry / exit	S-002 → S-004 on submit; back to S-002
Components	COMP-028 field cards (editable), COMP-029 confirm bar, COMP-049 duplicate banner (only own duplicate submission), COMP-044 inline alert
Primary action	ACT-008 "যাচাই করে জমা দিন" / Check and submit (enabled only after "আমি যাচাই করেছি" / I have checked)
Secondary	ACT-007 edit field; ACT-015 discard
Data required	Draft
API	**API-003**
States	Loading: "জমা হচ্ছে..." / Submitting.... Empty: n/a. Error: E-001 (field-level), E-006, E-019, E-024. Success: S-004
Permissions	Parent
Accessibility	Each card has a visible label and value; "অজানা" / unknown shown as text, not blank; error text tied with `aria-describedby`

SCREEN-004 Case submitted and outputs · Must

Attribute	Value
Purpose	Show case reference and get GD draft, poster and 13219 summary
Entry / exit	S-003, S-005 → S-006 (Should)
Components	Reference card, COMP-033 status badge, COMP-030 output cards, COMP-031 poster preview
Primary action	ACT-009 "জিডি খসড়া" / GD draft; ACT-010 "পোস্টার" / Poster
Secondary	ACT-011 "১৩২১৯ সারাংশ" / 13219 summary (Should); ACT-012 copy; ACT-013 download PNG; `tel:13219`
Data required	Case (own)
API	**API-006**
States	Loading per card: "তৈরি হচ্ছে..." / Preparing.... Empty: n/a. Error: E-016, E-020, E-024. Success: "কপি হয়েছে" / Copied
Permissions	Parent (owner)
Accessibility	Output text in a selectable `<pre>`-like block with lang="bn"; copy confirmation announced

Every output card carries the label "খসড়া — নিজে জমা দিন" / Draft — submit it yourself (D1 BR-002).

SCREEN-005 Manual case form · Must

Attribute	Value
Purpose	Fallback when AI fails or is unavailable (FR-026)
Entry / exit	S-002 (E-012, E-013, E-014 second time, E-023) or ACT-014 → S-003
Components	COMP-032 manual form (one field per row), COMP-044
Primary action	ACT-005 "পরের ধাপ" / Next (to S-003)
Secondary	ACT-015 discard
Data required	Any extracted fields pre-filled
States	Loading: none. Error: E-001 inline. Success: S-003
Permissions	Parent
Accessibility	Fieldset groups; district as searchable select with Bangla and English names; numeric fields accept Bangla digits

SCREEN-006 Parent case status and edit · Should

Attribute	Value
Purpose	Status only; edit while allowed (BR-025)
Entry / exit	Return visit on same device; S-004 → S-003 (edit)
Components	COMP-033 status badge, reference card, edit button
Primary action	ACT-016 view status; ACT-017 "তথ্য সংশোধন" / Correct details
Data required	API-018 (Should), API-005 (Should)
States	Loading: skeleton. Empty: "এই ফোনে কোনো রিপোর্ট নেই" / No reports on this phone. Error: E-018, E-019. Success: "সংশোধন জমা হয়েছে" / Correction submitted
Permissions	Parent (owner); never shows candidate or found data
Accessibility	Status uses icon + text + colour

Status labels: "খোলা" / Open; "যাচাই চলছে" / Being checked; "বন্ধ" / Closed.

SCREEN-007 Found-child log · Must

Attribute	Value
Purpose	Staff log a found child (FR-011)
Entry / exit	S-008 (staff home) → same screen with success card

Attribute	Value
Components	COMP-034 found-child form (free text + structured fields), COMP-025 (optional voice), COMP-028, COMP-029
Primary action	ACT-019 "সংরক্ষণ করুন" / Save
Secondary	ACT-003 type description; ACT-015 clear
API	API-002 (kind found), API-007
States	Loading: "সংরক্ষণ হচ্ছে..." / Saving.... Empty: blank form. Error: E-001, E-006, E-007, E-024. Success: "সংরক্ষিত। নম্বর F-...। মিল খোঁজা হয়েছে।" / Saved. Ref F-.... Matching state queued/ running/complete/failed is explicit. (no candidate details)
Permissions	Staff, officer
Accessibility	No photo field (D1 BR-006); time field defaults to now in BST

SCREEN-008 Sign-in · Must

Attribute	Value
Purpose	Staff, officer and admin sign-in (FR-037)
Entry / exit	`/staff`, `/officer`, `/admin`, or E-002 → role home
Components	COMP-047 sign-in form; demo account hint visible only when `DEMO_MODE=1`
Primary action	ACT-018 "সাইন ইন" / Sign in
API	**API-020**
States	Loading: button spinner. Error: "ইউজারনেম বা পাসওয়ার্ড ঠিক নয়" / Username or password is incorrect (E-002); E-007
Permissions	Public
Accessibility	Autocomplete attributes; password show/hide toggle

SCREEN-009 Review queue · Must

Attribute	Value
Purpose	Officer sees cases with pending candidates (FR-014)
Entry / exit	Officer home → S-010, S-011, S-012
Components	COMP-035 queue list, COMP-036 candidate rows, COMP-033, COMP-023, district filter
Primary action	ACT-020 open a candidate
Secondary	ACT-024 map; ACT-029 sign out
API	**API-008**

Attribute	Value
States	Loading: 5 skeleton rows. Empty: "এখনো কোনো সম্ভাব্য মিল নেই। নতুন রিপোর্ট এলে স্বয়ংক্রিয়ভাবে যাচাই হবে।" / No likely matches yet. New reports are checked automatically. Error: E-002, E-003, E-021
Permissions	Officer (verified)
Accessibility	Table semantics; keyboard: ↑/↓ moves rows, Enter opens

SCREEN-010 Match review (side by side) · Must

Attribute	Value
Purpose	Compare evidence and decide (FR-015 – FR-017)
Entry / exit	S-009 → S-012 (after confirm) or S-009 (after reject)
Components	COMP-037 evidence table, COMP-038 decision panel, COMP-049 duplicate banner, score with label "মিলের মাত্রা" / similarity
Primary action	ACT-021 "মিল নিশ্চিত করুন" / Confirm match
Secondary	ACT-022 "মিল নয়" / Not a match (reason required); ACT-025 audit
API	API-009, API-010
States	Loading: two-column skeleton. Error: E-004, E-005. Success (confirm): "মিল নিশ্চিত। শিশু সুরক্ষা সেবায় রেফারেল তৈরি হয়েছে।" / Match confirmed. Referral to child-protection services created. Success (reject): "বাতিল করা হয়েছে" / Rejected
Permissions	Officer (verified)
Accessibility	Agree / differ / unknown shown as icon + word ("মিল" / match, "অমিল" / differs, "অজানা" / unknown); confirm needs a second step dialog

Confirm dialog: "আপনি কি নিশ্চিত? এতে শিশুকে সরাসরি ফেরত দেওয়া হবে না; রেফারেল নথিভুক্ত হবে (সিমুলেশন)।" / Are you sure? This will not return the child directly; it creates a referral to child-protection services.

SCREEN-011 District dashboard · Must

Attribute	Value
Purpose	Map of Open/Matched/Closed counts by district (FR-019)
Entry / exit	S-009, S-013 → S-012 via district list
Components	COMP-040 district map, COMP-041 district table (always rendered), totals, "সিনথেটিক সহ" / include synthetic toggle
Primary action	ACT-024 select a district
API	**API-012**

Attribute	Value
States	Loading: totals skeleton. Empty district: counts "০". Error: tiles fail → markers on plain background + table (offline). Success: map rendered
Permissions	Officer, admin
Accessibility	The table is the accessible equivalent of the map; markers are focusable buttons with text counts

SCREEN-012 Case detail and audit trail · Must

Attribute	Value
Purpose	Full case, duplicates, close action, audit (FR-018, FR-021, FR-022)
Entry / exit	S-009, S-010, S-011
Components	Case fields, COMP-033, COMP-049, COMP-042 audit timeline, COMP-039 close dialog
Primary action	ACT-023 "কেস বন্ধ করুন" / Close case
Secondary	ACT-025 view audit
API	API-004, API-011, API-013
States	Loading: skeleton. Error: E-004, E-005, E-017. Success: "কেস বন্ধ হয়েছে" / Case closed
Permissions	Officer (actions); admin (read, payload-redacted fields hidden)
Accessibility	Timeline as an ordered list with times in BST

Close outcomes: "শনাক্ত, রেফারেল হয়েছে" / Located, referred; "ফেরার বাইরে নিষ্পত্তি" / Resolved outside Fera.

SCREEN-013 Admin data tools · Must

Attribute	Value
Purpose	Redact identifying payloads (Must), verify officer [MOCK] (Must), import (Should)
Entry / exit	Admin home
Components	COMP-043 admin actions, user list, import text area
Primary action	ACT-027 "বন্ধ কেস বেনামী করুন" / Redact identifying payloads from eligible terminal cases
Secondary	ACT-028 verify officer [MOCK]; ACT-026 import batch
API	API-015, API-016, API-017, API-021
States	Loading: progress text. Empty: "পরিচয় তথ্য অপসারণের জন্য উপযুক্ত রিপোর্ট নেই" / No eligible reports for payload redaction. Error: E-001, E-003. Success: "{n}টি রিপোর্টের পরিচয় তথ্য অপসারণ করা হয়েছে" / {n} cases payload-redacted
Permissions	Admin

Attribute	Value
Accessibility	Destructive action needs typed confirmation "বেনামী"

SCREEN-014 Duplicate review · Could

Attribute	Value
Purpose	Mark linked records as same child or not (FR-023)
Entry / exit	S-012 duplicate banner
Components	COMP-037 (two same-kind records), buttons
Primary action	ACT-032 "একই শিশু" / Same child or "আলাদা" / Different
States	Error: E-004, E-005
Permissions	Officer

SCREEN-015 Disaster mode · Could

Attribute	Value
Purpose	Shelter volunteer logs a separated child (FR-036)
Entry / exit	Link from S-007 footer
Components	COMP-034 with banner "পরিকল্পিত সম্প্রসারণ — বন্যা ও ঘূর্ণিঝড় আশ্রয়কেন্দ্র/ Planned extension — flood and cyclone shelters
Primary action	ACT-033 log separated child
Permissions	Staff

SCREEN-016 Not found / not allowed · Must

Attribute	Value
Purpose	Single response for missing or hidden records (E-004) and role errors (E-003)
Components	COMP-045 empty state with home link
Copy	"তথ্যটি পাওয়া যায়নি।" / We couldn't find that.
Accessibility	Focus moves to the heading

3 Main flow (core demo journey)

Step	Who	Screen	Action	API	Result
1	Parent	S-001	**ACT-001**	—	Intake opens
2	Parent	S-002	ACT-002 record 40 s note	**API-001**	Transcript shown
3	Parent	S-002	ACT-004 fix a word, ACT-005 send	**API-002**	Draft + district question
4	Parent	S-002	ACT-006 "মিরপুর, ঢাকা"	**API-002**	Draft complete
5	Parent	S-003	ACT-008 check and submit	**API-003**	Case M-… open
6	Parent	S-004	ACT-009, ACT-010	**API-006**	GD draft and poster
7	Staff	S-008 → S-007	ACT-018, ACT-019	API-020, API-007	Found case F-…; matching run
8	Officer	S-008 → S-009	ACT-018, ACT-020	**API-008**	Candidate rank 1
9	Officer	S-010	**ACT-021**	API-009, API-010	Missing confirmed; found referred; referral
10	Officer	S-012	**ACT-023**	**API-011**	Missing closed
11	Officer	S-011	**ACT-024**	**API-012**	Dhaka: Missing Closed +1, Missing Matched −1; found counts unchanged by missing closure

4 First-time flow

1. Parent opens a shared link → S-001 with COMP-050 consent notice.
2. ACT-001 → S-002 shows one example sentence: "যেমন: আমার মেয়ে মীম, বয়স ১১, আজ সকালে স্কুলে যাওয়ার পথে…" / For example: My daughter Mim, 11, this morning on the way to school….
3. First mic tap triggers the browser permission prompt (D-001).
4. Parent cookie is created on the first API call; no sign-up.

5 Returning flow

Role	Flow
Parent, unfinished draft	S-002 shows "আগের খসড়া পাওয়া গেছে — চালিয়ে যাবেন?" / Earlier draft found — continue? (ACT-031 continue or ACT-015 discard). Expired after 24 h → E-019.
Parent, submitted case	S-006 shows status (Should/Could). Other device → not supported (OQ-020).
Officer	S-008 → S-009, candidates created since last visit first.

6 Auth flow

Step	Detail
Parent	No sign-in. API-001/API-002 set `fera_ps` cookie (30 days). Draft token signed separately (24 h).
Staff/officer/admin	S-008 → API-020 credentials → JWT cookie 8 h → role home (§1).
D-008 role check	Every request: role + `verified` read from DB. Unverified officer → staff home.
Expiry	Staff/officer request after 8 h → E-002 → S-008 → return to the same URL.
Sign-out	ACT-029 clears cookie → S-008.

7 Error and recovery flows

#	Edge case	Trigger	System behaviour	User message (Bangla / English)	Error	Recovery
1	Invalid input	Bad phone, age 19, text < 20 chars	Field-level validation	"সঠিক মোবাইল নম্বর দিন (০১ দিয়ে শুরু, ১১ সংখ্যা)" / Enter a valid mobile number (starts 01, 11 digits)	E-001	Fix field
2	Unintelligible voice note	Empty or garbled transcript	No draft created	"কথাগুলো পরিষ্কার শোনা যায়নি। আবার বলুন বা লিখে দিন।" / We couldn't hear that clearly. Try again or type instead.	E-010	Re-record or type
3	Wrong language detected	< 30 % Bangla or Latin letters	Stop before extraction	"বাংলা বা ইংরেজিতে বলুন বা লিখুন।" / Please speak or write in Bangla or English.	E-015	Re-enter
4	Duplicate submission	Same principal + submission UUID; identical canonical payload	Return existing case	"এই রিপোর্ট আগেই জমা হয়েছে। নম্বর: M-7K3Q2A" / Already submitted. Ref: M-7K3Q2A	200 replay, not an error	Open S-004 for existing case
5	Same child reported from another source	D1 BR-021 rule fires	Both open; linked; shown once in candidates	Officer banner: "সম্ভাব্য একই শিশু: M-..." / Possible same child: M-...	—	S-012 / S-014
6	Network failure mid-upload	Fetch fails or offline event	Draft and audio kept in browser	"ইন্টারনেট সংযোগ নেই। আপনার তথ্য রাখা আছে — সংযোগ ফিরলে আবার পাঠান।" / No internet. Your details are saved — send again when back online.	E-024	ACT-031 retry
7	Expired session	Draft > 24 h; staff JWT > 8 h	Parent: draft cleared. Staff: redirect	"আপনার খসড়ার সময় শেষ। নতুন করে শুরু করুন।" / Your draft has expired. Please start again.	E-019 / E-002	S-002 new / S-008
8	Empty data	No candidates, no cases	Explicit empty state	"এখনো কোনো সম্ভাব্য মিল নেই।" / No likely matches yet.	—	Wait; auto-check
9	Unauthorised access to a matched record	Parent/staff/unverified opens a candidate or other case URL	404 body; `access.denied` audit	"তথ্যটি পাওয়া যায়নি।" / We couldn't find that.	E-004	S-016
10	Slow STT/LLM (> 10 s)	Elapsed 10 s	Message changes; abort at 25 s	"এখনও কাজ চলছে, একটু অপেক্ষা করুন" / Still working, please wait	E-014 at 25 s	Retry once, then S-005

#	Edge case	Trigger	System behaviour	User message (Bangla / English)	Error	Recovery
11	User cancellation	ACT-015 during recording or draft	Recording stops; confirm dialog; draft deleted	"খসড়া মুছে ফেলবেন?" / Delete this draft?	—	S-001
12	Parent edits after submission	**ACT-017**	Allowed in submitted/open/ matched_pending_review; new version; candidates voided	Locked: "মিল নিশ্চিত হওয়ার পর কেস বদলানো যায় না।" / Can't edit after a match is confirmed.	E-018 when locked	S-003 or `tel:13219`
13	Officer rejects a match	**ACT-022**	Reason required (3–500 chars); case back to open if none pending; pair not re-suggested	"কারণ লিখুন" / Enter a reason	E-001 if no reason	S-009
14	Unsupported browser / no mic permission	No MediaRecorder or permission denied	Text box focused; mic button disabled with reason	"মাইক্রোফোন চালু করা যায়নি। আপনি লিখে জানাতে পারেন।" / The microphone isn't available. You can type instead.	**E-022**	Text path
15	AI fails twice / unavailable	E-012, E-013	Manual form pre-filled	"সহকারী তথ্য গুছাতে পারেনি। ফর্মটি পূরণ করুন — আপনার লেখা রাখা আছে।" / The assistant couldn't organise this. Please fill the form — your text is kept.	E-012 / E-013	S-005

8 State transitions and recovery

Kind	Persisted states	Terminal meaning
Missing	submitted → open ↔ matched_pending_review → confirmed → closed; open → closed only for outside resolution	Closed coordination record with reason; no rescue proof
Found	open ↔ matched_pending_review → referred	Referral recorded (simulation), not agency receipt
Browser draft	Device payload plus server metadata; expires on access/metadata check	Not a submitted case

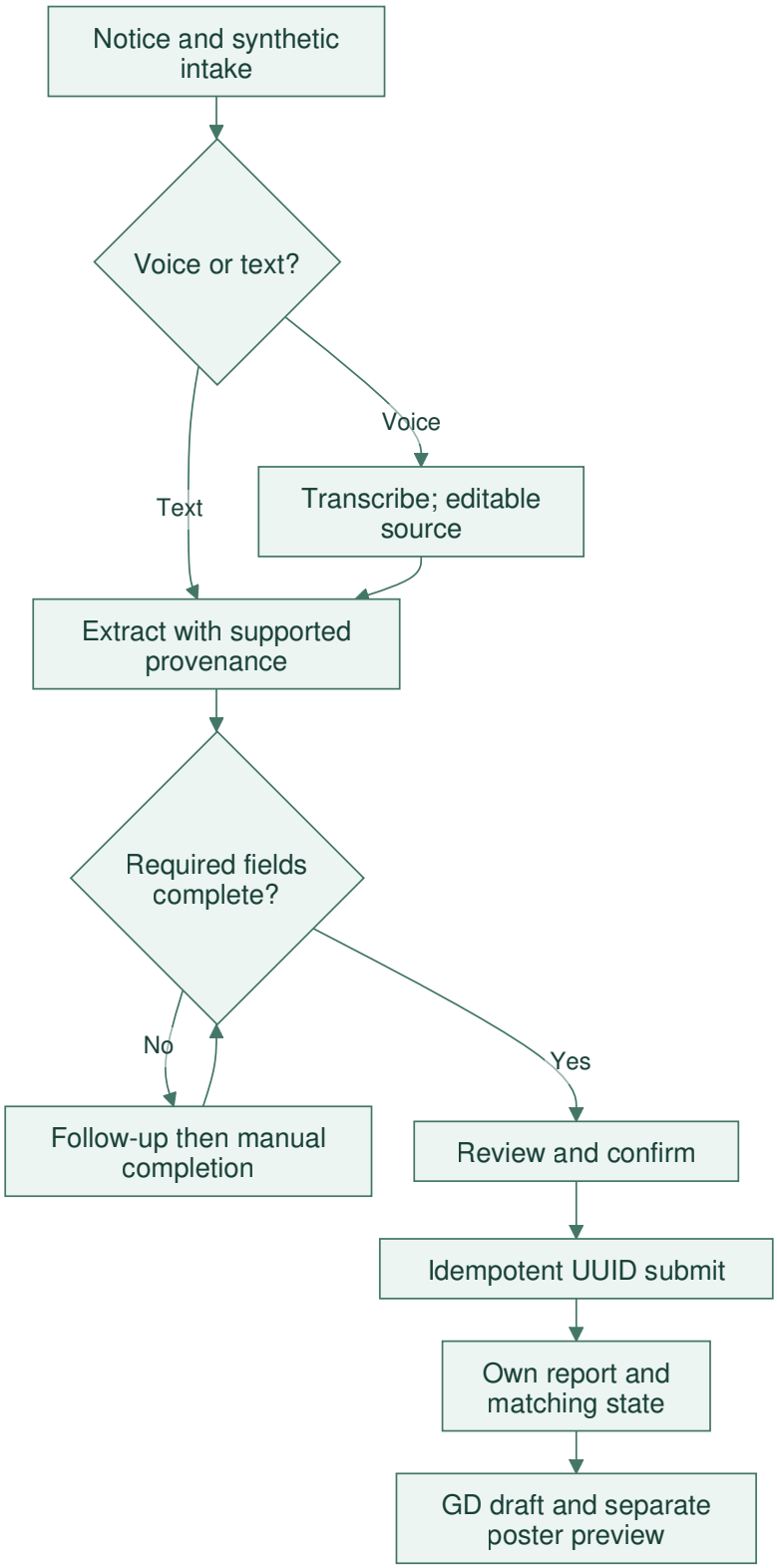
See D1 BR-012/013 for complete guards. Terminal correction/reopen is a pre-pilot two-reviewer procedure; no ordinary backward transition exists in the demo.

8.1 Recovery rules

Lost submission response: retry the same UUID and receive the same reference. Changed payload with same UUID: conflict. Matching failure: case remains saved with failed/queued status and retry control; do not claim no matches until matching completes. Stale candidate: server rejects and refreshes current evidence. Confirming/voiding competitors reconciles all affected case queues. Disconnected browser: retain device draft with an expiry notice and retry same UUID. Cached script: badge and fixture intake time remain visible. Paid-call failure: manual fields remain usable.

9 Flowcharts per role

9.1 Reporter

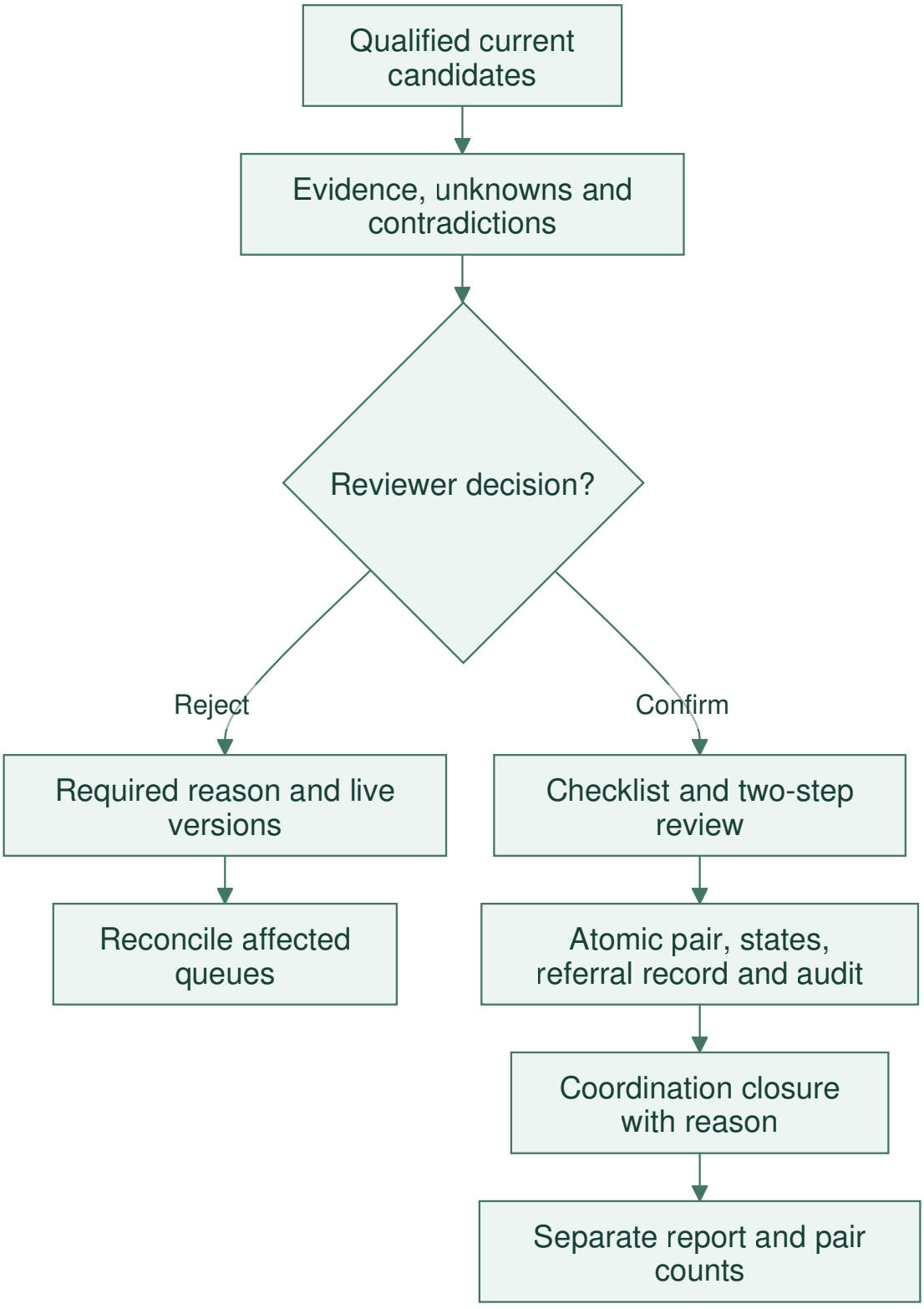


V2 specification diagram · see adjacent rules for authority and evidence limits

9.2 Staff

Sign in to a separate browser context; describe/review the synthetic found child; confirm required district/place/time/contact; use found-only speech/extraction if needed; save with scoped UUID; see own entry and matching state. Staff sees no candidate/other-party report payload. Queue access belongs to verified officers.

9.3 Officer



V2 specification diagram · see adjacent rules for authority and evidence limits

9.4 Admin

API-021 lists users for mock verification. API-016 redacts eligible payloads only through the restricted action, preserving active evidence. API-014 returns rematch status only. API-015 imports invented samples only. API-012/013 give aggregates/safe audit; API-004 gives metadata, never child or candidate detail. Duplicate resolution, if built, is officer-only API-023. District case drill-through uses officer-only API-022.

Actions (ACT)

ID	Action	Role
ACT-001	Tap Start	Parent
ACT-002	Record voice note	Parent, staff
ACT-003	Type report or description	Parent, staff
ACT-004	Edit transcript	Parent
ACT-005	Send / next	Parent
ACT-006	Answer follow-up question	Parent
ACT-007	Edit a field	Parent, staff
ACT-008	Check and submit	Parent
ACT-009	Request GD draft	Parent
ACT-010	Request poster	Parent
ACT-011	Request 13219 summary	Parent
ACT-012	Copy text	Parent
ACT-013	Download poster PNG	Parent
ACT-014	Open manual form	Parent
ACT-015	Discard / cancel	Parent, staff
ACT-016	View case status	Parent
ACT-017	Correct details after submit	Parent
ACT-018	Sign in	Staff, officer, admin
ACT-019	Save found child	Staff, officer
ACT-020	Open candidate	Officer
ACT-021	Confirm match	Officer

ID	Action	Role
ACT-022	Reject match with reason	Officer
ACT-023	Close case with outcome	Officer
ACT-024	Use map or district table	Officer, admin
ACT-025	View audit trail	Officer, admin
ACT-026	Import invented fixture texts	Admin
ACT-027	Redact identifying payloads from eligible terminal cases	Admin
ACT-028	Verify officer [MOCK]	Admin
ACT-029	Sign out	Staff, officer, admin
ACT-030	Switch language	All
ACT-031	Retry / continue draft	Parent, staff
ACT-032	Mark duplicate same/different	Officer
ACT-033	Log separated child (disaster mode)	Staff

Decisions (D)

ID	Decision
D-001	Microphone available and permitted?
D-002	Audio length 5–120 s and ≤ 2 MB?
D-003	Transcript intelligible and Bangla/English?
D-004	Required fields complete?
D-005	AI calls used < 6?
D-006	Extraction valid within 2 attempts?
D-007	Submission UUID retry or conflicting payload?
D-008	Role officer and verified?
D-009	Pending candidates exist?
D-010	Officer confirms or rejects?
D-011	Other pending candidates remain?

ID	Decision
D-012	Referral exists?
D-013	Case editable (BR-025)?
D-014	Device online?
D-015	Session valid?
D-016	Admin typed confirmation correct?

IDs introduced

- ACT-001 – ACT-033 (one line each in "Actions")
- D-001 – D-016 (one line each in "Decisions")

IDs referenced

- SCREEN-001 – SCREEN-016 (D1)
- API-001 – API-020, E-001 – E-024 (D2)
- COMP-009 (D2); COMP-022 – COMP-050 (D4)
- BR-002, BR-004, BR-006, BR-021, BR-024, BR-025, FR-011, FR-014 – FR-019, FR-021 – FR-023, FR-026, FR-036, FR-037, STORY-019 (D1)
- OQ-020 (D1)

V2 screen dependency acceptance

- SCREEN-006 uses Must owner read API-004 and Should API-018/005 together; missing optional history does not block intake.
- SCREEN-007 voice is authorised through API-001 for own found drafts. Device permission failure routes to text.
- SCREEN-011 drill-through uses API-022 with kind-specific report counts; admin has no child-detail drill-through.
- SCREEN-013 user listing uses API-021; seeded authority is explicitly mock.
- SCREEN-014 is Could with API-023 and reasoned profile review. Possible duplicate flags do not hide reports.
- If four follow-ups expire with mandatory unknowns, SCREEN-005 lists the missing fields and preserves known values. Submit is blocked clearly until completed.
- SCREEN-010 calls confirmation a reviewer decision and a simulated referral record; no child delivery claim.

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D4 / DESIGN / V2.0

UI & UX Design Brief

Bangla interface, component behaviour and accessibility · Authors: Sadman Shaharier and Obidit Islam

↗ Contents

Field	Value
Version	2.0
Date	11 Oct 2026 (Asia/Dhaka, UTC+6)
Owner	Sadman Shaharier (design shared across team)
Authors	Sadman Shaharier; Obidit Islam
Status	V2 team handover; implementation status UNVERIFIED
Companion	`tokens_v2.css` — byte-identical copy of §16
Inputs	D1, D2 Parts A/B and D3 V2.0; tokens_v2.css; validation report.

V2 release contract

Authors: Sadman Shaharier and Obidit Islam. **Version:** 2.0. **Issued:** 11 October 2026, Asia/Dhaka (UTC+6).

This is the revised team specification, not evidence that the application has been built. Actual repository progress, account configuration, interviews, evaluations, and spend remain **UNVERIFIED** until evidence is attached. V1 defaults are superseded by the V2 rules below; existing requirement IDs are retained.

- This release is a **synthetic-data prototype**. Officer verification and referral delivery are simulated and labelled. The schema rejects records marked non-synthetic; the application must also prevent real reports from entering this release.
- A candidate score is similarity, not identity probability. A reviewer decision does not establish guardian authority or safe reunification. A referral row means **referral recorded**, not delivered or acknowledged.
- Draft payload stays on the device before report confirmation, but providers process audio/text beforehand with a notice. Only expiry/counter/hash metadata and cost reservations are stored server-side before submission.
- Possible duplicates remain visible. Shared phone alone never establishes one child. No physical child-return action is performed by Fera.
- Counts distinguish missing-report rows, found-report rows, and confirmed pairs. They are not a census of unique children or proof of rescue.
- Sources, diagrams, tables, code blocks, and acceptance criteria are part of the handover. The companion code must match the embedded code exactly. Team handover records change closure, evidence gaps, and release gates.

Changes to earlier documents

V2 corrects border contrast, narrow officer layouts, shortcut controls, report-count labels and provider/ referral wording. ASCII sketches are replaced by structured layout specifications. The CSS companion is the exact embedded source.

1 Design objective

A frightened parent on a mid-range Android phone completes a reviewed Bangla report in ≤ 5 min (GOAL-001). An officer at a desk decides a match on evidence in ≤ 2 min (PERSONA-002). Nothing on screen sensationalises a child.

2 Principles

#	Principle	Test
1	Bangla first, plain words	Every parent string reads at a primary-school level; no legal terms without a gloss
2	One thing at a time	Parent screens show one question or one decision per view
3	Show what the AI understood	Every extracted field is visible and editable before submit
4	Calm, not alarming	No red banners, countdowns or flashing on parent screens
5	Evidence before decision	Officer never sees confirm without both records side by side
6	Honest labels	"খসড়া" (draft), "সিনথেটিক" (synthetic), "ক্যাশ" (cached), [MOCK] always visible
7	Status never by colour alone	Icon + text + colour on every status

3 Brand personality

Calm, trustworthy, urgent but not alarming. Voice of a steady neighbour who knows the process. Never: dramatic imagery, child photos, "runaway" language, exclamation-mark urgency, alert-style sirens.

4 User contexts

Context	Parent (PERSONA-001)	Officer (PERSONA-002)	Staff (PERSONA-003)
Device	Android phone, 360–412 px wide, 4G, often one hand	Laptop/desktop, 1280–1920 px, mouse + keyboard	Phone or shared desk PC
State	Distressed, time-pressured, may be outdoors	Focused, many cases, accountable	Busy, interrupted

Context	Parent (PERSONA-001)	Officer (PERSONA-002)	Staff (PERSONA-003)
Need	Reassurance, few steps, voice	Density, comparison, keyboard speed	Short form, quick save

5 Information architecture

Area	Screens
Parent (no account)	Welcome (SCREEN-001) → Intake (SCREEN-002) → Review (SCREEN-003) → Outputs (SCREEN-004); Manual form (SCREEN-005); Status (SCREEN-006)
Staff	Sign-in (SCREEN-008) → Found-child log (SCREEN-007); Disaster mode (SCREEN-015)
Officer	Queue (SCREEN-009) · Match review (SCREEN-010) · Map (SCREEN-011) · Case detail (SCREEN-012) · Duplicates (SCREEN-014)
Admin	Data tools (SCREEN-013); aggregate map/table and metadata-only case detail
Shared	Not found (SCREEN-016)

Officer top nav: "যাচাই তালিকা" / Review queue · "মানচিত্র" / Map · "সাইন আউট" / Sign out.

6 Visual direction

6.1 Palette (light)

Token	Hex	Use	Contrast
`--color-bg`	#FAFAF7	Page	text 15.20:1
`--color-surface`	#FFFFFF	Cards	text 15.90:1
`--color-surface-2`	#F1F3F2	Chat bubbles, table stripes	—
`--color-text`	#1B2421	Body text	15.90:1 on surface
`--color-text-muted`	#4E5A55	Secondary text	7.20:1 on surface; 6.88:1 on bg
`--color-border`	#6F7D77	Input borders	4.31:1 on surface; minimum 3.86:1 across all three surfaces
`--color-primary`	#0F5C55	Buttons, links (deep teal)	7.83:1 on surface
`--color-on-primary`	#FFFFFF	Text on primary	7.83:1
`--color-focus`	#2457C5	Focus ring	6.47:1 on surface
`--status-open-fg` / `-bg`	#7A4F00 / #FFF3D6	Open (amber)	6.46:1

Token	Hex	Use	Contrast
`--status-matched-fg` / `-bg`	#1E4E9C / #E3EDFC	Matched (blue)	6.78:1
`--status-closed-fg` / `-bg`	#1E6B3A / #E2F3E8	Closed (green)	5.66:1
`--color-danger-fg` / `-bg`	#B42318 / #FDECEA	Officer-side destructive and field errors only	5.75:1

6.2 Palette (dark)

Token	Hex	Contrast
`--color-bg`	#111816	text 15.32:1
`--color-surface`	#18211E	text 14.01:1
`--color-surface-2`	#212C28	—
`--color-text`	#E8EEEB	14.01:1 on surface
`--color-text-muted`	#A9B5B0	7.78:1 on surface
`--color-border`	#82938B	5.09:1 on surface; minimum 4.46:1 across all three surfaces
`--color-primary`	#5FC4B8	7.90:1 on surface
`--color-on-primary`	#0B1F1C	8.21:1 on primary
`--color-focus`	#8AB4F8	7.81:1 on surface
Open fg / bg	#F2C14E / #33290F	8.54:1
Matched fg / bg	#8DB4F5 / #172640	7.19:1
Closed fg / bg	#7FD39A / #14301F	7.94:1
Danger fg / bg	#F28B82 / #3A1714	6.70:1

V2 ratios are recalculated with the WCAG 2.x relative-luminance formula on 11 Oct 2026. These verify the listed colour pairs; full rendered accessibility, focus, zoom, reflow and screen-reader tests remain pending.

6.3 Typography

Item	Value
Bangla	Noto Sans Bengali (Google Fonts), weights 400, 600
Latin pair	Noto Sans (Google Fonts), weights 400, 600 — matched x-height and stroke with the Bangla face
Loading	Self-hosted woff2 subsets, `font-display: swap`; Bangla subset preloaded on parent routes

Item	Value
Scale	14 / 16 / 18 / 22 / 28 px (`--font-size-sm` … `--font-size-2xl`)
Parent body	18 px, line-height 1.7 (Bangla conjuncts and matras need room)
Officer body	16 px, line-height 1.55
Numerals	Bangla numerals on parent screens; officer screens show both where space allows ("১৩ / 13")

6.4 Spacing, radius, elevation, icons, illustration

Item	Value
Spacing	4-px base: 4, 8, 12, 16, 24, 32, 48, 64
Radius	6 px inputs, 10 px cards, 16 px sheets, 999 px pills
Elevation	Level 1 cards: `0 1px 2px rgba(0,0,0,.08)` ; level 2 dialogs: `0 8px 24px rgba(0,0,0,.16)`
Icons	Outline, 1.75 px stroke, 20/24 px (Lucide set); always paired with text on actions and statuses
Status icons	Open: circle-dot; Matched: git-compare; Closed: circle-check; Referral recorded: hand-heart with simulation text
Illustration	Simple flat line drawings of places (house, station, map pin). No children's faces, no stock photos, no silhouettes of lone children

7 Layout and responsive behaviour

Breakpoint	Layout
< 600 px	Parent screens single column, max width 480 px, sticky bottom action bar (safe-area aware)
600–1023 px	Staff form two-column for structured fields; officer screens stack columns
≥ 1024 px	Officer: 12-column grid, max 1280 px; match review two equal columns + 320 px decision panel; map 2/3 + table 1/3

Text scales to 200 % without horizontal scroll on parent screens. Tap targets ≥ 44 × 44 px.

8 Component inventory

ID	Component	Used on	Key states
COMP-022	App header with language toggle	All	bn / en
COMP-023	Synthetic badge "সিনথেটিক"	Any seeded record	—

ID	Component	Used on	Key states
COMP-024	Chat bubble (assistant / user)	SCREEN-002	sending, sent, error
COMP-025	Voice record button with timer	SCREEN-002, SCREEN-007	idle, requesting, recording, uploading, disabled
COMP-026	Transcript editor	SCREEN-002	editable, saved
COMP-027	Question prompt with quick-reply chips	SCREEN-002	—
COMP-028	Editable field card	SCREEN-003, SCREEN-007	filled, unknown, error, edited
COMP-029	Confirm bar (checkbox + submit)	SCREEN-003, SCREEN-007	disabled until checked, submitting
COMP-030	Output card with copy	SCREEN-004	loading, ready, copied, limit reached
COMP-031	Poster preview + PNG download	SCREEN-004	rendering, ready
COMP-032	Manual case form	SCREEN-005	prefilled, error
COMP-033	Status badge (icon + text + colour)	SCREEN-004, -006, -009, -011, -012	open, matched, closed, referred
COMP-034	Found-child form	SCREEN-007, SCREEN-015	—
COMP-035	Review queue list	SCREEN-009	loading, empty, populated
COMP-036	Candidate row (rank, score, reason)	SCREEN-009	focused, opened
COMP-037	Side-by-side evidence table	SCREEN-010, SCREEN-014	agree, differ, unknown per row
COMP-038	Decision panel (confirm / reject + reason)	SCREEN-010	idle, confirming dialog, submitting
COMP-039	Close-case dialog	SCREEN-012	outcome chosen, note entered
COMP-040	District map	SCREEN-011	tiles, no-tiles fallback
COMP-041	District table	SCREEN-011	sorted by open count
COMP-042	Audit timeline	SCREEN-012	—
COMP-043	Admin action panel	SCREEN-013	typed confirmation

ID	Component	Used on	Key states
COMP-044	Inline alert / toast	All	info, success, error
COMP-045	Empty state	SCREEN-006, -009, -013, -016	—
COMP-046	Loading state with 10 s message	SCREEN-002, -004, -007	< 10 s, ≥ 10 s
COMP-047	Sign-in form	**SCREEN-008**	error, rate-limited
COMP-048	Cached badge "ক্যাশ"	Any AI output from demo cache	—
COMP-049	Duplicate banner "সম্ভাব্য একই শিশু"	SCREEN-010, SCREEN-012	—
COMP-050	Consent notice	**SCREEN-001**	—

9 Interaction guidelines

Interaction	Rule
Voice record	Tap to start, tap to stop (no press-and-hold). Live timer "০:৩৮ / ২:০০". Auto-stop at 120 s. Haptic tick on start/stop where supported. After stop: "পাঠাচ্ছি..." / Sending.... Transcript appears in an editable box with "ঠিক আছে, পাঠান" / OK, send.
Review before register/ export; provider notice earlier	Cards in the order the parent spoke. Unknown fields read "অজানা — যোগ করুন" / Unknown — add. Required-field errors are visible; submit stays disabled until "আমি যাচাই করেছি" / I have checked is ticked.
Side-by-side evidence	Missing on the left, found on the right, same field order. Row verdict icon + word. Reason line and score on top; score labelled "মিলের মাত্রা ৮৭/১০০" / Similarity 87/100, never "%" or "probability".
Confirm / reject	Confirm opens a dialog restating the referral rule; focus starts on "ফিরে যান" / Go back. Reject requires a reason (min 3 chars) and quick chips: "বয়স মেলে না" / Age differs, "জেলা দূরে" / District far, "পোশাক আলাদা" / Clothing differs.
Copy and download	Copy shows "কপি হয়েছে" for 2 s and announces it. Poster PNG 1080 × 1350 px, white background, Bangla text, 13219 and the explicitly chosen contact option, "খসড়া" footer.
Keyboard (officer)	`J/K` or ↑/↓ move in queue; `Enter` open; `C` confirm (opens dialog); `R` reject; disabled while typing and switchable/remappable; `Esc` close dialog. Shortcuts listed under "?"

10 Microcopy guidelines

Rule	Bangla / English example
Address the parent with respect (আপনি)	"আপনার শিশুর কথা বলুন" / Tell us about your child

Rule	Bangla / English example
Say what happens next	"জমা দেওয়ার পর জিডি খসড়া ও পোস্টার পাবেন" / After submitting you'll get a GD draft and poster
Never promise outcomes	Not "আমরা খুঁজে দেব" / We will find. Use "মিল খোঁজা হবে" / Matches will be checked
State Fera's limits	"ফেরা পুলিশ বা সিআইডি-কে সতর্কবার্তা পাঠায় না। জরুরি হলে ১৩২১৯-এ কল করুন।" / Fera does not alert police or CID. In an emergency, call 13219.
Neutral words about the child	"শেষবার দেখা গিয়েছিল" / Last seen; never "পালিয়ে গেছে" / ran away
Errors: what happened + what to do	"কথাগুলো পরিষ্কার শোনা যায়নি। আবার বলুন বা লিখে দিন।" / We couldn't hear that clearly. Try again or type.
Officer copy may be denser	"৩টি সম্ভাব্য মিল · সর্বোচ্চ ৮৭" / 3 likely matches · top 87

11 Forms and validation

Field	Input	Validation	Message (Bangla / English)
Age	Numeric keypad; accepts ০–৯ and 0–9; range option	0–17 (missing)	"বয়স ০ থেকে ১৭-এর মধ্যে দিন" / Enter an age from 0 to 17
Sex	4 large buttons	Required	"একটি বেছে নিন" / Choose one
District	Searchable list, Bangla + English names, aliases ("চট্টগ্রাম / Chattogram / Chittagong")	One of 64	"তালিকা থেকে জেলা বেছে নিন" / Pick a district from the list
Last seen	Date + part-of-day chips (সকাল, দুপুর, বিকেল, সন্ধ্যা, রাত) or exact time	Not in future	"ভবিষ্যতের সময় দেওয়া যাবে না" / The time can't be in the future
Phone	`inputmode="tel"`; Bangla digits accepted	`^01[0-9]{9}$` after conversion	"০১ দিয়ে শুরু ১১ সংখ্যার নম্বর দিন" / Enter an 11-digit number starting 01
Free text	Auto-growing textarea	20–2,000 chars	"আরও একটু বিস্তারিত লিখুন" / Please add a little more detail

Validate on blur and on submit, never on each keystroke. Errors appear under the field, linked by `aria-describedby`, plus a summary at the top on submit.

12 Loading, empty, success and error states

State	Pattern
Loading < 300 ms	Nothing shown
Loading 300 ms – 10 s	COMP-046 spinner + verb ("বুঝে নিচ্ছি..." / Understanding...)

State	Pattern
Loading ≥ 10 s	"এখনও কাজ চলছে, একটু অপেক্ষা করুন" / Still working, please wait + "ফর্মে লিখুন" / Use the form link
Empty	COMP-045 line illustration + one sentence + one action
Success	Inline confirmation; parent screens use teal check, never confetti
Error	COMP-044 with message from D2 Part B §13; parent errors use neutral surface with icon, not red fill

13 Accessibility (WCAG 2.2 AA)

Requirement	Implementation
Keyboard access	All actions reachable by Tab; officer shortcuts (§9); dialogs trap focus and restore it
Visible focus	3 px `--color-focus` outline, 2 px offset; never hidden by sticky bars (2.4.11)
Contrast	Text ≥ 4.5:1; Control-identifying boundaries ≥ 3:1 against every adjacent surface; decorative separators excluded (§6)
Labels	Every input has a visible label; icons have `aria-label` in the current language
Alt text	Illustrations decorative (`alt=""`); poster PNG has a text equivalent shown beside it
Captions / transcripts	Every voice note has its editable transcript; audio is never the only record
Clear errors	Field + summary; error text says how to fix
Target size	≥ 44 × 44 px (exceeds 2.5.8 minimum of 24 px)
Responsive text	320 CSS px reflow, 200 % text zoom and 400 % viewport zoom without loss; `rem` units
No colour-only status	COMP-033 always icon + text + colour
Language	`lang="bn"` / `lang="en"` set per element for screen readers
Redundant entry	Draft reused across S-002 → S-003 → S-005; parent never retypes (3.3.7)
Motion	No auto-animation; respects `prefers-reduced-motion`

14 Screen-by-screen descriptions

Screen	Description
SCREEN-001	Logo wordmark "ফেরা", 2-line explanation, consent notice, Start button, small `tel:13219` link.
SCREEN-002	Chat column; assistant bubble left, parent right; large round mic button above a text box; transcript box after recording; one question at a time with quick-reply chips.

Screen	Description
SCREEN-003	Title "যা বুঝেছি" / What we understood; stacked field cards with edit icons; "অজানা" chips; checkbox + submit in sticky bar.
SCREEN-004	Reference "M-7K3Q2A" in large mono digits; status badge "খোলা"; three output cards; poster preview; 13219 call link.
SCREEN-005	Plain one-column form with the same fields as SCREEN-003; banner explains the assistant was unavailable.
SCREEN-006	Reference, status badge, last updated time, "তথ্য সংশোধন" button when editable.
SCREEN-007	Free-text box + structured fields side by side on desktop; save card shows ref "F-..." and "মিল খোঁজা হয়েছে".
SCREEN-008	Centered form, role named in title ("কর্মকর্তা সাইন ইন" / Officer sign-in).
SCREEN-009	Table: rank-1 score, missing ref, child summary, district, candidate count, newest time; filter by district.
SCREEN-010	Header with score, reason, duplicate banner; two columns of fields with verdict icons; right decision panel.
SCREEN-011	Map with district markers sized by total and labelled with three counts; adjacent table; totals strip; synthetic toggle.
SCREEN-012	Case fields, status badge, linked duplicates, referral card, close button, audit timeline.
SCREEN-013	Three panels: verify officers [MOCK], redact identifying payloads (typed confirm), import (Should).
SCREEN-014	Two same-kind records side by side; "একই শিশু" / "আলাদা" buttons.
SCREEN-015	Found-child form with "পরিকল্পিত সম্প্রসারণ" banner.
SCREEN-016	Line illustration, "তথ্যটি পাওয়া যায়নি", home link.

15 Layout specifications

These replace ASCII wireframes with precise responsive regions. PDF diagrams are vector-rendered; the source diagrams remain editable.

Screen/ layout	Regions in reading order	Behaviour
Parent intake	Header/language; processing notice; synthetic badge; conversation; editable transcript; text/voice controls; manual fallback	One column at 320px, voice never sole input; live status announced
Parent review	Required-field summary; child; event interval/place/ district; reporter contact; accuracy confirmation; submit	Known values preserved; unknown mandatory fields linked to manual completion

Screen/ layout	Regions in reading order	Behaviour
Submitted/ output	Reference and matching state; GD text; poster preview/ contact choice; share acknowledgement; download; clear-device	Fixed draft label; no police filing or alert claim
Found entry	Role and synthetic cue; description; optional age; place/ district/time/contact; review/save	Own found-only intake; no candidate payload in response
Officer queue	Role; district filter; missing/found report basis; current qualified candidates; empty/failed state	Tab navigation plus optional focus-scoped shortcuts; no auto hidden duplicates
Candidate review	Similarity label; versions; side-by-side supported values; unknown/contradiction rows; checklist; confirm/reject	Stack columns on narrow viewport; reason required on reject; score not identity proof
Decision dialog	Simulated-referral warning; reviewer checklist; Go back first; explicit decision	Focus trapped/restored; no keyboard shortcut triggers in note fields
Dashboard	Missing report counts; found report counts; confirmed pairs; district outlines; table; officer drill-through	Table alternative, zero districts, exact counting labels; no unique-child/recovery implication
Admin	Metadata; mock user list/verify; eligible redaction summary and typed confirmation	No child/candidate detail; external-copy redaction limitation stated

15.1 Required V2 microcopy

Moment	Bangla	English
Demo banner	সিনথেটিক তথ্য · কর্মকর্তা যাচাই সিমুলেশন	Synthetic data · officer verification simulated
Provider notice	কথা বা লেখা বোঝার জন্য বাইরের এআই সেবা ব্যবহার হতে পারে। এই সংস্করণে বাস্তব শিশুর তথ্য দেবেন না।	External AI may process audio/text. Do not enter real child information in this version.
Before submit	জমা দেওয়ার আগে তথ্য যাচাই করুন। ফেরা পুলিশে জিডি জমা দেয় না।	Review before submission. Fera does not file a police GD.
Confirm result	পর্যালোচকের সিদ্ধান্ত নথিভুক্ত। রেফারেল নথিভুক্ত (সিমুলেশন)।	Reviewer decision recorded. Referral recorded (simulation).
Score warning	মিলের মাত্রা পরিচয়ের প্রমাণ নয়।	Similarity is not proof of identity.
Duplicate cue	সম্ভাব্য একই শিশু · আলাদা প্রতিবেদন দৃশ্যমান	Possible duplicate · separate reports remain visible
Poster share	প্রকাশের আগে নম্বর ও স্থানের তথ্য দেখুন। ডাউনলোড করা কপি ফেরা মুছতে পারে না।	Review phone/location before sharing. Fera cannot remove downloaded copies.
Matching failure	প্রতিবেদন জমা হয়েছে। মিল খোঁজা সম্পূর্ণ হয়নি; আবার চেষ্টা করুন।	Report saved. Matching is incomplete; retry.
Cached fixture	ক্যাশ · পূর্বপ্রস্তুত সিনথেটিক উদাহরণ	Cached · prepared synthetic example

Moment	Bangla	English
Closure	সমন্বয়ের রেকর্ড বন্ধ হয়েছে; উদ্ধার নিশ্চিত করা হয়নি।	Coordination record closed; rescue is not established.

16 Design tokens

The exact companion is tokens_v2.css. Control borders have been strengthened against all light/dark surfaces; component usage still needs visual and keyboard checks.

```
/* Fera design tokens V2.0 - D4 section 16. Authors: Sadman Shaharier; Obidit Islam. Light default;
dark via prefers-color-scheme or [data-theme]. */
:root {
  /* Colour — light */
  --color-bg: #FAFAF7;
  --color-surface: #FFFFFF;
  --color-surface-2: #F1F3F2;
  --color-text: #1B2421;
  --color-text-muted: #4E5A55;
  --color-border: #6F7D77; /* ≥ 3:1 against all three light surfaces */
  --color-primary: #0F5C55;
  --color-primary-hover: #0B4843;
  --color-on-primary: #FFFFFF;
  --color-focus: #2457C5;
  --status-open-fg: #7A4F00;
  --status-open-bg: #FFF3D6;
  --status-matched-fg: #1E4E9C;
  --status-matched-bg: #E3EDFC;
  --status-closed-fg: #1E6B3A;
  --status-closed-bg: #E2F3E8;
  --color-danger-fg: #B42318;
  --color-danger-bg: #FDECEA;

  /* Typography */
  --font-bn: "Noto Sans Bengali", "Noto Sans", system-ui, sans-serif;
  --font-latin: "Noto Sans", "Noto Sans Bengali", system-ui, sans-serif;
  --font-mono: ui-monospace, "SFMono-Regular", Menlo, Consolas, monospace;
  --font-size-sm: 0.875rem;   /* 14px */
  --font-size-md: 1rem;       /* 16px */
  --font-size-lg: 1.125rem;   /* 18px — parent body */
  --font-size-xl: 1.375rem;   /* 22px */
  --font-size-2xl: 1.75rem;   /* 28px */
  --font-weight-regular: 400;
  --font-weight-semibold: 600;
  --line-height-bn: 1.7;
  --line-height-officer: 1.55;

  /* Spacing (4px base) */
  --space-1: 0.25rem;
  --space-2: 0.5rem;
  --space-3: 0.75rem;
  --space-4: 1rem;
  --space-6: 1.5rem;
  --space-8: 2rem;
  --space-12: 3rem;
  --space-16: 4rem;
```

```
  /* Radius */
  --radius-input: 6px;
  --radius-card: 10px;
  --radius-sheet: 16px;
  --radius-pill: 999px;

  /* Elevation */
  --shadow-1: 0 1px 2px rgba(0, 0, 0, 0.08);
  --shadow-2: 0 8px 24px rgba(0, 0, 0, 0.16);

  /* Interaction */
  --focus-ring: 0 0 0 2px var(--color-surface), 0 0 0 5px var(--color-focus);
  --tap-target-min: 44px;
  --report-flow-min-width: 320px;
  --duration-fast: 120ms;
  --duration-base: 200ms;

  /* Layout */
  --parent-max-width: 480px;
  --officer-max-width: 1280px;
  --decision-panel-width: 320px;
}

@media (prefers-color-scheme: dark) {
  :root:not([data-theme="light"]) {
    --color-bg: #111816;
    --color-surface: #18211E;
    --color-surface-2: #212C28;
    --color-text: #E8EEEB;
    --color-text-muted: #A9B5B0;
    --color-border: #82938B; /* ≥ 3:1 against all three dark surfaces */
    --color-primary: #5FC4B8;
    --color-primary-hover: #7FD3C9;
    --color-on-primary: #0B1F1C;
    --color-focus: #8AB4F8;
    --status-open-fg: #F2C14E;
    --status-open-bg: #33290F;
    --status-matched-fg: #8DB4F5;
    --status-matched-bg: #172640;
    --status-closed-fg: #7FD39A;
    --status-closed-bg: #14301F;
    --color-danger-fg: #F28B82;
    --color-danger-bg: #3A1714;
    --shadow-1: 0 1px 2px rgba(0, 0, 0, 0.4);
    --shadow-2: 0 8px 24px rgba(0, 0, 0, 0.5);
  }
}

:root[data-theme="dark"] {
  --color-bg: #111816;
  --color-surface: #18211E;
  --color-surface-2: #212C28;
  --color-text: #E8EEEB;
  --color-text-muted: #A9B5B0;
  --color-border: #82938B; /* ≥ 3:1 against all three dark surfaces */
  --color-primary: #5FC4B8;
  --color-primary-hover: #7FD3C9;
  --color-on-primary: #0B1F1C;
```

```
  --color-focus: #8AB4F8;
  --status-open-fg: #F2C14E;
  --status-open-bg: #33290F;
  --status-matched-fg: #8DB4F5;
  --status-matched-bg: #172640;
  --status-closed-fg: #7FD39A;
  --status-closed-bg: #14301F;
  --color-danger-fg: #F28B82;
  --color-danger-bg: #3A1714;
  --shadow-1: 0 1px 2px rgba(0, 0, 0, 0.4);
  --shadow-2: 0 8px 24px rgba(0, 0, 0, 0.5);
}

@media (prefers-reduced-motion: reduce) {
  :root {
    --duration-fast: 0ms;
    --duration-base: 0ms;
  }
}
```

17 Usability risks and mitigations

Risk	Mitigation
Parent is distressed and cannot read long text	One question per view; voice first; 18 px text; no jargon
Parent expects Fera to alert police	Consent notice and output labels state Fera does not alert; `tel:13219` always visible
Parent fears data exposure	Consent notice lists who sees what; matched records officer-only; no photos
Child photo privacy	No photo field anywhere; poster is text only; demo uses synthetic text only
AI misheard details shared wrongly	Mandatory review with "আমি যাচাই করেছি" checkbox
Officer over-trusts the score	Score labelled similarity; evidence rows with verdicts; 2-step confirm; referral reminder
Officer fatigue / mis-click	Confirm dialog focuses "Go back"; reject needs reason
Bangla rendering breaks on low-end phones	Self-hosted Noto Sans Bengali subset; tested on 2 Android phones (TASK-036)
Map unreadable or offline	Table always shown beside map; tile-less fallback
Demo shows cached outputs as live	"ক্যাশ" badge on every cached output

IDs introduced

- COMP-022 App header with language toggle

- COMP-023 Synthetic badge
- COMP-024 Chat bubble
- COMP-025 Voice record button
- COMP-026 Transcript editor
- COMP-027 Question prompt
- COMP-028 Editable field card
- COMP-029 Confirm bar
- COMP-030 Output card
- COMP-031 Poster preview
- COMP-032 Manual case form
- COMP-033 Status badge
- COMP-034 Found-child form
- COMP-035 Review queue list
- COMP-036 Candidate row
- COMP-037 Side-by-side evidence table
- COMP-038 Decision panel
- COMP-039 Close-case dialog
- COMP-040 District map
- COMP-041 District table
- COMP-042 Audit timeline
- COMP-043 Admin action panel
- COMP-044 Inline alert / toast
- COMP-045 Empty state
- COMP-046 Loading state
- COMP-047 Sign-in form
- COMP-048 Cached badge
- COMP-049 Duplicate banner
- COMP-050 Consent notice

IDs referenced

- SCREEN-001 – SCREEN-016 (D1)
- GOAL-001, PERSONA-001 – PERSONA-003 (00)
- TASK-036 (D6)

V2 accessibility and export acceptance

Required: 320 CSS-pixel reflow, 200% text zoom, 400% viewport zoom; keyboard and Bangla screen reader walk; focus not obscured; live/error announcements; shortcut disable/remap or focus scope; no shortcuts while typing. Test actual control backgrounds and all states, not tokens alone. axe is complementary, not a conformance certificate. Parent forms remain usable without maps/audio. Test generated poster after fonts load with long Bangla/unknown fields and a text equivalent. Export remains light and readable regardless of UI theme.

References: WCAG non-text contrast, shortcuts, reflow. No AA conformance claim is made before the implemented product is checked.

↗ Return to contents

D5 / CONTRACTS / V2.0

Background Schema & Registry

Shared TypeScript contracts, domain relationships and 537 IDs · Authors: Sadman Shaharier and Obidit Islam

↗ Contents

Field	Value
Version	2.0
Date	11 Oct 2026 (Asia/Dhaka, UTC+6)
Owner	Sadman Shaharier
Authors	Sadman Shaharier; Obidit Islam
Status	V2 team handover; implementation status UNVERIFIED
Companion	`fera_types_v2.ts` — byte-identical copy of §6
Covers	All nine V2 source documents and the three V2 companions; validation report.

V2 release contract

Authors: Sadman Shaharier and Obidit Islam. **Version:** 2.0. **Issued:** 11 October 2026, Asia/Dhaka (UTC+6).

This is the revised team specification, not evidence that the application has been built. Actual repository progress, account configuration, interviews, evaluations, and spend remain **UNVERIFIED** until evidence is attached. V1 defaults are superseded by the V2 rules below; existing requirement IDs are retained.

- This release is a **synthetic-data prototype**. Officer verification and referral delivery are simulated and labelled. The schema rejects records marked non-synthetic; the application must also prevent real reports from entering this release.
- A candidate score is similarity, not identity probability. A reviewer decision does not establish guardian authority or safe reunification. A referral row means **referral recorded**, not delivered or acknowledged.
- Draft payload stays on the device before report confirmation, but providers process audio/text beforehand with a notice. Only expiry/counter/hash metadata and cost reservations are stored server-side before submission.
- Possible duplicates remain visible. Shared phone alone never establishes one child. No physical child-return action is performed by Fera.
- Counts distinguish missing-report rows, found-report rows, and confirmed pairs. They are not a census of unique children or proof of rescue.
- Sources, diagrams, tables, code blocks, and acceptance criteria are part of the handover. The companion code must match the embedded code exactly. Team handover records change closure, evidence gaps, and release gates.

Changes to earlier documents

V2 consolidates 537 unique IDs, their corrected definition locations, new API/entity/test/task records, and the revised TypeScript contracts. All companion snippets match their delivered files. D2A and D5 consistently use Uint8Array at the speech adapter boundary.

1 Project metadata

Field	Value
Name	Fera (ফেরা)
One line	Bangla-first reviewed report and candidate coordination; no rescue guarantee
Event	Hack for Humanity 2026 (AI Collective Bangladesh, AIUB), Track 3 Smart Society & Public Services
Round 2 due	14 Oct 2026 confirmed date; exact time/deliverables/submission channel pending organiser email
Grand Finale	24 Oct 2026 at AIUB: 5–10 min presentation + live demo
Judging	Social/human impact, feasibility, scalability; weights UNKNOWN (OQ-025); no equal-weight claim
Team	Sadman Shaharier (lead/integration), Obidit Islam (intake & LLM), Masnun Kabir (data, matching & dashboard) — institutions/registration spelling pending verification
Platform	Responsive web app; mobile-first for parents, desktop-first for officers; no hardware

2 Problem context

Item	Value and evidence status
Reported missing, Jan–Jul 2026	15,717 children
Still untraced	2,038 (13 %), Jan–Jul cohort reported 3 Sep 2026; 13,679 were reported returned/rescued. No claim that existing agencies record no outcomes.
Aged 10–17	97 %
Rate	About 74 per day
Linked operational registry	Unverified hypothesis; no absence claim
Existing system	CID's MUN Alert, live since 13 Jan 2026, helpline 13219, joined Meta's alert network May 2026

Breaking points: lost first hours; scattered reports with no missing↔found link; unvalidated need for clearer case-level coordination follow-up.

3 Domain concepts

Concept	Definition in Fera	Fera's role
GD (General Diary)	Police-station record of a reported incident, here a missing child	Fera drafts the text; the family files it (FR-008)
MUN Alert	CID's missing-child alert system with Amber Alert for Bangladesh	Fera never issues or withdraws alerts (BR-002)
13219	MUN Alert's toll-free helpline	Fera shows the number and a summary to read out (FR-010)
CID Missing Children Cell	Unit named in sources as able to issue or withdraw alerts; public briefing names Missing Children Cell; operational partnership unverified	External; no integration
Missing case	Report of a child not found by family	`cases.kind = missing`
Found case	Report of a child found by staff, shelter, station or public source	`cases.kind = found`
Match	A ranked missing↔found pair; a candidate until an officer decides	ENTITY-005; BR-001
Referral	Internal simulated referral record after a reviewer decision; not proof of delivery, custody, identity or safe resolution	ENTITY-007; BR-005
Duplicate	Two same-kind records likely describing one child	ENTITY-008; BR-021
Closure	A missing case reaching `closed` with an outcome	FR-018; GOAL-006
Disaster mode	Planned extension for flood/cyclone shelters; no measured need	FR-036 (Could)
Synthetic data	Team-written records with known ground truth, labelled on screen	**BR-026**

4 Stakeholders

Stakeholder	Interest	Interaction
Parents and guardians (PERSONA-001)	Fast, complete report; updates	Parent screens
Verifying officers (PERSONA-002)	Accurate shortlist; accountable decisions	Officer screens
Shelter/station staff (PERSONA-003)	Found children matched	Staff screens
Child-protection caseworkers (PERSONA-004)	Safe placement after a match	Referral record only
Disaster-shelter volunteers (PERSONA-005)	Reunification after floods/cyclones	Could, SCREEN-015
CID / MUN Alert	Cleaner cases; alert authority	No integration; future data agreement

Stakeholder	Interest	Interaction
Organisers, judges, mentors	Impact, feasibility, scalability	Deck, video, demo
AI providers	Process audio and report text	Via adapters (TB4)
Team	Deliver by 14 and 24 Oct	All

5 Constraints

Constraint	Value
AI budget	USD 30 total, hard cap USD 25 in app (DECISION-017)
Time	Round 2 on 14 Oct; exact official cutoff pending; finale 24 Oct
Capacity	Actual remaining capacity pending; at most 120 h for a fresh four-day start at assumed 10 h/person/day (A-040)
Network	Venue internet may fail; demo-safe mode required
Devices	Laptop + Android phone browser
Language	Bangla text and voice end to end; mixed script and Bangla numerals
Data	No private data; no scraping of private accounts; synthetic only in this release; public real-child material requires a later lawful-use gate
Safeguarding	BR-001 – BR-011 non-negotiable
Law	Personal Data Protection Act, 2026, qualified interpretation and effective provisions review required before real-child use; no compliance claim (OQ-011)

6 TypeScript interfaces

`fera_types_v2.ts` is a byte-identical copy of this block. It type-checks with `tsc --strict` (TypeScript 5.4).

```
// Fera shared types V2.0 - D5 section 6. Exact companion of the V2 embedded block.
// Authors: Sadman Shaharier; Obidit Islam. 11 October 2026, Asia/Dhaka.
// Registry records (suite IDs) + domain records (D2 schema) + AI/API contracts (D2 §2, §5).

/* ---------- Shared primitives ---------- */
export type Uuid = string;
export type IsoDateTime = string; // ISO 8601 with +06:00
export type MoSCoW = "Must" | "Should" | "Could" | "Won't";
export type DocId = "00" | "D1" | "D2A" | "D2B" | "D3" | "D4" | "D6" | "D7";
export type Level = "L" | "M" | "H" | "Severe";
export type Role = "parent" | "staff" | "officer" | "admin";
export type AuthLabel = Role | "public";
export type Owner = "S" | "O" | "M";
```

```
/* ---------- Registry records ---------- */
export interface RegistryRecord {
  id: string;
  label: string;
  definedIn: DocId;
}
export interface Goal extends RegistryRecord {
  id: `GOAL-${string}`;
  metric: string;
  target: string;
  method: string;
  timeframe: string;
}
export interface Persona extends RegistryRecord {
  id: `PERSONA-${string}`;
  priority: MoSCoW;
}
export interface Story extends RegistryRecord {
  id: `STORY-${string}`;
  priority: MoSCoW;
  acceptance: string[];
  dependencyIds: string[];
  screenIds: string[];
}
export interface FunctionalRequirement extends RegistryRecord {
  id: `FR-${string}`;
  priority: MoSCoW;
  storyIds: string[];
  testIds: string[];
  mock: boolean;
}
export interface NonFunctionalRequirement extends RegistryRecord {
  id: `NFR-${string}`;
  category: string;
  measure: string;
}
export interface BusinessRule extends RegistryRecord {
  id: `BR-${string}`;
  safeguarding: boolean;
  frIds: string[];
  testIds: string[];
}
export interface Assumption extends RegistryRecord {
  id: `A-${string}`;
}
export interface Dependency extends RegistryRecord {
  id: `DEP-${string}`;
  owner: string;
  neededBy: string;
}
export interface Risk extends RegistryRecord {
  id: `RISK-${string}`;
  likelihood: Level;
  impact: Level;
  mitigation: string;
  owner: string;
}
export interface OpenQuestion extends RegistryRecord {
```

```
  id: `OQ-${string}`;
  defaultApplied: string | null;
  answeredBy: string;
  dueBy: string;
}
export interface Screen extends RegistryRecord {
  id: `SCREEN-${string}`;
  priority: MoSCoW;
  roles: Role[];
}
export interface Component extends RegistryRecord {
  id: `COMP-${string}`;
  layer: "server" | "client" | "ui";
}
export interface ApiEndpoint extends RegistryRecord {
  id: `API-${string}`;
  method: "GET" | "POST" | "PATCH" | "*";
  path: string;
  auth: AuthLabel[];
  priority: MoSCoW;
}
export interface EntityRecord extends RegistryRecord {
  id: `ENTITY-${string}`;
  table: string;
}
export interface Decision extends RegistryRecord {
  id: `DECISION-${string}`;
}
export interface TestCase extends RegistryRecord {
  id: `TC-${string}`;
  covers: string[];
  steps: string[];
  passIf: string;
}
export interface Task extends RegistryRecord {
  id: `TASK-${string}`;
  priority: MoSCoW;
  owners: Owner[];
  dependsOn: string[];
  hours: number;
  criticalPath: boolean;
}
export interface ErrorCode extends RegistryRecord {
  id: `E-${string}`;
  key: string;
  http: number | null;
  messageBn: string;
  messageEn: string;
}
export interface FlowAction extends RegistryRecord {
  id: `ACT-${string}`;
  roles: Role[];
}
export interface FlowDecision extends RegistryRecord {
  id: `D-${string}`;
}
export interface TraceLink {
  from: string;
  to: string;
```

```
  kind: "goal-story" | "story-fr" | "fr-screen" | "fr-test" | "br-fr" | "br-test" | "fr-task";
}

/* ---------- Domain enums (D2 §7) ---------- */
export type MatchingState = "queued" | "running" | "complete" | "failed";
export type MissingStatus = "submitted" | "open" | "matched_pending_review" | "confirmed" |
"closed";
export type FoundStatus = "open" | "matched_pending_review" | "referred";
export type CaseKind = "missing" | "found";
export type CaseStatus = "submitted" | "open" | "matched_pending_review" | "confirmed" | "closed" |
"referred";
export type StatusGroup = "open" | "matched" | "closed";
export type Sex = "male" | "female" | "other" | "unknown";
export type CaseSource = "parent" | "staff" | "public_import" | "synthetic";
export type UserRole = "staff" | "officer" | "admin";
export type CandidateStatus = "pending" | "confirmed" | "rejected" | "voided" | "superseded";
export type DecisionType = "confirm" | "reject";
export type Outcome = "coordination_recorded" | "resolved_outside_fera";
export type DuplicateStatus = "flagged" | "confirmed_same" | "dismissed";
export type OutputType = "gd" | "poster" | "helpline";
export type ActorType = "parent" | "staff" | "officer" | "admin" | "system";

/* ---------- Domain records (one per table) ---------- */
export interface User { // ENTITY-001
  id: Uuid;
  username: string;
  display_name: string;
  password_hash: string;
  role: UserRole;
  verified: boolean;
  district_code: number | null;
  is_active: boolean;
  created_at: IsoDateTime;
  updated_at: IsoDateTime;
}
export interface ParentSession { // ENTITY-002
  id: Uuid;
  created_at: IsoDateTime;
  last_seen_at: IsoDateTime;
  expires_at: IsoDateTime;
  revoked_at: IsoDateTime | null;
}
export interface Case { // ENTITY-003
  id: Uuid;
  ref_code: string;
  kind: CaseKind;
  status: CaseStatus;
  source: CaseSource;
  source_url: string | null;
  is_synthetic: boolean;
  reporter_session_id: Uuid | null;
  created_by_user_id: Uuid | null;
  raw_text: string | null;
  child_name: string | null;
  child_name_latin: string | null;
  age_years: number | null;
  age_min: number | null;
  age_max: number | null;
  sex: Sex;
```

```
  height_cm: number | null;
  clothing: string | null;
  marks: string | null;
  circumstances: string | null;
  event_at: IsoDateTime | null;
  event_at_approx: boolean;
  event_at_lower: IsoDateTime | null;
  event_at_upper: IsoDateTime | null;
  provenance: Record<string, EvidenceSpan>;
  embedding_model: string | null;
  embedding_revision: string | null;
  embedding_space: string | null;
  matching_state: MatchingState;
  matched_at: IsoDateTime | null;
  match_error_id: string | null;
  place_text: string | null;
  district_code: number | null;
  thana: string | null;
  reporter_relation: string | null;
  contact_phone: string | null;
  gd_number: string | null;
  embedding: number[] | null;
  client_submission_id: Uuid | null;
  submission_payload_hash: string | null;
  input_hash: string | null;
  version: number;
  outcome: Outcome | null;
  outcome_note: string | null;
  closed_at: IsoDateTime | null;
  anonymised_at: IsoDateTime | null;
  created_at: IsoDateTime;
  updated_at: IsoDateTime;
}
export interface CaseVersion { // ENTITY-004
  id: Uuid;
  case_id: Uuid;
  version: number;
  snapshot: Record<string, unknown> | null;
  edited_by: ActorType;
  edited_at: IsoDateTime;
}
export interface MatchCandidate { // ENTITY-005
  id: Uuid;
  missing_case_id: Uuid;
  found_case_id: Uuid;
  missing_version: number;
  found_version: number;
  score: number;
  rank: number;
  reason_bn: string;
  reason_en: string;
  components: MatchComponents;
  status: CandidateStatus;
  created_at: IsoDateTime;
  decided_at: IsoDateTime | null;
}
export interface MatchComponents {
  cosine: number;
  name: number;
```

```
  clothing: number;
  district: number;
  age: number;
}
export interface ReviewDecision { // ENTITY-006
  id: Uuid;
  candidate_id: Uuid;
  officer_id: Uuid;
  decision: DecisionType;
  officer_verified_at_decision: true;
  reason: string | null;
  decided_at: IsoDateTime;
}
export interface Referral { // ENTITY-007
  id: Uuid;
  candidate_id: Uuid;
  found_case_id: Uuid;
  missing_case_id: Uuid;
  agency_name: string;
  referred_by: Uuid;
  referred_at: IsoDateTime;
  record_state: "recorded";
  is_simulated: true;
  note: string | null;
}
export interface DuplicateLink { // ENTITY-008
  id: Uuid;
  case_a: Uuid;
  case_b: Uuid;
  score: number;
  rule: "profile" | "same_phone";
  status: DuplicateStatus;
  created_at: IsoDateTime;
  resolved_by: Uuid | null;
  resolved_at: IsoDateTime | null;
  reason: string | null;
}
export interface AuditEntry { // ENTITY-009
  id: number;
  at: IsoDateTime;
  actor_type: ActorType;
  actor_user_id: Uuid | null;
  actor_session_id: Uuid | null;
  action: string;
  case_id: Uuid | null;
  candidate_id: Uuid | null;
  details: Record<string, unknown>;
  request_id: string | null;
}
export interface District { // ENTITY-010
  code: number;
  name_en: string;
  name_bn: string;
  division_en: string;
  lat: number;
  lng: number;
}
export interface DistrictNeighbour { // ENTITY-011
  district_code: number;
```

```
  neighbour_code: number;
}
export interface GeneratedOutput { // ENTITY-012
  id: Uuid;
  case_id: Uuid;
  type: OutputType;
  text_bn: string | null;
  case_version: number;
  created_at: IsoDateTime;
}
export interface AiCall { // ENTITY-013
  id: number;
  at: IsoDateTime;
  draft_id: Uuid | null;
  case_id: Uuid | null;
  reservation_id: Uuid | null;
  request_id: string | null;
  operation: "transcribe" | "extract" | "embed";
  provider: string;
  model: string;
  duration_ms: number;
  input_units: number | null;
  output_units: number | null;
  est_cost_usd: number;
  cache_hit: boolean;
  ok: boolean;
  error_id: string | null;
}
export interface RateLimitBucket { // ENTITY-014
  key: string;
  window_start: IsoDateTime;
  count: number;
}

// New V2 metadata entities; these store no child text or audio before submission.
export interface IntakeDraft { // ENTITY-015
  id: Uuid;
  session_id: Uuid | null;
  user_id: Uuid | null;
  kind: CaseKind;
  created_at: IsoDateTime;
  expires_at: IsoDateTime;
  payload_hash: string | null;
  ai_calls_used: number;
  revision: number;
}
export interface AiBudget { // ENTITY-016
  id: true;
  cap_usd: number;
  committed_usd: number;
  reserved_usd: number;
}
export interface AiReservation { // ENTITY-017
  id: Uuid;
  draft_id: Uuid | null;
  reserved_usd: number;
  actual_usd: number | null;
  state: "reserved" | "settled";
  created_at: IsoDateTime;
```

```
  settled_at: IsoDateTime | null;
}
export interface EvidenceSpan {
  quote: string;
  start_utf16: number;
  end_utf16: number;
  derivation: "direct" | "digits" | "time_interval" | "district_mapping" | "romanisation";
}

/* ---------- AI adapter contracts (D2 §2.1–§2.3) ---------- */
export interface ExtractInput {
  kind: CaseKind;
  sourceText: string; // reviewed raw transcript/text; evidence quotes refer to this
  normalisedText: string; // deterministic normalisation; preserve span mapping
  nowIso: IsoDateTime;
}
export interface ExtractOutput {
  child: {
    name: string | null;
    age_years: number | null;
    age_min: number | null;
    age_max: number | null;
    sex: Sex;
    height_cm: number | null;
    clothing: string | null;
    marks: string | null;
  };
  event: {
    at: IsoDateTime | null;
    at_approx: boolean;
    at_lower: IsoDateTime | null;
    at_upper: IsoDateTime | null;
    place_text: string | null;
    district_text: string | null;
    thana: string | null;
    circumstances: string | null;
  };
  reporter: { relation: string | null; phone: string | null };
  evidence: Record<string, EvidenceSpan>;
}
export interface LLMProvider {
  readonly id: "gemini" | "openai" | "claude";
  extractCase(input: ExtractInput): Promise<ExtractOutput>;
}
export interface SpeechProvider {
  readonly id: "whisper" | "cache";
  transcribe(audio: Uint8Array, mime: string, langHint: "bn"): Promise<{ text: string; durationS:
number }>;
}
export interface EmbeddingProvider {
  readonly id: "local" | "api";
  readonly dim: 384;
  embed(texts: string[]): Promise<number[][]>;
}

/* ---------- API shapes (D2 §5) ---------- */
export interface CaseDraft extends ExtractOutput {
  kind: CaseKind;
  source_text: string; // browser payload; persist only after explicit report confirmation
```

```
  child: ExtractOutput["child"] & { name_latin: string | null };
  event: ExtractOutput["event"] & { district_code: number | null };
}
// Narrow DTOs, not ORM rows. Server projections are explicit allowlists.
export interface OwnerCaseView {
  id: Uuid; ref_code: string; kind: "missing"; status: MissingStatus;
  status_group: StatusGroup; is_synthetic: true; version: number;
  editable: boolean; matching_state: MatchingState;
  draft: CaseDraft | null; // own editable report only; null after terminal lock/redaction
  created_at: IsoDateTime; updated_at: IsoDateTime;
}
export interface StaffOwnCaseView {
  id: Uuid; ref_code: string; kind: "found"; status: FoundStatus;
  is_synthetic: true; version: number; matching_state: MatchingState;
  draft: CaseDraft | null; created_at: IsoDateTime;
}
export interface AdminCaseView {
  id: Uuid; ref_code: string; kind: CaseKind; status: CaseStatus;
  district_code: number | null; is_synthetic: true;
  anonymised_at: IsoDateTime | null;
  // No child payload, contact, candidate IDs/scores, or free-text notes.
}
export interface OfficerCaseView {
  id: Uuid;
  ref_code: string;
  kind: CaseKind;
  status: CaseStatus;
  status_group: StatusGroup;
  is_synthetic: boolean;
  source: CaseSource;
  source_url: string | null;
  child: {
    name: string | null;
    age_years: number | null;
    age_min: number | null;
    age_max: number | null;
    sex: Sex;
    height_cm: number | null;
    clothing: string | null;
    marks: string | null;
  };
  event: {
    at: IsoDateTime | null;
    at_approx: boolean;
    at_lower: IsoDateTime | null;
    at_upper: IsoDateTime | null;
    place_text: string | null;
    district_code: number | null;
    district_name_bn: string | null;
    thana: string | null;
  };
  contact_phone: string | null; // verified officer projection only; never logged
  contact_phone_masked: string | null;
  version: number;
  duplicate_of: string[]; // review cues; flagged links never suppress a distinct report
  matching_state: MatchingState;
  created_at: IsoDateTime;
  updated_at: IsoDateTime;
}
```

```
export interface NextQuestion {
  field: string;
  bn: string;
  en: string;
}
export interface ExtractResponse {
  draft_token: string; // opaque signed draft/principal/expiry binding; usage is server-owned
  draft_id: Uuid;
  draft_revision: number;
  expires_at: IsoDateTime;
  draft: CaseDraft;
  missing_fields: string[];
  next_question: NextQuestion | null;
  ai_calls_used: number;
  cached: boolean;
}
export interface TranscribeResponse {
  draft_token: string;
  draft_id: Uuid;
  ai_calls_used: number;
  transcript: string;
  language: "bn" | "mixed";
  duration_s: number;
  cached: boolean;
}
export interface OutputResponse {
  type: OutputType;
  text_bn: string;
  label_bn: string;
  word_count: number;
  generated_at: IsoDateTime;
}
export type DecisionRequest =
  | { decision: "reject"; reason: string; expected_missing_version: number; expected_found_version:
number }
  | { decision: "confirm"; reason?: string; agency_name?: string;
      expected_missing_version: number; expected_found_version: number;
      reviewer_checklist: { descriptions_reviewed: true; contradictions_reviewed: true;
        identity_not_proven_by_score: true; referral_is_simulated: true } };
export interface DecisionResponse {
  candidate_status: CandidateStatus;
  missing: { ref_code: string; status: CaseStatus };
  found: { ref_code: string; status: CaseStatus };
  referral_id: Uuid | null;
  referral_state: "recorded" | null;
  referral_is_simulated: true;
}
export interface ReportCounts { open: number; matched: number; closed: number; }
export interface DistrictStats {
  code: number; name_bn: string; name_en: string; lat: number; lng: number;
  missing_reports: ReportCounts;
  found_reports: { open: number; matched: number; referral_recorded: number };
  confirmed_pairs: number;
}
export interface DashboardStatsResponse {
  generated_at: IsoDateTime;
  counting_basis: "report_rows_separated_by_kind";
  totals: { missing_reports: ReportCounts;
    found_reports: { open: number; matched: number; referral_recorded: number };
```

```
    confirmed_pairs: number };
  districts: DistrictStats[];
}
export interface CreateMissingRequest {
  draft: CaseDraft & { kind: "missing" };
  confirmed: true;
  client_submission_id: Uuid;
  draft_token?: string; // absent for manual form; paid extraction requires its own token
}
export interface CandidateDecisionView {
  candidate_id: Uuid; missing: OfficerCaseView; found: OfficerCaseView;
  score: number; components: Record<string, number | null>; explanation_version: string;
  qualified: boolean; identity_probability: false;
}
export interface DuplicateDecisionRequest {
  status: "confirmed_same" | "dismissed";
  reason: string;
  profile_evidence_reviewed: true;
}
export interface RematchResponse {
  case_id: Uuid; matching_state: MatchingState;
  // Admin gets this status-only shape. Officer reads candidates via API-008/009.
}

export interface ErrorEnvelope {
  error: {
    id: string;
    key: string;
    message_bn: string;
    message_en: string;
    retryable: boolean;
    fallback?: "manual_form";
  };
}

export type TypedCaseRecord =
  | (Case & { kind: "missing"; status: MissingStatus })
  | (Case & { kind: "found"; status: FoundStatus });
// Uuid/IsoDateTime and number ranges need runtime validation. pg numeric/bigint/Date
// values must be explicitly mapped before returning these DTOs.
```

7 Registry

Every ID in the suite appears in exactly one row below. "Defined in" names the file that introduces it.

7.1 Goals (GOAL, 9)

ID	Label	Metric	Defined in
GOAL-001	A parent produces a reviewed Bangla report in the first hour	Time to complete report, assistant vs manual form	00_Gap_Check_and_Registry
GOAL-002	AI extracts case fields correctly from messy Bangla	Field-level extraction accuracy	00_Gap_Check_and_Registry
GOAL-003	The right found-child record appears in the officer's shortlist	Recall@5; precision@1; precision@5 reported	00_Gap_Check_and_Registry

ID	Label	Metric	Defined in
GOAL-004	Duplicate reports of one child are detected	Duplicate-detection pairwise F1	00_Gap_Check_and_Registry
GOAL-005	Bangla voice is usable for intake	Word error rate	00_Gap_Check_and_Registry
GOAL-006	Every demo report has an explicit recorded coordination status, visible by district	Seeded cases shown with correct status	00_Gap_Check_and_Registry
GOAL-007	The live demo does not fail	Rehearsals passing end to end	00_Gap_Check_and_Registry
GOAL-008	Humans make every match decision	Match-driven status changes with a logged officer action	00_Gap_Check_and_Registry
GOAL-009	Results are reported honestly	Metrics slide with failure cases	00_Gap_Check_and_Registry

7.2 Personas (PERSONA, 5)

ID	Label	Profile	Defined in
PERSONA-001	Parent/guardian ("Shirin")	Dhaka mother, 30s; son, 13, did not return from madrasa; Bangla-first, weak formal writing, And...	00_Gap_Check_and_Registry
PERSONA-002	Verifying officer ("SI Karim")	Thana duty officer, several missing-child GDs a week; reviews matches on a desktop.	00_Gap_Check_and_Registry
PERSONA-003	Shelter/station staff ("Rumana")	NGO volunteer at a railway-station children's desk; logs found children on a phone.	00_Gap_Check_and_Registry
PERSONA-004	Child-protection caseworker	Intended future referral recipient; V2 records a simulated referral only; no recipient UI or delivery integration.	00_Gap_Check_and_Registry
PERSONA-005	Disaster-mode shelter volunteer	Logs separated children after floods/cyclones; planned extension, no measured need.	00_Gap_Check_and_Registry

7.3 User stories (STORY, 23)

ID	Label	Priority	Defined in
STORY-001	Report by voice	Must	D1_PRD
STORY-002	Report by typed text	Must	D1_PRD
STORY-003	Answer follow-up questions	Must	D1_PRD
STORY-004	Review before anything is shared	Must	D1_PRD

ID	Label	Priority	Defined in
STORY-005	Get a GD draft	Must	D1_PRD
STORY-006	Get a shareable poster	Must	D1_PRD
STORY-007	Get a 13219-ready summary	Should	D1_PRD
STORY-008	Log a found child	Must	D1_PRD
STORY-009	See ranked candidates	Must	D1_PRD
STORY-010	Compare and confirm a match	Must	D1_PRD
STORY-011	Reject a candidate	Must	D1_PRD
STORY-012	Refer and close	Must	D1_PRD
STORY-013	See the district picture	Must	D1_PRD
STORY-014	Detect duplicate reports	Must	D1_PRD
STORY-015	Import synthetic public-format fixtures	Should	D1_PRD
STORY-016	Redact identifying payloads from eligible terminal cases	Must	D1_PRD
STORY-017	Audit decisions	Must	D1_PRD
STORY-018	Continue when AI fails	Must	D1_PRD
STORY-019	Edit after submission	Should	D1_PRD
STORY-020	See my case status	Should	D1_PRD
STORY-021	Disaster-mode logging	Could	D1_PRD
STORY-022	Measure and report results	Must	D1_PRD
STORY-023	Run the demo without internet	Must	D1_PRD

7.4 Functional requirements (FR, 37)

ID	Label	Priority	Defined in
FR-001	accept typed reports of 20–2,000 characters in Bangla, English or mixed script.	Must	D1_PRD
FR-002	accept voice notes of 5–120 s and return an editable Bangla transcript.	Must	D1_PRD
FR-003	normalise Bangla digits and derive explicit Asia/Dhaka event intervals anchored to the recorded intake clock; ambiguous time requires review.	Must	D1_PRD

ID	Label	Priority	Defined in
FR-004	extract case fields from text, leaving unstated fields empty and never guessing.	Must	D1_PRD
FR-005	ask at most four one-at-a-time follow-ups, then require accessible manual completion of unresolved mandatory fields.	Must	D1_PRD
FR-006	show every extracted field as editable and require an explicit confirm.	Must	D1_PRD
FR-007	keep drafts out of matching, staff/officer views and exports until confirmed.	Must	D1_PRD
FR-008	generate a Bangla GD draft from a fixed template using confirmed fields only.	Must	D1_PRD
FR-009	generate a reviewed no-photo poster with 13219, a separate sharing acknowledgement and optional contact inclusion.	Must	D1_PRD
FR-010	generate a 13219-ready summary of ≤ 80 words.	Should	D1_PRD
FR-011	let staff log a found child by form or free text and review before saving.	Must	D1_PRD
FR-012	rank up to five qualified candidates nationally with evidence reasons and similarity scores; allow an empty shortlist.	Must	D1_PRD
FR-013	exclude candidate pairs that fail the hard filters in BR-017.	Must	D1_PRD
FR-014	list cases with pending candidates for verified officers, newest candidate first.	Must	D1_PRD
FR-015	show missing and found records side by side with agree/ differ/unknown per field.	Must	D1_PRD
FR-016	atomically record a verified candidate decision, current endpoint states, simulated referral recorded and safe audit; no receipt or custody claim.	Must	D1_PRD
FR-017	on reject with a reason, remove the candidate and apply BR-012 and BR-019.	Must	D1_PRD
FR-018	record coordination_recorded closure after confirmed/ referral-recorded state, or resolved_outside_fera from open; neither proves safe reunification.	Must	D1_PRD
FR-019	show separate missing-report, found-report and confirmed-pair district counts with accessible icon, text and colour.	Must	D1_PRD
FR-020	restrict candidates and matched records to verified officers per BR-024.	Must	D1_PRD
FR-021	write an append-only audit entry for every action listed in STORY-017.	Must	D1_PRD

ID	Label	Priority	Defined in
FR-022	flag possible duplicates while keeping every unreviewed report independently discoverable; shared phone alone never suppresses a report.	Must	D1_PRD
FR-023	provide optional reviewed reversible duplicate grouping/ dismissal with profile evidence, reason and preserved original reports.	Could	D1_PRD
FR-024	import invented fixture texts in public-report format with a synthetic label and source metadata; real public-child import is disabled in V2.	Should	D1_PRD
FR-025	redact direct and related identifying payloads only from eligible terminal reports; retain active linked evidence.	Must	D1_PRD
FR-026	open a pre-filled manual form after 2 failed extractions or an AI service error.	Must	D1_PRD
FR-027	in demo-safe mode, return stored outputs for scripted demo inputs with no network.	Must	D1_PRD
FR-028	report frozen held-out metrics with explicit denominators, abstention/coverage, cache exclusion and failure examples.	Must	D1_PRD
FR-029	let parents edit a case after submission per BR-025, keeping all versions.	Should	D1_PRD
FR-030	show parents their own case status and authorised edit/ output availability without candidate or other-party evidence.	Should	D1_PRD
FR-031	idempotently return one case per principal and submission UUID; identical retry returns it, a changed payload conflicts, a new UUID remains a distinct report.	Must	D1_PRD
FR-032	load synthetic seed records and label them "সিনথেটিক" (synthetic) on every screen.	Must	D1_PRD
FR-033	default parent screens to Bangla with an English toggle on every screen.	Must	D1_PRD
FR-034	block generated or displayed text containing alert wording or "runaway" terms.	Must	D1_PRD
FR-035	treat officer accounts as verified only when an admin marks them so. [MOCK: seeded accounts]	Must	D1_PRD
FR-036	provide a disaster-mode logging screen labelled as a planned extension.	Could	D1_PRD
FR-037	sign in staff, officers and admins with a role; parents need no account.	Must	D1_PRD

7.5 Non-functional requirements (NFR, 17)

ID	Label	Requirement	Defined in
NFR-001	Performance	Parent intake screen usable on a mid-range Android over 4G	D1_PRD
NFR-002	Performance	Text → draft	D1_PRD
NFR-003	Performance	120 s voice note → transcript	D1_PRD
NFR-004	Performance	Matching for one case vs 500 records	D1_PRD
NFR-005	Performance	District map with 1,000 cases	D1_PRD
NFR-006	Performance	Concurrent users (A-020)	D1_PRD
NFR-007	Reliability	Scripted demo journey with network off	D1_PRD
NFR-008	Reliability	Draft survives network loss; fallback form after AI failure	D1_PRD
NFR-009	Security	Encrypted transport; no secrets reach the browser; rate limits 10/min/user, 30/min/IP on AI act...	D1_PRD
NFR-010	Privacy	No payload in logs/analytics; synthetic audio/text reaches declared providers before submit with notice; endpoint terms pending review	D1_PRD
NFR-011	Accessibility	WCAG 2.2 AA; tap targets ≥ 44 × 44 px; text contrast ≥ 4.5:1	D1_PRD
NFR-012	Usability	Parent stories completed unaided; any field corrected in 1 tap	D1_PRD
NFR-013	Compatibility	Android browser at 360 px width; current desktop browsers; Bangla renders with no missing glyphs	D1_PRD
NFR-014	Maintainability	AI provider switched by configuration only; setup from README	D1_PRD
NFR-015	Scalability	Register supports all 64 districts and 30,000 cases/year (74/day × 365 ≈ 27,000) without data-m...	D1_PRD
NFR-016	Cost	Total external AI spend from 9 Oct to 24 Oct	D1_PRD
NFR-017	Auditability	Audit entries immutable through the product	D1_PRD

7.6 Business rules (BR, 26)

ID	Label	Linked FRs (safeguarding rules)	Defined in
BR-001	AI retrieves/ranks only. Active verified officers record candidate decisions; identity and guardian verification are external pilot gates.	FR-016, FR-017, FR-020	D1_PRD
BR-002	No issuing, withdrawing or broadcasting alerts. Outputs are reviewed drafts; no CID integration is claimed.	FR-008, FR-009, FR-010, FR-034	D1_PRD
BR-003	No runaway labels, inferred absence reasons, or rescue promises.	FR-004, FR-034	D1_PRD
BR-004	No submitted child payload, matching, other-party visibility, or export before reporter review/confirmation. Providers process intake after notice; pre-submit DB holds metadata only.	FR-006, FR-007	D1_PRD
BR-005	Confirm creates a simulated internal referral record. referred means referral recorded; no receipt, custody, return, or safe reunification is inferred.	FR-016, FR-018	D1_PRD
BR-006	No face search, photos, identity documents, exact coordinates, or physical return action in this release.	FR-009, FR-011, FR-020	D1_PRD
BR-007	Candidates and other-party matched evidence are officer-only. Parent gets own scoped editable payload/status; staff gets own found payload/status. Admin gets aggregate/redacted metadata only.	FR-020, FR-030, FR-035	D1_PRD
BR-008	Only specified report fields are collected. Evidence quotes refer to reviewed source text; deterministic derivations retain source spans; unknowns remain unknown.	FR-004, FR-011	D1_PRD
BR-009	Admin redacts identifying payloads only after closure; found evidence is retained while linked missing case is unclosed. Retained IDs/date/district are pseudonymous, not irreversible anonymity.	FR-025	D1_PRD
BR-010	Decisions, state changes, edits, payload redaction and denied access have safe append-only audit events. Denials after rollback are logged separately; unknown case FK is null.	FR-021	D1_PRD
BR-011	No unsupported legal compliance claim. Accurate primary-law references are permitted. Real-data processing requires qualified partner/legal/provider review.	FR-025, NFR-010	D1_PRD

ID	Label	Linked FRs (safeguarding rules)	Defined in
BR-012	Missing report: Browser draft is not a persisted case. Initial persisted state is submitted; submitted → open; open → matched_pending_review when at least one qualified candidate exists; matched_pending_review → open when none remain; matched_pending_review...		D1_PRD
BR-013	Found report: Initial state open; open ↔ matched_pending_review; matched_pending_review → referred only in the confirmation transaction. The UI says “রেফারেল নথিভুক্ত (সিমুলেশন)” / “Referral recorded (simulation)”. closed_at on referred is the coordination ...		D1_PRD
BR-014	Counts: Missing Open = submitted/open; Missing Matched = matched_pending_review/confirmed; Missing Closed = closed. Found Open = open; Found Matched = matched_pending_review; Found Referral recorded = referred. Confirmed pairs are counted once from confirme...		D1_PRD
BR-015	Required submission fields: Missing: age or complete age range, sex selection (unknown allowed), district, non-empty last-seen place, event interval, and reporter phone. Found: description, district, place, event interval, staff contact; age may be unknown....		D1_PRD
BR-016	Field rules: Missing ages 0–17. Exact age yields age_min = age_max = age_years; a range has both endpoints and no conflicting exact age. Found estimated ranges may extend to 25 to represent uncertainty; wholly adult profiles are not automatically treated as...		D1_PRD
BR-017	Eligibility: Opposite kinds; sex equal or unknown; age ranges overlap within ±2 years when both known, otherwise unknown is allowed with less evidence; found upper time ≥ missing lower time; live open/matched versions; neither record payload-redacted/confir...		D1_PRD
BR-018	Ranking: Retrieve eligible records nationally using exact filtered search for the ~200-record baseline. District/adjacency is a soft bonus, never a gate after five weak local results. L2-normalised embeddings share a pinned space/revision/prefix policy. Use...		D1_PRD
BR-019	Rejections: A rejected pair is not re-suggested at the same report versions. Editing creates new versions and invalidates old pending candidates. Display prior rejection history to the officer.		D1_PRD
BR-020	One-to-one by report: At most one confirmed candidate per endpoint. The transaction locks endpoints in sorted UUID order, rechecks versions/authorisation, confirms one pair, voids other pending pairs, and reconciles each affected report's queue state. Dupli...		D1_PRD

ID	Label	Linked FRs (safeguarding rules)	Defined in
BR-021	Duplicates: Shared phone creates only a weak flagged cue and cannot independently yield confirmed_same or suppress a report. Compatible profiles may create a possible-duplicate cue using dev-calibrated similarity, district and time evidence. Human review ma...		D1_PRD
BR-022	Outputs: Fixed templates use only reporter-reviewed fields plus fixed instructions and 13219. Poster export has a separate preview/share acknowledgement and contact inclusion choice. Downloads cannot be remotely recalled; no photo does not mean no sensitive...		D1_PRD
BR-023	Limits: Audio 5–120 s and ≤2 MB; codec/bitrate must satisfy both, and duration is verified server-side. Text 20–2,000 characters; at most four conversational follow-ups, then manual completion. At most six dispatched paid transcribe/ extract calls per draft,...		D1_PRD
BR-024	Permissions: Parent: own missing creation/read/edit/output only. Staff/unverified officer: own found creation/read and found-only intake. Verified officer: candidate review/decision/ closure, operational case list, aggregate dashboard and safe audit. Admin: ...		D1_PRD
BR-025	Edits: Owner edits only submitted/open/ matched_pending_review with expected_version. Increment version, snapshot the old reviewed payload, void pending pairs, update affected states, and queue bounded reliable rematching. Terminal edits are rejected; payloa...		D1_PRD
BR-026	Synthetic release: Every demo/seed/import record is synthetic, with a visible label; real public child reports are not imported in V2. Model-quality evaluation disables fixtures and caches. Mock authority, cached outputs, compressed video time, and lack of ...		D1_PRD

7.7 Assumptions (A, 40)

ID	Label	Defined in
A-001	OQ-001 – OQ-007 unanswered; Turn 0 defaults apply: voice and poster Must, 13219 Should; 7-state...	D1_PRD
A-002	Numeric targets are provisional and revised after the first evaluation run on 11 Oct.	D1_PRD
A-003	Personas are fictional, built from published reporting.	D1_PRD
A-004	No real police, CID or MUN Alert data is available before the finale.	D1_PRD

ID	Label	Defined in
A-005	Parents use a smartphone browser with a microphone; feature phones are out of scope.	D1_PRD
A-006	Officer verification is a flag set by an admin on seeded accounts [MOCK].	D1_PRD
A-007	A referral is a recorded entry with an agency name; no external integration.	D1_PRD
A-008	The GD draft uses a generic Bangla template; the official online GD field list is UNKNOWN (OQ-0…	D1_PRD
A-009	The parent reads the 13219 summary on their own call; Fera never contacts 13219.	D1_PRD
A-010	Usability testers are adult volunteers role-playing a parent, not real affected parents (OQ-019…	D1_PRD
A-011	The manual-form baseline is the same testers filling a plain form with the same fields.	D1_PRD
A-012	The speech test set is 20 scripted Bangla prompts recorded by ≥ 3 speakers.	D1_PRD
A-013	The team compiles a neighbouring-district list; if absent, search goes same district, then nati…	D1_PRD
A-014	Duplicate thresholds (±1 year, 72 h) are provisional and tuned on the synthetic set.	D1_PRD
A-015	A 0–100 similarity score is understandable to officers if labelled as similarity.	D1_PRD
A-016	Drafts expire after 24 h.	D1_PRD
A-017	Demo-safe mode covers only scripted inputs for ≥ 3 named synthetic cases.	D1_PRD
A-018	Hosting and database run on free tiers; the USD 30 budget covers AI services only.	D1_PRD
A-019	Public reference data gives names and centroids for all 64 districts (DEP-004).	D1_PRD
A-020	50 concurrent users is a PRD v2 target; realistic demo load is ≤ 10.	D1_PRD
A-021	A structured draft shortens the time to file a GD.	D1_PRD
A-022	Parents accept AI-drafted text if they can review and edit it.	D1_PRD
A-023	Low-literacy parents prefer speaking to typing.	D1_PRD
A-024	Public found-child notices exist in enough volume to be worth matching.	D1_PRD

ID	Label	Defined in
A-025	Officers use a ranked queue if each candidate shows evidence side by side.	D1_PRD
A-026	The candidate local model (multilingual-e5-small, 384-d) gives usable Bangla similarity; verifi…	D2_TRD_PartA
A-027	The local model loads within the hosting function's size and time limits; if not, EMBED_PROVIDE…	D2_TRD_PartA
A-028	Match score = 100 × (0.55 cosine + 0.15 name + 0.15 clothing + 0.10 district + 0.05 age closene…	D2_TRD_PartA
A-029	Duplicate profile threshold is a development-only hypothesis, calibrated on hard negatives and frozen before held-out test; phone alone never confirms a duplicate.	D2_TRD_PartA
A-030	Android Chrome MediaRecorder output (webm/opus) is accepted by the speech provider; verified 10…	D2_TRD_PartA
A-031	Browsers allow microphone access only on HTTPS or localhost. The finale phone reaches the deplo…	D2_TRD_PartA
A-032	District codes 1–64 are Fera-internal, not official geocodes.	D2_TRD_PartA
A-033	The team builds a Bangla→Latin romanisation table for name similarity.	D2_TRD_PartA
A-034	A free-tier managed Postgres with pgvector >= 0.8.1 (HNSW) is available (OQ-021).	D2_TRD_PartA
A-035	Time words map to fixed hours: সকাল 08:00, দুপুর 13:00, বিকেল 16:00, সন্ধ্যা 18:30, রাত 21:00, …	D2_TRD_PartA
A-036	2 MB is enough for a 120 s webm/opus note from Android Chrome.	D2_TRD_PartA
A-037	Usage volumes in §12.2 (190 audio minutes, 1,100 extraction calls, ~1,200 input + 400 output to…	D2_TRD_PartB
A-038	Four-times output weighting is a budget assumption; verify actual input/output rates at the 11 Oct contract gate.	D2_TRD_PartB
A-039	200 total reports: 80 missing, 80 found, 40 additional decoys; 60 canonical true pairs, 20 no-match missing queries, 25 duplicate links within the total.	D2_TRD_PartB
A-040	Actual availability and completed work must be inventoried on 11 Oct. A fresh 11–14 Oct start at 10 h/person/day gives at most 120 team-hours; this is an assumption, not verified capacity.	D6_Implementation_Plan

7.8 Dependencies (DEP, 11)

ID	Label	Owner	Defined in
DEP-001	Speech-to-text service with Bangla support	Obidit	D1_PRD
DEP-002	At least two interchangeable language-model services	Obidit	D1_PRD
DEP-003	Text-similarity capability that also works offline for scripted inputs	Masnun	D1_PRD
DEP-004	District reference data: 64 districts, Bangla/ English names, centroids	Masnun	D1_PRD
DEP-005	Synthetic set, ~200 records, known pairs, duplicates, decoys	Masnun	D1_PRD
DEP-006	50 hand-labelled reports	Masnun, Obidit	D1_PRD
DEP-007	Public news and alert texts (public pages only)	Masnun	D1_PRD
DEP-008	Free-tier hosting and database	Sadman	D1_PRD
DEP-009	Finale kit: laptop, Android phone, mobile hotspot	Sadman	D1_PRD
DEP-010	Mentor or organiser answers to OQ-003 and OQ-010	Sadman	D1_PRD
DEP-011	Base map display; offline fallback without map tiles	Masnun	D1_PRD

7.9 Risks (RISK, 17)

ID	Label	Likelihood	Defined in
RISK-001	Core journey unfinished by 14 Oct official cutoff (time pending)	M	D1_PRD
RISK-002	Bangla speech recognition too poor	H	D1_PRD
RISK-003	AI invents or misreads fields	M	D1_PRD
RISK-004	AI service outage, quota or spend over USD 30	M	D1_PRD
RISK-005	Venue internet fails at the finale	M	D1_PRD
RISK-006	Matching below targets	M	D1_PRD
RISK-007	Synthetic set too easy, inflating metrics	M	D1_PRD
RISK-008	Stalker or trafficker misuse	L	D1_PRD
RISK-009	False match sends a child into danger	L	D1_PRD

ID	Label	Likelihood	Defined in
RISK-010	Judges favour environment-themed projects	M	D1_PRD
RISK-011	"Why not just MUN Alert?"	H	D1_PRD
RISK-012	Overclaiming on metrics, interviews or law	M	D1_PRD
RISK-013	Bangla text renders badly on low-end phones	L	D1_PRD
RISK-014	Instructions hidden in imported public text steer the AI	M	D1_PRD
RISK-015	Local embedding model too large or slow for serverless hosting	M	D2_TRD_PartA
RISK-016	Phone microphone blocked when the app is served over plain HTTP in local mode	M	D2_TRD_PartA
RISK-017	V2 Round 2 estimates total 184 h and may exceed actual remaining capacity	M	D6_Implementation_Plan

7.10 Open questions (OQ, 26)

ID	Label	Context / default / owner	Defined in
OQ-001	Are **voice intake** and **poster + 13219 summary** Must for Round 2?	PRD v2 marks both Should (v2 FR7, FR8). PROJECT CONTEXT lists "Bangla intake → poster + GD" as ...	00_Gap_Check_and_Registry
OQ-002	Adopt the separate missing/found state machine from PROJECT CONTEXT (draft → submitted → open → matched-pending-review → confirmed → closed/referred)?	PRD v2 uses open / matched / closed / reopened and closes both cases on confirm. Changes BRs, s...	00_Gap_Check_and_Registry
OQ-003	What did mentors say in session 1 (7–8 Oct) about the environment theme and disaster mode?	PRD v2 rates "judges favour environmental projects" Medium/High risk. The answer decides whethe...	00_Gap_Check_and_Registry
OQ-004	Which accounts and API keys does the team already hold (LLM providers, OpenAI for Whisper, Postgres host, Vercel)? Any free credits?	Sets the default LLM provider, the embedding source and the USD 30 budget estimate.	00_Gap_Check_and_Registry
OQ-005	Finale demo topology: deployed app over venue Wi-Fi, or laptop-local (localhost DB, phone on laptop hotspot)?	Venue internet may fail. Local-first changes DB hosting, embeddings and the fallback design.	00_Gap_Check_and_Registry

ID	Label	Context / default / owner	Defined in
OQ-006	What exists today (9 Oct)? Repo skeleton, synthetic set, 50 labelled reports, any user conversations held?	Actual repository progress and user/ evaluation evidence are UNVERIFIED; inventory now.	00_Gap_Check_and_Registry
OQ-007	Demo auth and roles: real login or a role switcher? Keep PRD v2's extra "helper" role?	PRD v2 roles: parent, helper, staff, officer, admin. PROJECT CONTEXT: parent, officer, shelter ...	00_Gap_Check_and_Registry
OQ-008	Evidence scope	PROJECT CONTEXT says use only 5 figures. Deck and PRD v2 also cite age 0–9 split, MUN Alert "20...	00_Gap_Check_and_Registry
OQ-009	Accessibility level	PRD v2 targets WCAG 2.1 AA; D4 spec asks 2.2 AA.	00_Gap_Check_and_Registry
OQ-010	Alert authority wording	Deck slide 4 says only CID's Missing Children Cell can issue alerts, and also that a proposed C...	00_Gap_Check_and_Registry
OQ-011	Personal Data Protection Act (2026)	PRD v2 says "passed April 2026". No clauses are in any source.	00_Gap_Check_and_Registry
OQ-012	Retention period for closed cases	PRD v2: "to be set with a partner". Safeguarding requires delete/redact identifying payloads.	00_Gap_Check_and_Registry
OQ-013	LLM/STT provider data terms	PRD v2: "confirm per provider" no-training and retention terms.	00_Gap_Check_and_Registry
OQ-014	Persona count	PRD v2 has 4 personas incl. child-protection officer; D1 spec asks for 3.	00_Gap_Check_and_Registry
OQ-015	Dedupe priority	PRD v2 puts duplicate merge at Could (v2 FR10). Deck promises a duplicate-detection quality metric...	00_Gap_Check_and_Registry
OQ-016	Official online GD field list and format?	UNKNOWN; generic template (A-008)	D1_PRD
OQ-017	Source for neighbouring-district adjacency?	Team-compiled (A-013)	D1_PRD
OQ-018	Which child-protection agency name should a referral show?	UNKNOWN; demo shows "শিশু সুরক্ষা সেবা" (child-protection services)	D1_PRD
OQ-019	Can usability tests ethically include real affected parents?	Default: no (A-010)	D1_PRD
OQ-020	Should parents return to a case from another device?	Default: Won't this release	D1_PRD

ID	Label	Context / default / owner	Defined in
OQ-021	Managed Postgres host	Pick on 10 Oct; must pass CREATE EXTENSION vector and HNSW index creation	D2_TRD_Part A
OQ-022	Speech provider limits and price	Verify from provider docs on 10 Oct; record in Part B §12	D2_TRD_Part A
OQ-023	Hosting limits for local embeddings	10 Oct spike; fallback A-027	D2_TRD_Part A
OQ-024	Do classes, exams or labs reduce team hours before 14 Oct?	UNKNOWN; inventory actual availability and finished work at the 11 Oct review	D6_Implementation_Plan
OQ-025	Judging weights for impact, feasibility, scalability	Organisers	D7_Closing_Pack
OQ-026	Is synthetic data without real public-child news acceptable for Round 2 technical evaluation?	Organisers / mentors	D7_Closing_Pack

7.11 Screens (SCREEN, 16)

ID	Label	Priority	Defined in
SCREEN-001	Welcome and language	Must	D3_App_Flow
SCREEN-002	Intake chat (voice/text)	Must	D3_App_Flow
SCREEN-003	Case review	Must	D3_App_Flow
SCREEN-004	Case submitted and outputs	Must	D3_App_Flow
SCREEN-005	Manual case form	Must	D3_App_Flow
SCREEN-006	Parent case status and edit	Should	D3_App_Flow
SCREEN-007	Found-child log	Must	D3_App_Flow
SCREEN-008	Sign-in	Must	D3_App_Flow
SCREEN-009	Review queue	Must	D3_App_Flow
SCREEN-010	Match review (side by side)	Must	D3_App_Flow
SCREEN-011	District dashboard	Must	D3_App_Flow
SCREEN-012	Case detail and audit trail	Must	D3_App_Flow
SCREEN-013	Admin data tools	Must	D3_App_Flow
SCREEN-014	Duplicate review	Could	D3_App_Flow

ID	Label	Priority	Defined in
SCREEN-015	Disaster mode	Could	D3_App_Flow
SCREEN-016	Not found / not allowed	Must	D3_App_Flow

7.12 Components (server COMP-001–021; UI COMP-022–050) (COMP, 50)

ID	Label	Detail	Defined in
COMP-001	Parent client	Screens SCREEN-001–006; draft in localStorage	D2_TRD_PartA
COMP-002	Staff/officer/admin client	SCREEN-007–016	D2_TRD_PartA
COMP-003	Voice recorder	Capture 5–120 s via MediaRecorder; show timer	D2_TRD_PartA
COMP-004	API layer	Route handlers; Zod; auth guard; rate limit; request ID	D2_TRD_PartA
COMP-005	Auth and session	Auth.js credentials; parent cookie	D2_TRD_PartA
COMP-006	Speech adapter	Transcribe; script check; cache lookup	D2_TRD_PartA
COMP-007	LLM adapter	Extraction via contract §2.1; retry; fallback	D2_TRD_PartA
COMP-008	Normaliser	Bangla numerals → ASCII; dates/time words → BST ISO; phone; district lookup (bn/en/ alias); roma...	D2_TRD_PartA
COMP-009	Intake orchestrator	Merge extraction + answers; evidence check; required-field check; templated question; AI call c...	D2_TRD_PartA
COMP-010	Output generator	GD, poster text, 13219 summary from templates	D2_TRD_PartA
COMP-011	Embedding service	Build case description string; embed	D2_TRD_PartA
COMP-012	Matching engine	Hard filters (SQL), cosine (pgvector), bonuses, top 5, template reason	D2_TRD_PartA
COMP-013	Duplicate detector	Apply D1 BR-021; write duplicate_links	D2_TRD_PartA
COMP-014	Case service + state guard	Create, edit (versions), transitions, referral, close, redact identifying payloads	D2_TRD_PartA

ID	Label	Detail	Defined in
COMP-015	Audit logger	Append audit_log rows inside the same transaction	D2_TRD_PartA
COMP-016	Wording guard	Block alert wording and "runaway" terms in Fera-generated text and labels	D2_TRD_PartA
COMP-017	Demo-safe cache	Fixtures for scripted inputs: transcript, extraction, embeddings	D2_TRD_PartA
COMP-018	Dashboard and map	Aggregate counts; Leaflet map; table view	D2_TRD_PartA
COMP-019	Admin tools	Seed loader (CLI), public import, redact identifying payloadsr, verify officer [MOCK]	D2_TRD_PartA
COMP-020	Evaluation harness	Compute 5 metrics; write report	D2_TRD_PartA
COMP-021	Rate limiter	Fixed-window counters in Postgres	D2_TRD_PartA
COMP-022	App header with language toggle	All	D4_Design_Brief
COMP-023	Synthetic badge "সিনথেটিক"	Any seeded record	D4_Design_Brief
COMP-024	Chat bubble (assistant / user)	SCREEN-002	D4_Design_Brief
COMP-025	Voice record button with timer	SCREEN-002, SCREEN-007	D4_Design_Brief
COMP-026	Transcript editor	SCREEN-002	D4_Design_Brief
COMP-027	Question prompt with quick-reply chips	SCREEN-002	D4_Design_Brief
COMP-028	Editable field card	SCREEN-003, SCREEN-007	D4_Design_Brief
COMP-029	Confirm bar (checkbox + submit)	SCREEN-003, SCREEN-007	D4_Design_Brief
COMP-030	Output card with copy	SCREEN-004	D4_Design_Brief
COMP-031	Poster preview + PNG download	SCREEN-004	D4_Design_Brief
COMP-032	Manual case form	SCREEN-005	D4_Design_Brief
COMP-033	Status badge (icon + text + colour)	SCREEN-004, -006, -009, -011, -012	D4_Design_Brief

ID	Label	Detail	Defined in
COMP-034	Found-child form	SCREEN-007, SCREEN-015	D4_Design_Brief
COMP-035	Review queue list	SCREEN-009	D4_Design_Brief
COMP-036	Candidate row (rank, score, reason)	SCREEN-009	D4_Design_Brief
COMP-037	Side-by-side evidence table	SCREEN-010, SCREEN-014	D4_Design_Brief
COMP-038	Decision panel (confirm / reject + reason)	SCREEN-010	D4_Design_Brief
COMP-039	Close-case dialog	SCREEN-012	D4_Design_Brief
COMP-040	District map	SCREEN-011	D4_Design_Brief
COMP-041	District table	SCREEN-011	D4_Design_Brief
COMP-042	Audit timeline	SCREEN-012	D4_Design_Brief
COMP-043	Admin action panel	SCREEN-013	D4_Design_Brief
COMP-044	Inline alert / toast	All	D4_Design_Brief
COMP-045	Empty state	SCREEN-006, -009, -013, -016	D4_Design_Brief
COMP-046	Loading state with 10 s message	SCREEN-002, -004, -007	D4_Design_Brief
COMP-047	Sign-in form	SCREEN-008	D4_Design_Brief
COMP-048	Cached badge "ক্যাশ"	Any AI output from demo cache	D4_Design_Brief
COMP-049	Duplicate banner "সম্ভাব্য একই শিশু"	SCREEN-010, SCREEN-012	D4_Design_Brief
COMP-050	Consent notice	SCREEN-001	D4_Design_Brief

7.13 API endpoints (API, 23)

ID	Label	Priority	Defined in
API-001	POST /api/intake/transcribe	Must	D2_TRD_PartA
API-002	POST /api/intake/extract	Must	D2_TRD_PartA
API-003	POST /api/cases	Must	D2_TRD_PartA
API-004	GET /api/cases/:id	Must	D2_TRD_PartA
API-005	PATCH /api/cases/:id	Should	D2_TRD_PartA

ID	Label	Priority	Defined in
API-006	POST /api/cases/:id/outputs	Must (gd, poster) / Should (helpline)	D2_TRD_PartA
API-007	POST /api/found-cases	Must	D2_TRD_PartA
API-008	GET /api/review-queue	Must	D2_TRD_PartA
API-009	GET /api/candidates/:id	Must	D2_TRD_PartA
API-010	POST /api/candidates/:id/decision	Must	D2_TRD_PartA
API-011	POST /api/cases/:id/close	Must	D2_TRD_PartA
API-012	GET /api/dashboard/stats	Must	D2_TRD_PartA
API-013	GET /api/cases/:id/audit	Must	D2_TRD_PartA
API-014	POST /api/match/:caseId	Must	D2_TRD_PartA
API-015	POST /api/admin/import	Should	D2_TRD_PartA
API-016	POST /api/admin/anonymise	Must	D2_TRD_PartA
API-017	POST /api/admin/users/:id/verify	Must [MOCK]	D2_TRD_PartA
API-018	GET /api/my-case	Could	D2_TRD_PartA
API-019	GET /api/health	Must	D2_TRD_PartA
API-020	/api/auth/*	Must	D2_TRD_PartA
API-021	GET /api/admin/users	Must	D2_TRD_PartA
API-022	GET /api/cases (officer district list)	Must	D2_TRD_PartA
API-023	POST /api/duplicates/:id/decision	Could	D2_TRD_PartA

7.14 Entities (ENTITY, 17)

ID	Label	Table	Defined in
ENTITY-001	User	users	D2_TRD_PartA
ENTITY-002	ParentSession	parent_sessions	D2_TRD_PartA
ENTITY-003	Case	cases	D2_TRD_PartA
ENTITY-004	CaseVersion	case_versions	D2_TRD_PartA

ID	Label	Table	Defined in
ENTITY-005	MatchCandidate	match_candidates	D2_TRD_PartA
ENTITY-006	ReviewDecision	review_decisions	D2_TRD_PartA
ENTITY-007	Referral	referrals	D2_TRD_PartA
ENTITY-008	DuplicateLink	duplicate_links	D2_TRD_PartA
ENTITY-009	AuditEntry	audit_log	D2_TRD_PartA
ENTITY-010	District	districts	D2_TRD_PartA
ENTITY-011	DistrictNeighbour	district_neighbours	D2_TRD_PartA
ENTITY-012	GeneratedOutput	generated_outputs	D2_TRD_PartA
ENTITY-013	AiCall	ai_calls	D2_TRD_PartA
ENTITY-014	RateLimitBucket	rate_limits	D2_TRD_PartA
ENTITY-015	IntakeDraft	`intake_drafts`	D2_TRD_PartA
ENTITY-016	AiBudget	`ai_budget`	D2_TRD_PartA
ENTITY-017	AiReservation	`ai_reservations`	D2_TRD_PartA

7.15 Decisions (DECISION, 17)

ID	Label	Section	Defined in
DECISION-001	Single Next.js App Router app (UI + API), TypeScript strict	§2	D2_TRD_PartA
DECISION-002	PostgreSQL 16 + pgvector; raw SQL via `pg`; `schema_v2.sql` is the migration	§2, §7	D2_TRD_PartA
DECISION-003	Auth.js credentials for staff/officer/admin; signed anonymous cookie for parents	§9	D2_TRD_PartA
DECISION-004	LLM used for extraction only, behind a fixed JSON contract, temp 0, 1 retry, 1 fallback provider	§2.1	D2_TRD_PartA
DECISION-005	Speech via `SpeechProvider`; Whisper API default; cache for scripted notes	§2.2	D2_TRD_PartA
DECISION-006	Local embeddings (Transformers.js, 384-d) by default; API embeddings as switchable fallback	§2.3	D2_TRD_PartA
DECISION-007	Follow-up questions, GD, poster, 13219 summary and match reasons from templates, not LLM	§4	D2_TRD_PartA
DECISION-008	Matching = SQL hard filters + pgvector cosine + rule bonuses	§4	D2_TRD_PartA

ID	Label	Section	Defined in
DECISION-009	Leaflet with OSM tiles online; tile-less markers plus table offline	§2	D2_TRD_Part A
DECISION-010	Vercel for Round 2; laptop-local copy for the finale; phone uses its own mobile data	§3.6	D2_TRD_Part A
DECISION-011	Rate limits stored in Postgres (serverless-safe)	§4	D2_TRD_Part A
DECISION-012	No optimistic updates for any state-changing action	§8	D2_TRD_Part A
DECISION-013	No per-case coordinates; district centroid only	§6	D2_TRD_Part A
DECISION-014	Draft held in the browser; nothing persisted before confirm	§3.4, §8	D2_TRD_Part A
DECISION-015	Poster PNG rendered in the browser via canvas	§2	D2_TRD_Part A
DECISION-016	Dev/test split for matching evaluation		D2_TRD_Part B
DECISION-017	Hard AI spend cap in app		D2_TRD_Part B

7.16 Error codes (E, 24)

ID	Label	HTTP	Defined in
E-001	VALIDATION_FAILED	400	D2_TRD_PartA
E-002	UNAUTHENTICATED	401	D2_TRD_PartA
E-003	FORBIDDEN_ACTION	403	D2_TRD_PartA
E-004	NOT_FOUND	404	D2_TRD_PartA
E-005	ILLEGAL_TRANSITION	409	D2_TRD_PartA
E-006	SUBMISSION_PAYLOAD_CONFLICT	409	D2_TRD_PartA
E-007	RATE_LIMITED	429	D2_TRD_PartA
E-008	AUDIO_TOO_LARGE	413	D2_TRD_PartA
E-009	AUDIO_LENGTH_OUT_OF_RANGE	422	D2_TRD_PartA
E-010	SPEECH_UNINTELLIGIBLE	422	D2_TRD_PartA
E-011	SPEECH_UNAVAILABLE	503	D2_TRD_PartA

ID	Label	HTTP	Defined in
E-012	LLM_INVALID_OUTPUT	422	D2_TRD_PartA
E-013	LLM_UNAVAILABLE	503	D2_TRD_PartA
E-014	AI_TIMEOUT	504	D2_TRD_PartA
E-015	UNSUPPORTED_LANGUAGE	422	D2_TRD_PartA
E-016	OUTPUT_BLOCKED	422	D2_TRD_PartA
E-017	REFERRAL_REQUIRED	409	D2_TRD_PartA
E-018	EDIT_LOCKED	409	D2_TRD_PartA
E-019	SESSION_EXPIRED	401	D2_TRD_PartA
E-020	OUTPUT_REQUEST_LIMIT_RESERVED	429	D2_TRD_PartA
E-021	INTERNAL	500	D2_TRD_PartA
E-022	MIC_UNAVAILABLE	—	D2_TRD_PartA
E-023	AI_BUDGET_CAP	429	D2_TRD_PartA
E-024	NETWORK_OFFLINE	client only	D2_TRD_PartB

7.17 Test cases (TC, 66)

ID	Label	Defined in
TC-001	Voice intake	D6_Implementation_Plan
TC-002	Text intake, mixed script, Bangla numerals	D6_Implementation_Plan
TC-003	Follow-up questions	D6_Implementation_Plan
TC-004	Review before share	D6_Implementation_Plan
TC-005	GD draft fields, no invented facts	D6_Implementation_Plan
TC-006	Poster, no photo	D6_Implementation_Plan
TC-007	13219 summary ≤ 80 words	D6_Implementation_Plan
TC-008	Found-child log	D6_Implementation_Plan
TC-009	Top-5 candidates, hard filters	D6_Implementation_Plan
TC-010	Confirm → referral → close	D6_Implementation_Plan
TC-011	Reject → open, not re-suggested	D6_Implementation_Plan

ID	Label	Defined in
TC-012	Map counts equal register	D6_Implementation_Plan
TC-013	Duplicate flag and link	D6_Implementation_Plan
TC-014	Public import with source URL	D6_Implementation_Plan
TC-015	Related-payload redaction	D6_Implementation_Plan
TC-016	Audit completeness	D6_Implementation_Plan
TC-017	Manual form fallback	D6_Implementation_Plan
TC-018	Edit after submission	D6_Implementation_Plan
TC-019	Parent status view hides match data	D6_Implementation_Plan
TC-020	Evaluation report	D6_Implementation_Plan
TC-021	Offline scripted demo	D6_Implementation_Plan
TC-022	Unauthorised access to matched record	D6_Implementation_Plan
TC-023	Non-officer cannot confirm	D6_Implementation_Plan
TC-024	No alert or "runaway" wording	D6_Implementation_Plan
TC-025	Outputs labelled as drafts	D6_Implementation_Plan
TC-026	No photo upload or face search	D6_Implementation_Plan
TC-027	Data minimisation	D6_Implementation_Plan
TC-028	No personal data in logs	D6_Implementation_Plan
TC-029	One-to-one confirm lock	D6_Implementation_Plan
TC-030	Field validation	D6_Implementation_Plan
TC-031	Scoped submission UUID idempotency	D6_Implementation_Plan
TC-032	Illegal state transitions rejected	D6_Implementation_Plan
TC-033	Accessibility audit	D6_Implementation_Plan
TC-034	Performance targets	D6_Implementation_Plan
TC-035	Disaster-mode screen	D6_Implementation_Plan
TC-036	AI spend tracking	D6_Implementation_Plan
TC-037	No legal compliance claims	D6_Implementation_Plan
TC-038	Shared-phone distinct children	D6_Implementation_Plan

ID	Label	Defined in
TC-039	Unknown found age	D6_Implementation_Plan
TC-040	Overlapping time uncertainty	D6_Implementation_Plan
TC-041	No plausible candidate	D6_Implementation_Plan
TC-042	Concurrent submission UUID	D6_Implementation_Plan
TC-043	Draft replay/concurrency	D6_Implementation_Plan
TC-044	Admin rematch projection	D6_Implementation_Plan
TC-045	Owner editable read	D6_Implementation_Plan
TC-046	Stale candidate decision	D6_Implementation_Plan
TC-047	Wrong referral endpoints	D6_Implementation_Plan
TC-048	Reject without reason	D6_Implementation_Plan
TC-049	Terminal report edit	D6_Implementation_Plan
TC-050	Related note redaction	D6_Implementation_Plan
TC-051	Audit truncate/privilege	D6_Implementation_Plan
TC-052	Unknown-resource denial	D6_Implementation_Plan
TC-053	Last competitor invalidated	D6_Implementation_Plan
TC-054	Matching fails after submit	D6_Implementation_Plan
TC-055	One pair dashboard lifecycle	D6_Implementation_Plan
TC-056	Evaluation fixture contamination	D6_Implementation_Plan
TC-057	Single-DB offline scenario	D6_Implementation_Plan
TC-058	Reflow/keyboard/Bangla	D6_Implementation_Plan
TC-059	Embedding space switch	D6_Implementation_Plan
TC-060	Provider notice/metadata	D6_Implementation_Plan
TC-061	Encoded audio limits	D6_Implementation_Plan
TC-062	PostgreSQL16/runtime host	D6_Implementation_Plan
TC-063	Correction/reopen pilot gate	D6_Implementation_Plan
TC-064	Manifest/group split	D6_Implementation_Plan
TC-065	Concurrent AI budget	D6_Implementation_Plan
TC-066	Legal claims/source accuracy	D6_Implementation_Plan

7.18 Flow actions (ACT, 33)

ID	Label	Role	Defined in
ACT-001	Tap Start	Parent	D3_App_Flow
ACT-002	Record voice note	Parent, staff	D3_App_Flow
ACT-003	Type report or description	Parent, staff	D3_App_Flow
ACT-004	Edit transcript	Parent	D3_App_Flow
ACT-005	Send / next	Parent	D3_App_Flow
ACT-006	Answer follow-up question	Parent	D3_App_Flow
ACT-007	Edit a field	Parent, staff	D3_App_Flow
ACT-008	Check and submit	Parent	D3_App_Flow
ACT-009	Request GD draft	Parent	D3_App_Flow
ACT-010	Request poster	Parent	D3_App_Flow
ACT-011	Request 13219 summary	Parent	D3_App_Flow
ACT-012	Copy text	Parent	D3_App_Flow
ACT-013	Download poster PNG	Parent	D3_App_Flow
ACT-014	Open manual form	Parent	D3_App_Flow
ACT-015	Discard / cancel	Parent, staff	D3_App_Flow
ACT-016	View case status	Parent	D3_App_Flow
ACT-017	Correct details after submit	Parent	D3_App_Flow
ACT-018	Sign in	Staff, officer, admin	D3_App_Flow
ACT-019	Save found child	Staff, officer	D3_App_Flow
ACT-020	Open candidate	Officer	D3_App_Flow
ACT-021	Confirm match	Officer	D3_App_Flow
ACT-022	Reject match with reason	Officer	D3_App_Flow
ACT-023	Close case with outcome	Officer	D3_App_Flow
ACT-024	Use map or district table	Officer, admin	D3_App_Flow
ACT-025	View audit trail	Officer, admin	D3_App_Flow
ACT-026	Import invented fixture texts	Admin	D3_App_Flow

ID	Label	Role	Defined in
ACT-027	Redact identifying payloads from eligible terminal cases	Admin	D3_App_Flow
ACT-028	Verify officer [MOCK]	Admin	D3_App_Flow
ACT-029	Sign out	Staff, officer, admin	D3_App_Flow
ACT-030	Switch language	All	D3_App_Flow
ACT-031	Retry / continue draft	Parent, staff	D3_App_Flow
ACT-032	Mark duplicate same/ different	Officer	D3_App_Flow
ACT-033	Log separated child (disaster mode)	Staff	D3_App_Flow

7.19 Flow decisions (D, 16)

ID	Label	Defined in
D-001	Microphone available and permitted?	D3_App_Flow
D-002	Audio length 5–120 s and ≤ 2 MB?	D3_App_Flow
D-003	Transcript intelligible and Bangla/ English?	D3_App_Flow
D-004	Required fields complete?	D3_App_Flow
D-005	AI calls used < 6?	D3_App_Flow
D-006	Extraction valid within 2 attempts?	D3_App_Flow
D-007	Submission UUID retry or conflicting payload?	D3_App_Flow
D-008	Role officer and verified?	D3_App_Flow
D-009	Pending candidates exist?	D3_App_Flow
D-010	Officer confirms or rejects?	D3_App_Flow
D-011	Other pending candidates remain?	D3_App_Flow
D-012	Referral exists?	D3_App_Flow
D-013	Case editable (BR-025)?	D3_App_Flow
D-014	Device online?	D3_App_Flow
D-015	Session valid?	D3_App_Flow
D-016	Admin typed confirmation correct?	D3_App_Flow

7.20 Tasks (TASK, 64)

ID	Label	Priority	Defined in
TASK-001	Repo: Next.js App Router, TS strict, Tailwind, ESLint, CI (lint, typecheck, test)	Must	D6_Implementation_Plan
TASK-002	Pick DB host (OQ-021); apply `schema_v2.sql`; local Docker DB	Must	D6_Implementation_Plan
TASK-003	Provider keys, `.env.example`, record real prices in `docs/costs.md` (OQ-004, OQ-022)	Must	D6_Implementation_Plan
TASK-004	Spike: local embedding model on Vercel (OQ-023, RISK-015)	Must	D6_Implementation_Plan
TASK-005	Spike: Android MediaRecorder → STT (A-030)	Must	D6_Implementation_Plan
TASK-006	Auth.js credentials, parent cookie, `requireRole`, seeded users	Must	D6_Implementation_Plan
TASK-007	API layer: error envelope E-001 – E-024, Zod helpers, request ID, rate limiter (COMP-021)	Must	D6_Implementation_Plan
TASK-008	Layout shell, `tokens_v2.css`, Noto Sans Bengali, language toggle (COMP-022)	Must	D6_Implementation_Plan
TASK-009	District data: 64 rows bn/en/centroids + neighbours (DEP-004, A-013)	Must	D6_Implementation_Plan
TASK-010	Synthetic set generator: ~200 records, ground truth, duplicates, decoys (A-039)	Must	D6_Implementation_Plan
TASK-011	50 hand-labelled reports	Must	D6_Implementation_Plan
TASK-012	Normaliser: Bangla digits, time words, phone, district aliases, romaniser (COMP-008)	Must	D6_Implementation_Plan
TASK-013	LLM adapter: 2 providers, Zod contract, evidence check, retry, fallback (COMP-007)	Must	D6_Implementation_Plan
TASK-014	Speech adapter + API-001 (COMP-006)	Must	D6_Implementation_Plan
TASK-015	Case service, state guard, audit, versions (COMP-014, COMP-015)	Must	D6_Implementation_Plan
TASK-016	Embedding service + `npm run reembed` (COMP-011)	Must	D6_Implementation_Plan
TASK-017	Matching engine: filters, cosine, bonuses, template reasons (COMP-012)	Must	D6_Implementation_Plan
TASK-018	Duplicate detector (COMP-013)	Must	D6_Implementation_Plan

ID	Label	Priority	Defined in
TASK-019	Seed loader with synthetic labels and 3 demo cases	Must	D6_Implementation_Plan
TASK-020	Intake orchestrator + API-002 + question templates (COMP-009)	Must	D6_Implementation_Plan
TASK-021	SCREEN-001/002 intake chat + recorder (COMP-024 – COMP-027)	Must	D6_Implementation_Plan
TASK-022	SCREEN-003 review + API-003 + draft persistence	Must	D6_Implementation_Plan
TASK-023	Output templates GD/poster + wording guard + API-006 (COMP-010, COMP-016)	Must	D6_Implementation_Plan
TASK-024	SCREEN-004 outputs + poster PNG (COMP-030, COMP-031)	Must	D6_Implementation_Plan
TASK-025	SCREEN-005 manual form fallback (COMP-032)	Must	D6_Implementation_Plan
TASK-026	SCREEN-007 found-child log + API-007 (COMP-034)	Must	D6_Implementation_Plan
TASK-027	SCREEN-009 queue + API-008 (COMP-035, COMP-036)	Must	D6_Implementation_Plan
TASK-028	SCREEN-010 match review + API-009/010 (COMP-037, COMP-038)	Must	D6_Implementation_Plan
TASK-029	SCREEN-012 case detail, close, audit + API-011/013 (COMP-039, COMP-042)	Must	D6_Implementation_Plan
TASK-030	SCREEN-011 map + table + API-012 (COMP-040, COMP-041)	Must	D6_Implementation_Plan
TASK-031	SCREEN-008 sign-in + SCREEN-016 (COMP-047, COMP-045)	Must	D6_Implementation_Plan
TASK-032	SCREEN-013 admin: redact identifying payloads + verify [MOCK] + API-016/017 (COMP-043)	Must	D6_Implementation_Plan
TASK-033	Demo-safe cache + fixtures for 3 cases + cached badge (COMP-017, COMP-048)	Must	D6_Implementation_Plan
TASK-034	Accessibility pass: axe + keyboard + 320 CSS px reflow, 200 % text zoom and 400 % viewport zoom	Must	D6_Implementation_Plan
TASK-035	Bangla copy review; all empty/error/loading states	Must	D6_Implementation_Plan
TASK-036	Test on 2 Android phones (Bangla rendering, mic, 360 px)	Must	D6_Implementation_Plan

ID	Label	Priority	Defined in
TASK-037	Unit tests: normaliser, matcher, state guard, wording guard	Must	D6_Implementation_Plan
TASK-038	Playwright E2E: essential journey + safeguarding TCs	Must	D6_Implementation_Plan
TASK-039	Evaluation harness: smoke/dev run first; held-out run after manifest/model freeze	Must	D6_Implementation_Plan
TASK-040	WER recordings (60 clips) + run	Must	D6_Implementation_Plan
TASK-041	Timing study, ≥ 5 testers, assistant vs manual form	Must	D6_Implementation_Plan
TASK-042	Production deploy + managed DB seed + smoke test	Must	D6_Implementation_Plan
TASK-043	Final evaluation run on freeze build	Must	D6_Implementation_Plan
TASK-044	Round 2 deck, 5–8 slides	Must	D6_Implementation_Plan
TASK-045	Round 2 demo video, 1–2 min	Must	D6_Implementation_Plan
TASK-046	README, Drive folder (links TXT, deck, video), reply to selection email	Must	D6_Implementation_Plan
TASK-047	13219 summary (FR-010)	Should	D6_Implementation_Plan
TASK-048	Synthetic fixture import + API-015 (FR-024)	Should	D6_Implementation_Plan
TASK-049	Edit after submit + API-005 + SCREEN-006 (FR-029)	Should	D6_Implementation_Plan
TASK-050	Parent status API-018 (FR-030)	Should	D6_Implementation_Plan
TASK-051	Duplicate review SCREEN-014 (FR-023)	Could	D6_Implementation_Plan
TASK-052	Disaster mode SCREEN-015 (FR-036)	Could	D6_Implementation_Plan
TASK-053	Finale local stack: Docker DB, model cache, fixtures, `DEMO_MODE=1`	Must	D6_Implementation_Plan
TASK-054	5 rehearsals, ≥ 1 internet-off	Must	D6_Implementation_Plan
TASK-055	Finale deck, 5–10 min	Must	D6_Implementation_Plan

ID	Label	Priority	Defined in
TASK-056	Fixes from mentor session 2 (12–13 Oct)	Should	D6_Implementation_Plan
TASK-057	5–10 user conversations, only if OQ-019 allows	Should	D6_Implementation_Plan
TASK-058	Backup recording of the finale demo	Must	D6_Implementation_Plan
TASK-059	`npm run spend` + spend cap check (DECISION-017)	Must	D6_Implementation_Plan
TASK-060	Performance check against D2 Part B §12.1	Must	D6_Implementation_Plan
TASK-061	Apply V2 required fields, DTOs, token/ idempotency and referral/count semantics	Must	D6_Implementation_Plan
TASK-062	V2 schema invariants, restricted roles and full related-payload redaction	Must	D6_Implementation_Plan
TASK-063	Atomic paid-call counters/reservations, replay handling and price sheet	Must	D6_Implementation_Plan
TASK-064	Identity-grouped manifest, no-match/ duplicate hard negatives, frozen metrics	Must	D6_Implementation_Plan

8 Relationships

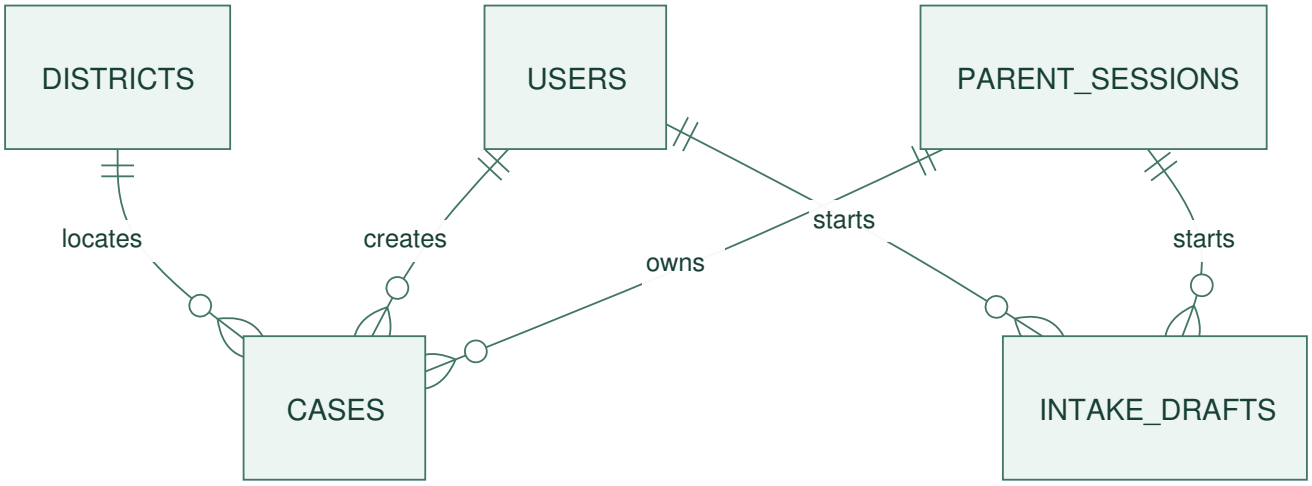
8.1 Link types

From	To	Meaning	Where recorded
GOAL	STORY	Story contributes to goal	D1 §19
STORY	FR	Requirement implements story	D1 §11 "Story" column
FR	SCREEN	Requirement visible on screen	D1 §10 stories, D3 §2
FR / BR / NFR	TC	Test verifies requirement	D1 §11, §13.1; D6 §8
BR	FR	Rule enforced by requirement	D1 §13.1
FR	API	Endpoint serves requirement	D2 §5
API	COMP	Component handles endpoint	D2 §4
COMP	ENTITY	Component reads/writes table	D2 §4, §6
SCREEN	COMP (UI)	Screen uses UI component	D3 §2, D4 §8
SCREEN	ACT / D / E	Flow actions, decisions, errors	D3 §7, §9

From	To	Meaning	Where recorded
TASK	FR / COMP / TC	Task builds and proves	D6 §5
RISK	TASK / OQ	Mitigation and open question	D1 §16, D6, D7
A	OQ	Assumption awaiting answer	D1 §14, §17

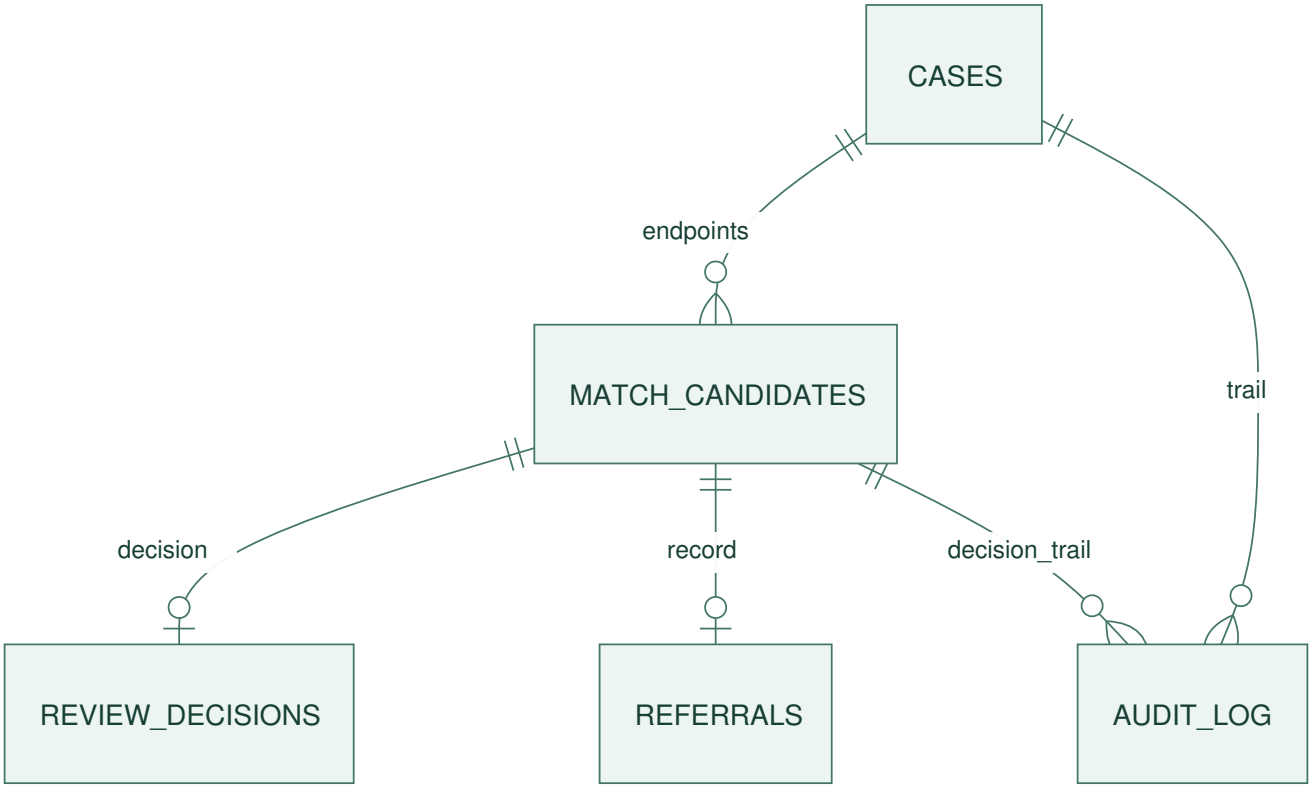
8.2 Domain ER diagram

Intake and ownership relationships



V2 specification diagram · see adjacent rules for authority and evidence limits

Candidate decision and audit relationships



V2 specification diagram · see adjacent rules for authority and evidence limits

Additional relationship	Cardinality / rule
District → district_neighbours	One-to-many outgoing rows; no self-neighbour
Case → case_versions / generated_outputs	One-to-many; versioned snapshots/templates, cleared on redaction
Case ↔ duplicate_links	Each link references two ordered same-kind endpoints; reports retained
User → review_decisions / referrals	One-to-many authors; decision author must be active verified officer at decision time
User / parent_session → audit_log	Optional actor references; safe audit remains after payload redaction
Intake draft → ai_reservations / ai_calls	One-to-many metadata and usage links; no child payload in draft metadata
Reservation → ai_calls	At most one usage row per reservation
Case → ai_calls	Optional case link after submission
ai_budget singleton	Global reserved/committed cap enforced transactionally; no per-child identity

8.3 Requirement traceability (essential journey)

Goal	Story	Requirement / business rule	Screen	Acceptance
GOAL-001 reviewed report	STORY-001 voice	**FR-002**	**SCREEN-002**	**TC-001**
GOAL-001 reviewed report	STORY-004 review	FR-006; BR-004 review before register sharing	**SCREEN-003**	**TC-004**
GOAL-003 shortlist quality	STORY-009 candidates	FR-012 qualification/abstention	**SCREEN-009**	TC-009, TC-041
GOAL-008 human decision	STORY-010 confirm	FR-016; BR-001 human review; BR-005 referral recorded	**SCREEN-010**	TC-010, TC-046/047
GOAL-006 report status	STORY-013 district picture	FR-019; BR-014 counting basis	**SCREEN-011**	TC-012, TC-055

9 Glossary

Bangla	English	Note
ফেরা	Fera ("return")	Product name
নিখোঁজ	Missing	Case kind
পাওয়া গেছে	Found	Case kind
সাধারণ ডায়েরি (জিডি)	General Diary (GD)	Police record
থানা	Thana (police station area)	Optional field
জেলা	District	64 districts
শেষ দেখা	Last seen	Event time/place
খসড়া	Draft	Label on every output
পোস্টার	Poster	Text-only output
মিল	Match	Candidate pair
সম্ভাব্য মিল	Likely match	Queue label
মিলের মাত্রা	Similarity score	0–100, not a probability
মিল নিশ্চিত করুন	Confirm match	Officer action
মিল নয়	Not a match	Officer action
সম্ভাব্য একই শিশু	Possible same child	Duplicate banner
রেফারেল	Referral	After confirm

Bangla	English	Note
শিশু সুরক্ষা সেবা	Child-protection services	Default referral agency text (OQ-018)
কর্মকর্তা	Officer	Role
স্বেচ্ছাসেবক	Volunteer	Staff persona
আশ্রয়কেন্দ্র	Shelter	Disaster mode
অজানা	Unknown	Empty field
সিনথেটিক	Synthetic	Seed data label
ক্যাশ	Cached	Demo-safe output label
পরিচয় তথ্য অপসারণ	Redact identifying payloads	Admin action
খোলা / যাচাই চলছে / বন্ধ	Open / Being checked / Closed	Parent status labels
পরিকল্পিত সম্প্রসারণ	Planned extension	Disaster mode banner
দুর্যোগ মোড	Disaster mode	Could

10 V2 integrity and evidence

The companion code is generated once and embedded exactly. Registry counts and uniqueness are mechanically checked; definitions and references must be checked against the delivered source. Requirement-to-test links describe planned acceptance, not achieved application results. The verification folder distinguishes executed companion checks from pending application/PG16/host/partner evidence.

Prefix	Count
A	40
ACT	33
API	23
BR	26
COMP	50
D	16
DECISION	17
DEP	11
E	24
ENTITY	17
FR	37

Prefix	Count
GOAL	9
NFR	17
OQ	26
PERSONA	5
RISK	17
SCREEN	16
STORY	23
TASK	64
TC	66
Total	537

Authoritative locations: GOAL/PERSONA/OQ baseline in 00/D1; business requirements in D1; API/ENTITY in D2; SCREEN/ACT in D3; UI COMP in D4; shared interfaces and index here; TC/TASK execution specifications in D6. TASK-036 attribution is corrected. Reserved IDs are reservations, not duplicate definitions. The V2 delta adds API-021–023, ENTITY-015–017, TC-038–066 and TASK-061–064. Existing IDs remain stable.

IDs introduced

- None. D5 is a registry only.

IDs referenced

- All IDs listed in §7 (GOAL-001 – GOAL-009, PERSONA-001 – PERSONA-005, STORY-001 – STORY-023, FR-001 – FR-037, NFR-001 – NFR-017, BR-001 – BR-026, A-001 – A-040, DEP-001 – DEP-011, RISK-001 – RISK-017, OQ-001 – OQ-026, SCREEN-001 – SCREEN-016, COMP-001 – COMP-050, API-001 – API-020, ENTITY-001 – ENTITY-014, DECISION-001 – DECISION-017, E-001 – E-024, TC-001 – TC-037, ACT-001 – ACT-033, D-001 – D-016, TASK-001 – TASK-060)

↗ Return to contents

D6 / DELIVERY / V2.0

Implementation Plan

Task dependencies, acceptance tests and release gates · Authors: Sadman Shaharier and Obidit Islam

↗ Contents

Field	Value
Version	2.0
Date	11 Oct 2026 (Asia/Dhaka, UTC+6)
Owner	Sadman Shaharier
Authors	Sadman Shaharier; Obidit Islam
Status	V2 team handover; implementation status UNVERIFIED
Inputs	D1–D5 V2.0; validation report; verified progress and availability still pending.
Key dates	Round 2 submission 14 Oct 2026; exact cutoff/channel pending official email. Mentor dates require team confirmation. Finale 24 Oct 2026 at AIUB.

V2 release contract

Authors: Sadman Shaharier and Obidit Islam. **Version:** 2.0. **Issued:** 11 October 2026, Asia/Dhaka (UTC+6).

This is the revised team specification, not evidence that the application has been built. Actual repository progress, account configuration, interviews, evaluations, and spend remain **UNVERIFIED** until evidence is attached. V1 defaults are superseded by the V2 rules below; existing requirement IDs are retained.

- This release is a **synthetic-data prototype**. Officer verification and referral delivery are simulated and labelled. The schema rejects records marked non-synthetic; the application must also prevent real reports from entering this release.
- A candidate score is similarity, not identity probability. A reviewer decision does not establish guardian authority or safe reunification. A referral row means **referral recorded**, not delivered or acknowledged.
- Draft payload stays on the device before report confirmation, but providers process audio/text beforehand with a notice. Only expiry/counter/hash metadata and cost reservations are stored server-side before submission.
- Possible duplicates remain visible. Shared phone alone never establishes one child. No physical child-return action is performed by Fera.
- Counts distinguish missing-report rows, found-report rows, and confirmed pairs. They are not a census of unique children or proof of rescue.

- Sources, diagrams, tables, code blocks, and acceptance criteria are part of the handover. The companion code must match the embedded code exactly. Team handover records change closure, evidence gaps, and release gates.

Changes to earlier documents

V2 rebases gates from actual remaining work, corrects workload/dependencies, adds TASK-061–064 and TC-038–066, and removes unsupported delivery guarantees. The task estimates and resource totals are computed in the handover.

1 MVP definition

Smallest complete demoable product: one parent, one station volunteer and one officer move one synthetic missing report through a reviewed candidate decision and coordination closure on the map, in Bangla, with a human deciding the match.

The one essential journey

#	Step	Screen	FR
1	Parent records a Bangla voice note	**SCREEN-002**	**FR-002**
2	Fera builds a case draft and asks for the district	**SCREEN-002**	FR-003 – FR-005
3	Parent reviews and submits	**SCREEN-003**	FR-006, FR-007
4	Parent gets the GD draft and poster	**SCREEN-004**	FR-008, FR-009
5	Station volunteer logs a found boy	**SCREEN-007**	**FR-011**
6	Matching surfaces a qualified candidate in the scripted scenario	**SCREEN-009**	FR-012 – FR-014
7	Officer compares evidence and confirms; simulated referral record created	**SCREEN-010**	FR-015, FR-016
8	Officer closes the case; the Dhaka marker updates	SCREEN-012, SCREEN-011	FR-018, FR-019

Included (Must): FR-001 – FR-009, FR-011 – FR-022, FR-025 – FR-028, FR-031 – FR-035, FR-037.

Excluded from MVP: FR-010 13219 summary (Should), FR-024 import (Should), FR-029 edit after submit (Should), FR-023 duplicate screen, FR-030 parent status (Should), FR-036 disaster mode (Could). Alerts, face matching, telecom targeting and live police data are Won't; shown only as [MOCK] on slides.

2 Goals

Type	Goal	Measure	Due
Product	Essential journey works on the deployed app	TC-001 – TC-012 pass	14 Oct 18:00
Product	Safeguarding rules enforced	TC-010, TC-019, TC-022 – TC-027 pass	14 Oct 18:00
Product	Report time beats manual form	Median ≤ 5 min and below manual (GOAL-001)	13 Oct
Technical	Extraction accuracy measured	≥ 85 % target; actual reported (GOAL-002)	13 Oct
Technical	Matching measured	Recall@5 ≥ 80 %, P@1 ≥ 60 % targets; actual reported (GOAL-003)	13 Oct
Technical	Dedupe and WER measured	F1 ≥ 0.8; WER ≤ 30 % targets; actual reported (GOAL-004, GOAL-005)	13 Oct
Technical	AI spend	≤ USD 25 at freeze, ≤ USD 30 at finale (NFR-016)	14 Oct, 24 Oct
Demo	Round 2 video	1–2 min, shows steps 1–8 of §1	14 Oct 20:00
Demo	Finale reliability	5 of 5 rehearsals pass, ≥ 1 with internet off (GOAL-007)	23 Oct
Demo	Honest metrics slide	All 5 metrics + ≥ 2 failures (GOAL-009)	14 Oct

Working times in the preceding table are internal planning targets, not verified organiser requirements or delivery promises. Confirm the official cutoff and submission channel before setting the final schedule.

3 Non-goals

Category	Not doing
Features	Alerts of any kind; face matching; telecom location targeting; live police/CID data; SMS/IVR; native app; parent cross-device access; dashboard filters
Technical work deferred	Phone OTP for parents; migrations framework; background job queue; multi-region hosting; automated retention scheduler; LLM re-ranking; real officer verification
Production concerns	Load beyond 50 users; 24/7 uptime; backups beyond the managed host's defaults; formal legal review; security audit by a third party; data-sharing agreements

4 Rebased gates and capacity

The previous 9–14 October empty-repository assumption is historical. At issue time (11 October, Asia/Dhaka), actual completion and team availability are UNVERIFIED. Do not mark work done merely because a specification exists.

Gate	Target working window	Exit evidence
A Contract/inventory	11 Oct	Repo/deploy and task status; V2 fields/states/permissions approved; provider/audio/embedding evidence; official cutoff confirmed
B Complete typed journey	11–12 Oct	One synthetic case on one DB: manual/review/submit/output/found/rank/decision/referral record/close/counts/audit
C Quality and evaluation	12–13 Oct	Target PG16/role and concurrent tests; frozen manifest; live cache-disabled metrics and failures; device/a11y checks
D Record and submit	13–14 Oct	Tested script/topology, honest labels/timings/spend, export/upload/link checks, organiser submission evidence
E Finale hardening	15–24 Oct	Same-backend local rehearsal, five recorded runs including internet-off, backup recording, final test/limitations

If starting from zero on 11 October, four full days at 10 hours/person/day yield about 120 team-hours. Actual remaining work is the sum of unfinished estimates, not the full historic workload. The V2 added tasks below and dependency path are calculated in the handover. Estimates are planning inputs, not delivery promises. Team lead must reassign work/cut optional scope or report missing features if remaining work exceeds available hours. Preserve correctness, honest evaluation and submission buffer.

5 Task board

Owners: **S** = Sadman (lead/integration), **O** = Obidit (intake & LLM), **M** = Masnun (data, matching & dashboard). CP labels inherited from V1 are removed; computed V2 dependency path is in the handover. Owner capacity can change the actual schedule.

ID	Task	Priority	Owner	Dependencies	Hours	Completion evidence	Plan note
TASK-001	Repo: Next.js App Router, TS strict, Tailwind, ESLint, CI (lint, typecheck, test)	Must	S	—	3	CI green on `main`; preview deploy URL	
TASK-002	Pick DB host (OQ-021); apply `schema_v2.sql`; local Docker DB	Must	S	TASK-001	2	HNSW index created on host and locally	
TASK-003	Provider keys, `.env.example`, record real prices in `docs/costs.md` (OQ-004, OQ-022)	Must	O	—	1	Both LLMs and STT answer a test call	
TASK-004	Spike: local embedding model on Vercel (OQ-023, RISK-015)	Must	M	TASK-001	3	Report: cold start, size; decision local vs API	
TASK-005	Spike: Android MediaRecorder → STT (A-030)	Must	O	TASK-003	2	3 phone recordings transcribed	
TASK-006	Auth.js credentials, parent cookie, `requireRole`, seeded users	Must	S	TASK-002	5	TC-023 passes	
TASK-007	API layer: error envelope E-001 – E-024, Zod helpers, request ID, rate limiter (COMP-021)	Must	S	TASK-001	4	Unit tests for envelope and limiter	
TASK-008	Layout shell, `tokens_v2.css`, Noto Sans Bengali, language toggle (COMP-022)	Must	S	TASK-001	3	Bangla renders on 1 Android phone	
TASK-009	District data: 64 rows bn/en/centroids + neighbours (DEP-004, A-013)	Must	M	TASK-002	3	`npm run seed` loads 64 districts	
TASK-010	Synthetic set generator: ~200 records, ground truth, duplicates, decoys (A-039)	Must	M	TASK-009	6	Ground-truth file hashed and committed	
TASK-011	50 hand-labelled reports	Must	M	—	5	`eval/data/labelled.jsonl` reviewed by O	

ID	Task	Priority	Owner	Dependencies	Hours	Completion evidence	Plan note
TASK-012	Normaliser: Bangla digits, time words, phone, district aliases, romaniser (COMP-008)	Must	O	TASK-009	5	30 unit tests pass	
TASK-013	LLM adapter: 2 providers, Zod contract, evidence check, retry, fallback (COMP-007)	Must	O	TASK-003, TASK-012	7	Extracts 10 sample texts; fallback test passes	
TASK-014	Speech adapter + API-001 (COMP-006)	Must	O	TASK-005, TASK-007	3	TC-001 steps 1–3 pass	
TASK-015	Case service, state guard, audit, versions (COMP-014, COMP-015)	Must	S	TASK-007, TASK-061, TASK-062	6	TC-032 and TC-016 pass	
TASK-016	Embedding service + `npm run reembed` (COMP-011)	Must	M	TASK-004	3	200 records embedded	
TASK-017	Matching engine: filters, cosine, bonuses, template reasons (COMP-012)	Must	M	TASK-015, TASK-016	7	TC-009 passes; first recall@5 printed	
TASK-018	Duplicate detector (COMP-013)	Must	M	TASK-016	3	TC-013 passes	
TASK-019	Seed loader with synthetic labels and 3 demo cases	Must	M	TASK-010	2	`npm run seed -- --set demo` < 2 min	
TASK-020	Intake orchestrator + API-002 + question templates (COMP-009)	Must	O	TASK-013	4	TC-003 passes	
TASK-021	SCREEN-001/002 intake chat + recorder (COMP-024 – COMP-027)	Must	O	TASK-008, TASK-014, TASK-020	7	Voice and text paths work on phone	
TASK-022	SCREEN-003 review + API-003 + draft persistence	Must	O	TASK-015, TASK-021	5	TC-004, TC-031 pass	

ID	Task	Priority	Owner	Dependencies	Hours	Completion evidence	Plan note
TASK-023	Output templates GD/poster + wording guard + API-006 (COMP-010, COMP-016)	Must	O	**TASK-015**	4	TC-005, TC-024 pass	
TASK-024	SCREEN-004 outputs + poster PNG (COMP-030, COMP-031)	Must	S	**TASK-023**	3	TC-006 passes	
TASK-025	SCREEN-005 manual form fallback (COMP-032)	Must	S	**TASK-022**	2	TC-017 passes	
TASK-026	SCREEN-007 found-child log + API-007 (COMP-034)	Must	M	TASK-015, TASK-020	3	TC-008 passes	
TASK-027	SCREEN-009 queue + API-008 (COMP-035, COMP-036)	Must	M	**TASK-017**	3	Queue lists seeded candidates	
TASK-028	SCREEN-010 match review + API-009/010 (COMP-037, COMP-038)	Must	M	TASK-017, TASK-027	6	TC-010 (confirm part), TC-011 pass	
TASK-029	SCREEN-012 case detail, close, audit + API-011/013 (COMP-039, COMP-042)	Must	S	**TASK-028**	5	TC-010 full passes	
TASK-030	SCREEN-011 map + table + API-012 (COMP-040, COMP-041)	Must	M	TASK-009, TASK-029	5	TC-012 passes	
TASK-031	SCREEN-008 sign-in + SCREEN-016 (COMP-047, COMP-045)	Must	S	**TASK-006**	2	TC-022 passes	
TASK-032	SCREEN-013 admin: redact identifying payloads + verify [MOCK] + API-016/017 (COMP-043)	Must	S	**TASK-015**	3	TC-015 passes	
TASK-033	Demo-safe cache + fixtures for 3 cases + cached badge (COMP-017, COMP-048)	Must	O	TASK-020, TASK-019	3	TC-021 passes on laptop	

ID	Task	Priority	Owner	Dependencies	Hours	Completion evidence	Plan note
TASK-034	Accessibility pass: axe + keyboard + 320 CSS px reflow, 200 % text zoom and 400 % viewport zoom	Must	S	TASK-030	3	TC-033 passes	
TASK-035	Bangla copy review; all empty/error/loading states	Must	O	TASK-030	3	Every E-xxx shows bn + en text	
TASK-036	Test on 2 Android phones (Bangla rendering, mic, 360 px)	Must	M	TASK-030	2	Checklist signed in `docs/device-test.md`	
TASK-037	Unit tests: normaliser, matcher, state guard, wording guard	Must	S	TASK-017	4	≥ 60 tests, CI green	
TASK-038	Playwright E2E: essential journey + safeguarding TCs	Must	S	TASK-030, TASK-032, TASK-059	5	TC-001–066 app/service checks where applicable; companion-only results kept separate	
TASK-039	Evaluation harness: smoke/dev run first; held-out run after manifest/model freeze	Must	M	TASK-011, TASK-017, TASK-018, TASK-064	5	TC-020 passes; report committed	
TASK-040	WER recordings (60 clips) + run	Must	O	TASK-014	3	WER in report	
TASK-041	Timing study, ≥ 5 testers, assistant vs manual form	Must	S	TASK-025	3	`eval/data/timing.csv`; median in report	
TASK-042	Production deploy + managed DB seed + smoke test	Must	S	TASK-038	2	`npm run smoke` passes on prod	

ID	Task	Priority	Owner	Dependencies	Hours	Completion evidence	Plan note
TASK-043	Final evaluation run on freeze build	Must	M	TASK-042, TASK-039, TASK-040, TASK-041, TASK-060	2	Frozen cache-disabled report with all metrics, denominators, failures, commit/model/manifest hashes	
TASK-044	Round 2 deck, 5–8 slides	Must	S	**TASK-043**	4	Working 5–8-slide deck; exact organiser deliverables confirmed; honest measured/ pending labels	
TASK-045	Round 2 demo video, 1–2 min	Must	O	**TASK-042**	4	Working 1–2-minute MP4; confirm official format; compressed/cached timings labelled	
TASK-046	README, Drive folder (links TXT, deck, video), reply to selection email	Must	S	TASK-044, TASK-045	1	Receipt via organiser-confirmed channel before confirmed cutoff; links accessible	
TASK-047	13219 summary (FR-010)	Should	O	**TASK-023**	1	TC-007 passes	
TASK-048	Synthetic fixture import + API-015 (FR-024)	Should	M	**TASK-020**	3	TC-014 passes	
TASK-049	Edit after submit + API-005 + SCREEN-006 (FR-029)	Should	O	**TASK-022**	4	TC-018 passes	
TASK-050	Parent status API-018 (FR-030)	Should	O	**TASK-049**	1	TC-019 passes	
TASK-051	Duplicate review SCREEN-014 (FR-023)	Could	M	**TASK-018**	3	Mark same/different works	
TASK-052	Disaster mode SCREEN-015 (FR-036)	Could	M	**TASK-026**	2	TC-035 passes	

ID	Task	Priority	Owner	Dependencies	Hours	Completion evidence	Plan note
TASK-053	Finale local stack: Docker DB, model cache, fixtures, `DEMO_MODE=1`	Must	S	**TASK-033**	4	Laptop runs journey offline	
TASK-054	5 rehearsals, ≥ 1 internet-off	Must	S, O, M	**TASK-053**	5	Rehearsal log 5/5	
TASK-055	Finale deck, 5–10 min	Must	S	**TASK-043**	4	Timed run ≤ 10 min	
TASK-056	Fixes from mentor session 2 (12–13 Oct)	Should	S, O, M	—	5	Agreed fixes merged by 14 Oct 18:00	
TASK-057	5–10 user conversations, only if OQ-019 allows	Should	S	—	4	Notes with consent; findings slide	
TASK-058	Backup recording of the finale demo	Must	O	**TASK-054**	2	MP4 on laptop and Drive	
TASK-059	`npm run spend` + spend cap check (DECISION-017)	Must	O	TASK-013, TASK-063	1	TC-036 passes	
TASK-060	Performance check against D2 Part B §12.1	Must	M	**TASK-042**	2	TC-034 results in report	
TASK-061	Apply V2 required fields, DTOs, token/ idempotency and referral/count semantics	Must	S	**TASK-001**	3	Runtime schemas/examples and role projections agree; FV-01–05/10 closed	
TASK-062	V2 schema invariants, restricted roles and full related-payload redaction	Must	S	TASK-002, TASK-061	4	Negative and positive lifecycle/redaction checks; target PG16 CI pending until run	
TASK-063	Atomic paid-call counters/reservations, replay handling and price sheet	Must	O	TASK-003, TASK-062	2	Draft replay/concurrency/ budget checks; manual path survives paid refusal	

ID	Task	Priority	Owner	Dependencies	Hours	Completion evidence	Plan note
TASK-064	Identity-grouped manifest, no-match/ duplicate hard negatives, frozen metrics	Must	M	TASK-010	3	Manifest counts/split/hash asserted before held-out evaluation	

Critical path: The V1 printed chain was invalid (67 hours, not ~59; four undeclared links). V2 dependency/resource summary is generated from the task table in TEAM_HANDOVER_V2. A dependency longest path is a lower bound, not a resource-feasible finish date.

6 Folder structure

```
fera/
├─ app/                          # Next.js App Router
│  ├─ (parent)/                  # SCREEN-001–006, no auth
│  │  ├─ page.tsx                # SCREEN-001 welcome
│  │  ├─ report/page.tsx         # SCREEN-002 intake chat
│  │  ├─ review/page.tsx         # SCREEN-003
│  │  ├─ case/[ref]/page.tsx     # SCREEN-004 / SCREEN-006
│  │  └─ form/page.tsx           # SCREEN-005 manual form
│  ├─ staff/                     # SCREEN-007, SCREEN-015
│  ├─ officer/                   # SCREEN-009–012, SCREEN-014
│  ├─ admin/                     # SCREEN-013
│  ├─ signin/page.tsx            # SCREEN-008
│  ├─ not-found.tsx              # SCREEN-016
│  └─ api/                       # API-001 – API-020 route handlers
├─ components/                   # COMP-022 – COMP-050 (UI)
├─ lib/
│  ├─ ai/                        # COMP-006 speech, COMP-007 llm, COMP-011 embeddings, providers/
│  ├─ intake/                    # COMP-008 normaliser, COMP-009 orchestrator, question templates
│  ├─ outputs/                   # COMP-010 templates, COMP-016 wording guard
│  ├─ cases/                     # COMP-014 case service, COMP-015 audit
│  ├─ matching/                  # COMP-012 matcher, COMP-013 dedupe
│  ├─ auth/                      # COMP-005
```

```
│   ├─ http/                    # COMP-004 envelope, COMP-021 rate limiter, errors E-001–E-024
│   └─ db.ts                    # pg pool
├─ fixtures/demo/               # COMP-017 cached transcripts/extractions/embeddings
├─ scripts/                     # seed, reembed, model:fetch, spend, smoke
├─ eval/                        # COMP-020 metric scripts, data/, report-YYYY-MM-DD.md
├─ tests/                       # vitest unit + playwright e2e
├─ styles/tokens.css            # D4 §16
├─ schema.sql                   # D2 §7
├─ docs/                        # D1–D7, costs.md, device-test.md
├─ docker-compose.yml           # local Postgres + pgvector
└─ .env.example
```

7 Testing strategy

Level	Tool	Scope	When
Unit	Vitest	Normaliser, evidence check, matcher scoring, dedupe rule, state guard, wording guard, templates	Every push (CI)
Integration	Vitest + local Postgres	API handlers against a seeded DB; trigger and constraint behaviour	Every push
End-to-end	Playwright (mobile 360 px + desktop 1280 px)	Essential journey; safeguarding TCs; error flows	Nightly + before freeze
Accessibility	@axe-core/playwright + manual keyboard and screen-reader spot check	All Must screens, light and dark	13 Oct, 23 Oct
Manual demo checks	Checklist on 2 Android phones + laptop	Bangla rendering, mic, copy, PNG, map	13 Oct, each rehearsal
Failure-state tests	Playwright with network throttling/offline; provider keys removed	E-010 – E-014, E-022 – E-024, fallback form, demo cache	13 Oct
AI evaluation	`/eval` scripts	matching/duplicate/extraction/speech metrics, abstention/coverage, timing and audit coverage	11 Oct (first), 13 Oct, 14 Oct (final), 23 Oct

8 Test cases

TC	Covers	Preconditions	Steps	Expected result	Pass if
TC-001	STORY-001, FR-002	Phone, mic allowed, preview deploy	1 Open SCREEN-002. 2 Record 38 s scripted Rafi note. 3 Stop.	Editable Bangla transcript appears	Shown ≤ 20 s; contains "রাফি" and "১৩" or "13"
TC-002	STORY-002, FR-001, FR-003, FR-004, FR-033	—	Submit "আমার ছেলে রাফি, বয়স ১৩, গতকাল বিকেলে মাদ্রাসা থেকে ফেরেনি। নম্বর ০১৭১১২২৩৩৪৪"	Draft: age 13, sex male, explicit fixture-clock yesterday afternoon interval with 16:00 display point, phone 01711223344; district empty	Required fields and supported time bounds correct; unknowns explicit; draft ≤ 15 s; UI in Bangla
TC-003	STORY-003, FR-005	TC-002 draft	Read question; answer "মিরপুর, ঢাকা"	One question about district; then draft complete	Exactly 1 question per missing field; district code = Dhaka
TC-004	STORY-004, FR-006, FR-007, BR-004	Complete draft	1 Query DB and officer queue before submit. 2 Edit clothing. 3 Tick check. 4 Submit.	Before submit: 0 child case rows; metadata allowed. After: case open with edited clothing	0 child case rows pre-submit; metadata/ cost rows allowed; stored = edited values
TC-005	STORY-005, FR-008, BR-022	Submitted case	Tap "জিডি খসড়া"	GD text with name, age, place, time, phone; "খসড়া" label	All 5 fields present; no token absent from the case fields except template text
TC-006	STORY-006, FR-009, BR-006	Submitted case	Tap "পোস্টার"; download PNG	Text + PNG with 13219 and phone; no photo	PNG renders Bangla; no image input exists
TC-007	STORY-007, FR-010	Submitted case	Tap "১৩২১৯ সারাংশ"	Bangla summary	≤ 80 words
TC-008	STORY-008, FR-011	Staff signed in	Log Kamalapur found boy (demo data)	Ref F-... shown; "মিল খোঁজা হয়েছে"	Case open or matched_pending_review ≤ 15 s; no candidate details shown to staff
TC-009	STORY-009, FR-012 – FR-014, BR-017, BR-018	Seeded set + TC-008	Officer opens queue	Rafi case with ≤ 5 candidates, scores, reasons	Scripted fixture true pair rank recorded; held-out quality reported separately; no candidate violates eligibility (SQL check)
TC-010	STORY-010, STORY-012, FR-015, FR-016, FR-018, BR-001, BR-005	**TC-009**	1 Open candidate. 2 Confirm (2 steps). 3 Close with "সমন্বয়ের সিদ্ধান্ত ও রেফারেল নথিভুক্ত(সিমুলেশন)".	Missing confirmed → closed; found referred; 1 referral; audit rows	All states correct; referral exists; 3 audit entries

TC	Covers	Preconditions	Steps	Expected result	Pass if
TC-011	STORY-011, FR-017, BR-019	Candidate pending	Reject with reason "পোশাক আলাদা"; re-run API-014	Candidate rejected; case open; pair absent after re-run	Pair not re-suggested
TC-012	STORY-013, FR-019, BR-014	Seeded + TC-010	Open map and table	Counts per district	Every district count = SQL count by group
TC-013	STORY-014, FR-022, FR-023	Mim parent case + public duplicate	Open either case	Banner "সম্ভাব্য একই শিশু"; both flagged reports remain independently visible	Link exists; distinct-child shared-phone controls stay visible
TC-014	STORY-015, FR-024	Admin	Import 5 invented synthetic sample texts with example.invalid URLs, one containing "ignore previous instructions"	5 synthetic cases tagged simulated public_import with invented URLs	Injected text yields only schema fields; no server fetch of URLs
TC-015	STORY-016, FR-025, BR-009	3 closed/referred, 1 open	Run redact identifying payloads all eligible	eligible payloads redacted, active linked evidence skipped	direct and related identifying payloads cleared; pseudonymous counts unchanged
TC-016	STORY-017, FR-021, BR-010, NFR-017	TC-010 done	1 Read audit. 2 Try SQL UPDATE/ DELETE on audit_log	Entries for confirm, referral, close	All present; UPDATE/DELETE/TRUNCATE raise error; runtime cannot alter guards
TC-017	STORY-018, FR-026	LLM keys invalidated	Submit text	Manual form opens pre-filled with any parsed fields	Form ≤ 2 s after error; text kept
TC-018	STORY-019, FR-029, BR-025	Case open with pending candidate	Edit age; confirm	Version 2; candidates voided; re-match run	Version row exists; confirmed case returns E-018
TC-019	STORY-020, FR-030, BR-007	Parent case matched	Parent opens status and own scoped edit read	"যাচাই চলছে" plus own fields if editable	Response JSON has no found-case or candidate fields
TC-020	STORY-022, FR-028, GOAL-002 – GOAL-005, GOAL-009	Labelled sets	`npm run eval:all`	Report with 5 metrics, sizes, SHA, ≥ 2 failures	All sections present

TC	Covers	Preconditions	Steps	Expected result	Pass if
TC-021	STORY-023, FR-027, FR-032, NFR-007	`DEMO_MODE=1`, laptop offline	Run Rafi journey with scripted inputs	Journey completes; "ক্যাশ" badges; "সিনথেটিক" labels	Completes with network off
TC-022	BR-007, FR-020	Parent and staff sessions	Request API-009 and API-004 for another case	404 body	E-004 returned; `access.denied` audit row
TC-023	BR-001, FR-035, FR-037	Staff and unverified officer	POST API-010 confirm	Refused	E-003/E-004; candidate still pending
TC-024	BR-002, BR-003, FR-034	Template with "পালিয়ে" injected in test	Generate outputs	Blocked	E-016; text not shown
TC-025	BR-002	Submitted case	Inspect all outputs and UI	Draft labels; no "send alert" control	Every output labelled "খসড়া"; 0 alert actions in UI
TC-026	BR-006	—	Search UI and API schema for image inputs	None	0 file inputs except audio
TC-027	BR-008	—	Compare `cases` columns and forms with D1 BR-015 + optional list	Match	No extra personal fields
TC-028	NFR-010	Full E2E run	Regex scan logs for `01[0-9]{9}` and seeded names	None found	0 matches
TC-029	BR-020	Found case with 2 pending candidates	Confirm one	Other superseded; second confirm on same found fails	Unique index blocks second confirm
TC-030	BR-015, BR-016	—	Submit age 19, future time, phone 0171, district "XYZ"	4 field errors	E-001 with 4 paths
TC-031	FR-031	Submitted case	Concurrently replay the same submission UUID and canonical payload	Existing ref shown	200 stable retry reference; 1 row; changed payload conflicts

TC	Covers	Preconditions	Steps	Expected result	Pass if
TC-032	BR-012, BR-013	Seeded cases	SQL: open→confirmed; found→closed	Rejected by trigger	Both raise `check_violation`
TC-033	NFR-011	Prod build	axe on Must screens, light and dark; keyboard walk	No serious/critical	0 serious/critical; all actions keyboard-reachable
TC-034	NFR-001 – NFR-005	Prod build	Lighthouse mobile; `eval/perf.ts`	Values recorded	Each target met or reported as missed
TC-035	STORY-021, FR-036	Staff	Open disaster screen; log child	Banner "পরিকল্পিত সম্প্রসারণ"	Saved as found case
TC-036	NFR-016, DECISION-017	Spend cap set to current total	Call API-002	Refused	E-023; manual form opens
TC-037	BR-011	Deck, UI, README	Search for compliance claims	None	0 unsupported compliance claims; cited legal facts have primary sources

8.1 V2 additional acceptance cases

TC	Case	Covers	Pass condition
TC-038	Shared-phone distinct children	BR-021, FV-02	Two siblings/shelter reports remain searchable; phone-only link cannot confirm_same
TC-039	Unknown found age	BR-017, FV-03	Eligible with unknown cue; no fabricated age agreement
TC-040	Overlapping time uncertainty	BR-016/017	Interval-compatible found report is not removed by midpoint
TC-041	No plausible candidate	BR-018	Qualified shortlist may be empty; no misleading confidence
TC-042	Concurrent submission UUID	FR-031, FV-05	One case; stable response; different payload conflicts
TC-043	Draft replay/concurrency	BR-023, FV-05	Server usage cannot reset; six dispatched calls maximum
TC-044	Admin rematch projection	BR-007/024	Status only, no candidate IDs/scores/child payload

TC	Case	Covers	Pass condition
TC-045	Owner editable read	API-004, SCREEN-006	Own fields before lock; no other-party payload
TC-046	Stale candidate decision	**API-010**	Expected-version conflict, no new decision/referral
TC-047	Wrong referral endpoints	FV-06	Constraint rejects mismatched candidate pair
TC-048	Reject without reason	**BR-001**	Service and DB reject NULL/short reason
TC-049	Terminal report edit	**BR-025**	Blocked; privileged redaction separate; no silent history rewrite
TC-050	Related note redaction	**BR-009**	Outcome/referral/decision/candidate/provenance/output payloads cleared; IDs remain pseudonymous
TC-051	Audit truncate/privilege	**BR-010**	UPDATE/DELETE/TRUNCATE rejected; runtime no DDL/ownership
TC-052	Unknown-resource denial	**API-004**	Safe audit recorded with null FK and no information leak
TC-053	Last competitor invalidated	**BR-020**	Other affected report returns to open if no pending candidate
TC-054	Matching fails after submit	API-003/014	Saved report, explicit failed/queued state, reliable retry
TC-055	One pair dashboard lifecycle	**BR-014**	Missing/found categories change separately; pair count one
TC-056	Evaluation fixture contamination	**GOAL-009**	Quality run rejects/excludes cache/fixture hits and labels result
TC-057	Single-DB offline scenario	FV-08	All local roles see same new report; assets preloaded
TC-058	Reflow/keyboard/Bangla	**NFR-011**	320px/400%/200% checks; shortcuts disabled during typing and switchable
TC-059	Embedding space switch	FV-09/15	No mixed vectors; matching paused until validated rebuild
TC-060	Provider notice/metadata	BR-004, FV-12	Notice before dispatch; no child payload in pre-submit DB metadata
TC-061	Encoded audio limits	**BR-023**	Real phone codec/duration/size tested; WAV >2 MB rejected or compressed
TC-062	PostgreSQL16/runtime host	D2B acceptance	Original V2 SQL applies and meaningful constraints tested on target PG16/roles; portable engine result alone not enough
TC-063	Correction/reopen pilot gate	**BR-025**	Documented two distinct verified reviewers, reason/evidence/history; real data blocked until implemented/approved
TC-064	Manifest/group split	FV-11	200 count, identities/no-match/duplicates declared; no connected-group leakage

TC	Case	Covers	Pass condition
TC-065	Concurrent AI budget	FV-12/13	Reservations cannot exceed cap; uncertain usage retained; duplicate settlement idempotent
TC-066	Legal claims/source accuracy	BR-011	No unsupported compliance or integration claim; accurate primary-law citations allowed

9 Demo plan

9.1 Setup

Item	Detail
Devices	Online: all roles on deployed origin/one managed DB. Local: all roles on laptop stack/one local DB, in separate browser profiles; phone only with tested secure LAN access. Spare laptop preloaded.
Build	Round 2: production deploy. Finale: production deploy + laptop-local copy with `DEMO_MODE=1`
Accounts	`officer.karim` (verified), `staff.rumana`, `admin` — passwords in the team vault
Reset	`npm run seed -- --reset --set demo` 10 min before each run
Checks	`/api/health` green; `npm run spend` < cap; phone font check

9.2 Demo data (synthetic, labelled "সিনথেটিক")

Name	Record	Role in demo
রাফি (Rafi)	Missing boy, 13, blue shirt, last seen 8 Oct afternoon interval, display reference 16:00; fixture intake clock explicitly 9 Oct near madrasa, Mirpur, Dhaka	Live parent voice report
Kamalapur boy	Found boy, est. 12–14, blue shirt, scar on left hand, Kamalapur railway station, Dhaka, 9 Oct 07:30	Logged live by staff; true match
Narayanganj decoy	Found boy, 13, red shirt, Narayanganj	Rank 2–3; rejected in finale version
মীম (Mim)	Missing girl, 11, Tongi, Gazipur; parent report + public-post duplicate	Shows duplicate banner
সাব্বির (Sabbir)	Missing boy, 15, Chattogram; no found record	Shows open case and empty candidates
~195 others	Seeded synthetic set	Fills map

9.3 Actions and expected results

#	Action	Expected on screen
1	Phone: SCREEN-001 → Start	Intake chat in Bangla

#	Action	Expected on screen
2	Record the Rafi script (38 s)	Transcript appears
3	Send; answer district "মিরপুর, ঢাকা"	Review cards filled
4	Tick check; submit	Ref M-...; status "খোলা"
5	Tap GD draft and poster	Bangla GD text; poster preview
6	Laptop: staff logs Kamalapur boy	Ref F-...; "মিল খোঁজা হয়েছে"
7	Officer: queue	Rafi at top, score shown
8	Open; side-by-side; confirm (2 steps)	"পর্যালোচকের সিদ্ধান্ত... রেফারেল নথিভুক্ত (সিমুলেশন)"
9	Close case	"কেস বন্ধ হয়েছে"
10	Map	Dhaka missing reports: Closed +1, Matched −1; found referral-recorded unchanged; confirmed pairs unchanged
Finale extras	Reject Narayanganj decoy with reason; show Mim duplicate banner; show audit trail; show metrics slide	As in D3

9.4 Backup paths

Failure	Backup
Speech provider slow/ down	Type the scripted Rafi text; or `DEMO_MODE=1` scripted note → cached transcript (badge "ক্যাশ")
LLM down	Fallback provider; then cached extraction; then manual form
Venue internet down	Restart the whole labelled scenario on the local stack, with all roles sharing its DB. Never mix a new cloud report with local officer state. If local fails, play labelled backup recording (TASK-058).
Laptop failure	Spare laptop with same local setup
Map tiles fail	Bundled district outlines + table; no bulk offline caching of public OSM tiles

9.5 Versions

Version	Length	Content
Round 2 video	1:45	Steps 1–10, sped-up waits, English captions over Bangla UI
Finale live	7–8 min of a 10 min slot	Problem (1 min) → live steps 1–10 (4 min) → extras (1.5 min) → metrics, safeguards, scale (1.5 min)

Spoken scripts and timings are in D7 §3.

10 Hackathon checklist

Area	Item	Done
Product	Essential journey on prod (TC-001 – TC-012)	☐
Product	Should items only if Must is green	☐
UI	Bangla default, English toggle, synthetic/cached/draft labels	☐
Backend	State trigger, audit append-only, spend cap	☐
Security	`npm audit` clean; no secrets in repo; rate limits; E-004 for hidden records	☐
Accessibility	TC-033 green, light and dark	☐
Testing	Unit + E2E green; failure-state tests run	☐
Evaluation	Report with 5 metrics + failures, dated 14 Oct	☐
Deployment	Prod smoke test; `r2-freeze` tag	☐
Presentation	5–8 slides incl. metrics slide and [MOCK] items marked	☐
Submission	GitHub repo link	☐
Submission	5–8 slide PowerPoint deck	☐
Submission	1–2 min demo video	☐
Submission	Drive folder with TXT/DOC of links + deck + video	☐
Submission	Reply to selection email with folder link before 14 Oct; exact cutoff pending organiser confirmation	☐
Finale	Local stack, 5 rehearsals, backup recording, spare laptop	☐

11 Recommendation

Topic	Advice
Build first	Critical path in §5: case service + state guard, extraction, intake screens, found log, matcher, match review, close, map. Text path before voice.
Cut if short (in order)	1 FR-036 disaster screen → 2 FR-010 13219 summary → 3 FR-024 import → 4 FR-029 edit after submit → 5 live voice (keep cached scripted note, label it) → 6 poster PNG (keep poster text). Never cut review-before-submit, officer confirm, referral, audit or redact identifying payloads.
Impact (judges)	Show one child's case reaching a known status, and the closure count on the map; tie to "2,038 still untraced" (pitch deck).
Feasibility	Show measured metrics with failures, the USD spend so far, and the offline demo.

Topic	Advice
Scalability	Show 64-district map, 30,000-cases/year design (NFR-015), provider adapter, and that CID integration is a data agreement, not new infrastructure.
Do not attempt	Face matching, alert sending, scraping social media, real child data, real officer verification, a native app.

Assumptions (D6)

ID	Assumption
A-040	Actual availability and completed work must be inventoried on 11 Oct. A fresh 11–14 Oct start at 10 h/person/day gives at most 120 team-hours; this is an assumption, not verified capacity.

Open questions (D6)

ID	Question	Default
OQ-024	Do classes, exams or labs reduce team hours before 14 Oct?	UNKNOWN; inventory actual availability and finished work at the 11 Oct review

Risks (D6)

ID	Risk	L	I	Mitigation	Owner
RISK-017	V2 Round 2 estimates total 184 h and may exceed actual remaining capacity	M	H	Daily stand-up; inventory completed work and actual owner capacity, sequence dependencies, reassign and cut optional scope on 11 Oct	Sadman

IDs introduced

- TASK-001 – TASK-060 (one line each in §5)
- A-040 Team capacity
- OQ-024 Team availability
- RISK-017 Capacity overrun

IDs referenced

- FR-001 – FR-037, STORY-001 – STORY-023, BR-001 – BR-022, BR-025, NFR-001 – NFR-017 (D1)
- TC-001 – TC-037 (D1, written here)
- GOAL-001 – GOAL-009 (00)
- SCREEN-001 – SCREEN-016 (D1)
- API-001 – API-020, COMP-004 – COMP-021, E-001 – E-024, DECISION-017, A-013, A-030, A-039 (D2)
- COMP-022 – COMP-050 (D4)
- OQ-004, OQ-019, OQ-021 – OQ-023, RISK-015 (00, D1, D2)
- DEP-004 (D1)

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D7 / MEETING / V2.0

Closing Pack

Priorities, unresolved decisions and demonstration narrative · Authors: Sadman Shaharier and Obidit Islam

↗ Contents

Field	Value
Version	2.0
Date	11 Oct 2026 (Asia/Dhaka, UTC+6)
Owner	Sadman Shaharier
Authors	Sadman Shaharier; Obidit Islam
Status	V2 team handover; implementation status UNVERIFIED
Inputs	V2 registry and D1–D6; team handover, verification evidence and meeting decisions.

V2 release contract

Authors: Sadman Shaharier and Obidit Islam. **Version:** 2.0. **Issued:** 11 October 2026, Asia/Dhaka (UTC+6).

This is the revised team specification, not evidence that the application has been built. Actual repository progress, account configuration, interviews, evaluations, and spend remain **UNVERIFIED** until evidence is attached. V1 defaults are superseded by the V2 rules below; existing requirement IDs are retained.

- This release is a **synthetic-data prototype**. Officer verification and referral delivery are simulated and labelled. The schema rejects records marked non-synthetic; the application must also prevent real reports from entering this release.
- A candidate score is similarity, not identity probability. A reviewer decision does not establish guardian authority or safe reunification. A referral row means **referral recorded**, not delivered or acknowledged.
- Draft payload stays on the device before report confirmation, but providers process audio/text beforehand with a notice. Only expiry/counter/hash metadata and cost reservations are stored server-side before submission.
- Possible duplicates remain visible. Shared phone alone never establishes one child. No physical child-return action is performed by Fera.
- Counts distinguish missing-report rows, found-report rows, and confirmed pairs. They are not a census of unique children or proof of rescue.
- Sources, diagrams, tables, code blocks, and acceptance criteria are part of the handover. The companion code must match the embedded code exactly. Team handover records change closure, evidence gaps, and release gates.

Changes to earlier documents

V2 replaces unsupported metrics and receipt/return claims, identifies pending meeting decisions, revises the compressed demo narrative and aligns release evidence with the V2 contracts.

1 Top 10 implementation priorities

#	Priority	TASK	FR	Why first
1	Schema, case service, state guard, audit	TASK-002, TASK-015	FR-016, FR-021	Every safeguarding rule depends on them
2	Extraction adapter with evidence check and fallback	**TASK-013**	FR-004, FR-026	Core AI; critical path
3	Text intake + follow-up questions + review-before-submit	TASK-020 – TASK-022	FR-001, FR-003, FR-005 – FR-007	Guaranteed demo path; BR-004
4	Synthetic set with ground truth, frozen before matcher code	TASK-010, TASK-019	**FR-032**	Honest metrics need it first
5	Matcher: filters, cosine, bonuses, template reasons	TASK-016, TASK-017	FR-012, FR-013	The "match surfaced" moment
6	Found-child log + review queue + side-by-side confirm/reject	TASK-026 – TASK-028	FR-011, FR-014 – FR-017	Human-confirms moment; BR-001
7	Close with referral + district map	TASK-029, TASK-030	FR-018, FR-019	Closure visible to judges
8	Voice intake + demo-safe cache	TASK-014, TASK-021, TASK-033	FR-002, FR-027	Hero moment, offline-safe
9	Evaluation harness, WER, timing study	TASK-039 – TASK-041, TASK-043	**FR-028**	Deck promised all 5 metrics
10	Prod deploy, video, submission	TASK-042, TASK-045, TASK-046	—	Nothing counts unless submitted by the official 14 Oct cutoff, exact time pending

2 Top 10 risks

Separate execution risks from pilot safety gates. Synthetic-demo risk scores do not waive real-case safeguards. Score = likelihood (L 1, M 2, H 3) × impact (L 1, M 2, H 3, Severe 4). Ties broken by nearness of the deadline it threatens.

Rank	RISK	Risk	L × I	Score	First action	Owner
1	RISK-017	Round 2 work exceeds capacity	M × H	6	21:00 stand-up; cut order on 11 Oct	Sadman
2	RISK-001	Core journey unfinished by freeze	M × H	6	Text path first; voice after	Sadman
3	RISK-003	AI invents or misreads fields	M × H	6	Evidence check; review screen	Obidit
4	RISK-005	Venue internet fails	M × H	6	One backend per run; restart complete local scenario; labelled backup recording	Sadman
5	RISK-004	AI outage or spend over USD 30	M × H	6	Fallback provider; spend cap 25	Obidit
6	RISK-012	Overclaiming metrics, interviews or law	M × H	6	Metrics slide with failures; "0 interviews" stated	Sadman
7	RISK-002	Bangla speech recognition too poor	H × M	6	Typed path; editable transcript; report WER	Obidit
8	RISK-011	"Why not just MUN Alert?"	H × M	6	One slide: Potential future coordination support; no integration or alert authority	Sadman
9	RISK-010	Judges favour environment-themed work	M × H	6	Ask mentors (OQ-003); disaster-mode slide	Sadman
10	RISK-009	False match sends a child into danger	L × Severe	4	No physical return; external identity/guardian procedure and acknowledged partner hand-off required before a pilot	Sadman

3 Demonstration script

All cases are synthetic. Officer verification is mocked; referral is an internal simulated record. No integration, custody transfer, rescue, or legal compliance is demonstrated. Display actual measured results or “not measured”; never present a target as a result.

3.1 Round 2 video: proposed 1:45

Time	Screen/action	Spoken line
0:00–0:12	Title, sourced January–July figures	“The September report recorded 15,717 missing-child reports and 2,038 remaining untraced for January–July 2026. Fera explores better report coordination.”
0:12–0:20	Synthetic and processing notice	“This is a synthetic prototype. A parent can speak or type in Bangla.”
0:20–0:38	Compressed voice capture/ transcript; time-compressed label	“The transcript is editable. External AI processing is explained before intake.”
0:38–0:52	Required field question/manual completion/review/confirm	“The parent reviews the fields before a report enters the register.”
0:52–1:02	GD draft; separate poster contact preview	“Fera drafts text. It does not file a GD or issue alerts.”
1:02–1:12	Staff own found report	“Staff records an invented found-child description.”
1:12–1:28	Qualified candidates, evidence and checklist	“Similarity supplies candidates, not identity proof. A reviewer records the decision and a simulated referral.”
1:28–1:38	Coordination closure and separate counts	“The missing-report coordination status changes. Found reports and confirmed pairs are counted separately.”
1:38–1:45	Measured metrics, sample counts, failures	“These are held-out synthetic measurements, with failures and limitations.”

A 38-second recording is compressed in the 18-second segment; label the time compression and report measured live latency separately. If the required video length differs in the organiser email, adapt this script without hiding delays or cache use.

3.2 Finale: 8 minutes plus 2-minute Q&A buffer

Time	Speaker	Action and content
0:00–1:00	Sadman	Sourced scale, existing recorded outcomes; unmet coordination hypothesis, no national-registry absence claim
1:00–1:45	Sadman	Fera's proposed reviewed intake/coordination role; no MUN/CID integration or alert authority
1:45–3:15	Obidit	Processing notice, synthetic Bangla voice/text, editable transcript, source evidence, unknown/manual completion, reviewed submission
3:15–3:45	Obidit	Fixed GD draft, explicit contact choice and poster sharing limits
3:45–5:00	Masnun	Staff entry, officer evidence, no-match/uncertainty cue, rejection reason and simulated confirmation/referral record
5:00–5:45	Masnun	Coordination closure, distinct report/pair counts, visible duplicate cues and safe audit

Time	Speaker	Action and content
5:45–6:45	Sadman	Actual held-out dataset/model/denominators, baselines, failures, live latency and measured spend
6:45–7:30	Sadman	Synthetic-only restriction; no face/photo; real-case verification/legal/provider/partner/ correction gates
7:30–8:00	Sadman	64-district data model and next evidence needed; no unsupported scalability or rescue promise
8:00–10:00	All	MUN overlap, AI contribution, false candidates, data availability, offline scope, correction and operating capacity

Failure line: “We are restarting the labelled synthetic scenario on our local stack.” All roles switch to the same local DB; never continue a cloud report on a separate local officer DB. If local fails, show the labelled backup recording. Cache and mock badges remain visible. Cost lines use measured spend at recording time, not the USD 30 target.

4 Clarifications still needed

OQ	Question	Who can answer	By when
OQ-001	Voice and poster Must for Round 2?	Team (Sadman)	11 Oct contract gate
OQ-002	separate missing/found state machine confirmed?	Team (Sadman)	11 Oct contract gate
OQ-003	Environment theme / disaster mode weight	Mentors	Session 2, 12–Before recording or official submission
OQ-004	Keys held; default provider; prices	Obidit	11 Oct evidence gate
OQ-005	Finale topology confirmed	Team; organisers (venue Wi-Fi)	Before recording or official submission
OQ-006	Build status and interviews held	Team	11 Oct evidence gate
OQ-007	Demo login and roles	Team	11 Oct evidence gate
OQ-010	Correct CID unit name for alerts	Public CID briefing identifies Missing Children Cell; partner workflow still pending	11 Oct
OQ-011	Personal Data Protection Act specifics	Legal adviser	Before any pilot (synthetic prototype makes no compliance claim)
OQ-012	Retention period for closed cases	Partner agency	Before any pilot
OQ-013	Provider training and retention terms	Obidit (provider docs)	11 Oct evidence gate

OQ	Question	Who can answer	By when
OQ-016	Official online GD field list	Mentors or a police contact	Before claiming template/agency acceptance; use draft/simulation labels now
OQ-017	District adjacency source	Masnun	11 Oct
OQ-018	Referral agency name to show	Child-protection NGO or mentors	Before claiming template/agency acceptance; use draft/simulation labels now
OQ-019	Ethics of testing with real parents	Mentors / faculty	12 Oct
OQ-020	Cross-device parent access	Team	After the finale
OQ-021	Managed Postgres host	Sadman	11 Oct evidence gate
OQ-022	Speech provider limits and price	Obidit	11 Oct evidence gate
OQ-023	Hosting limits for local embeddings	Masnun (spike)	11 Oct evidence gate
OQ-024	Team availability before freeze	Team	11 Oct evidence gate
OQ-025	Judging weights for impact, feasibility, scalability	Organisers	Before recording or official submission
OQ-026	Is synthetic data without real public-child news acceptable for Round 2 technical evaluation?	Organisers / mentors	Before recording or official submission

Resolved by default and needing only team sign-off: OQ-008, OQ-009, OQ-014, OQ-015.

IDs introduced

- OQ-025 Judging weights
- OQ-026 Acceptability of synthetic data for Round 2

IDs referenced

- TASK-002, TASK-010, TASK-013 – TASK-017, TASK-019 – TASK-022, TASK-026 – TASK-030, TASK-033, TASK-039 – TASK-043, TASK-045, TASK-046 (D6)
- FR-001 – FR-007, FR-011 – FR-019, FR-021, FR-026 – FR-028, FR-032 (D1)
- RISK-001 – RISK-005, RISK-009 – RISK-012, RISK-017 (D1, D6)
- BR-001, BR-004 (D1)
- SCREEN-001 (D1)
- OQ-001 – OQ-024 (00, D1, D2, D6)

V2 meeting outputs

The next meeting closes: actual build/interviews/availability; state/referral/closure meanings; duplicate visibility; required fields/unknown/time rules; DTOs and idempotency; exact provider/model/account terms/prices; single-DB topology; manifest/metric freeze; official deliverables/cutoff and responsible submitter. Track unresolved partner/legal questions as pilot gates. Track actual institution/name spelling from registration; do not infer affiliation.

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